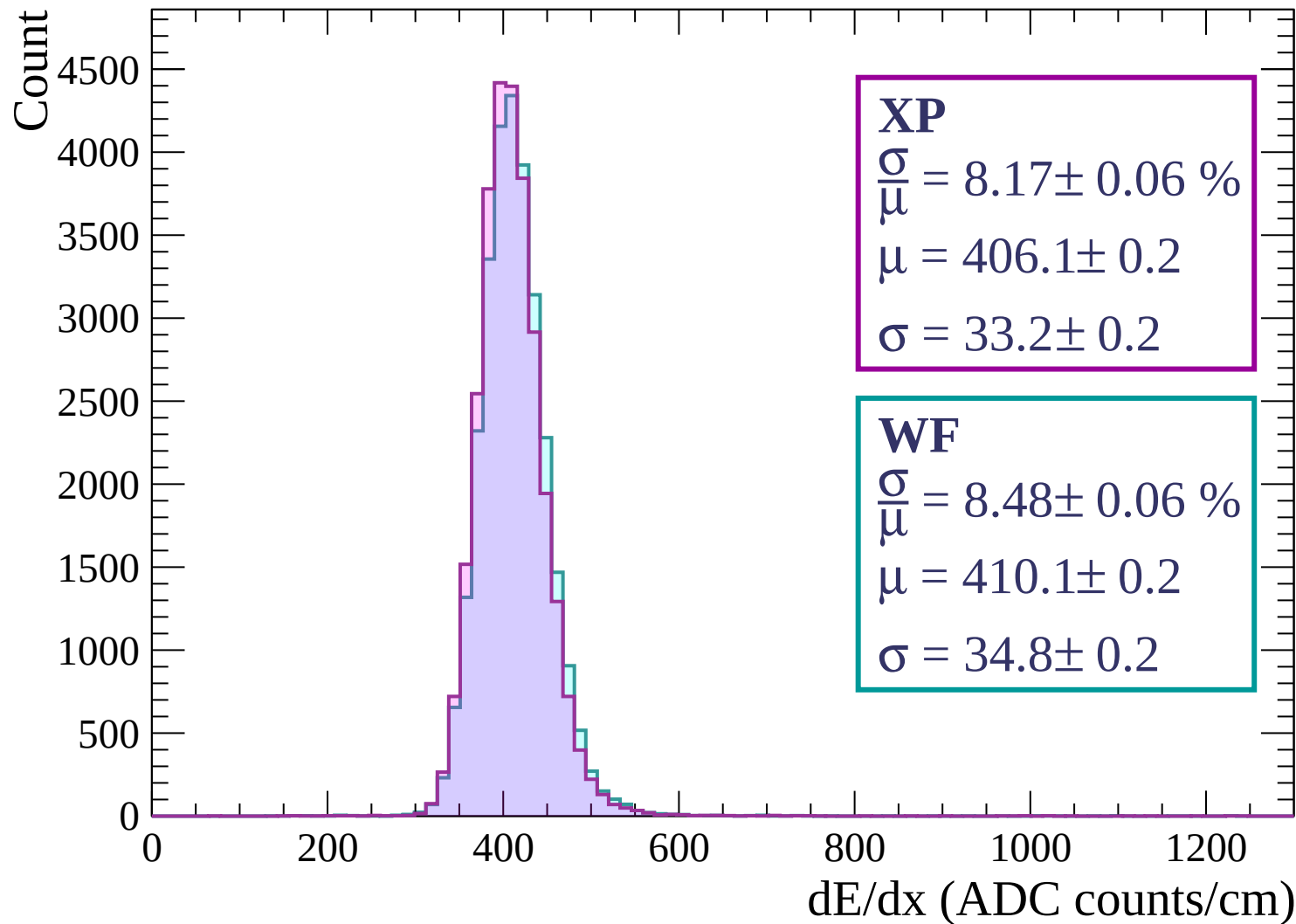
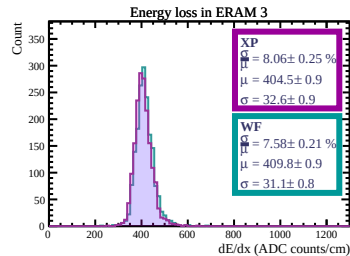
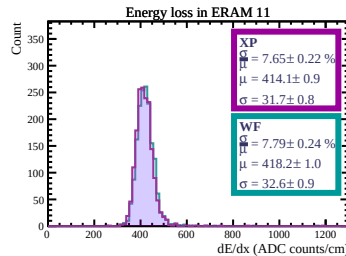
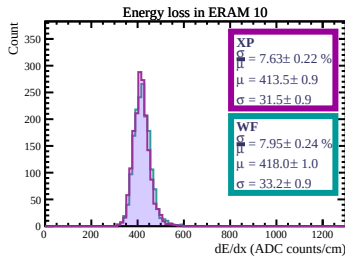
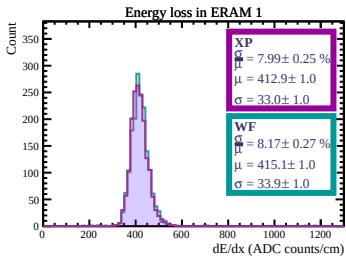
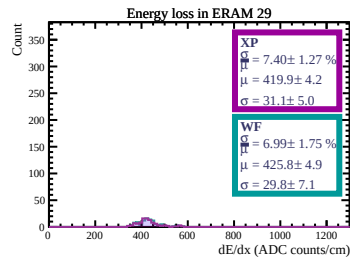
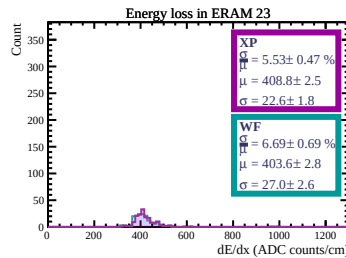
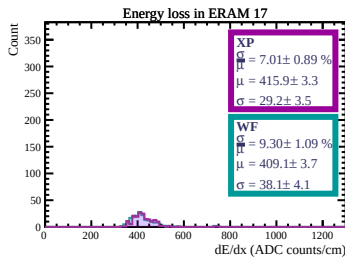
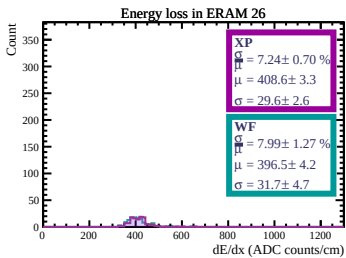
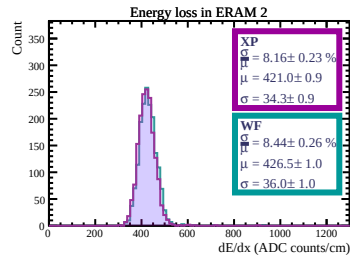
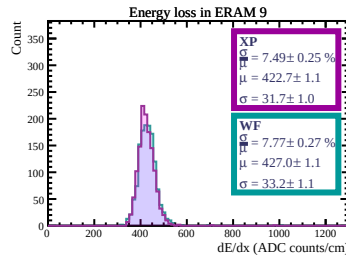
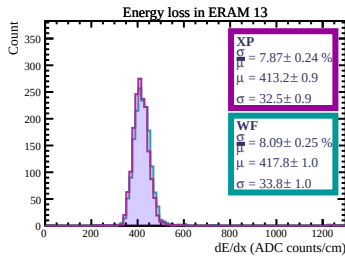
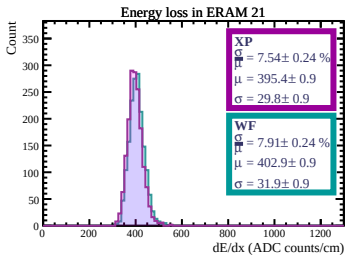
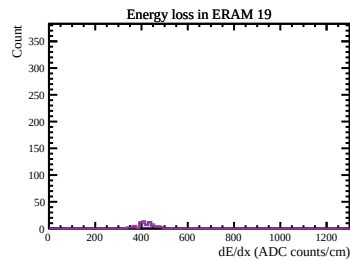
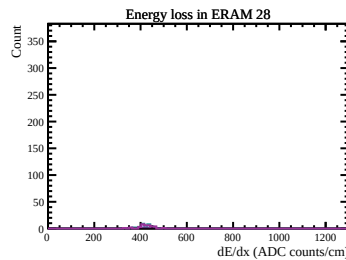
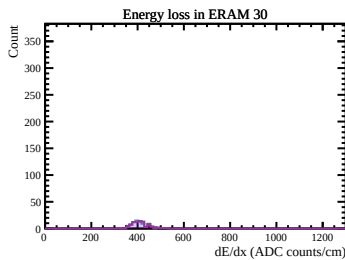
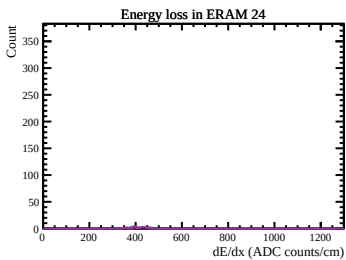
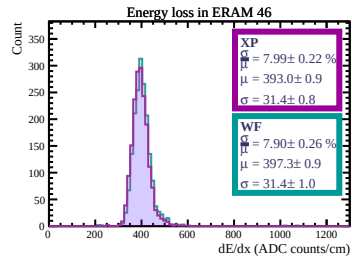
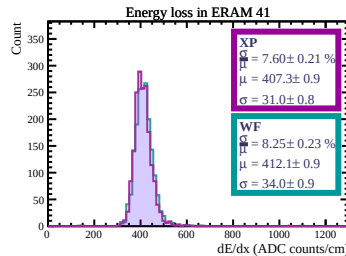
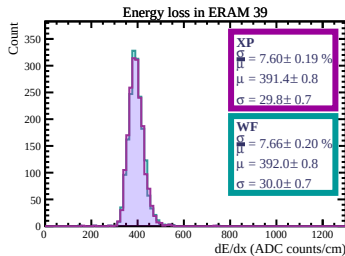
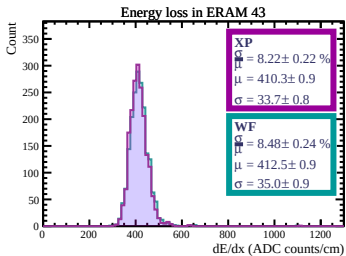
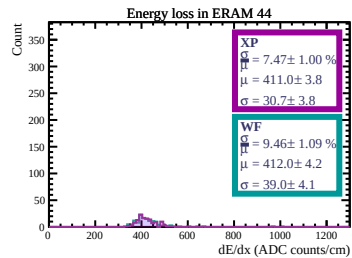
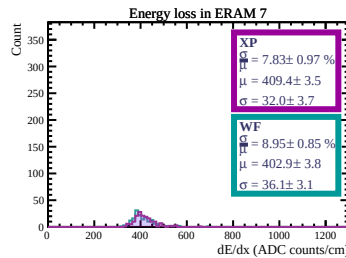
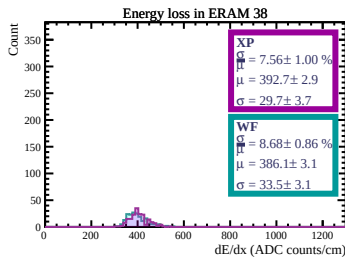
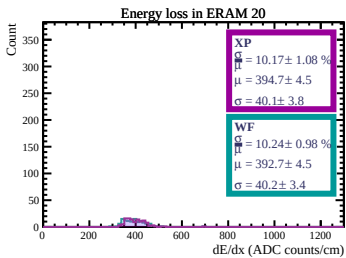
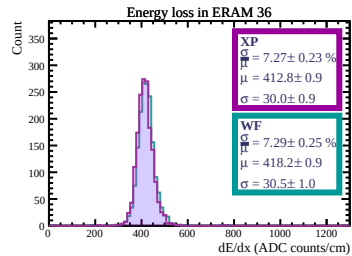
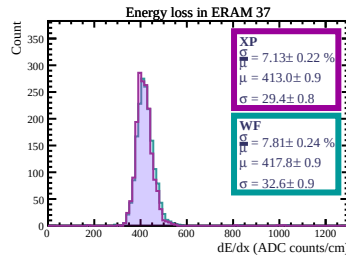
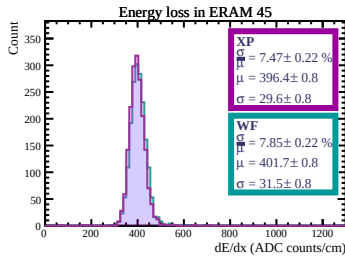
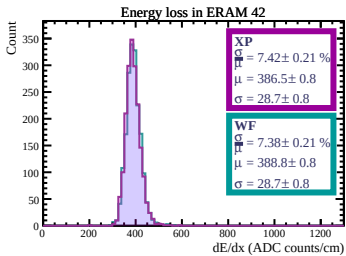
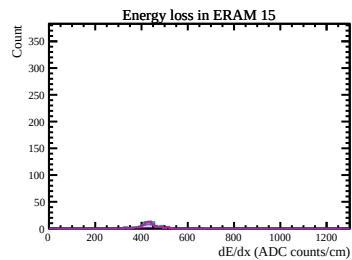
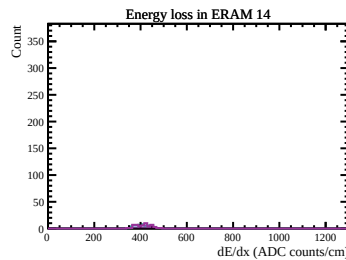
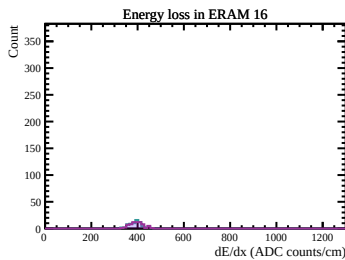
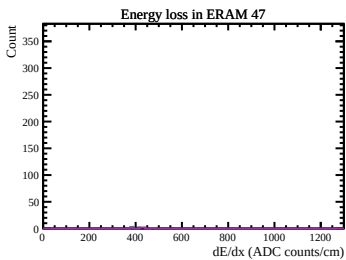
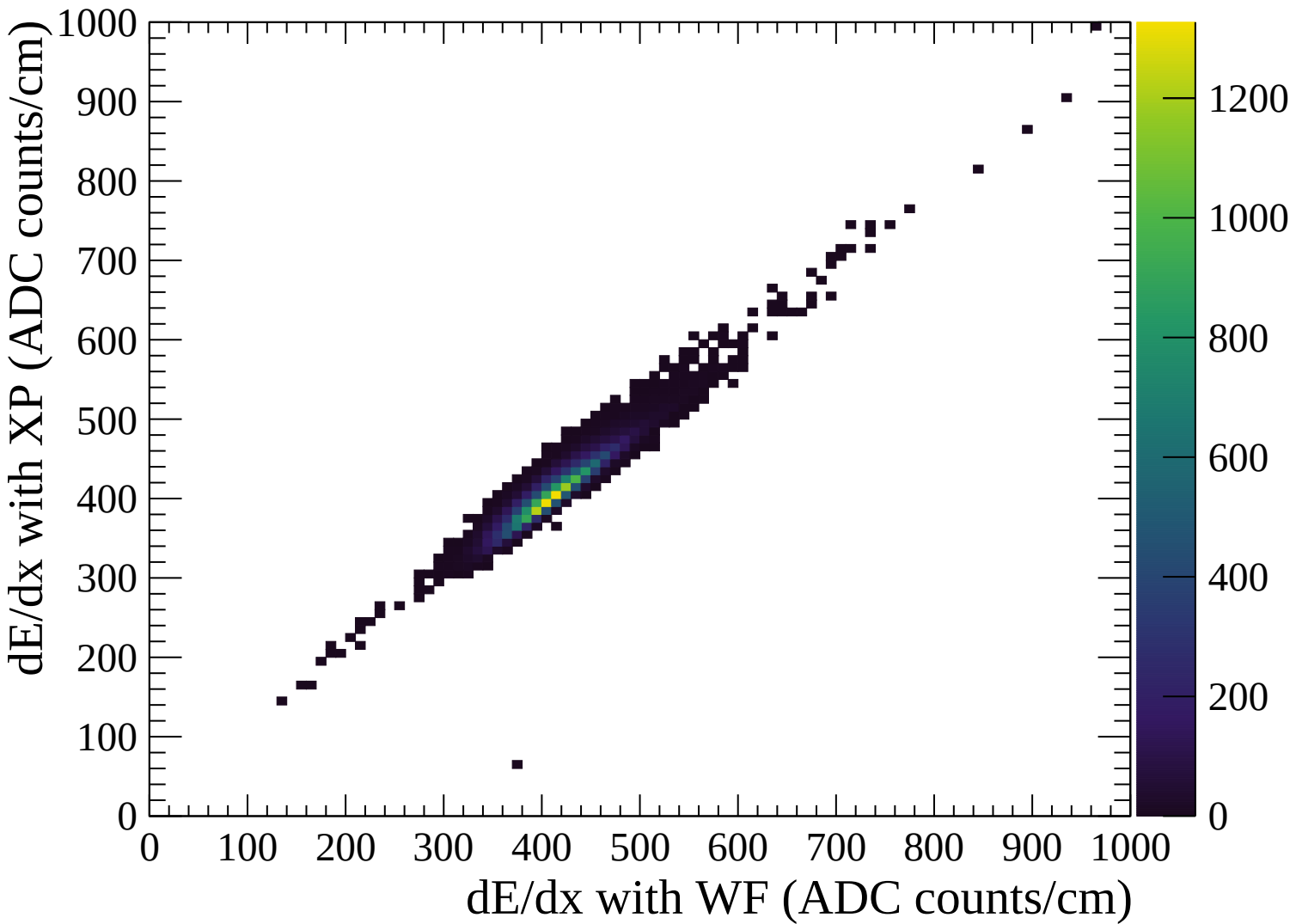
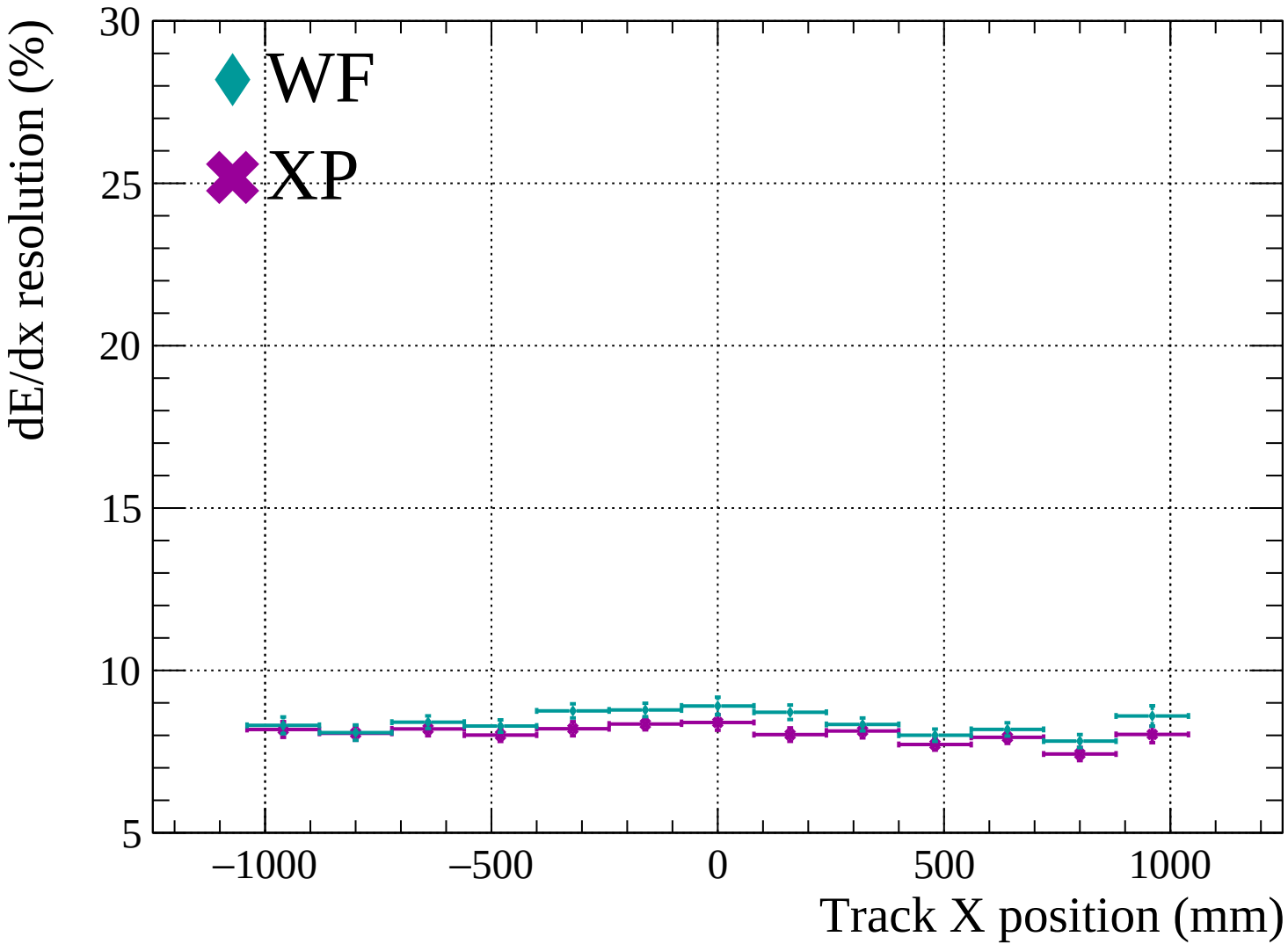


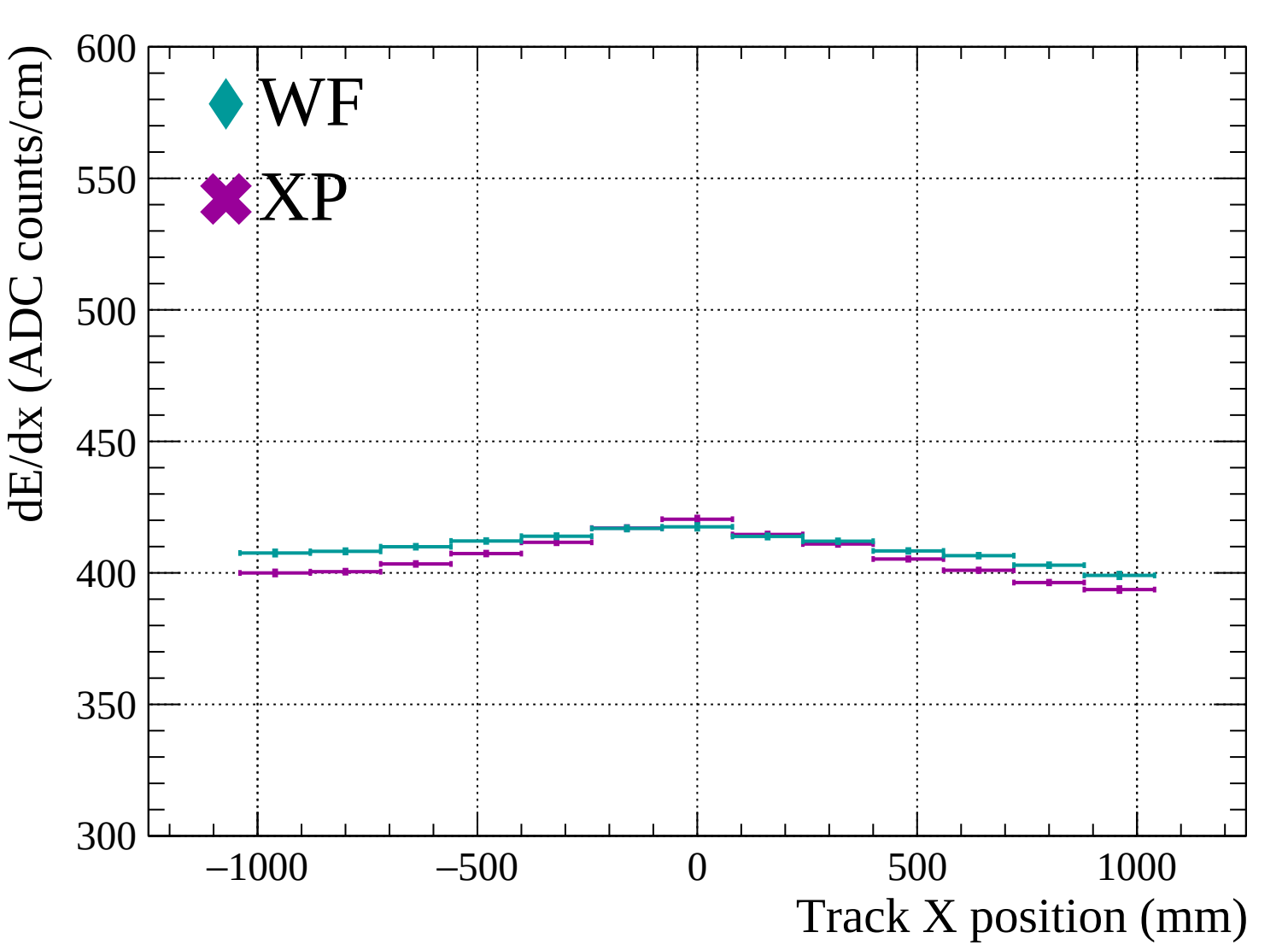


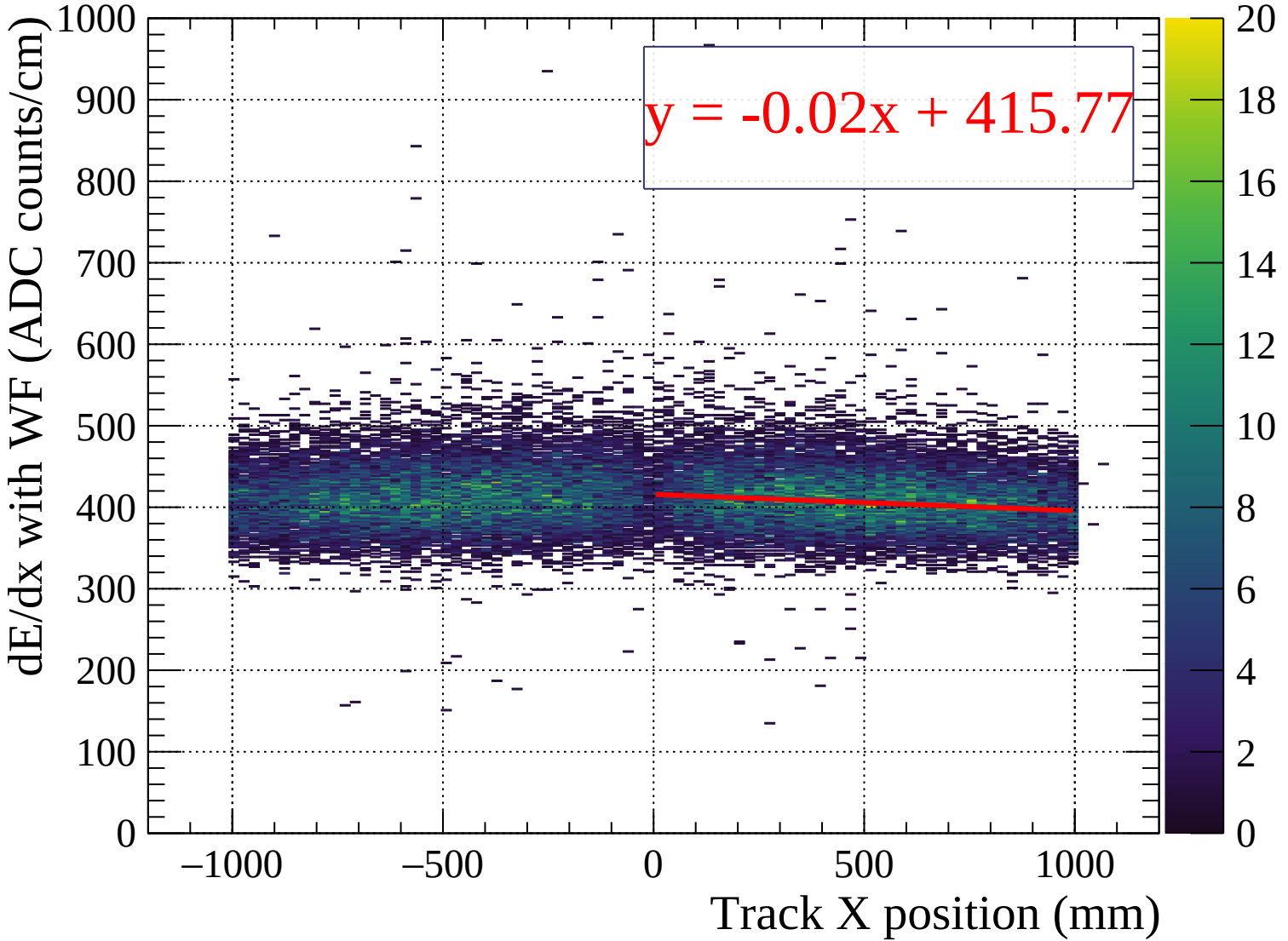


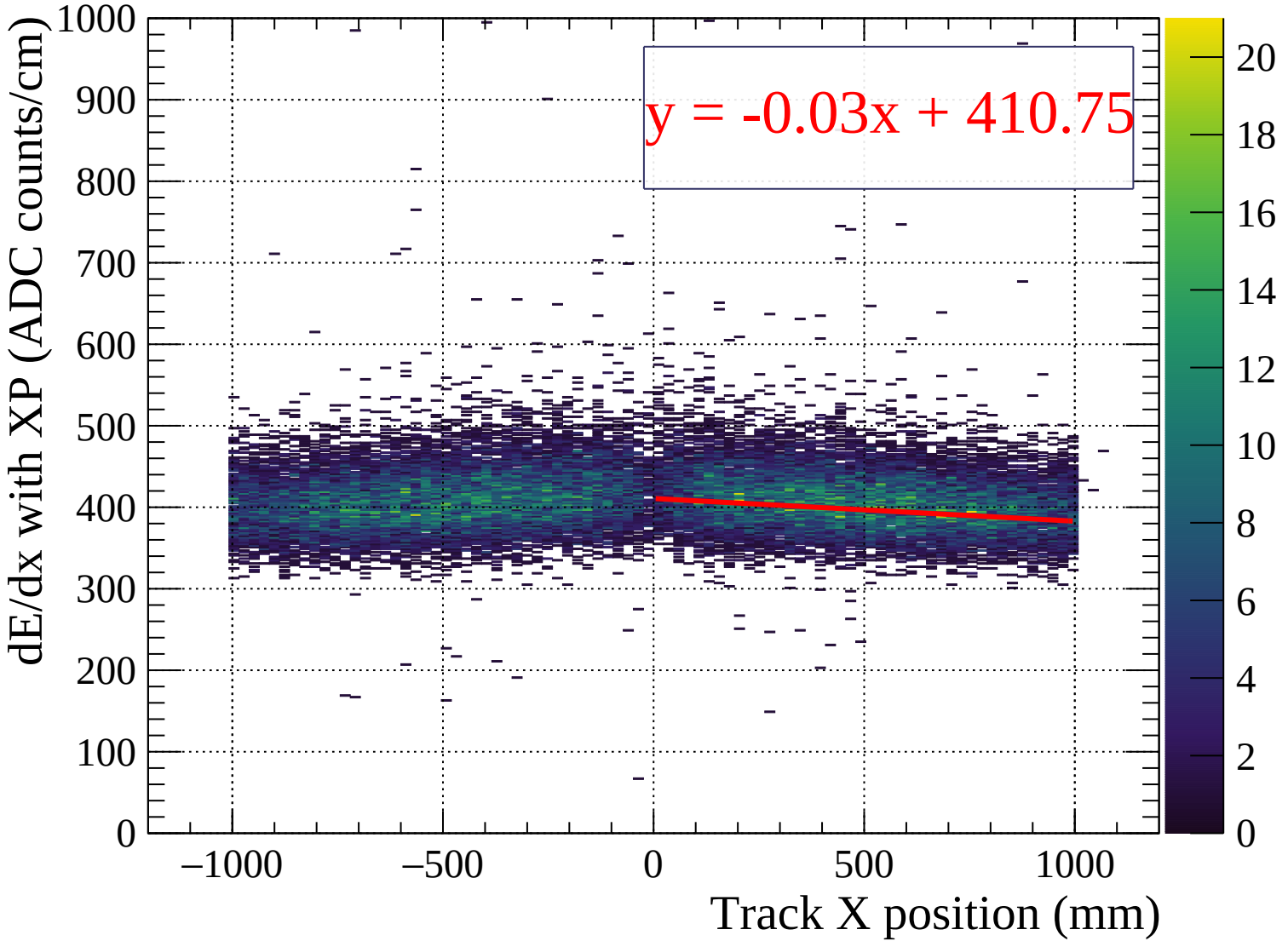


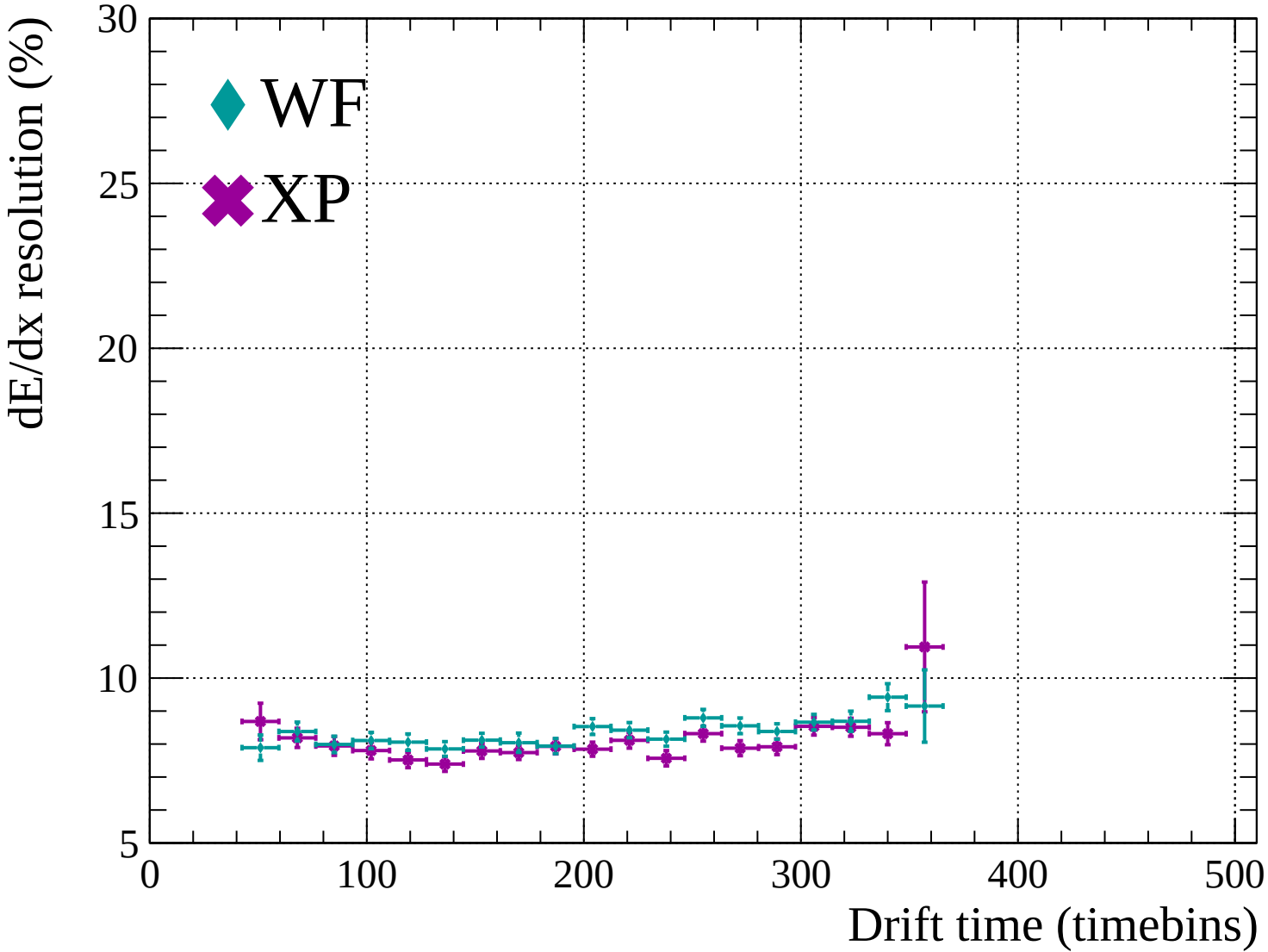


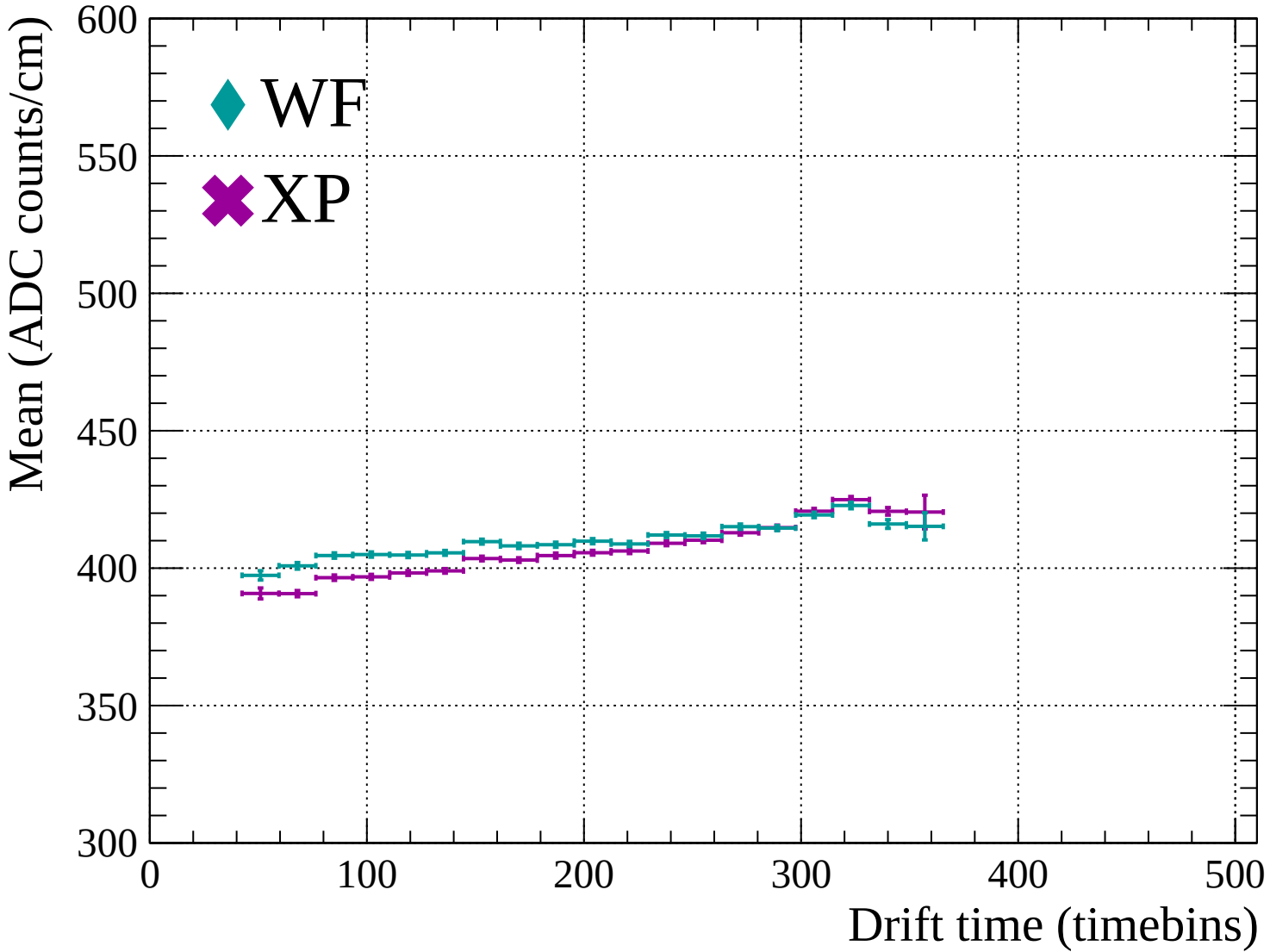


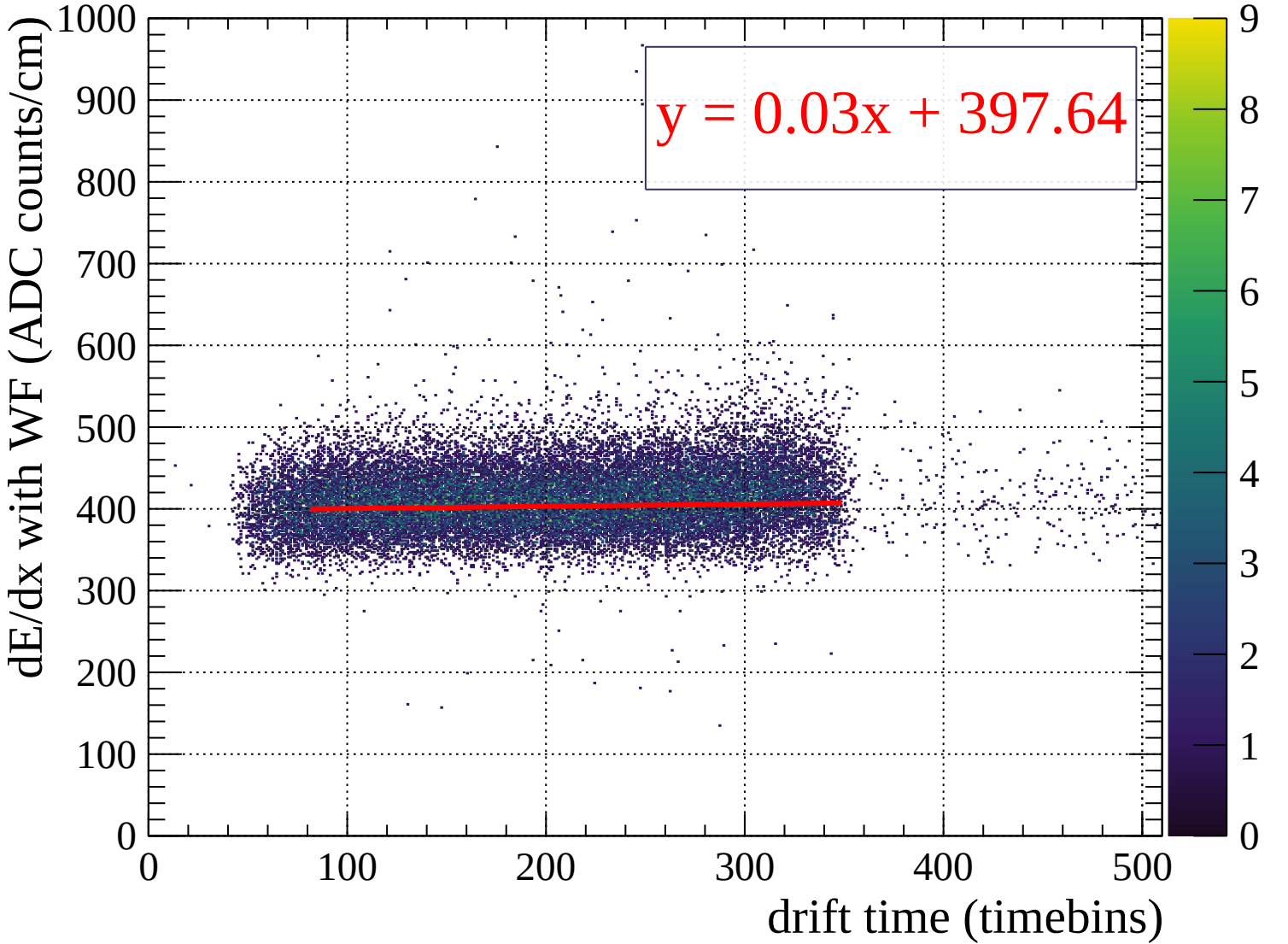


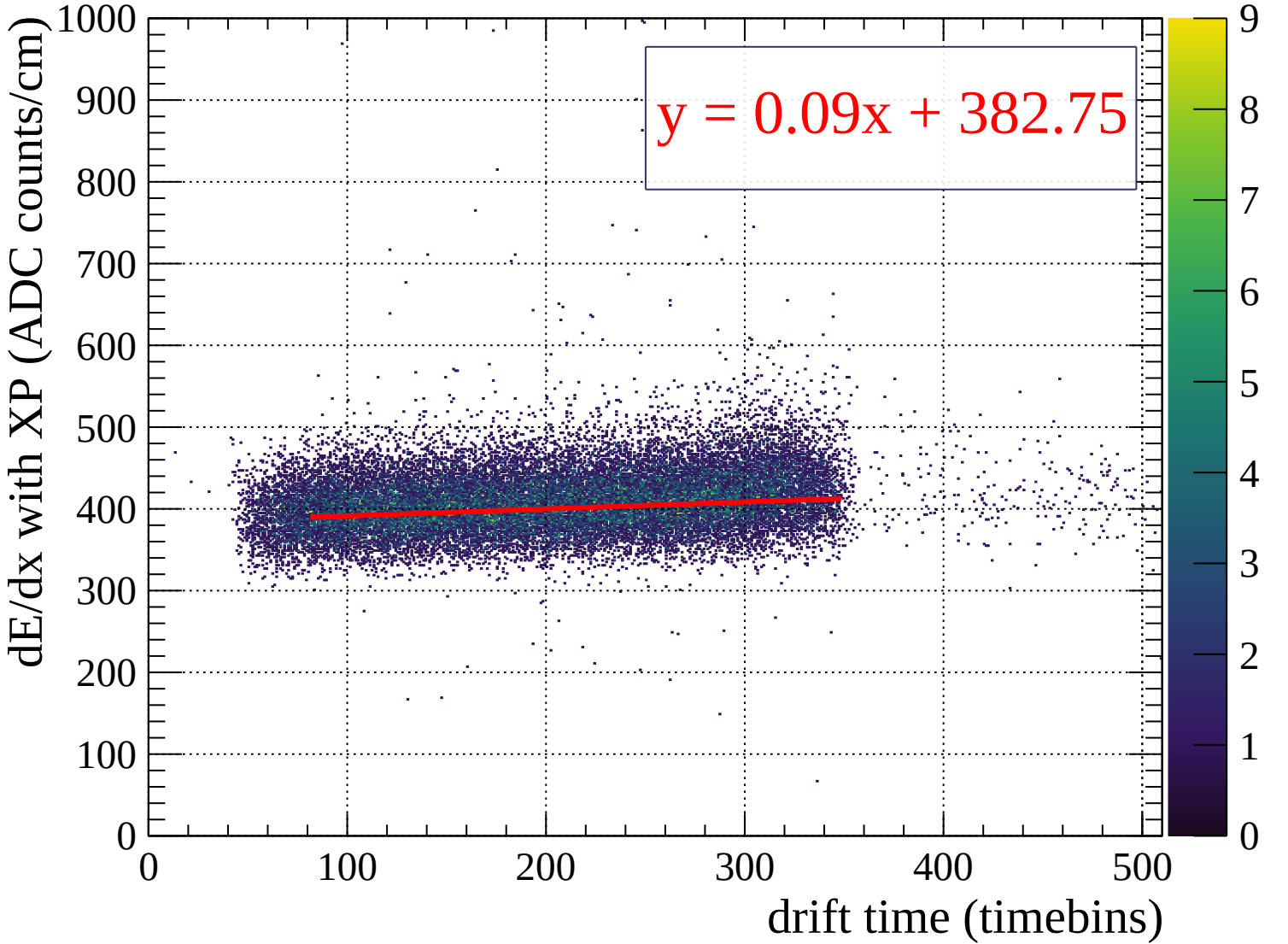


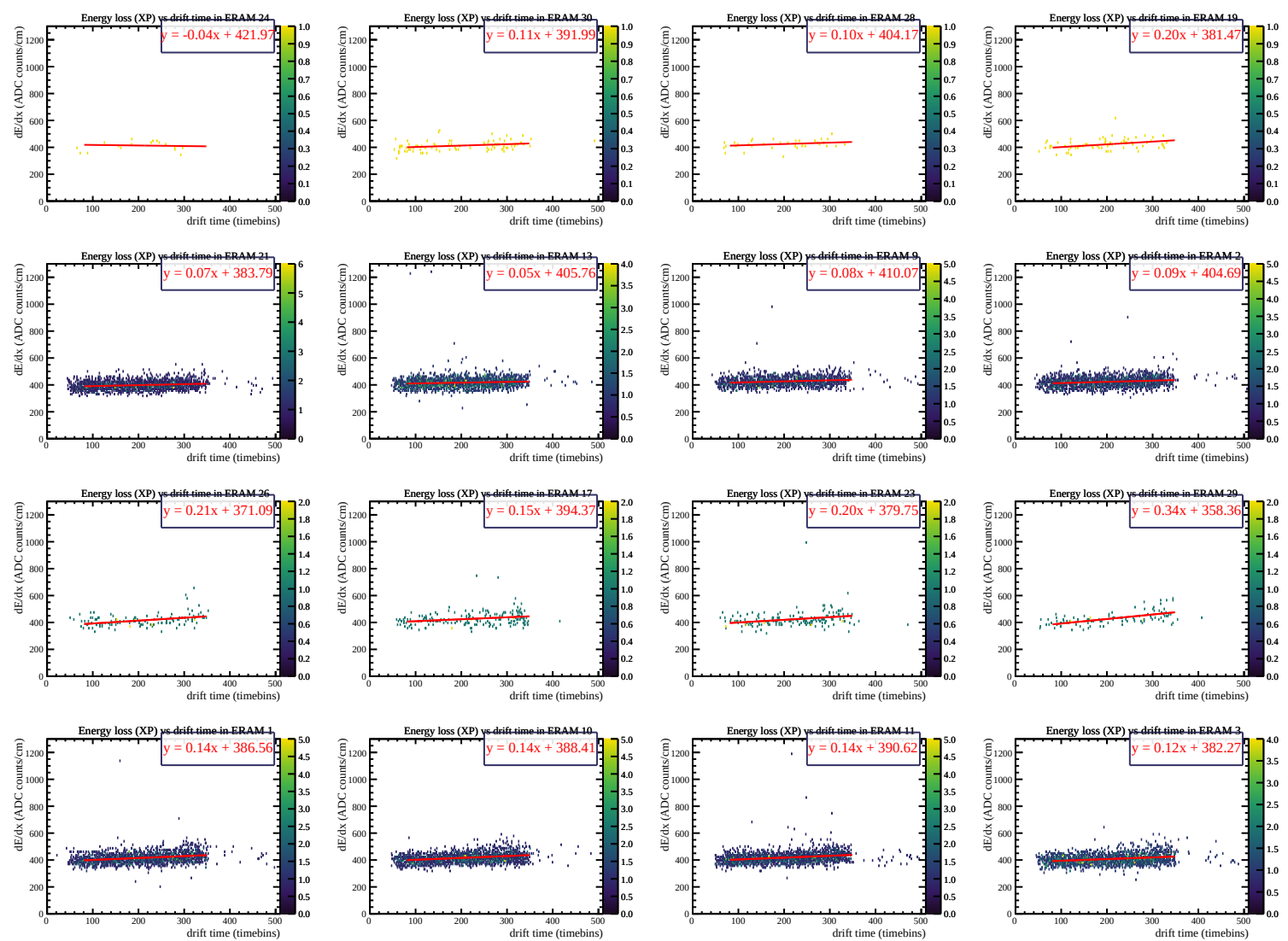


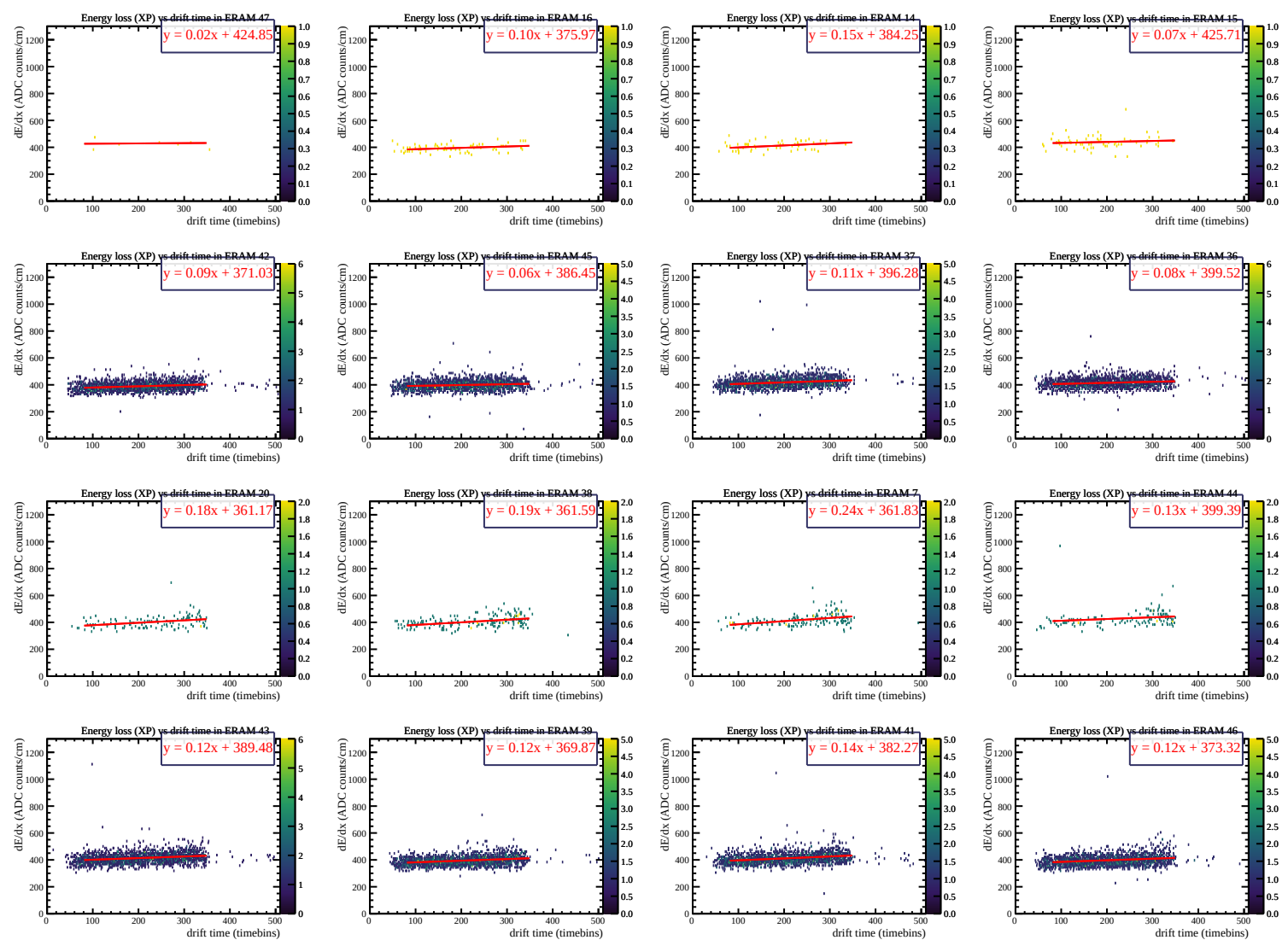


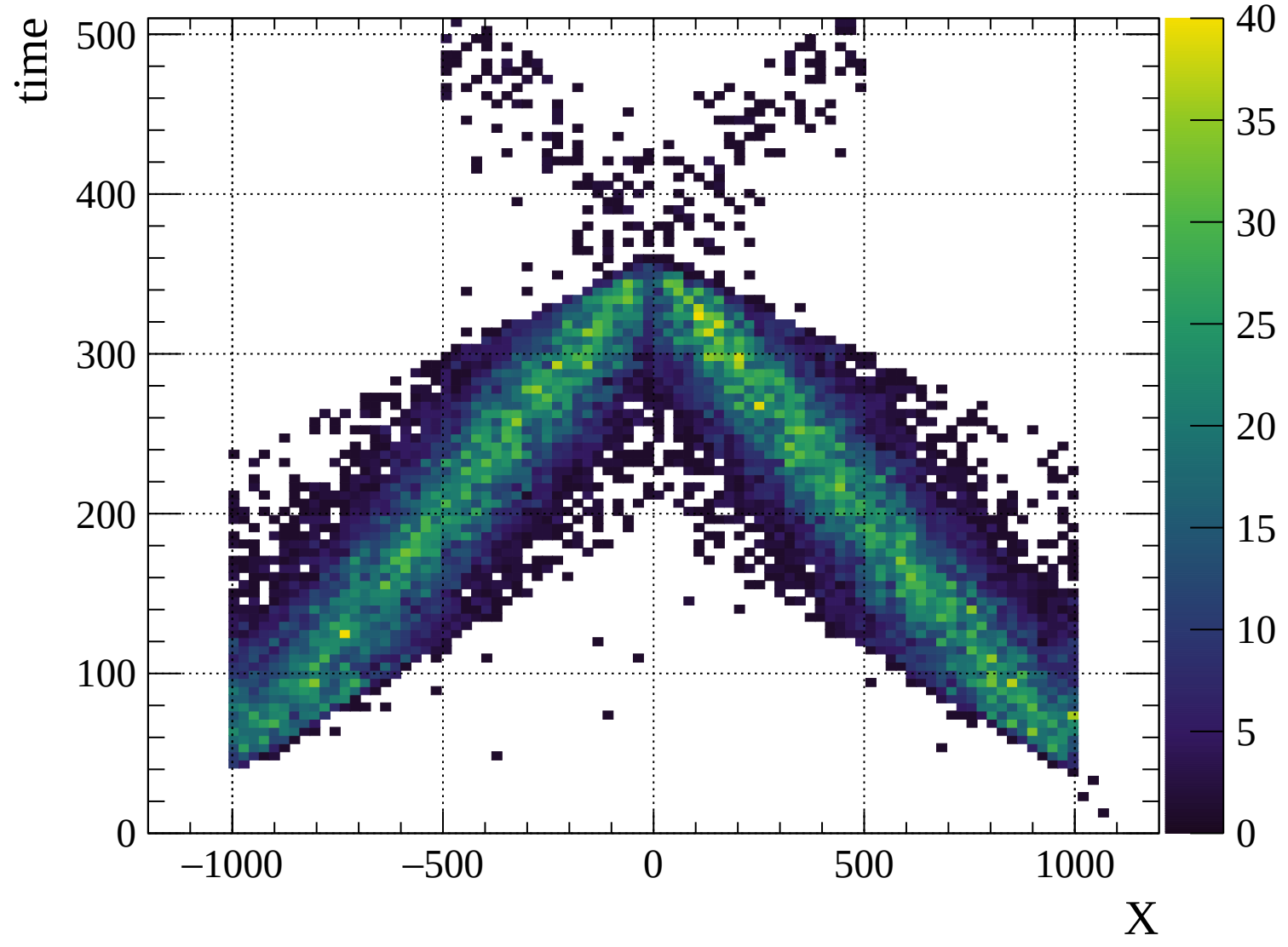


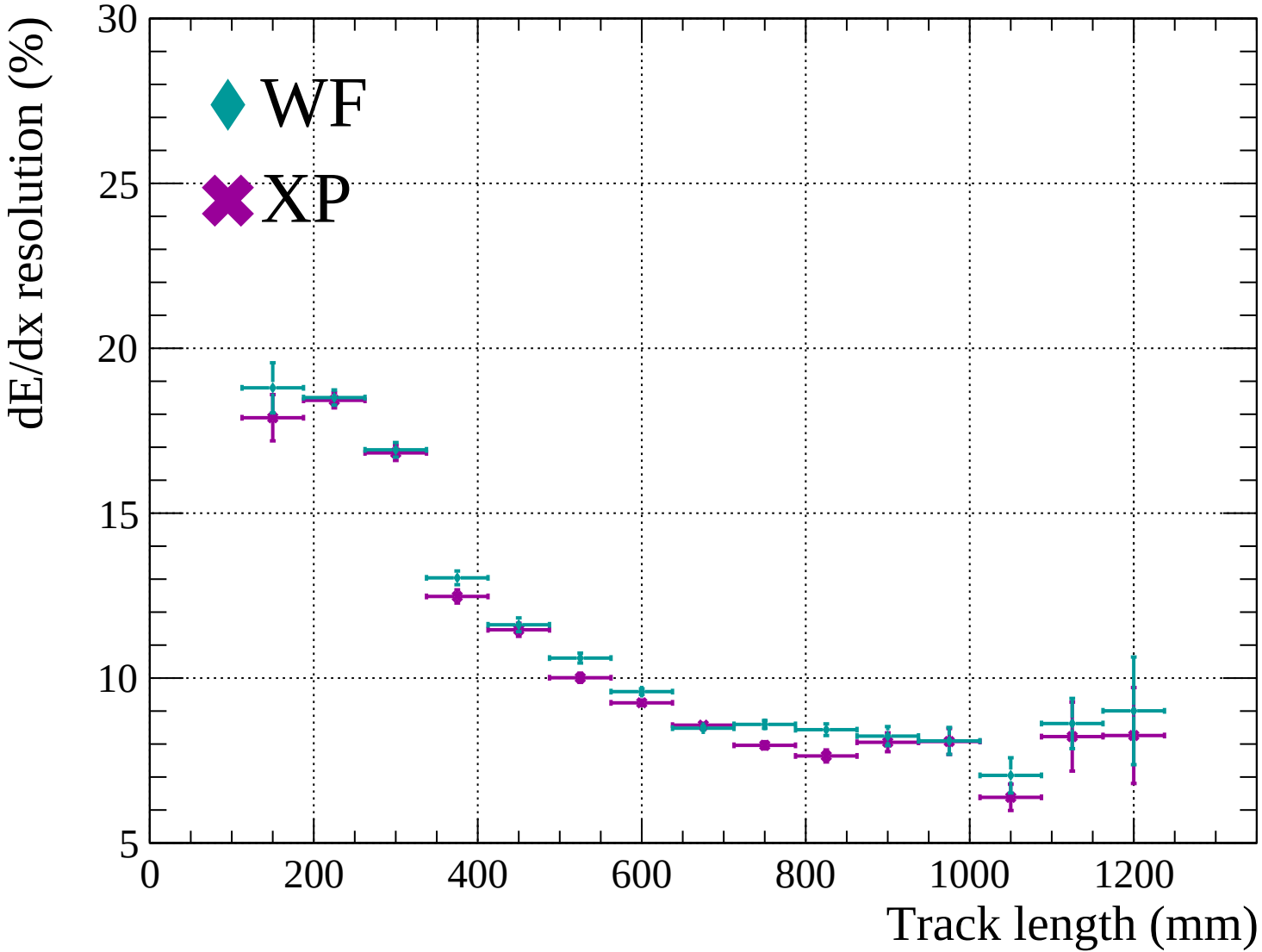


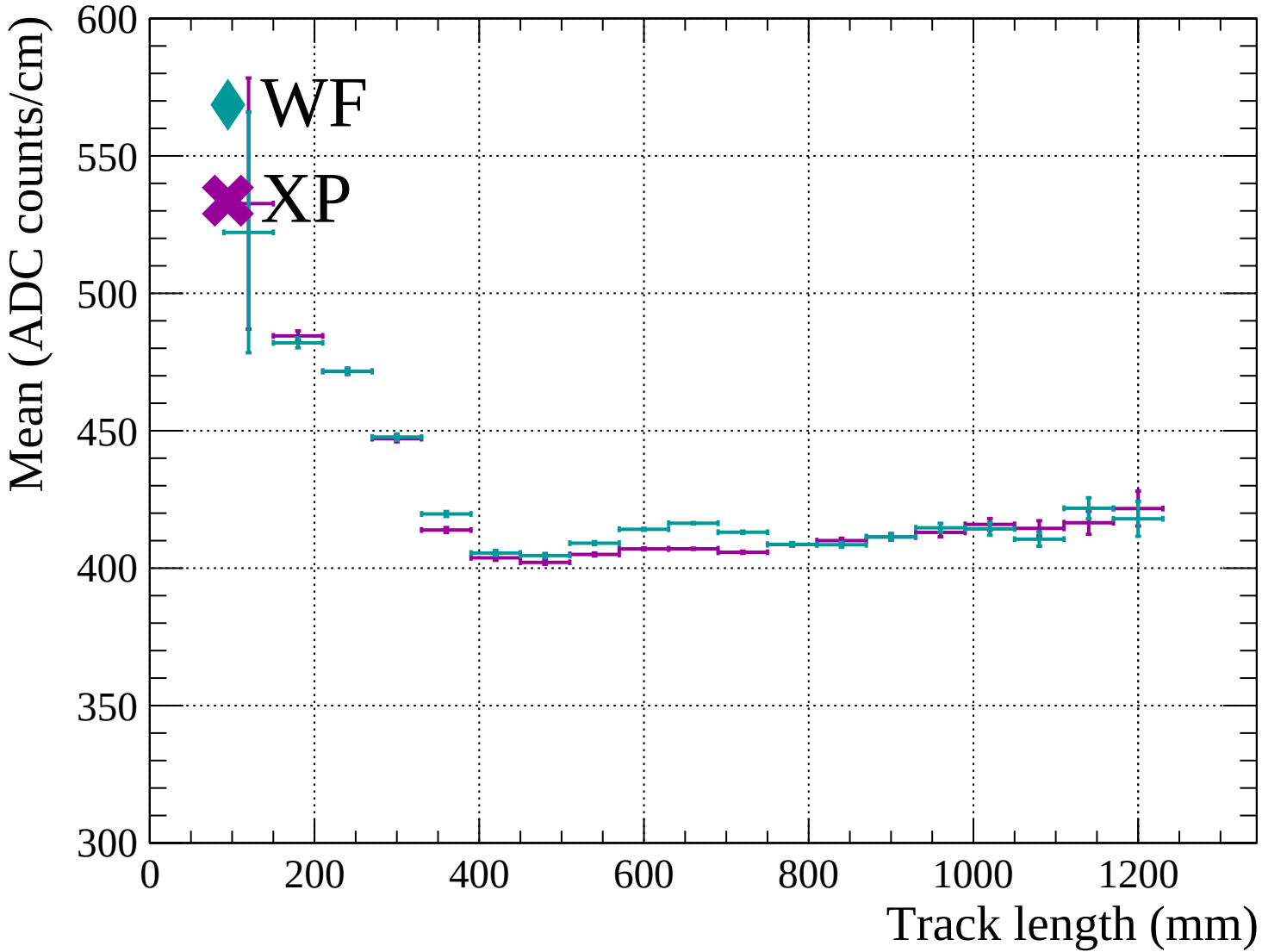


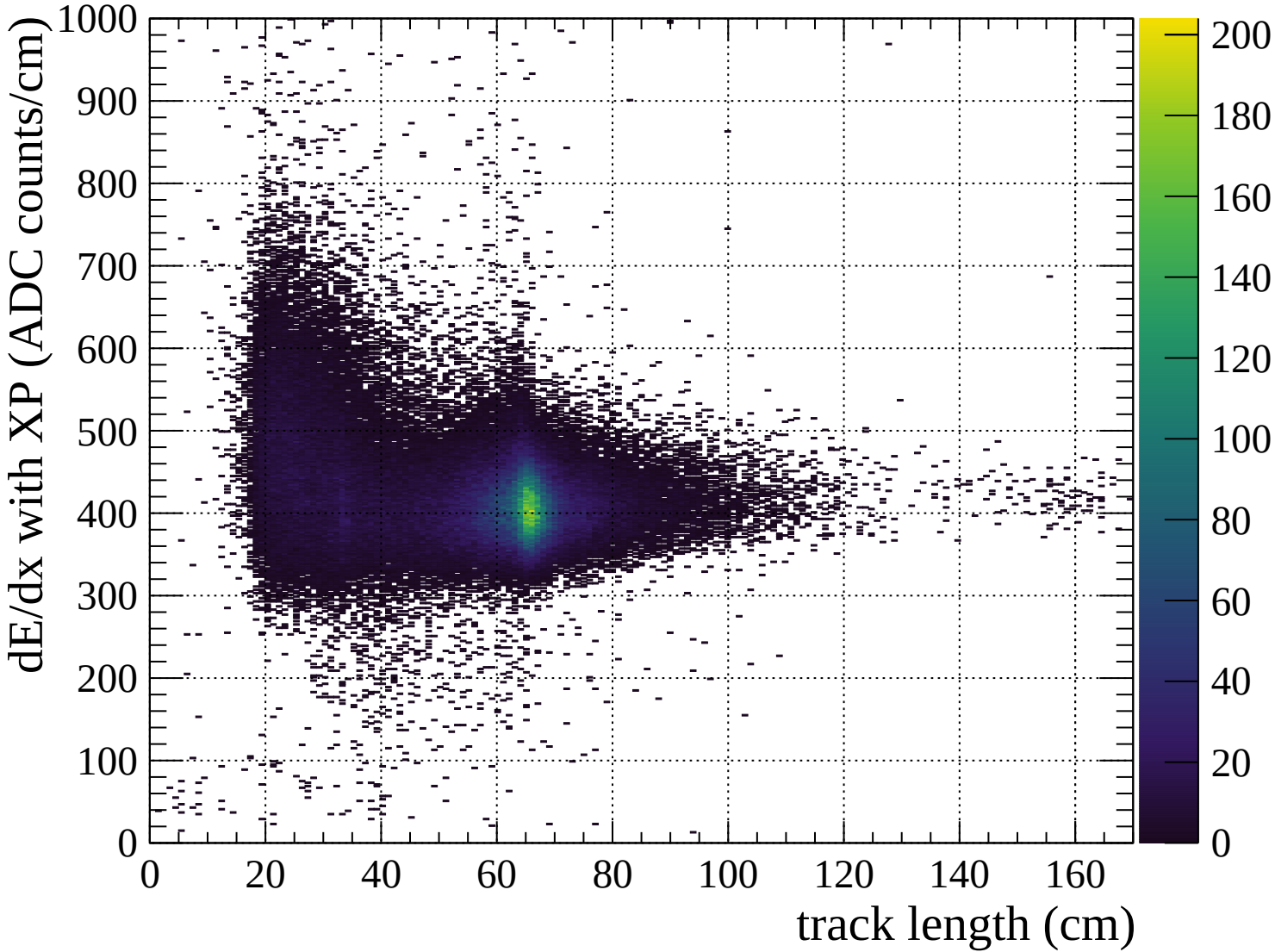


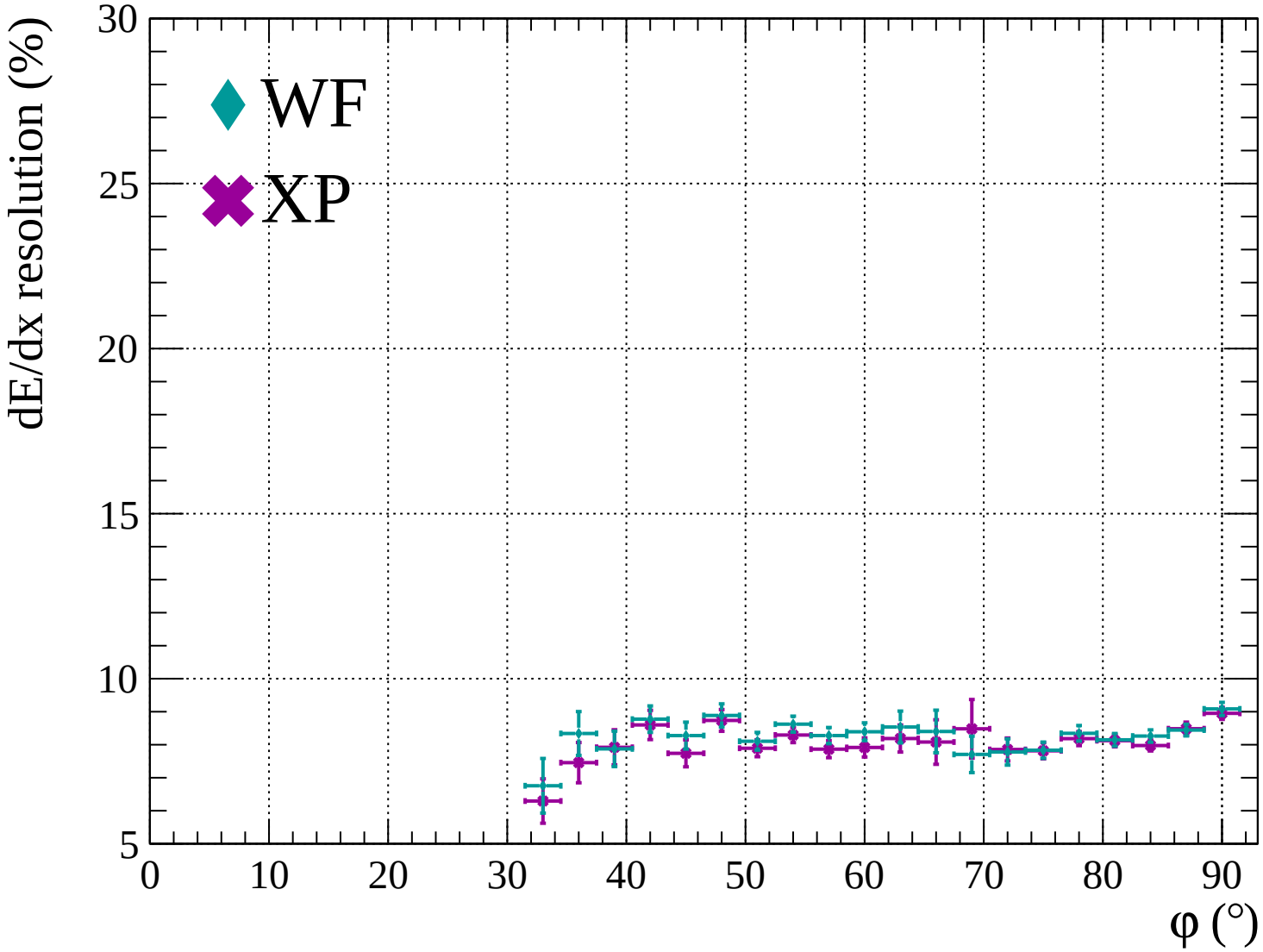


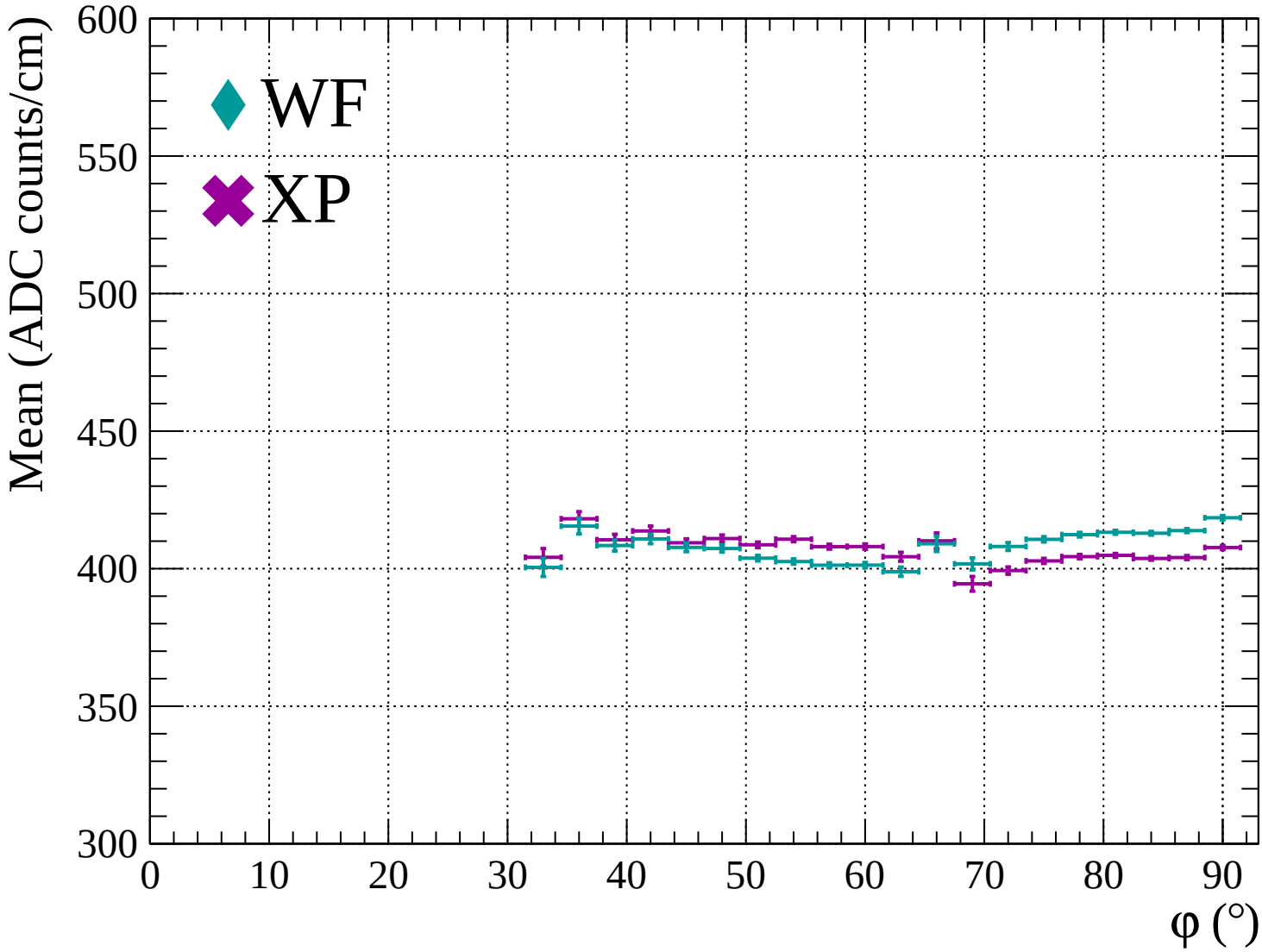


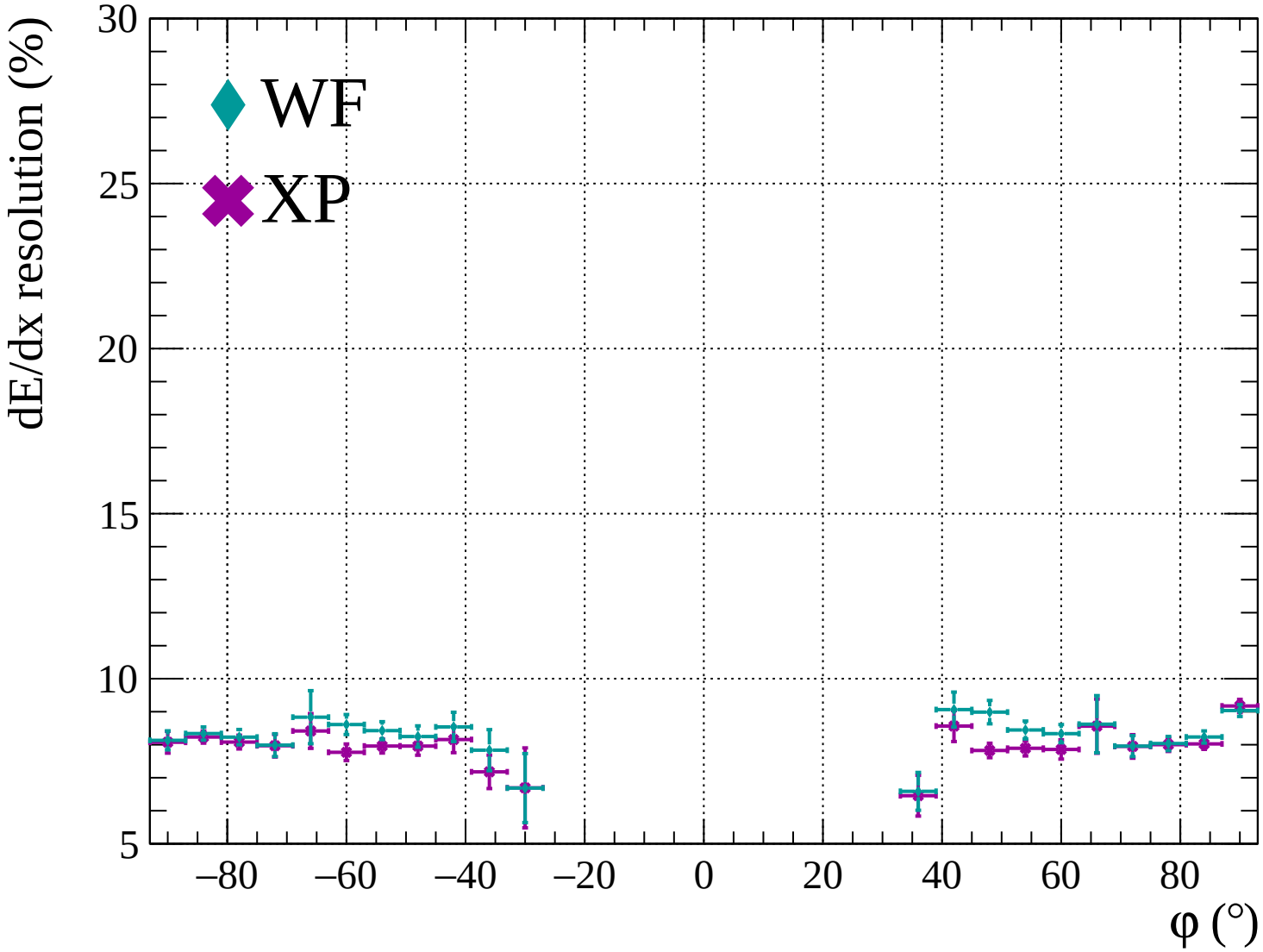


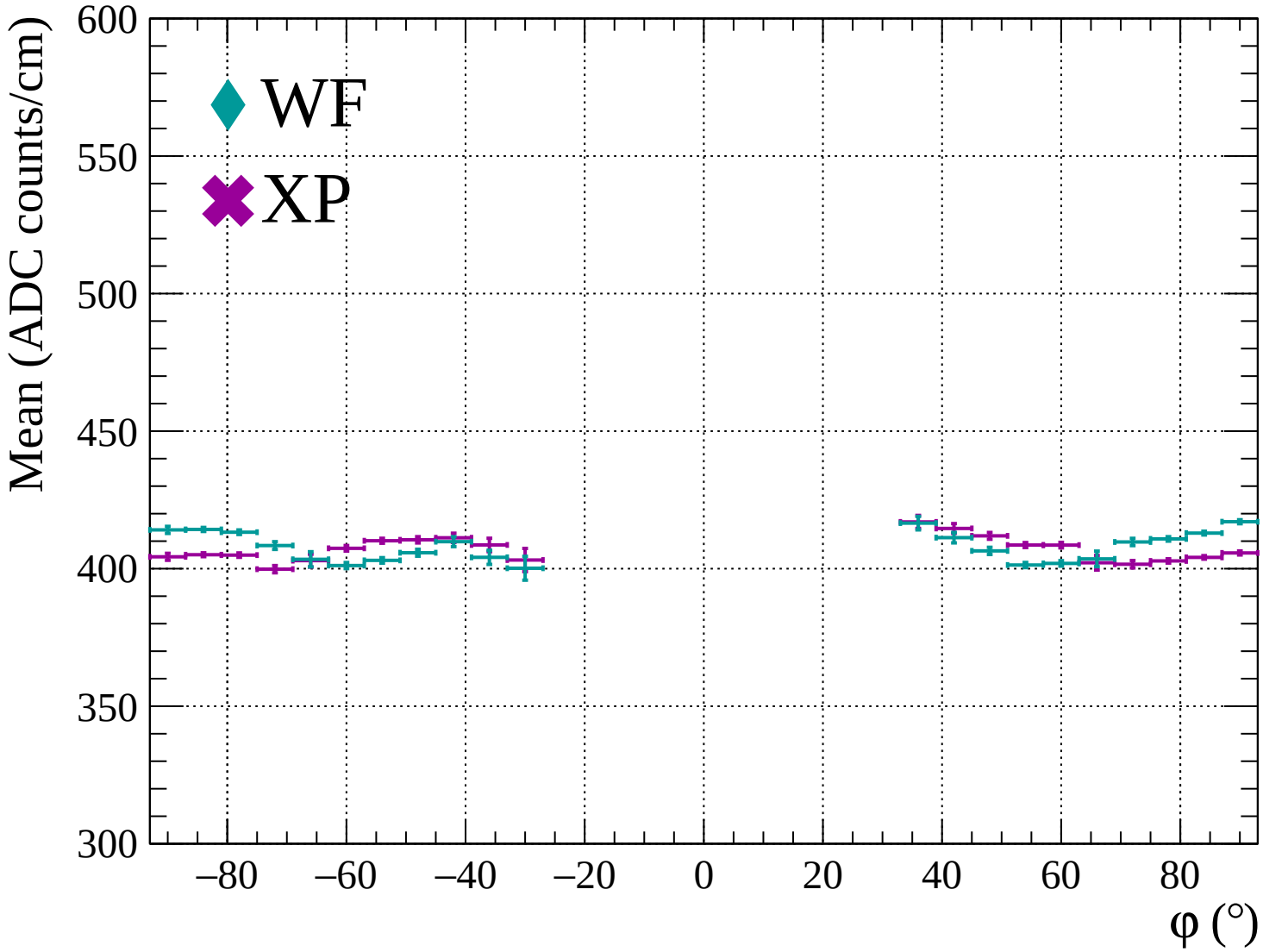


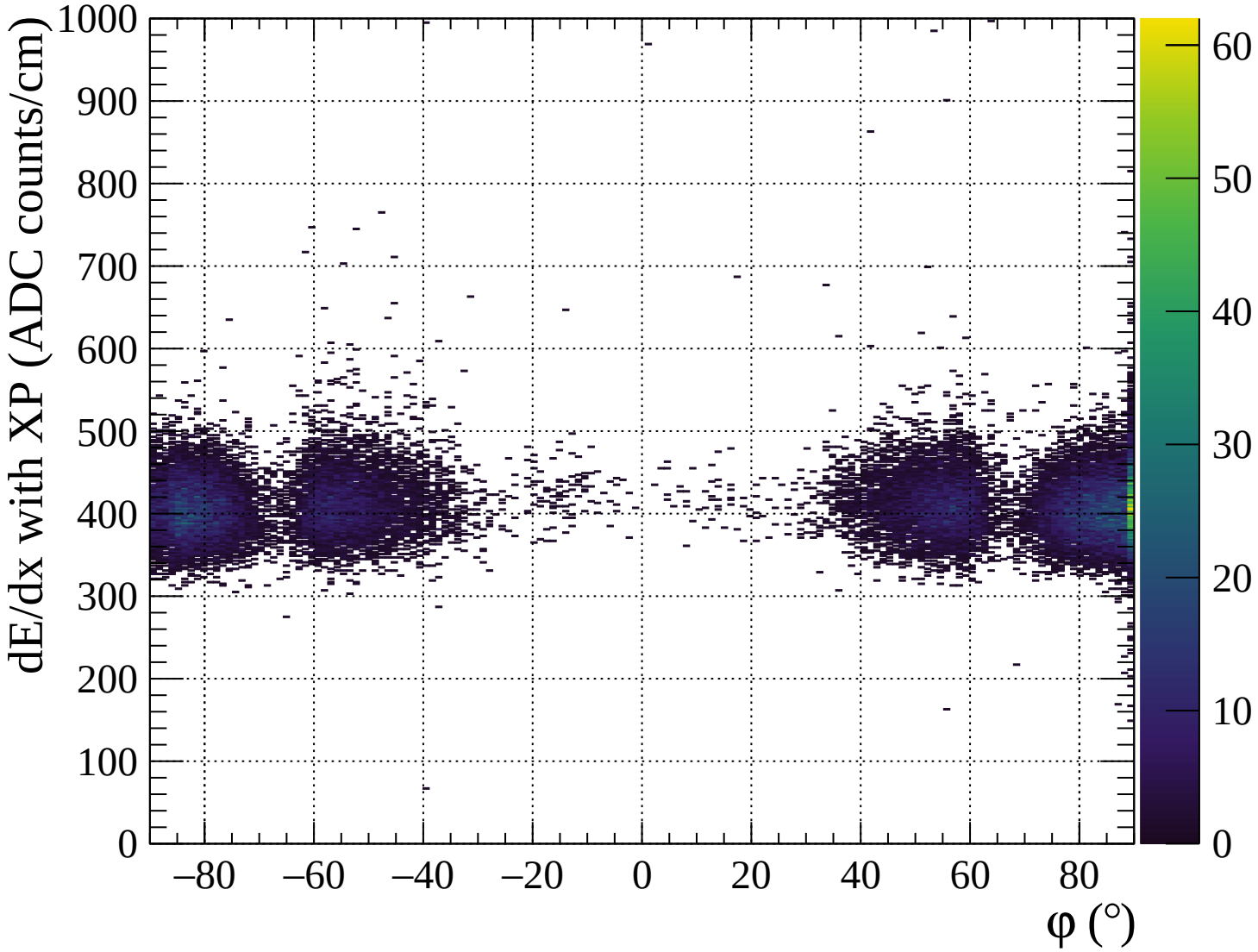


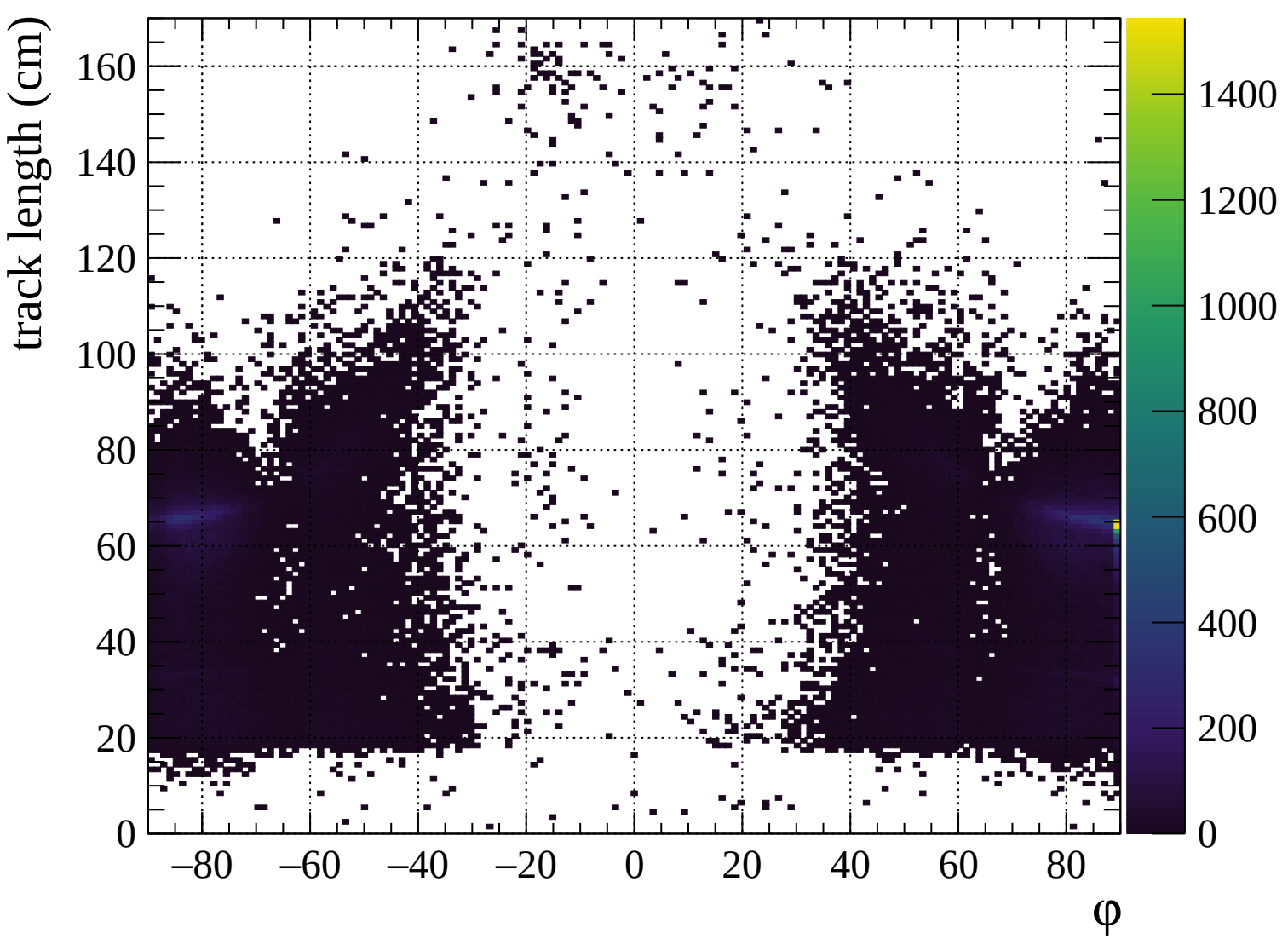


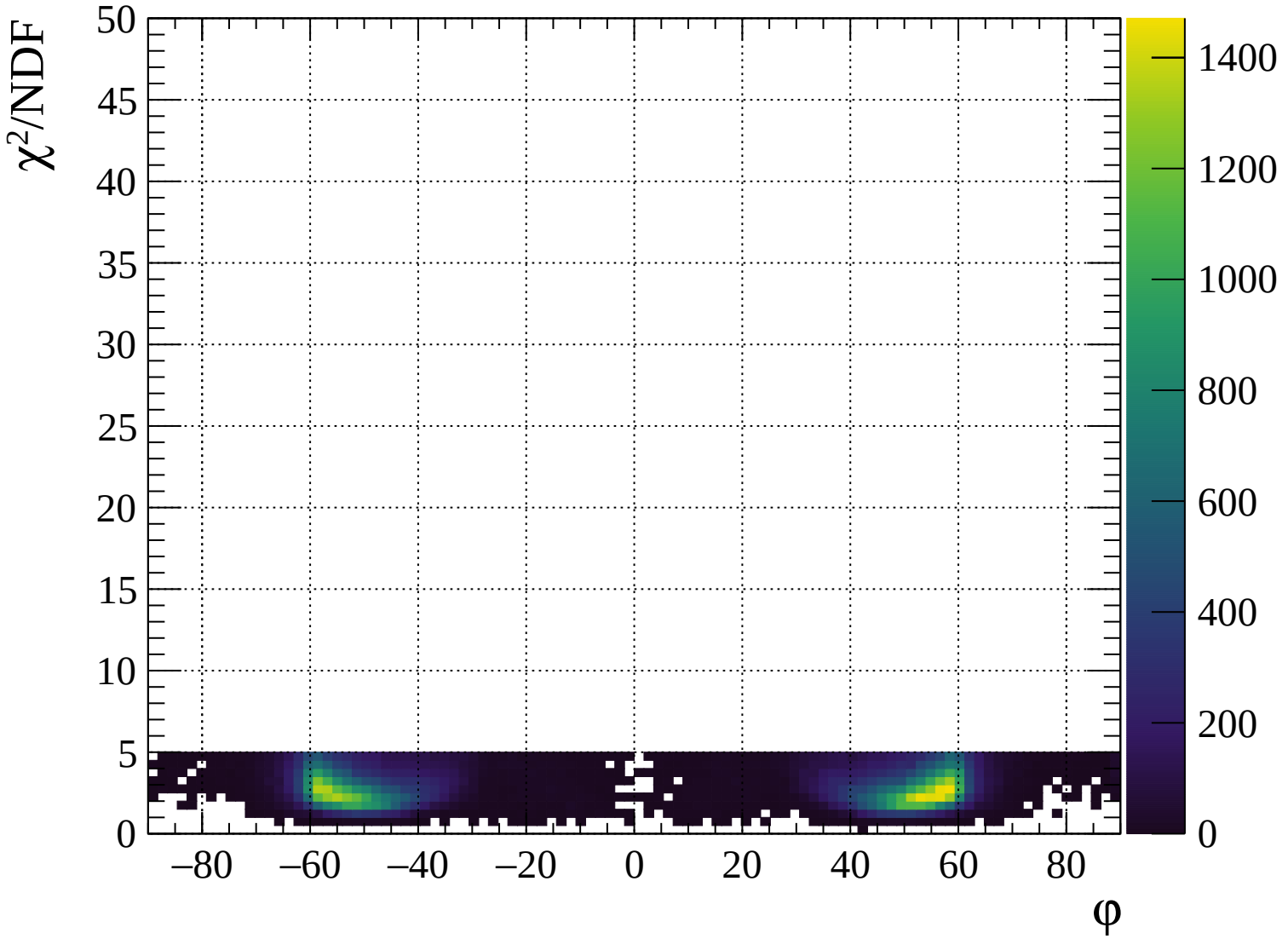


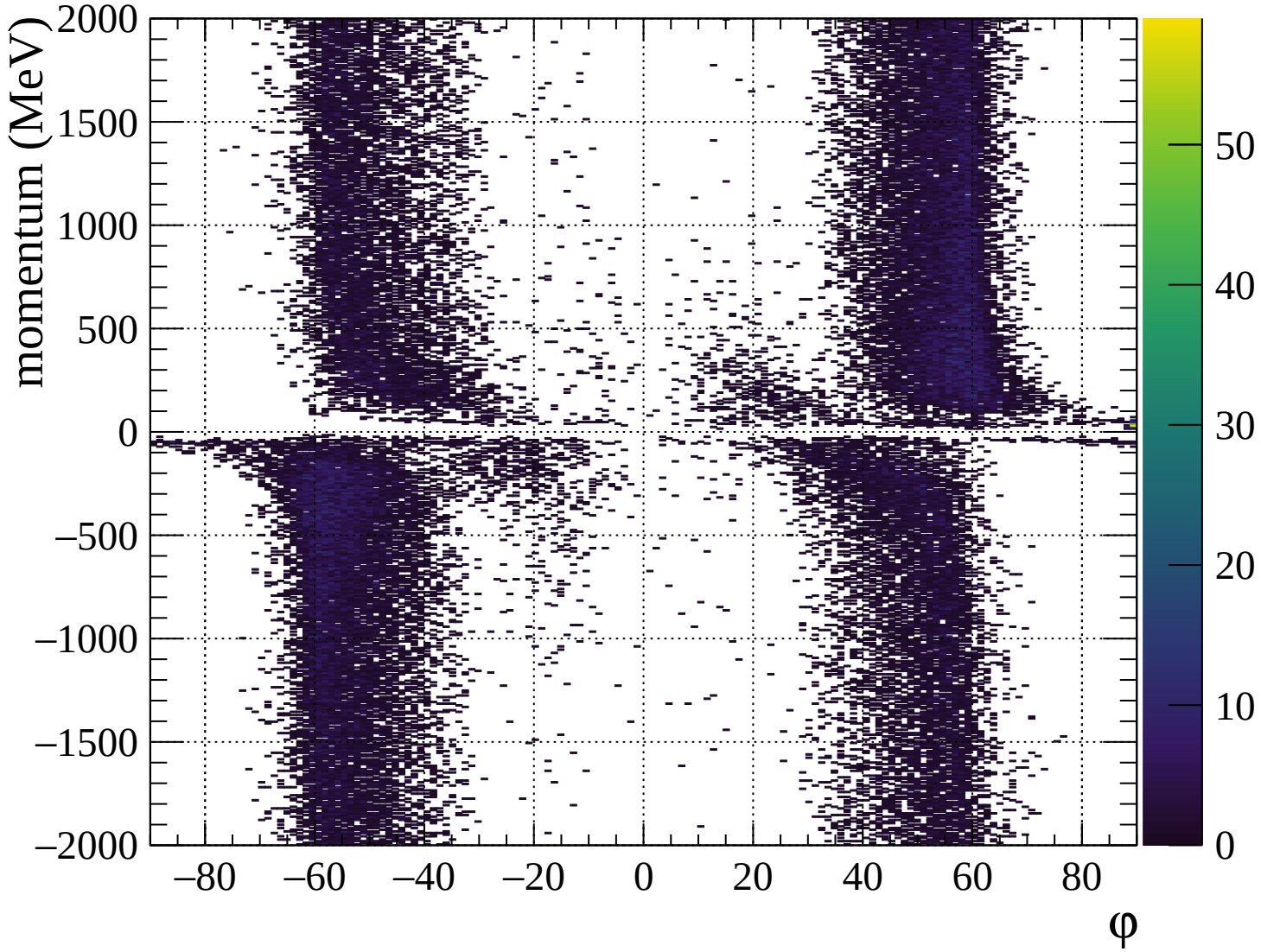


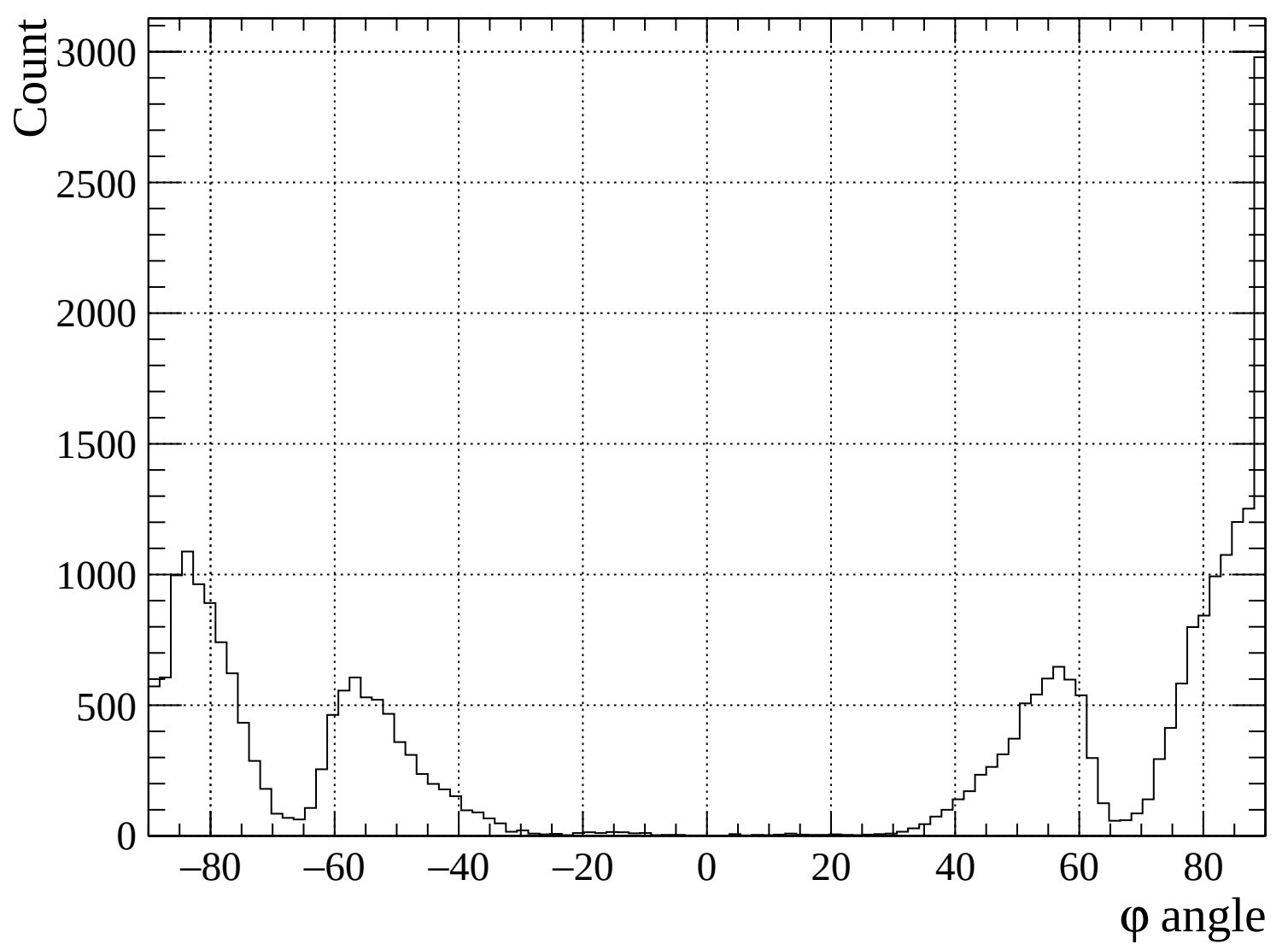


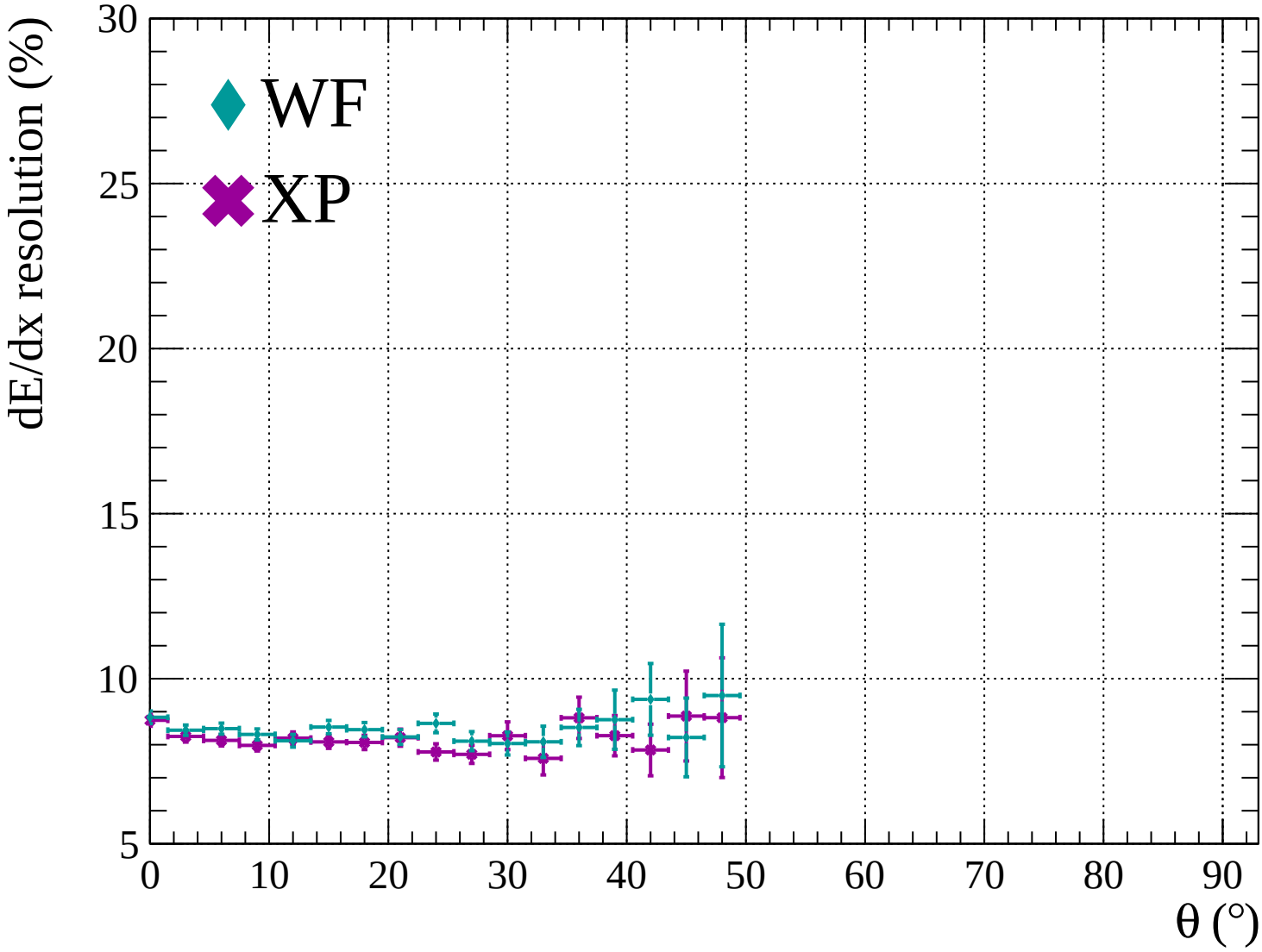


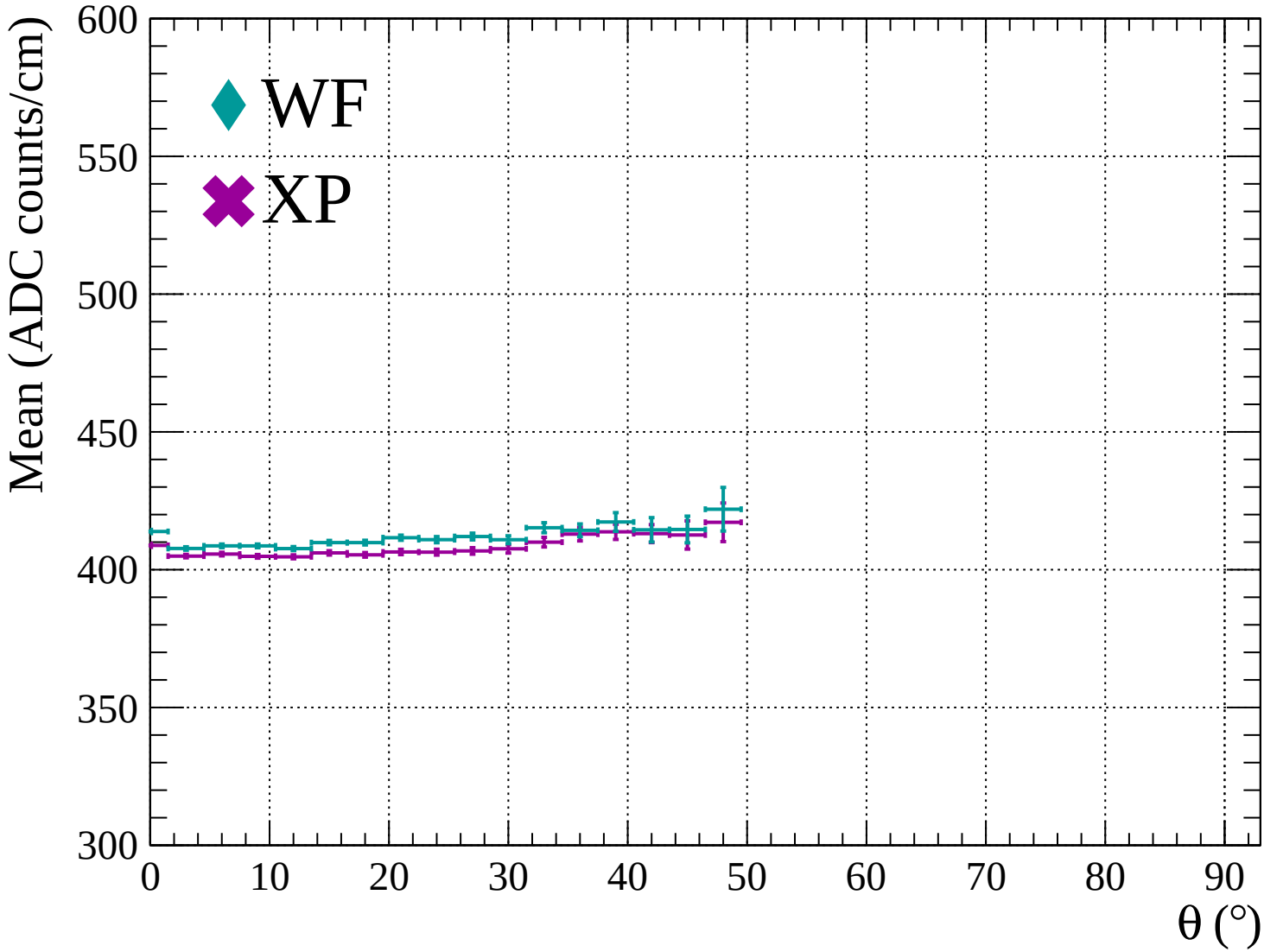


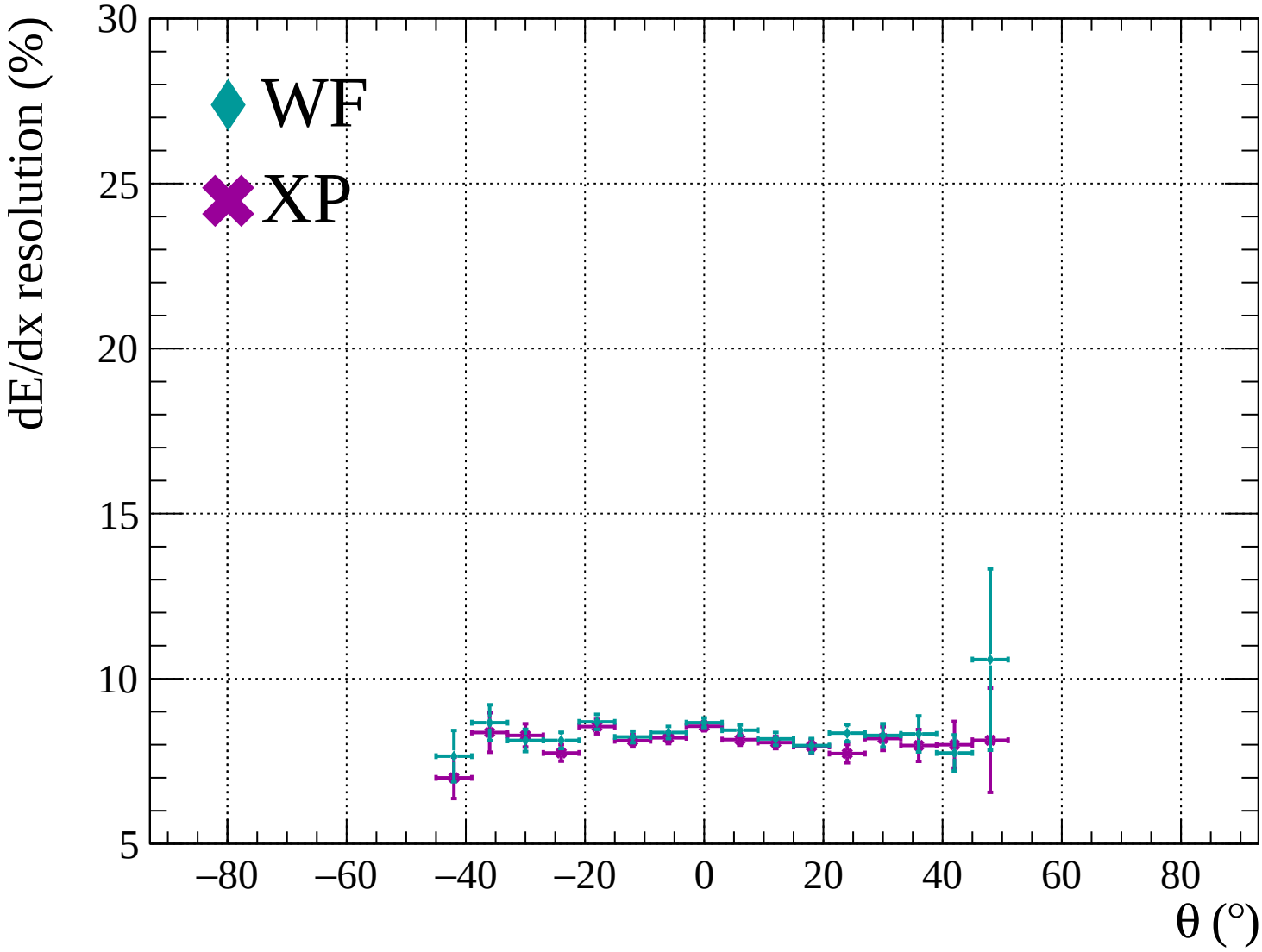


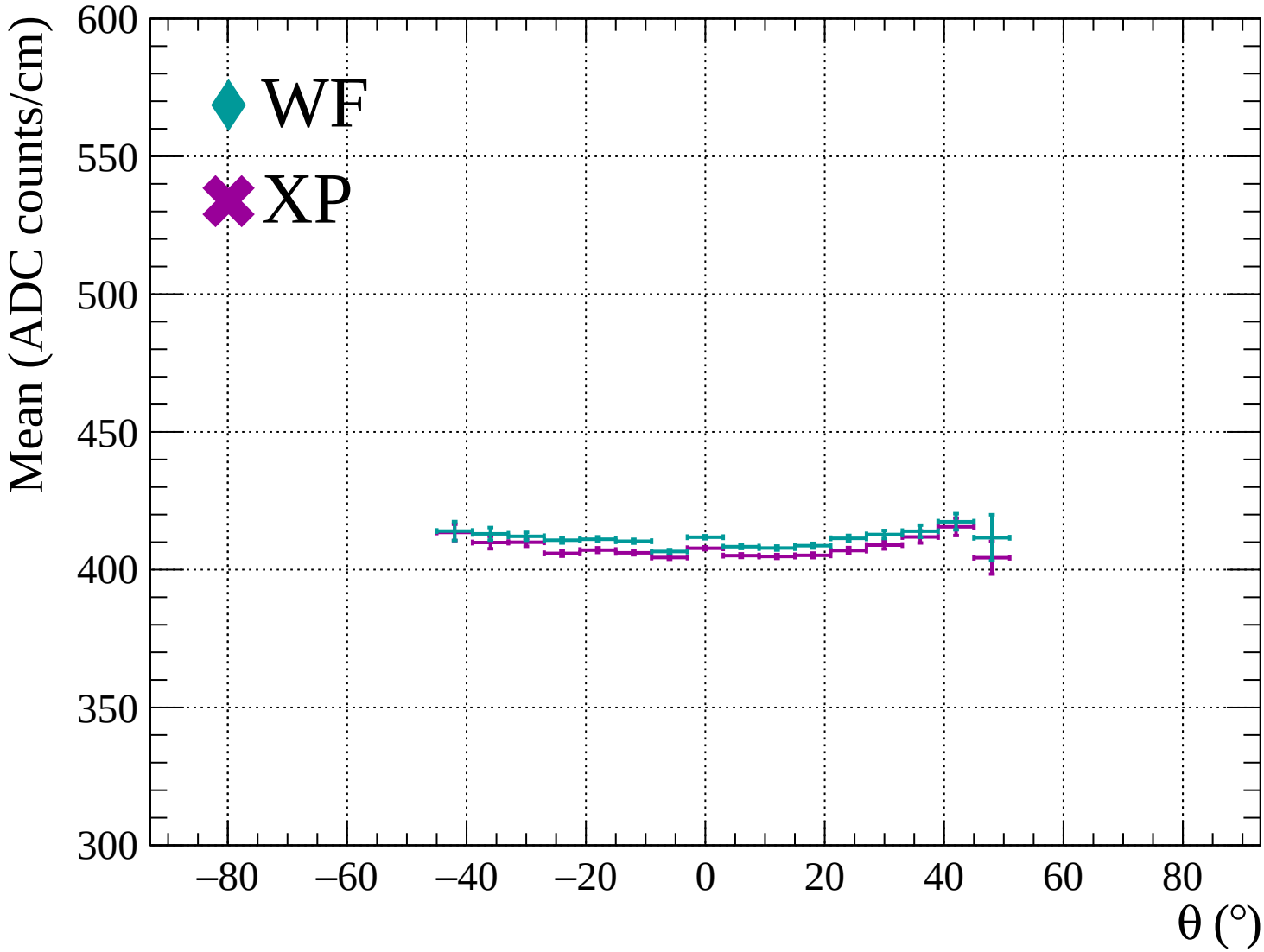


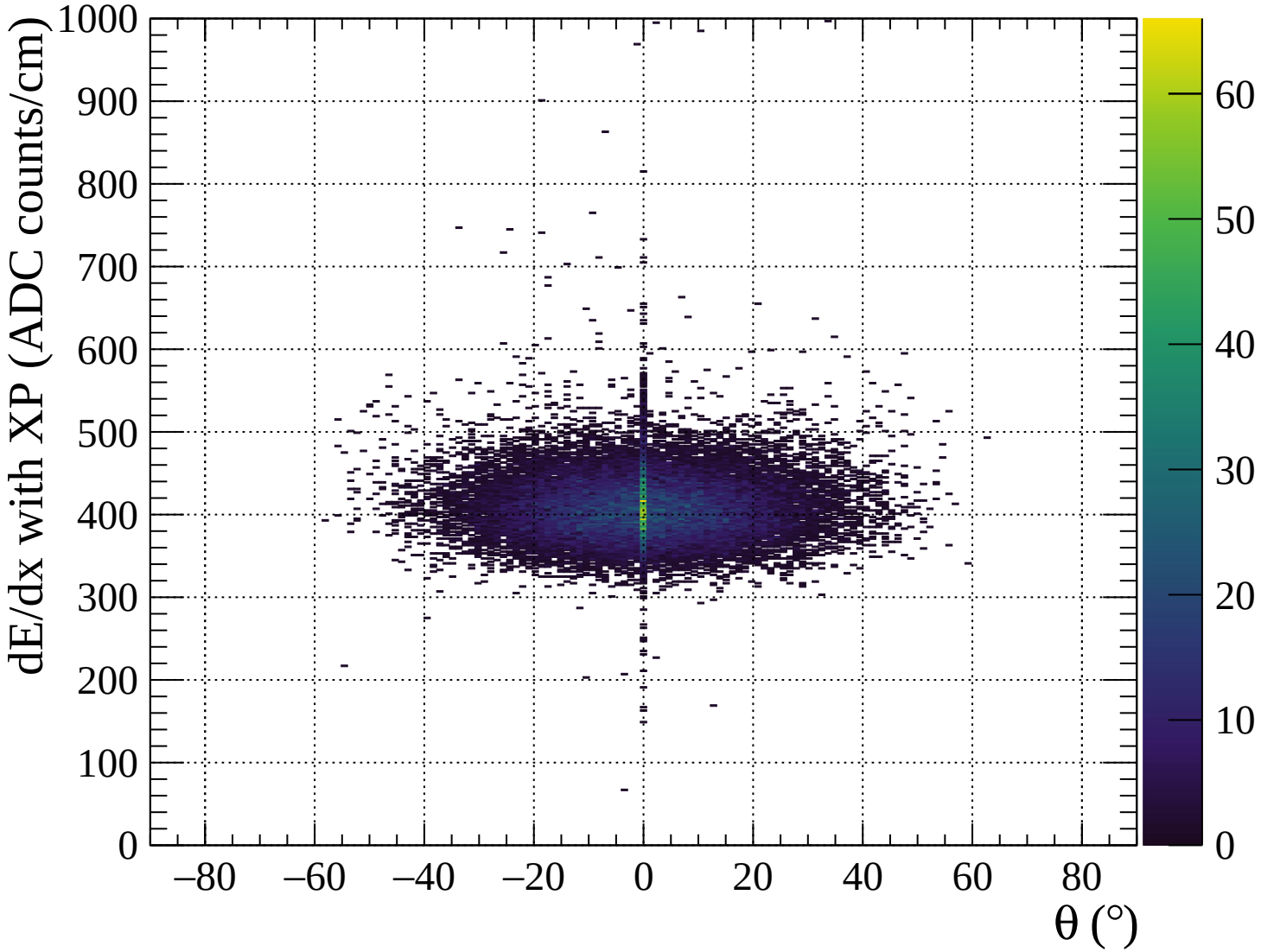


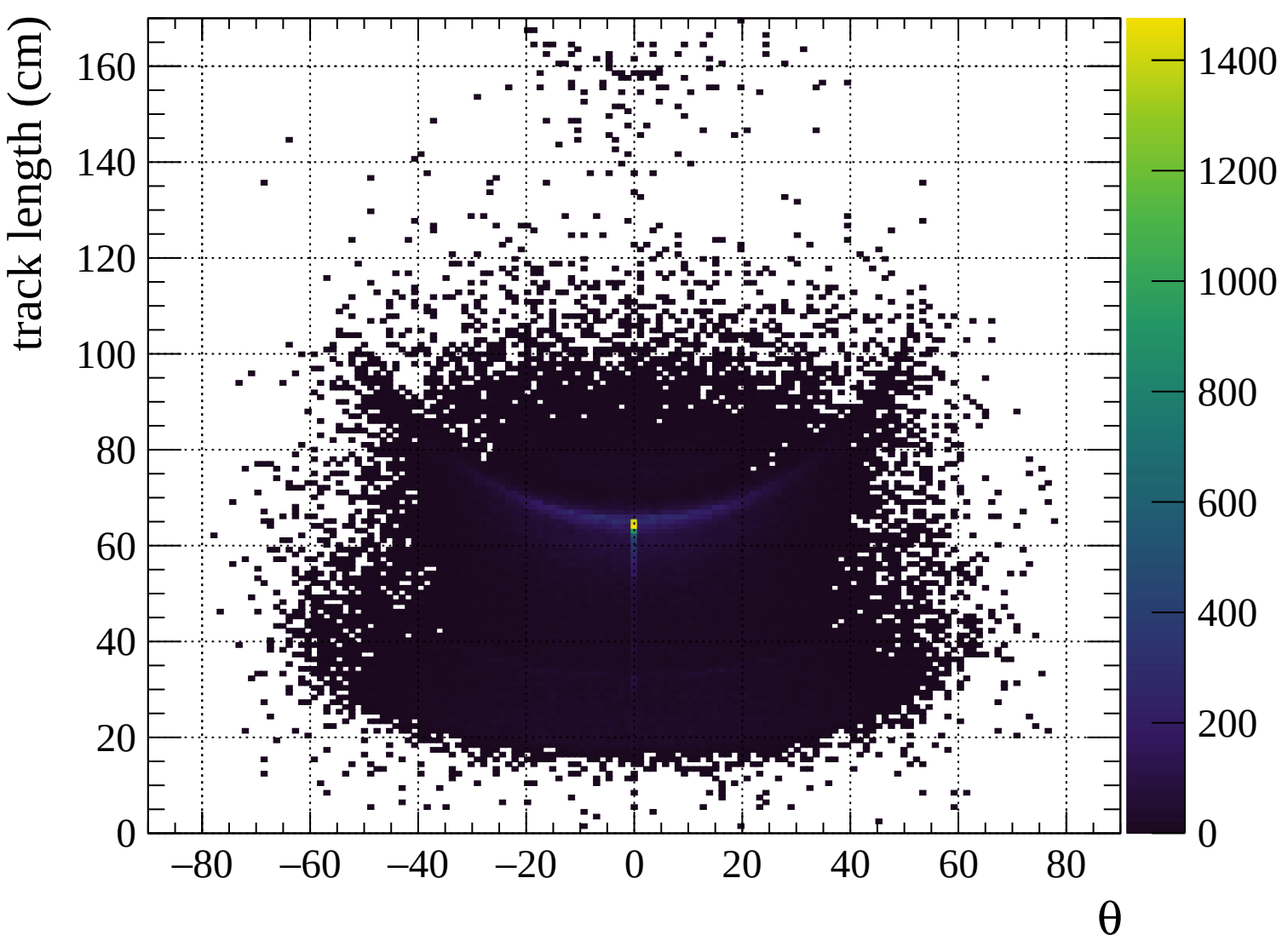


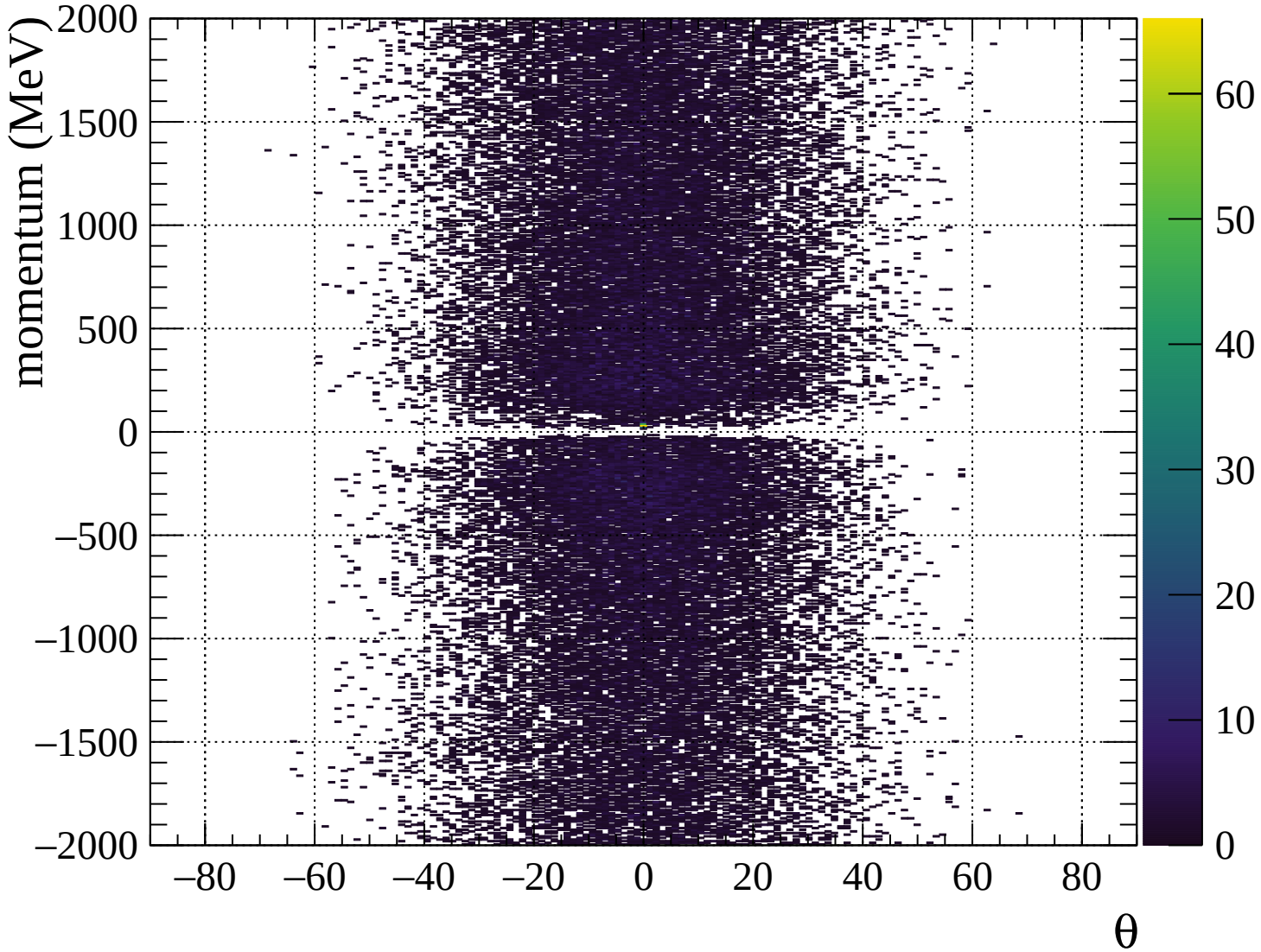


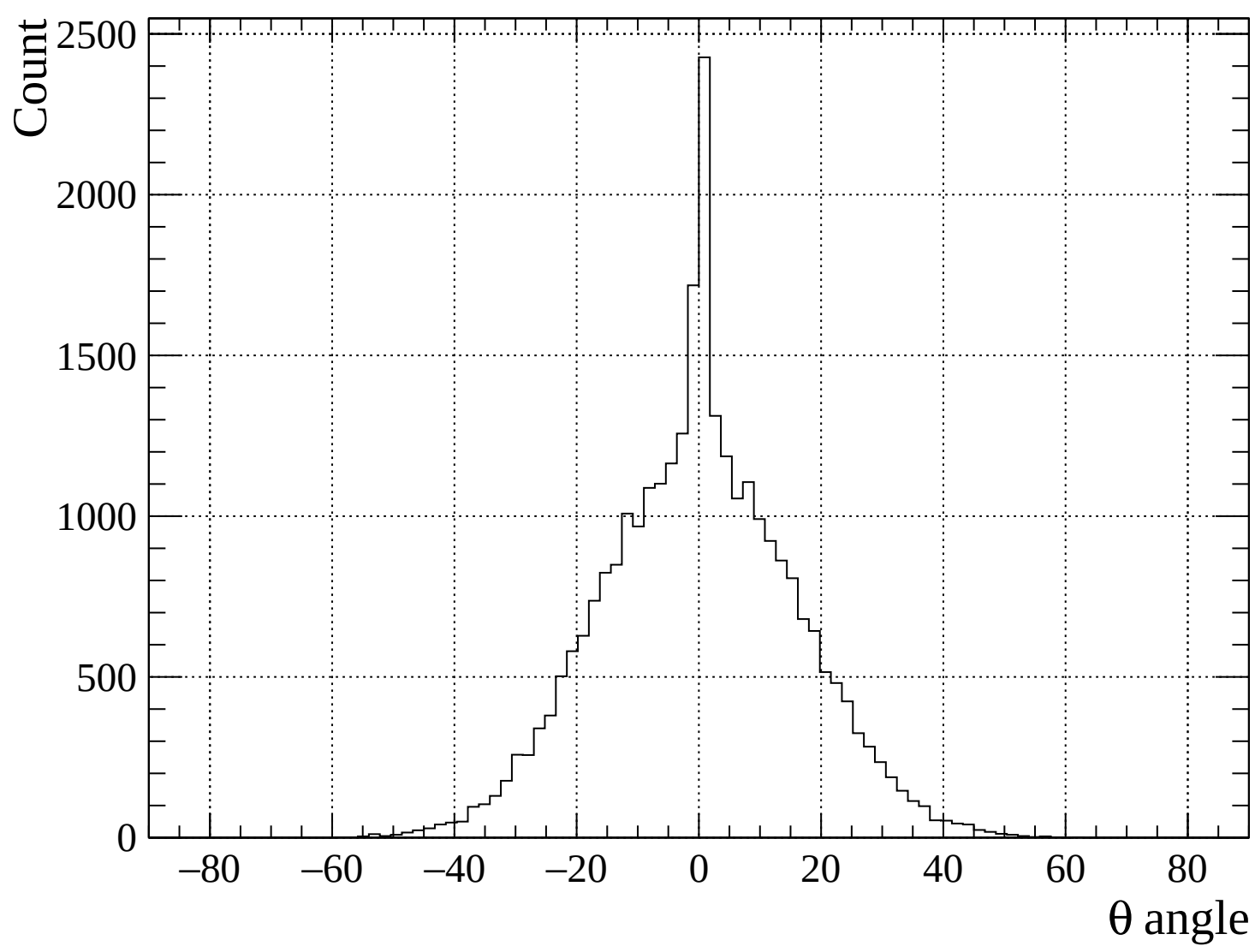


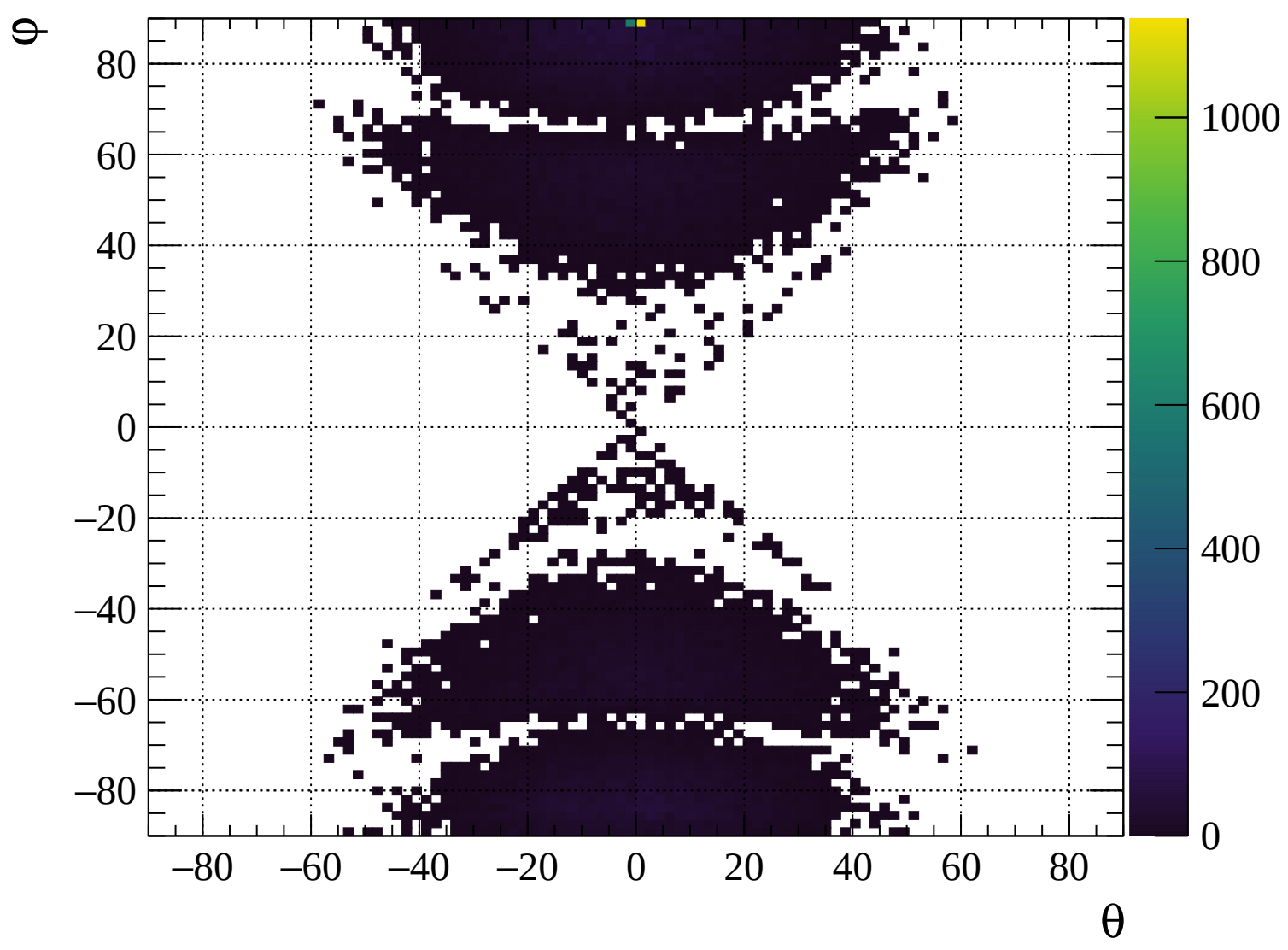


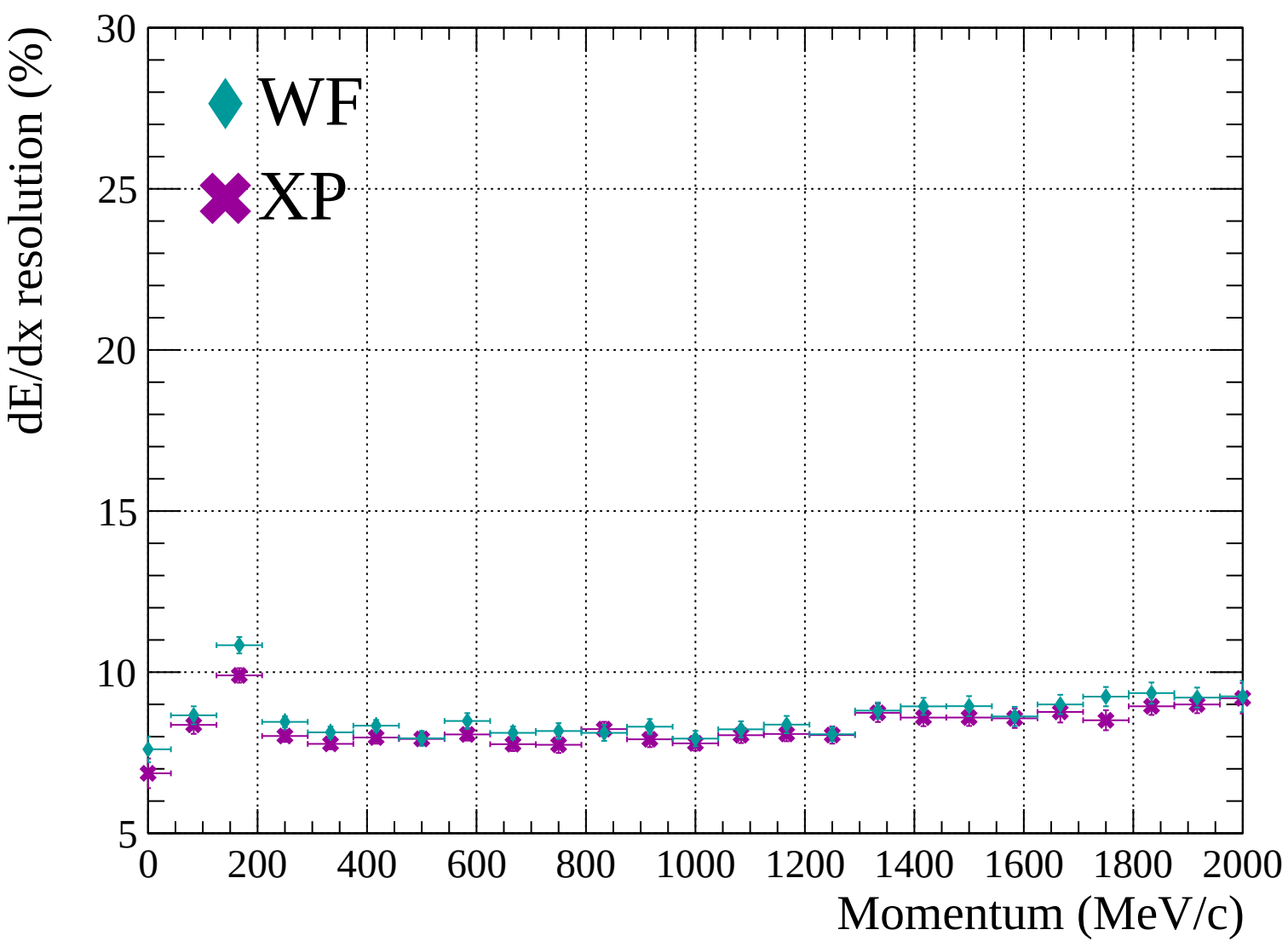


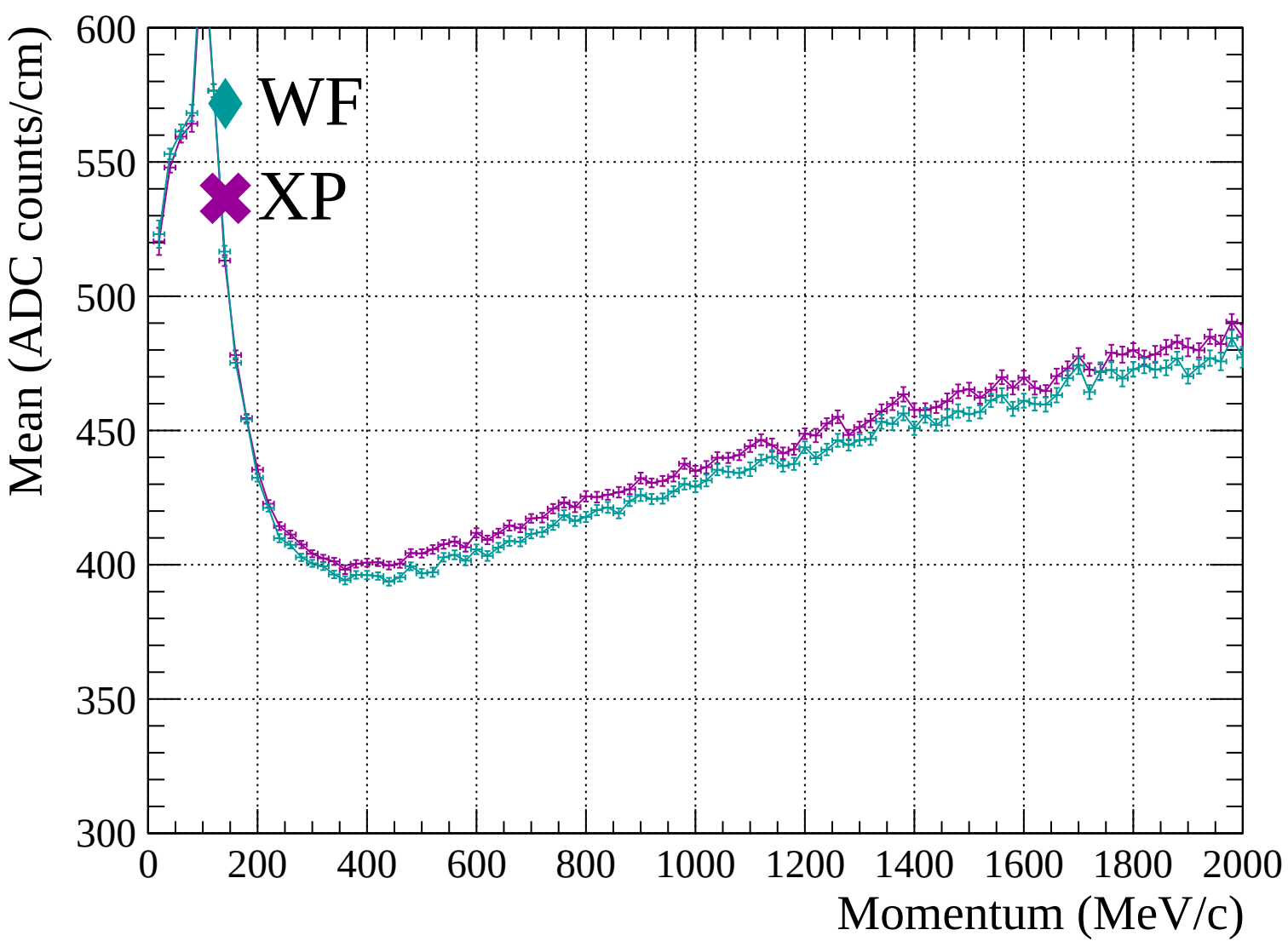


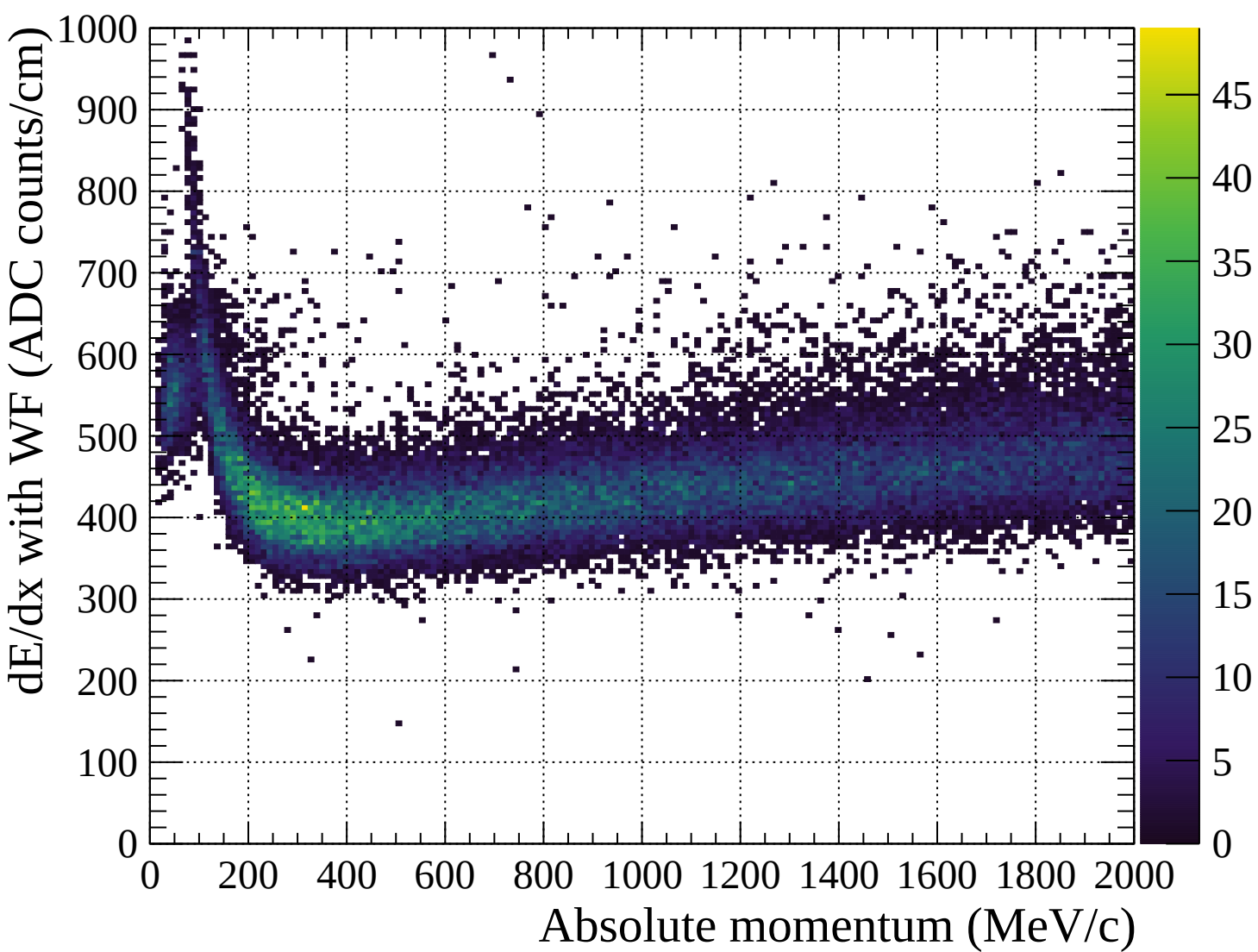


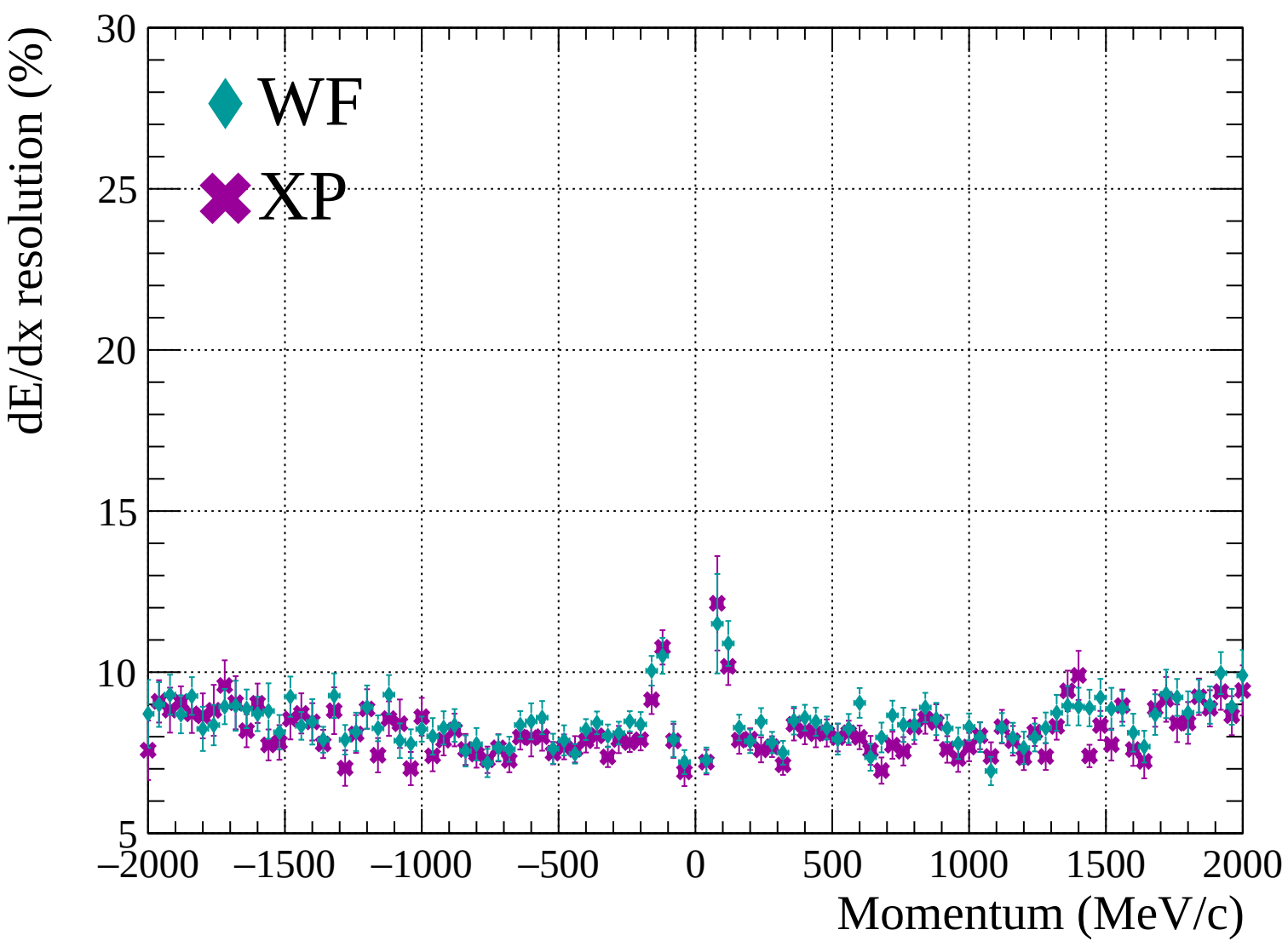


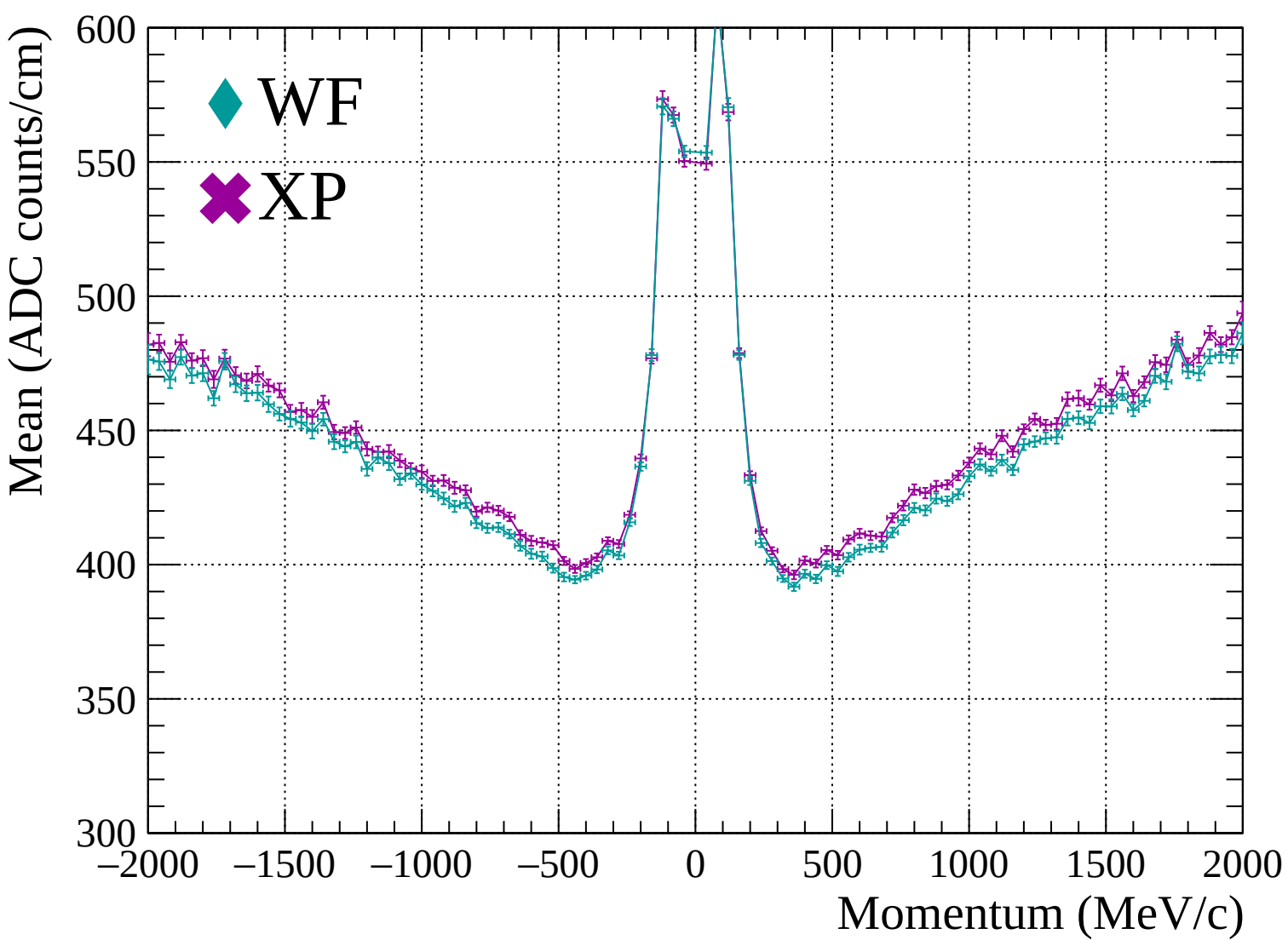


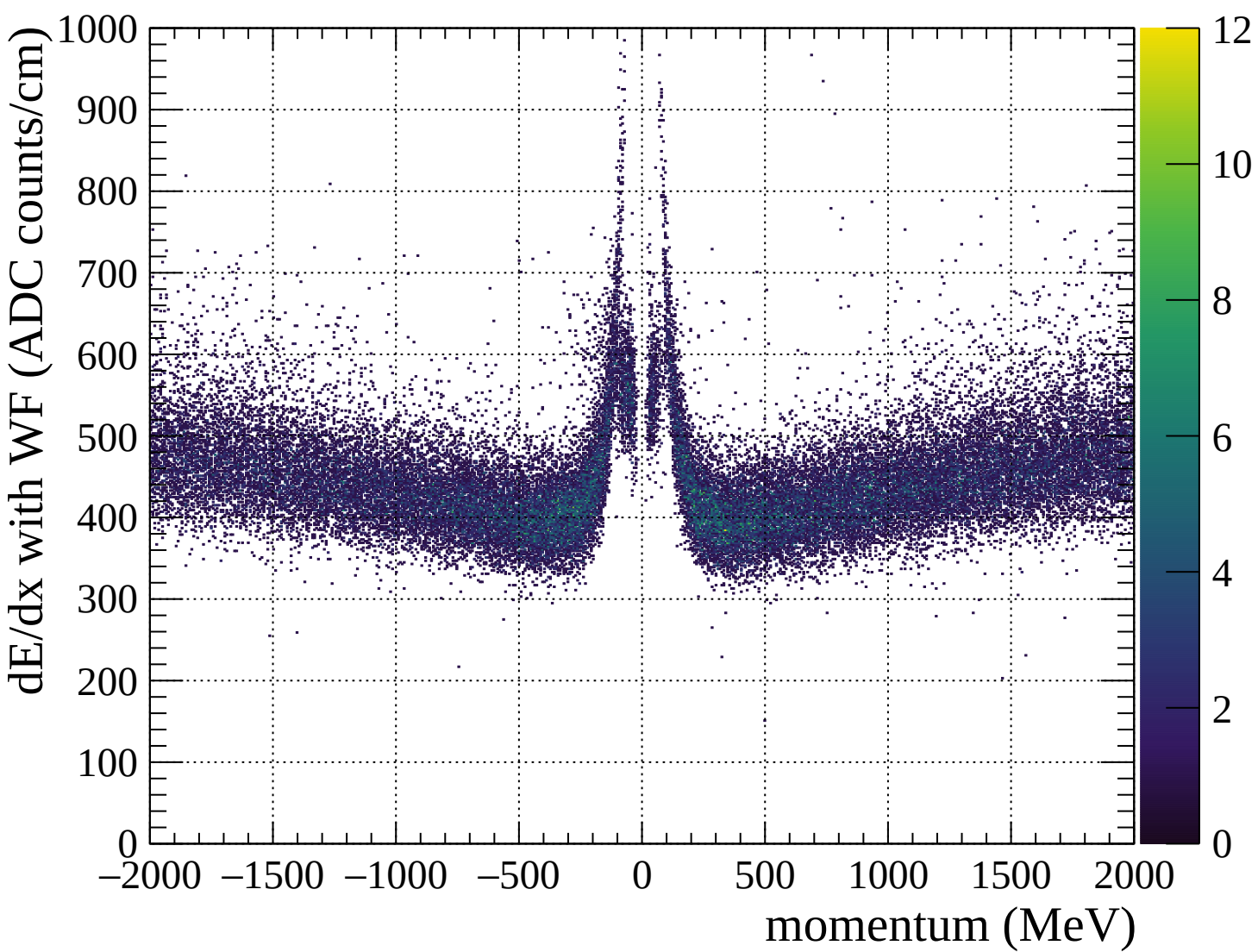


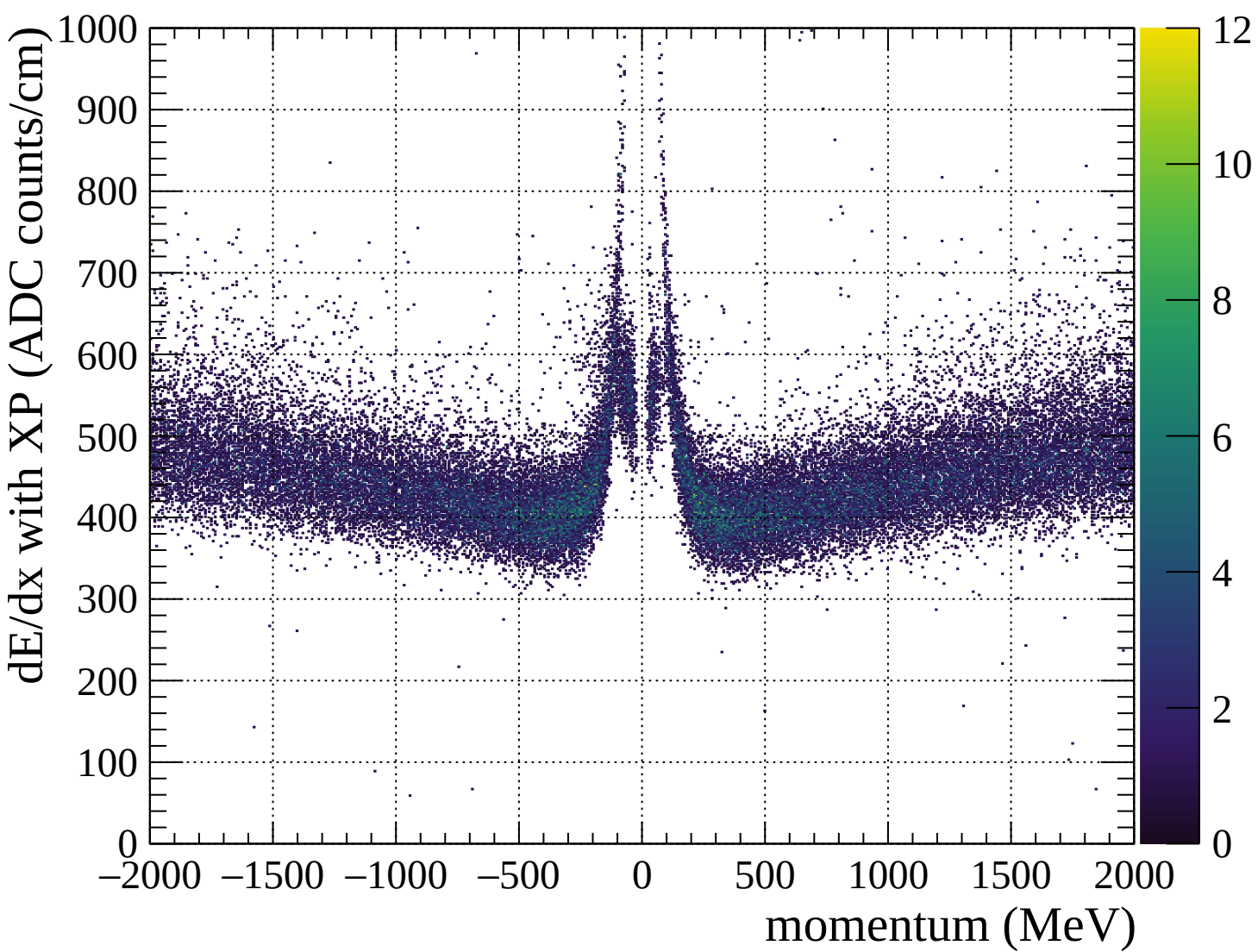


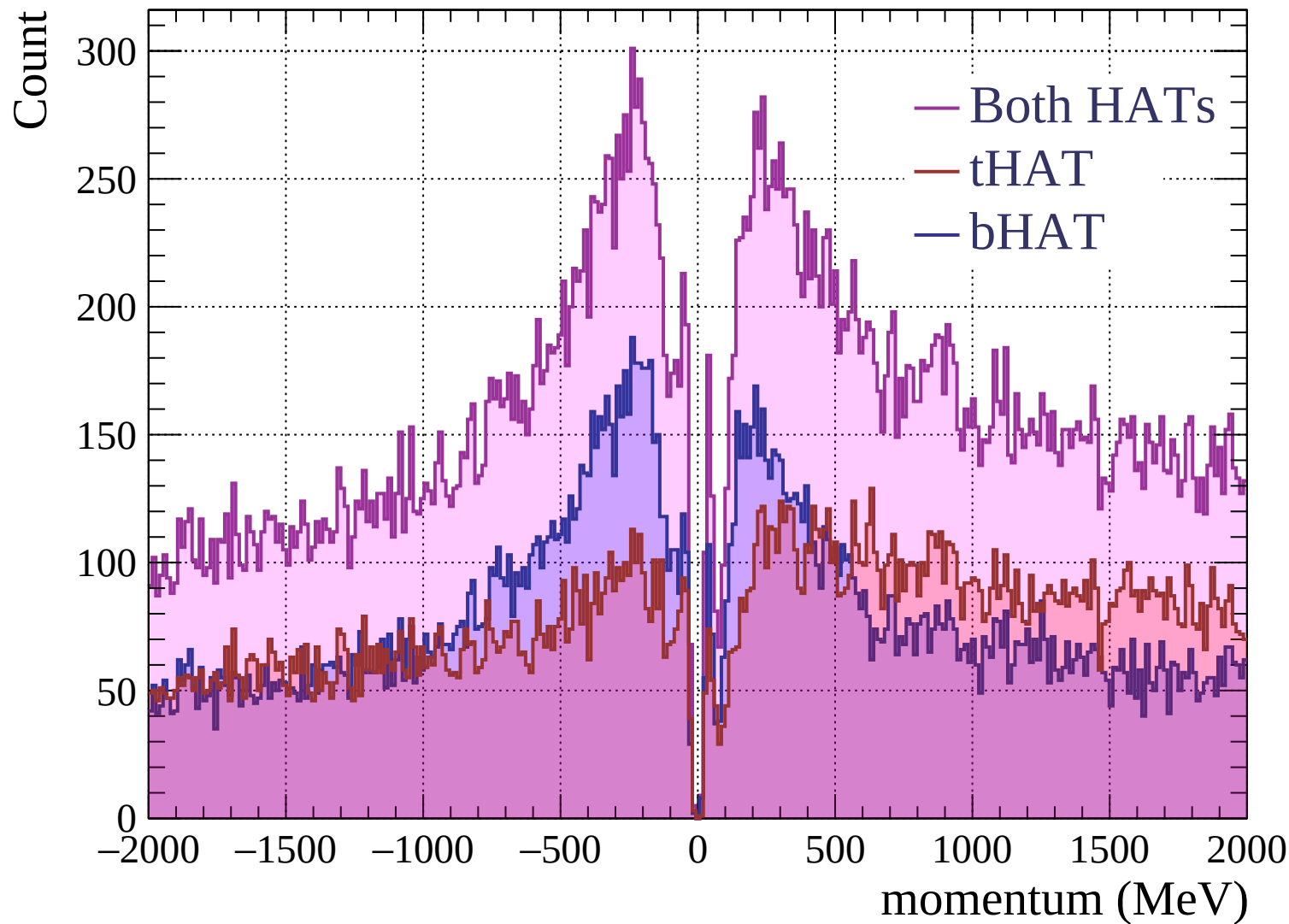


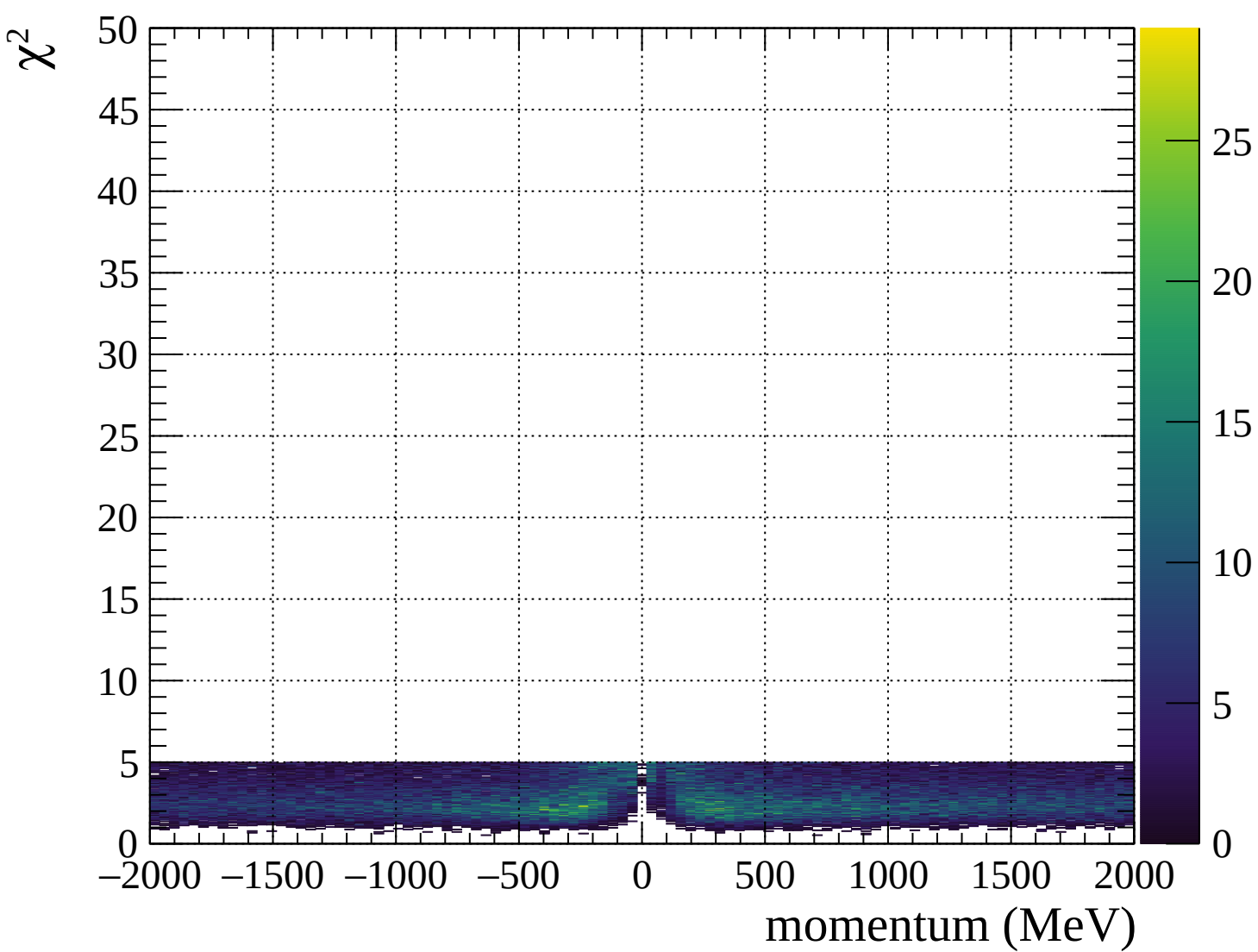




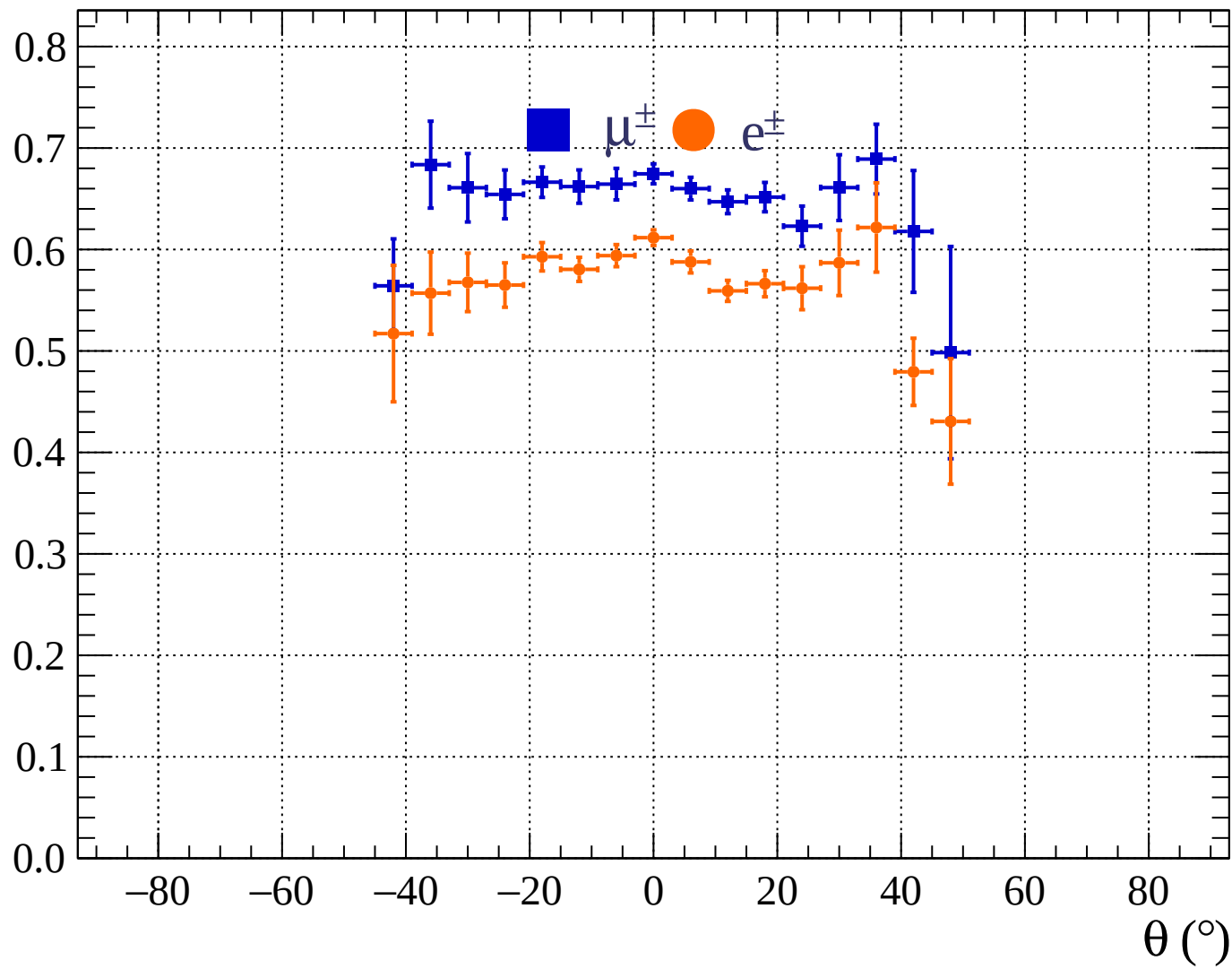


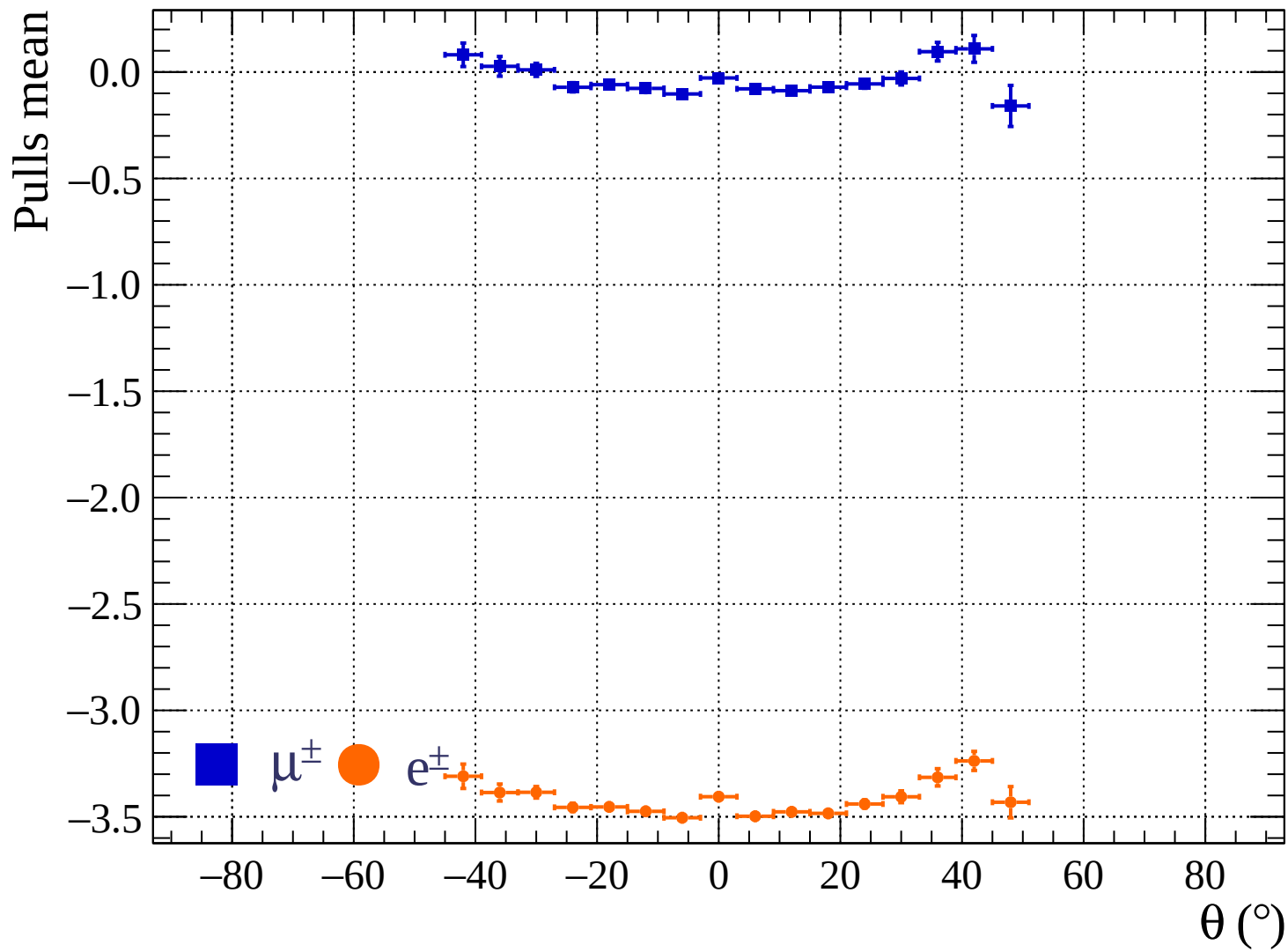




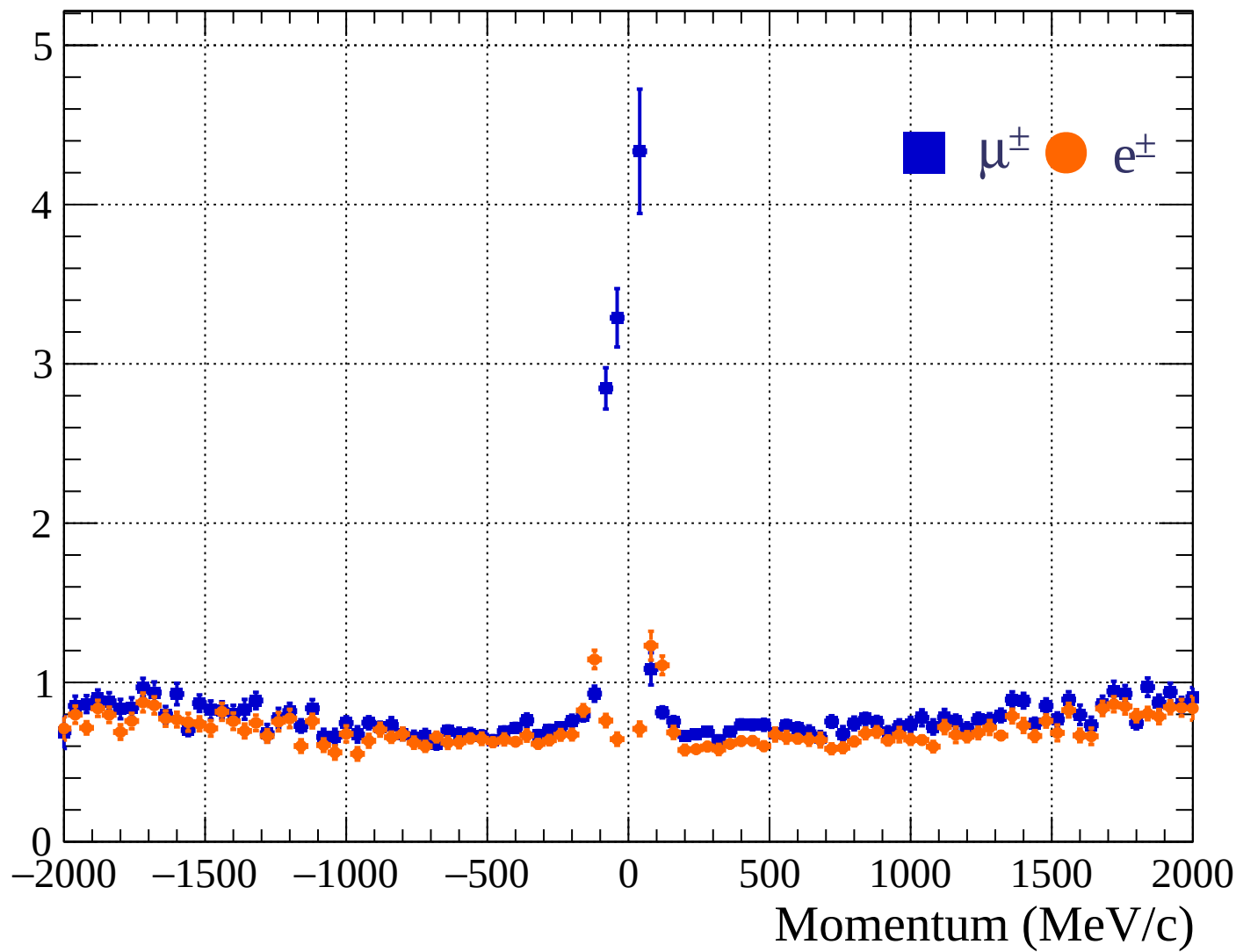


Pulls standard deviation

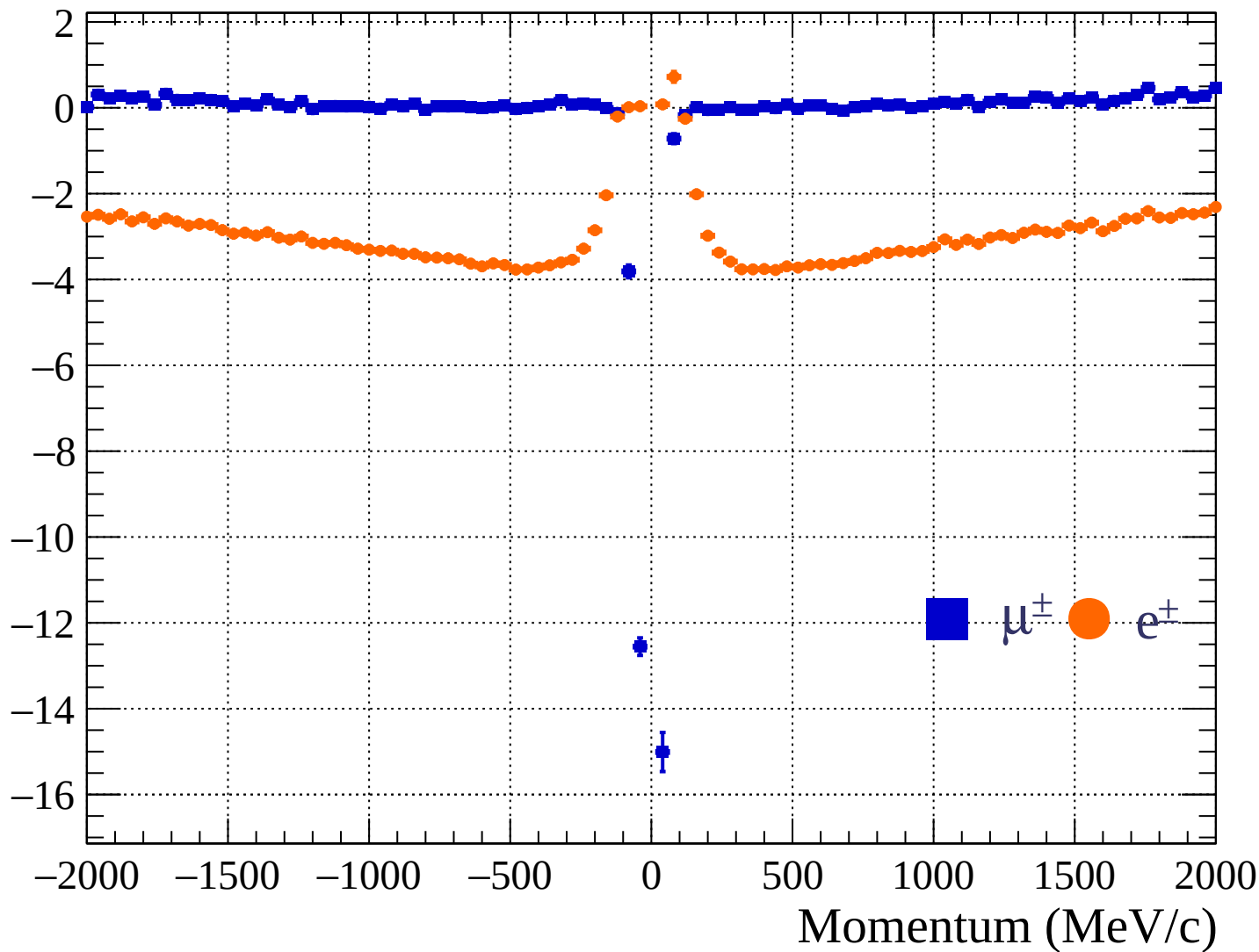


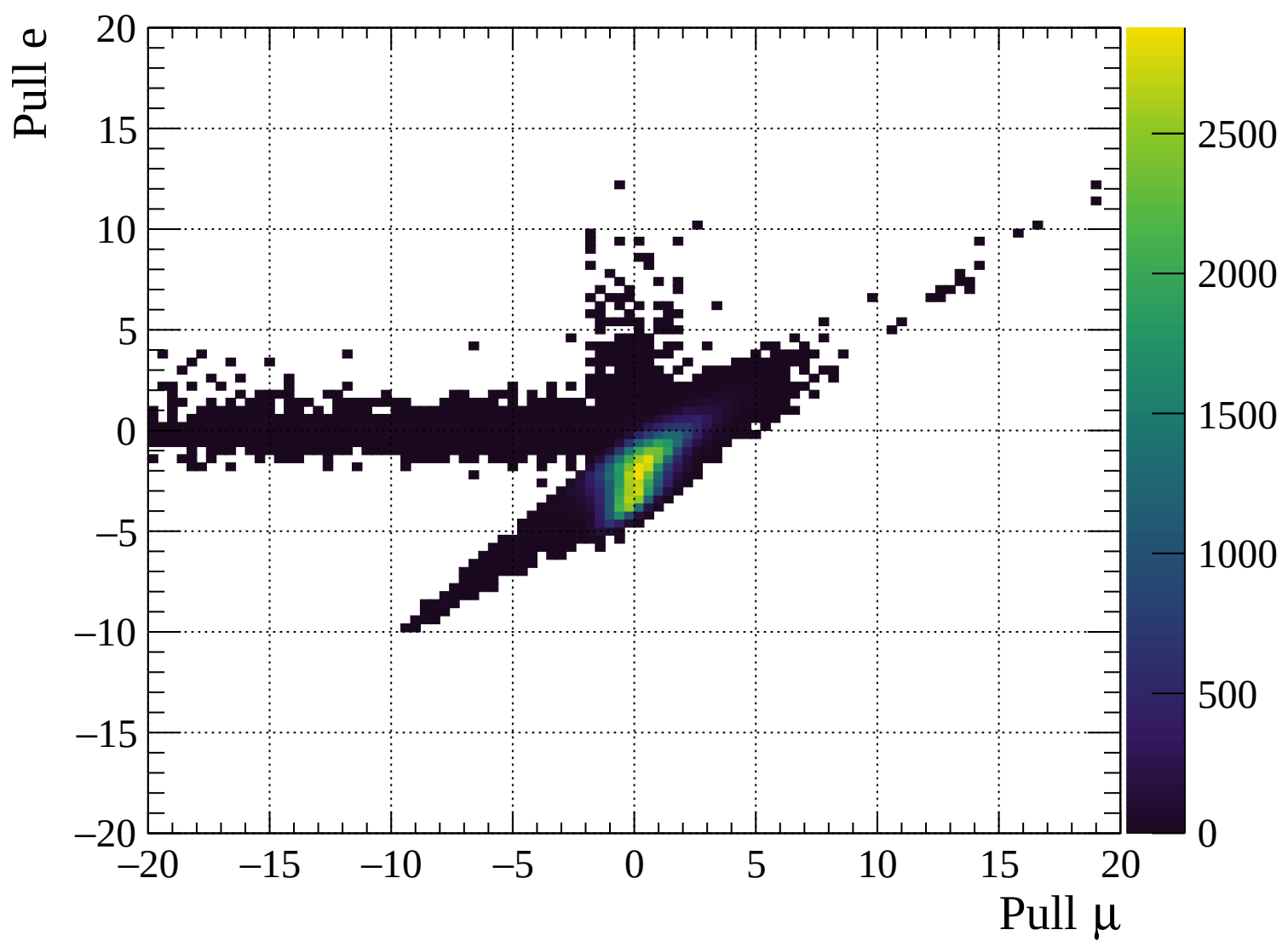


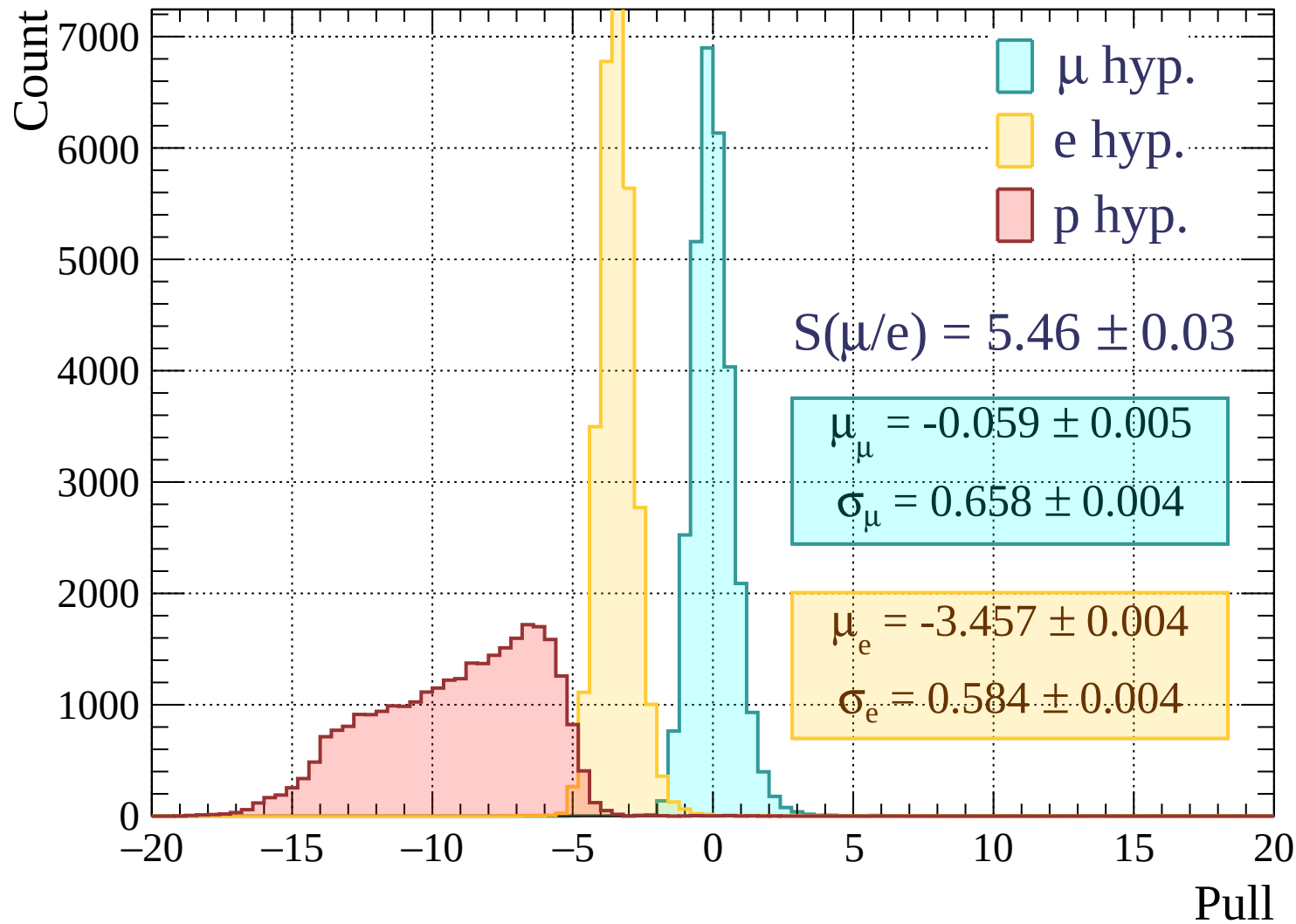
Pulls standard deviation

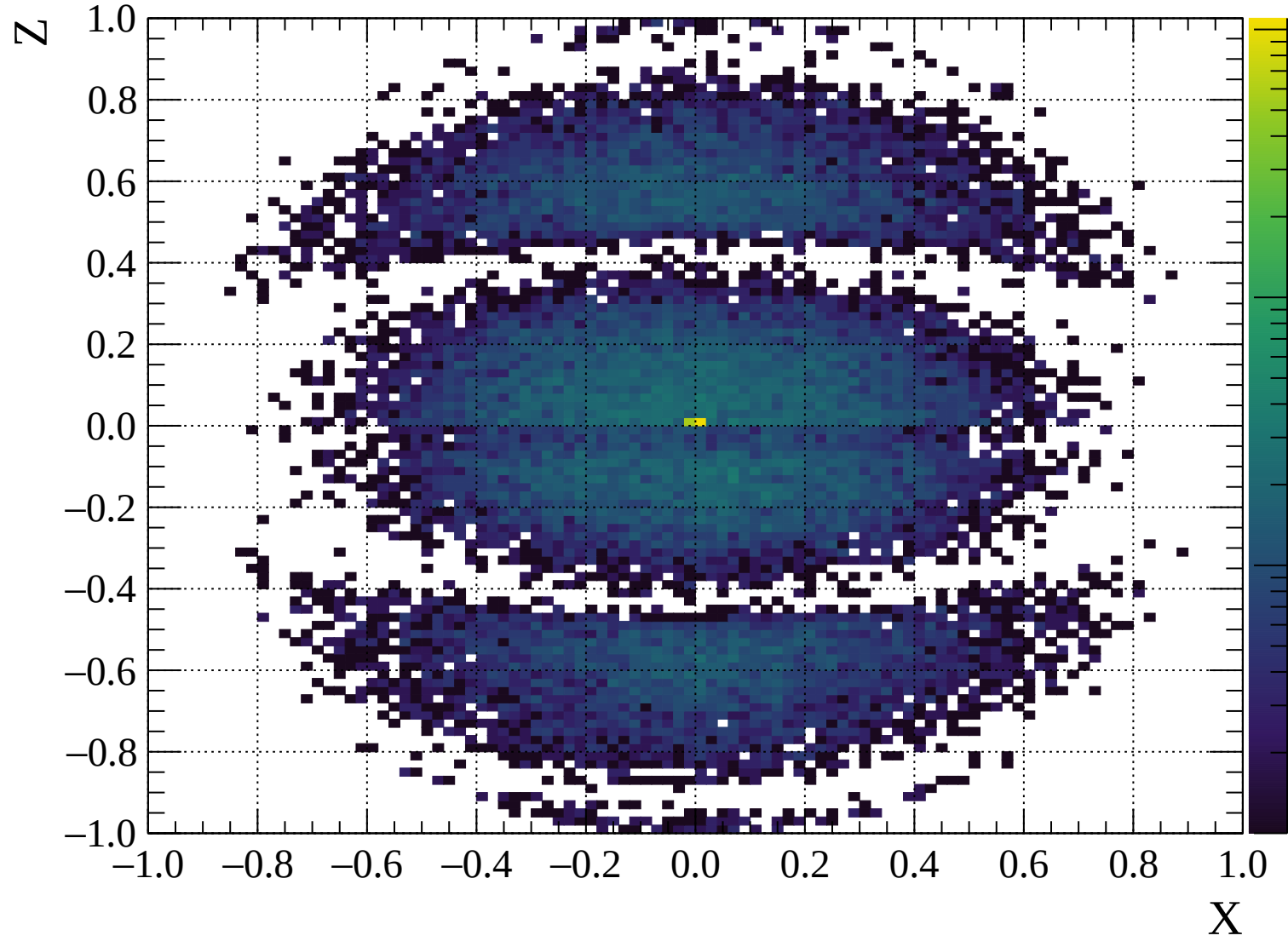


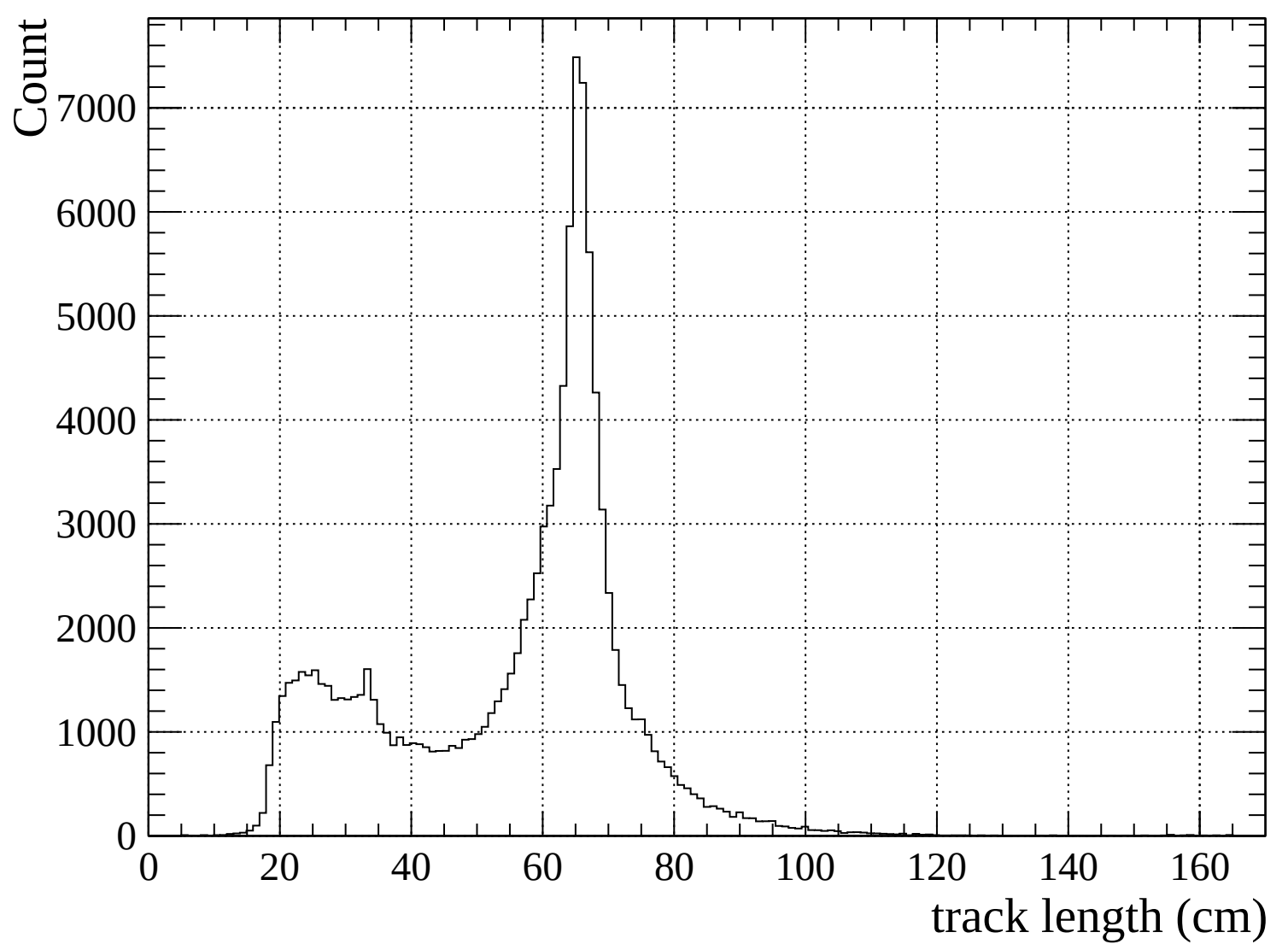
Pulls mean

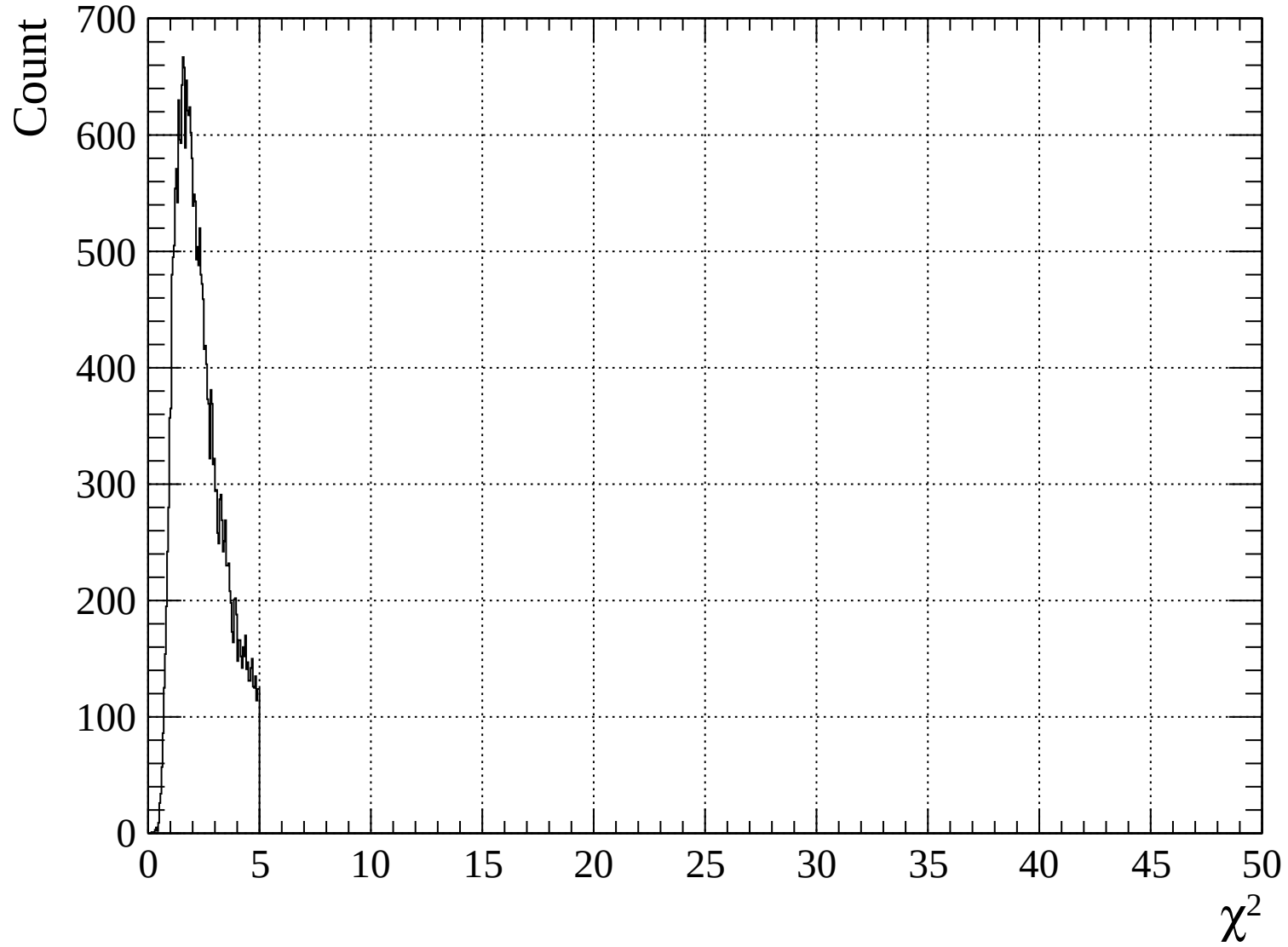


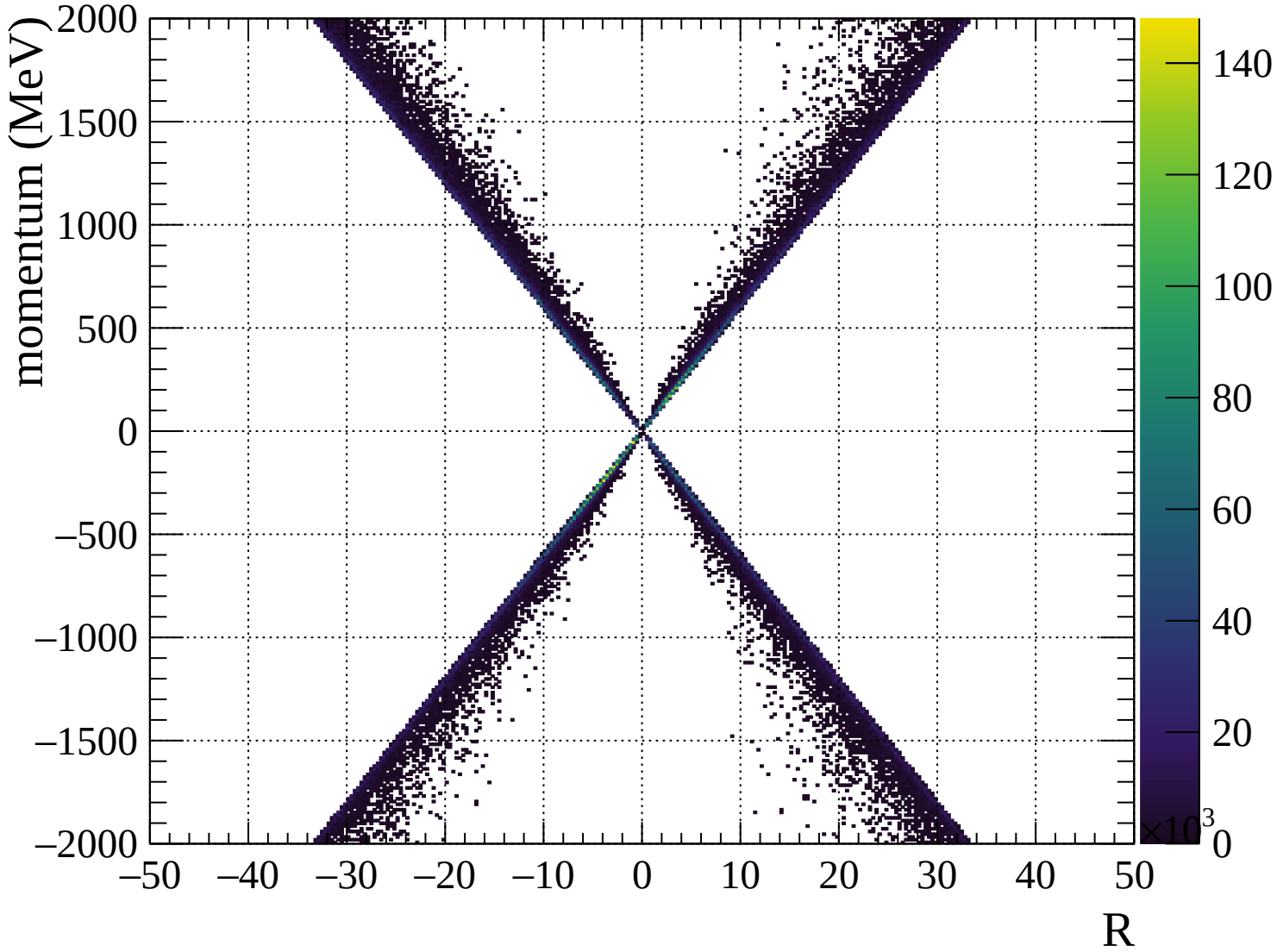


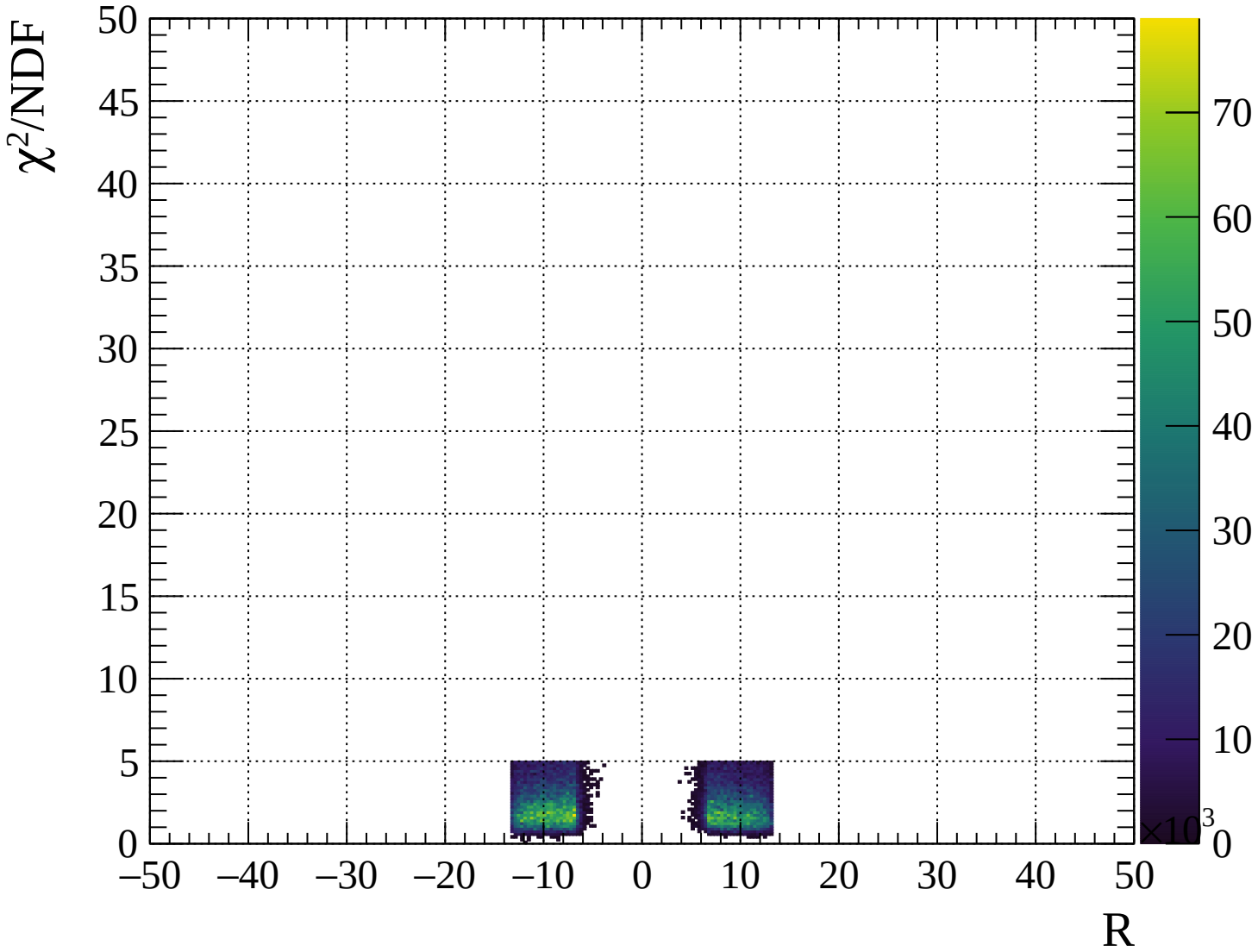






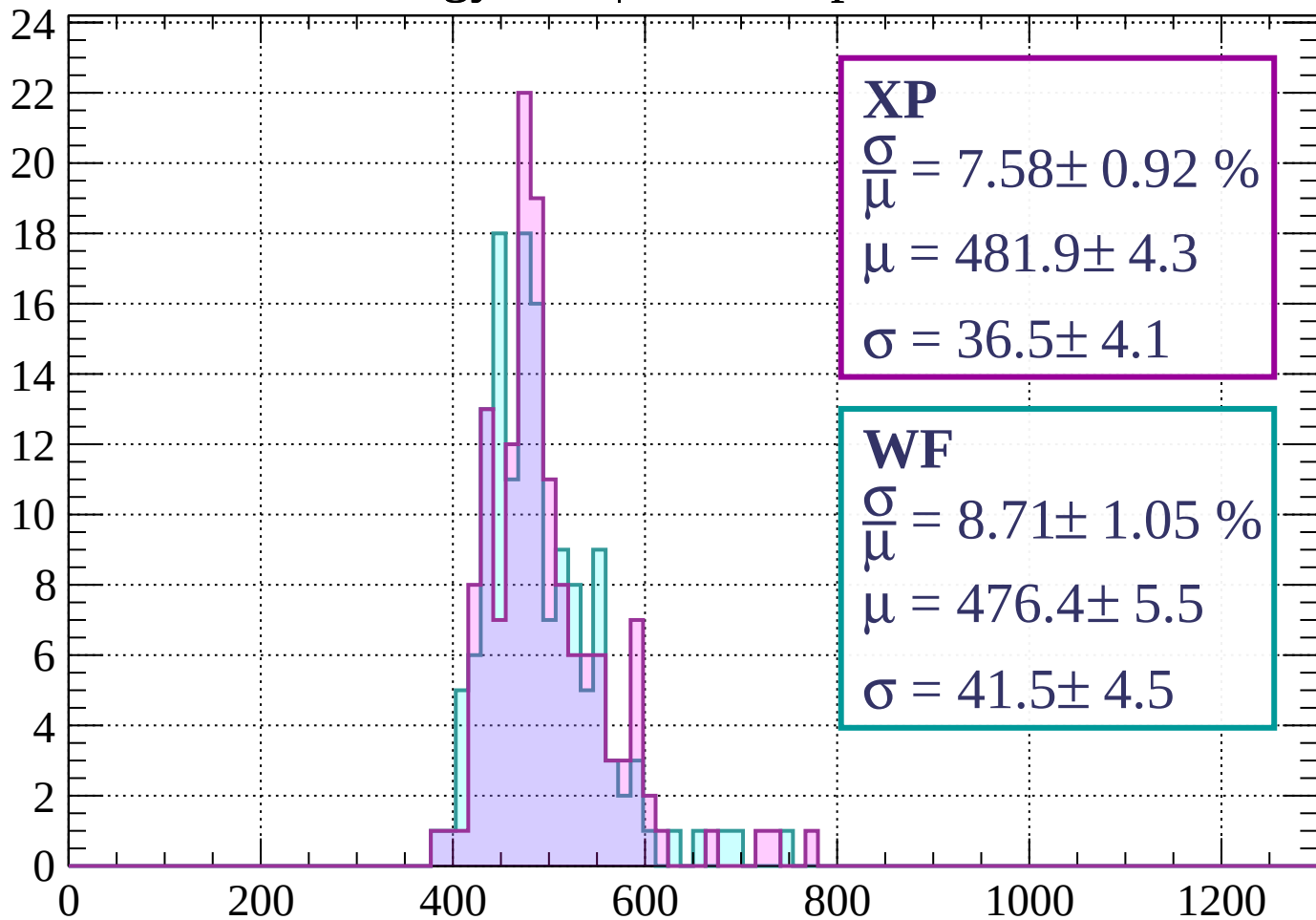






Energy loss | -2000 < p < -1960

Count



XP

$$\frac{\sigma}{\mu} = 7.58 \pm 0.92 \%$$

$$\mu = 481.9 \pm 4.3$$

$$\sigma = 36.5 \pm 4.1$$

WF

$$\frac{\sigma}{\mu} = 8.71 \pm 1.05 \%$$

$$\mu = 476.4 \pm 5.5$$

$$\sigma = 41.5 \pm 4.5$$

dE/dx (ADC counts/cm)

Energy loss | $-1960 < p < -1920$

Count

XP

$$\frac{\sigma}{\mu} = 9.09 \pm 0.66 \%$$

$$\mu = 482.6 \pm 3.1$$

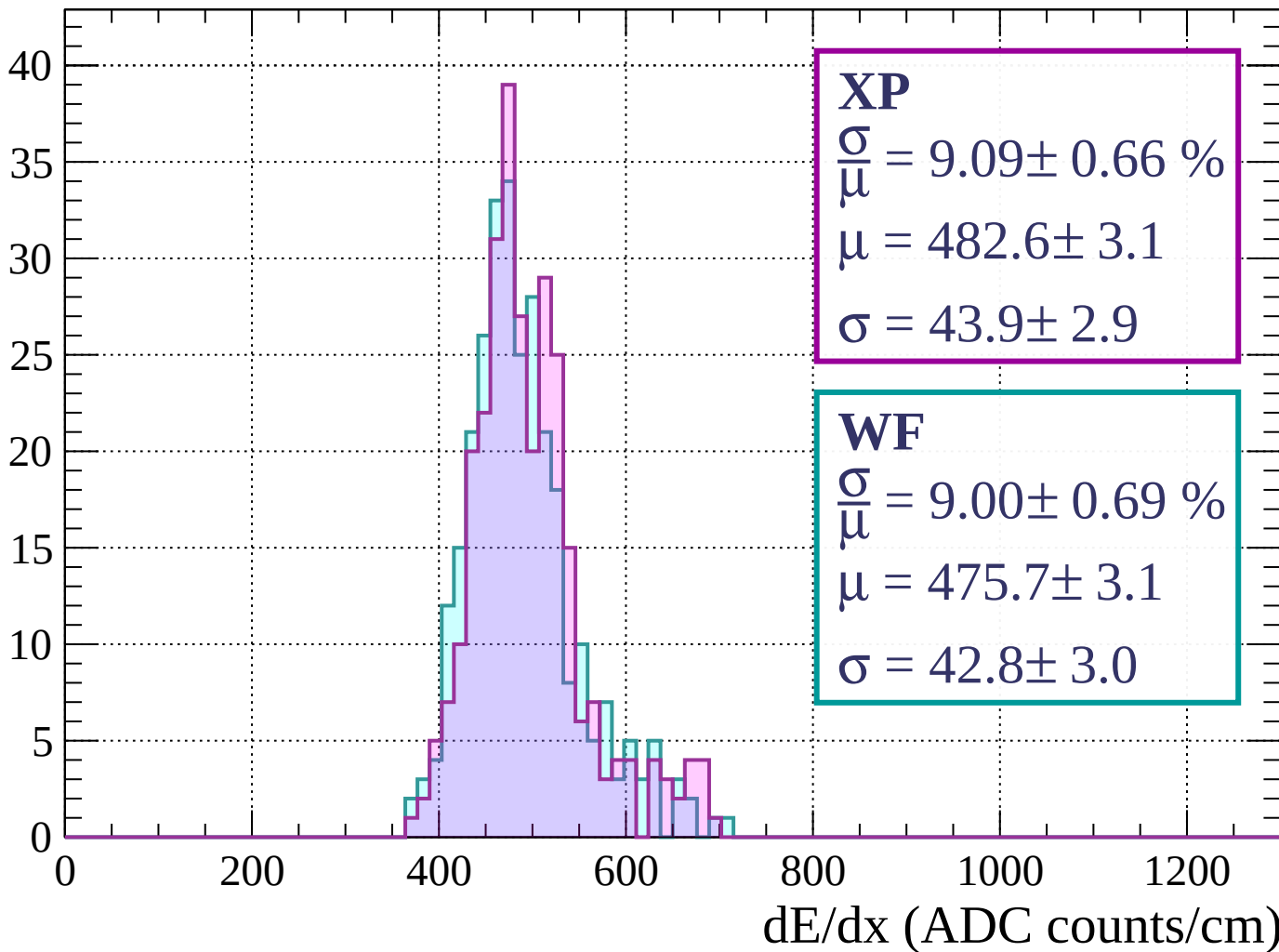
$$\sigma = 43.9 \pm 2.9$$

WF

$$\frac{\sigma}{\mu} = 9.00 \pm 0.69 \%$$

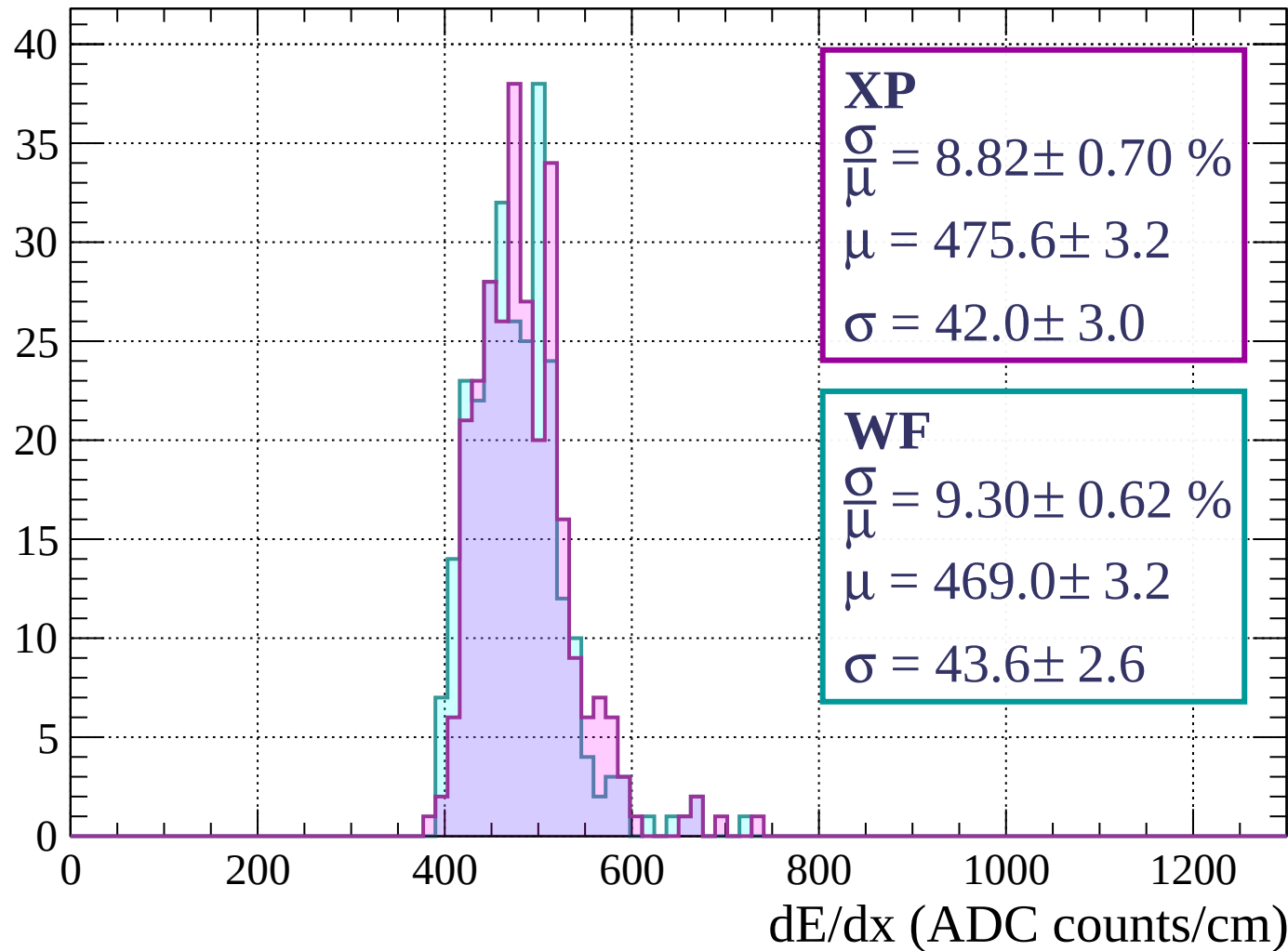
$$\mu = 475.7 \pm 3.1$$

$$\sigma = 42.8 \pm 3.0$$



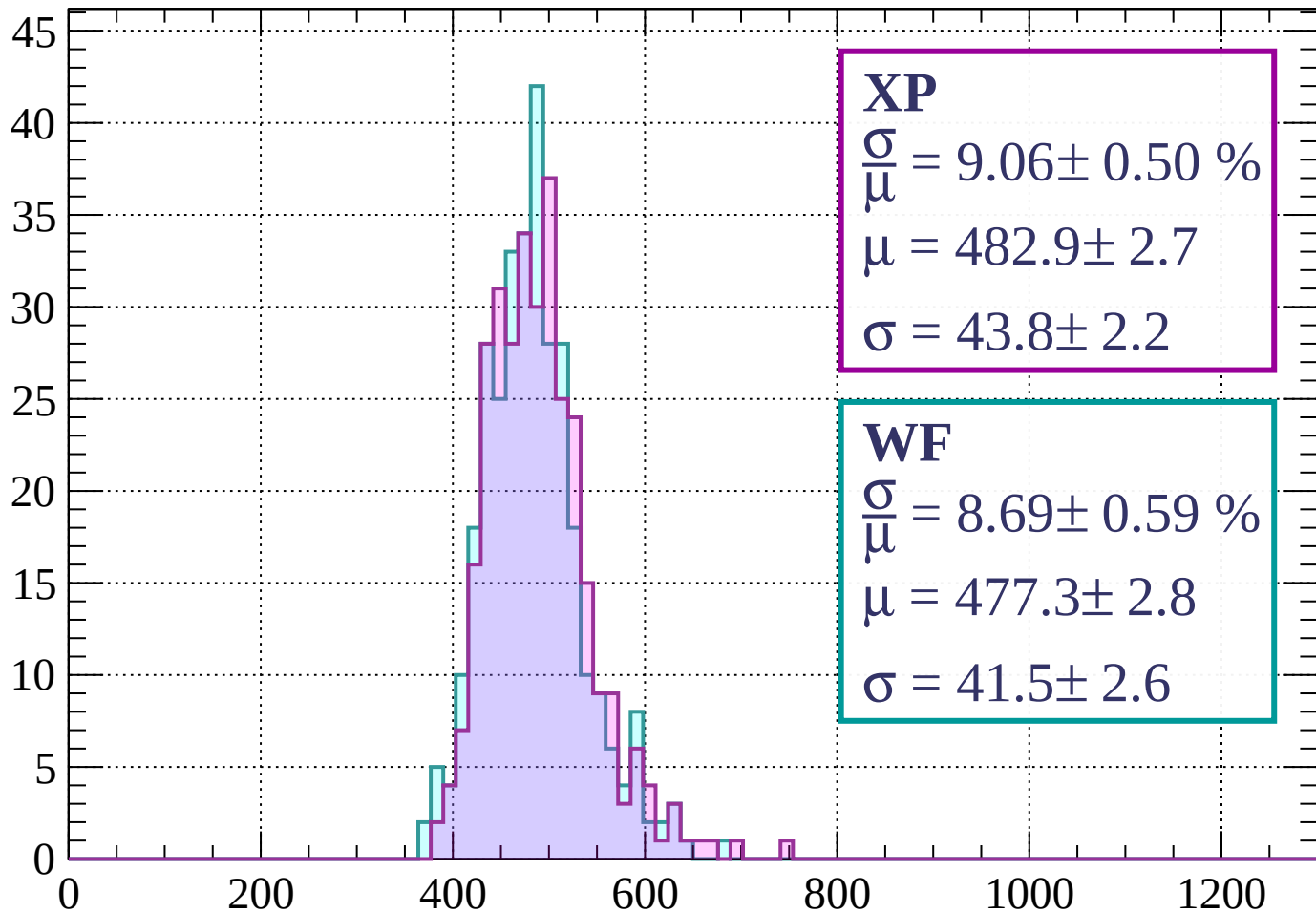
Energy loss | $-1920 < p < -1880$

Count



Energy loss | $-1880 < p < -1840$

Count



XP

$$\frac{\sigma}{\bar{\mu}} = 9.06 \pm 0.50 \%$$

$$\mu = 482.9 \pm 2.7$$

$$\sigma = 43.8 \pm 2.2$$

WF

$$\frac{\sigma}{\bar{\mu}} = 8.69 \pm 0.59 \%$$

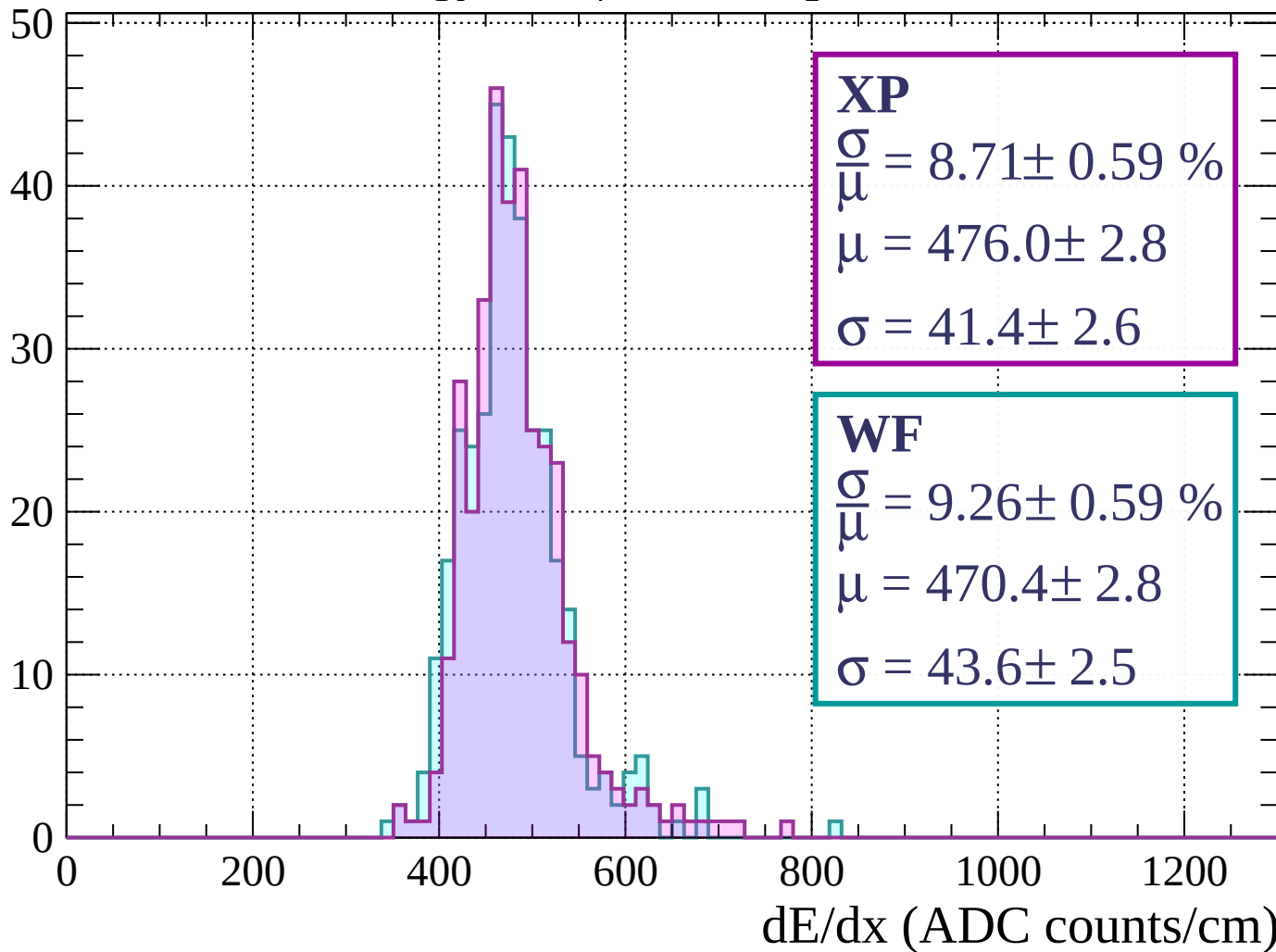
$$\mu = 477.3 \pm 2.8$$

$$\sigma = 41.5 \pm 2.6$$

dE/dx (ADC counts/cm)

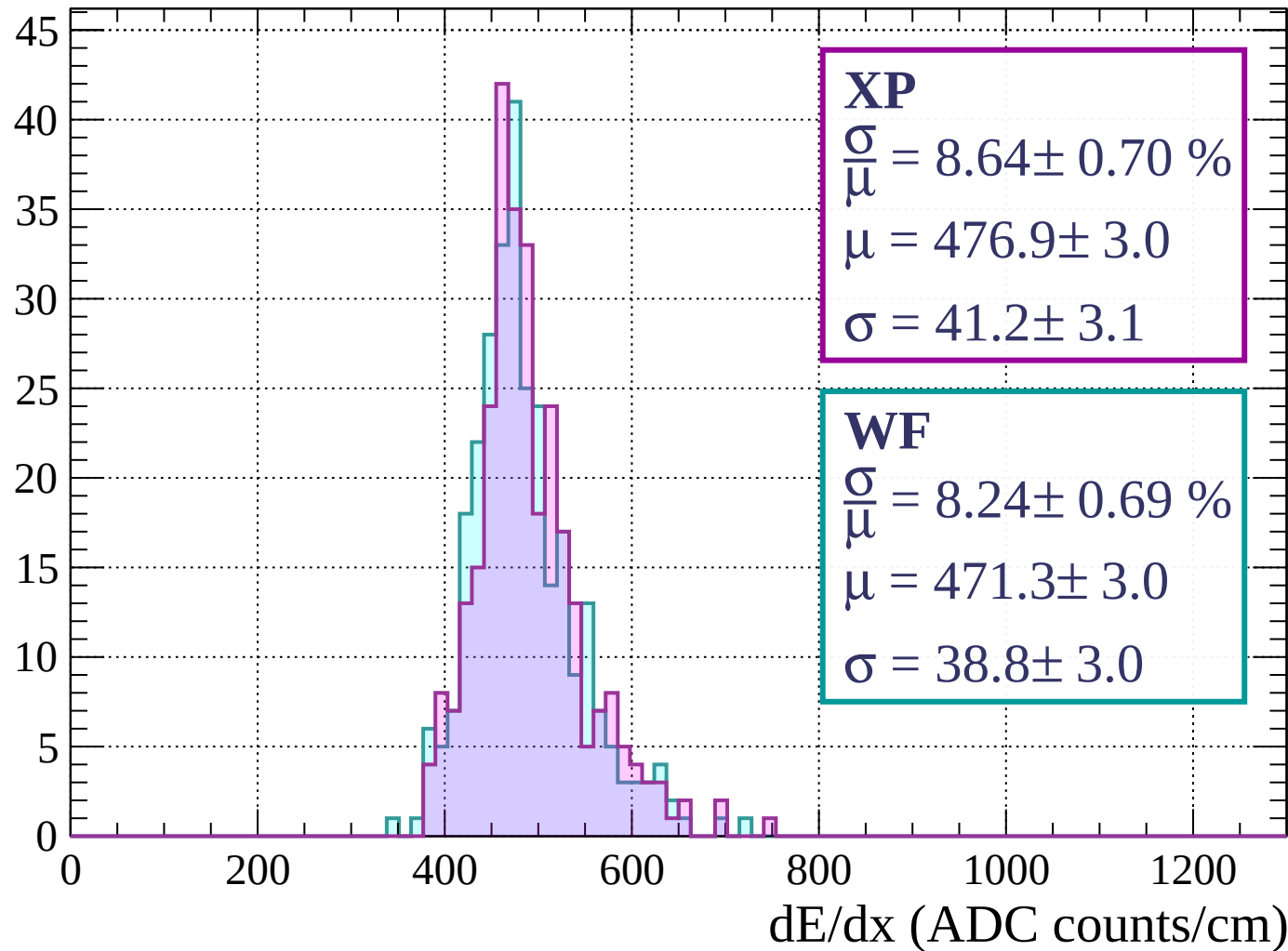
Energy loss | $-1840 < p < -1800$

Count



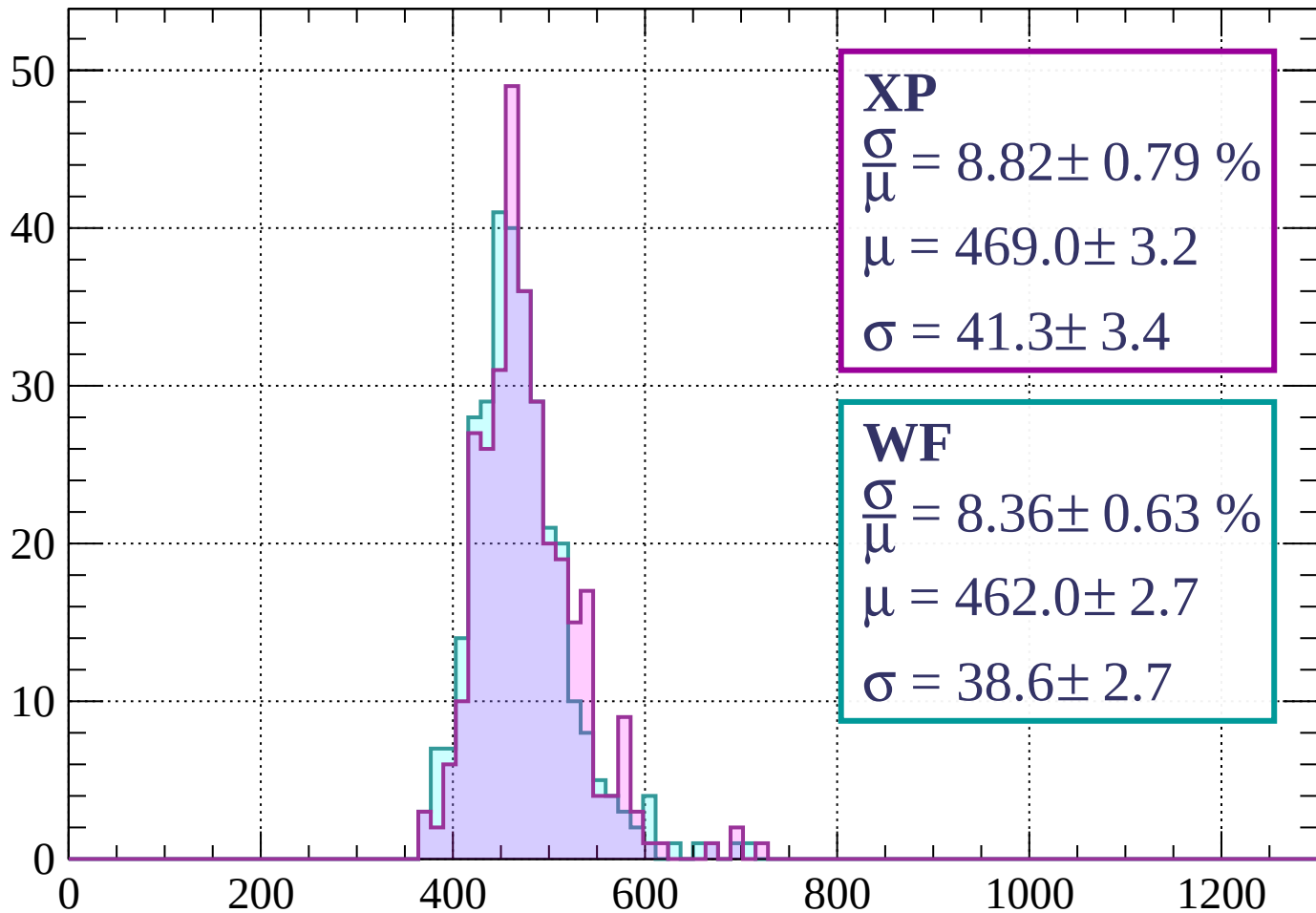
Energy loss | $-1800 < p < -1760$

Count



Energy loss | $-1760 < p < -1720$

Count



XP

$$\frac{\sigma}{\mu} = 8.82 \pm 0.79 \%$$

$$\mu = 469.0 \pm 3.2$$

$$\sigma = 41.3 \pm 3.4$$

WF

$$\frac{\sigma}{\mu} = 8.36 \pm 0.63 \%$$

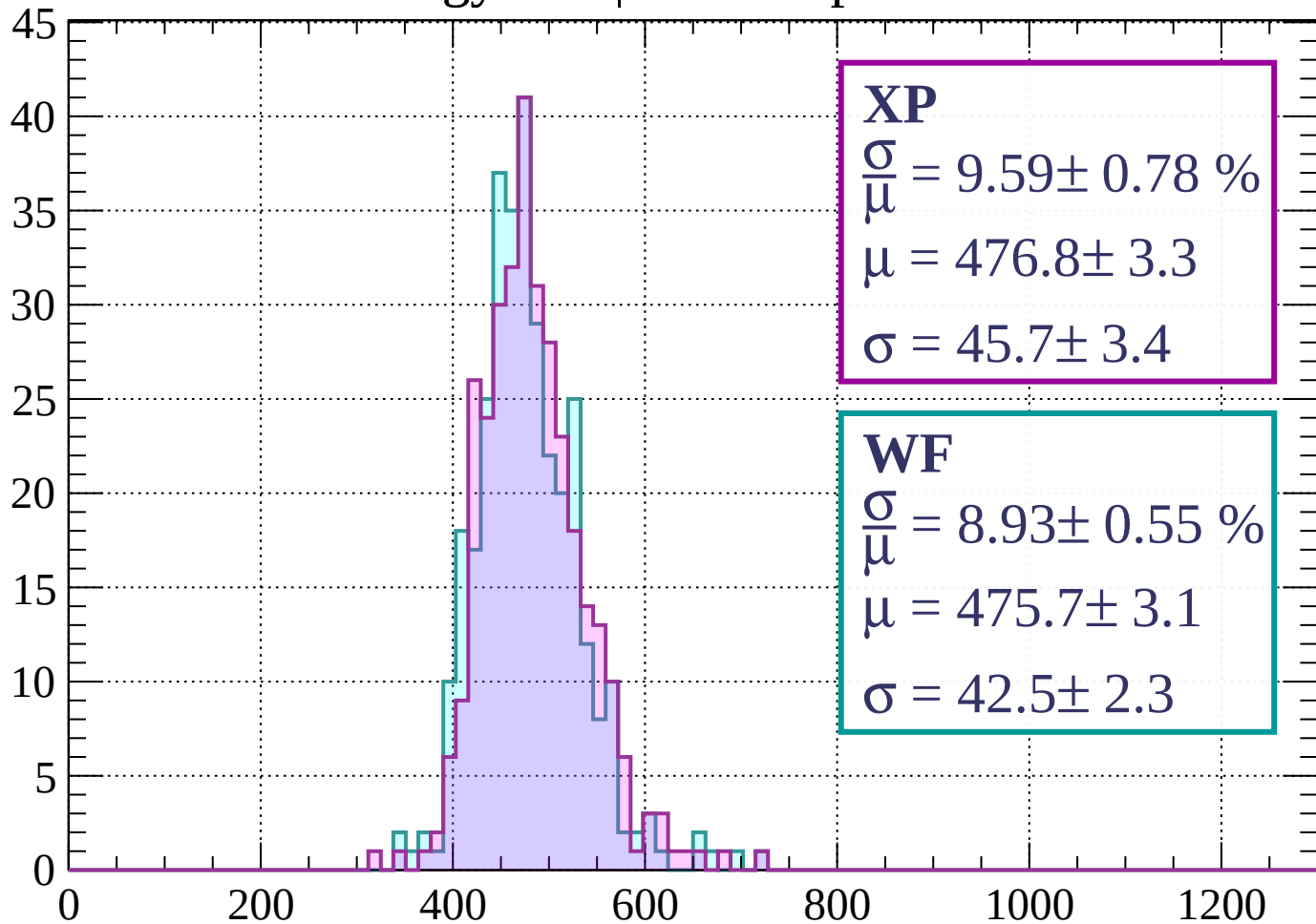
$$\mu = 462.0 \pm 2.7$$

$$\sigma = 38.6 \pm 2.7$$

dE/dx (ADC counts/cm)

Energy loss | $-1720 < p < -1680$

Count



XP

$$\frac{\sigma}{\mu} = 9.59 \pm 0.78 \%$$

$$\mu = 476.8 \pm 3.3$$

$$\sigma = 45.7 \pm 3.4$$

WF

$$\frac{\sigma}{\mu} = 8.93 \pm 0.55 \%$$

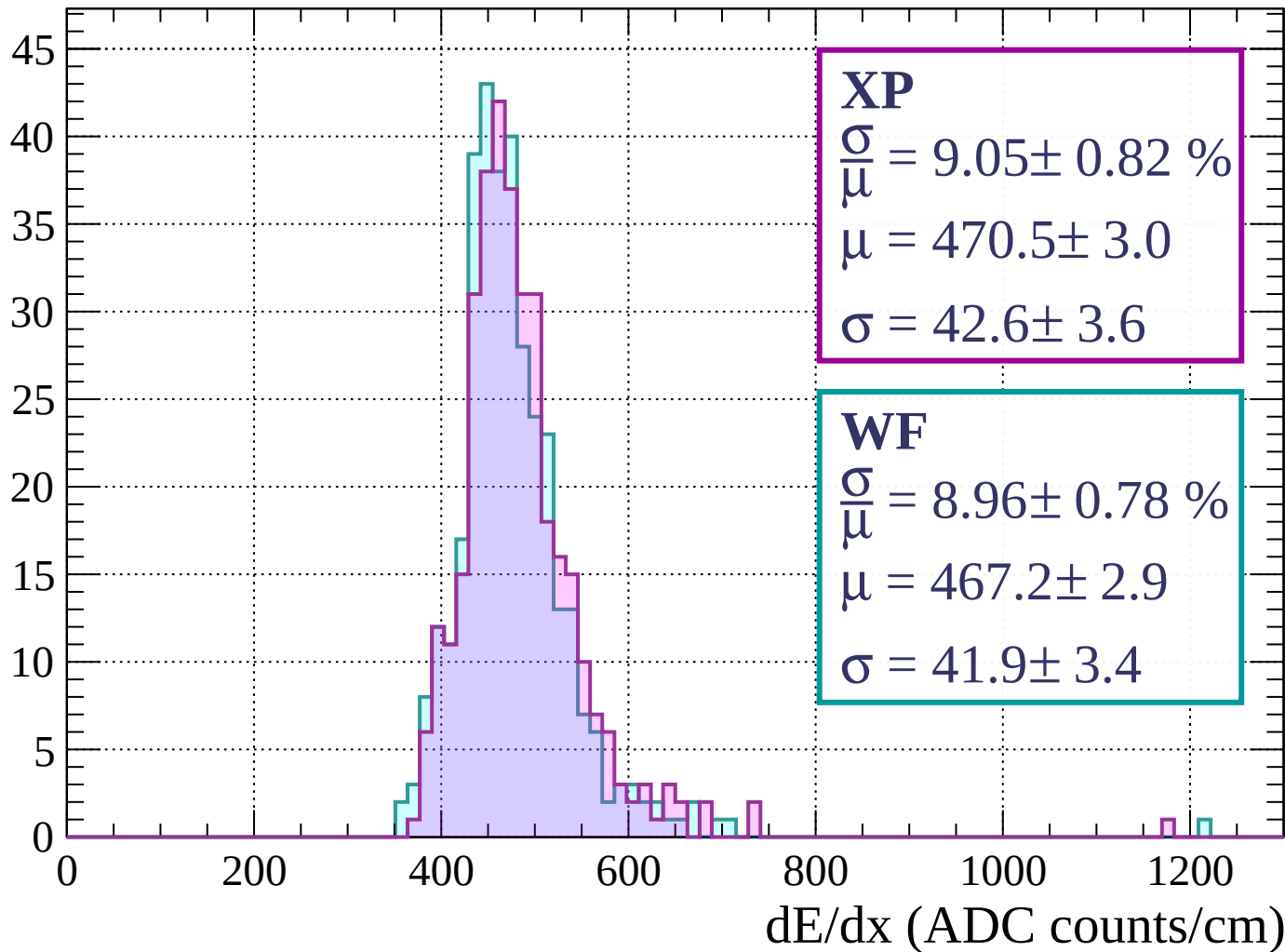
$$\mu = 475.7 \pm 3.1$$

$$\sigma = 42.5 \pm 2.3$$

dE/dx (ADC counts/cm)

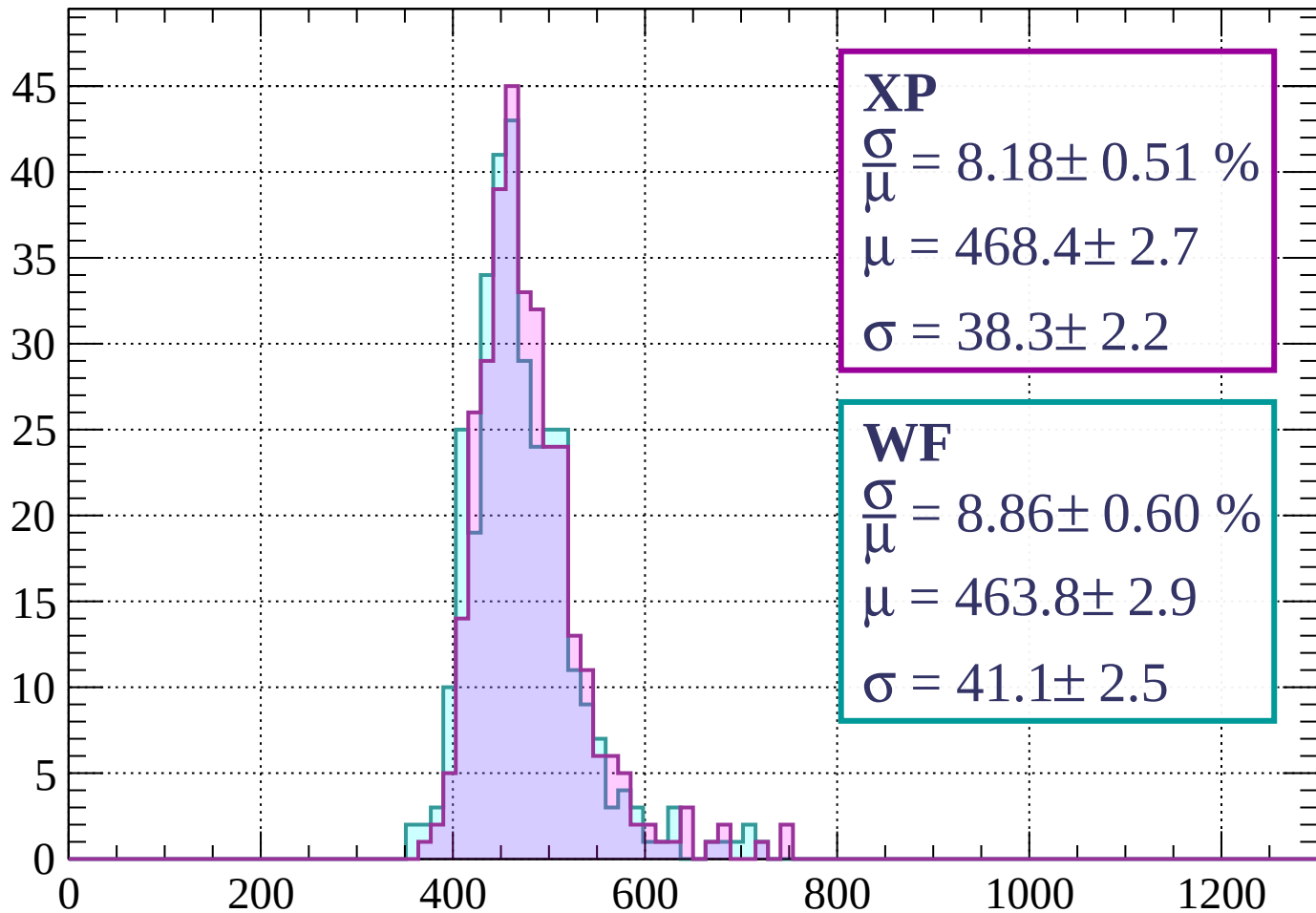
Energy loss | $-1680 < p < -1640$

Count



Energy loss | $-1640 < p < -1600$

Count



XP

$$\frac{\sigma}{\mu} = 8.18 \pm 0.51 \%$$

$$\mu = 468.4 \pm 2.7$$

$$\sigma = 38.3 \pm 2.2$$

WF

$$\frac{\sigma}{\mu} = 8.86 \pm 0.60 \%$$

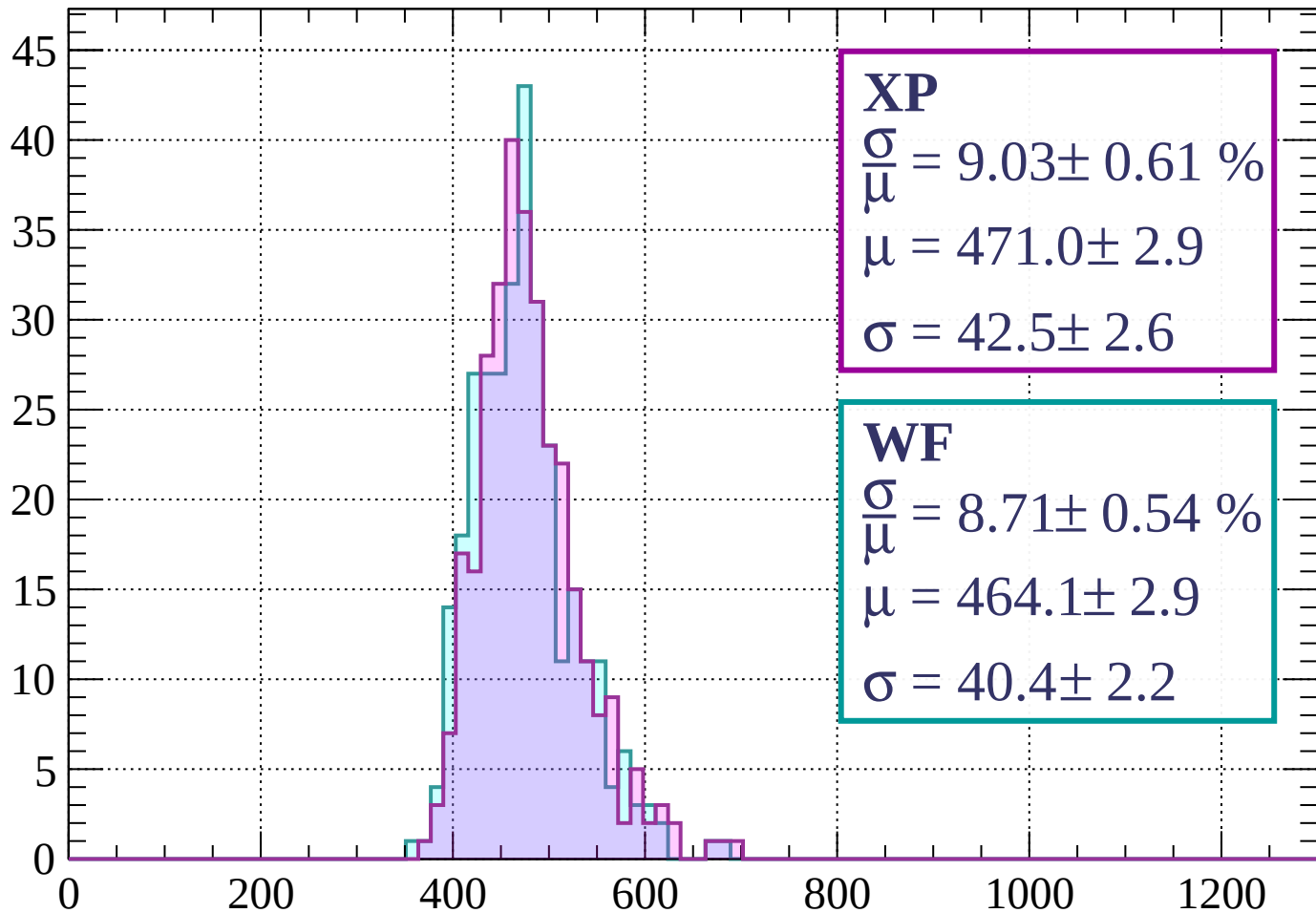
$$\mu = 463.8 \pm 2.9$$

$$\sigma = 41.1 \pm 2.5$$

dE/dx (ADC counts/cm)

Energy loss | $-1600 < p < -1560$

Count



XP

$$\frac{\sigma}{\mu} = 9.03 \pm 0.61 \%$$

$$\mu = 471.0 \pm 2.9$$

$$\sigma = 42.5 \pm 2.6$$

WF

$$\frac{\sigma}{\mu} = 8.71 \pm 0.54 \%$$

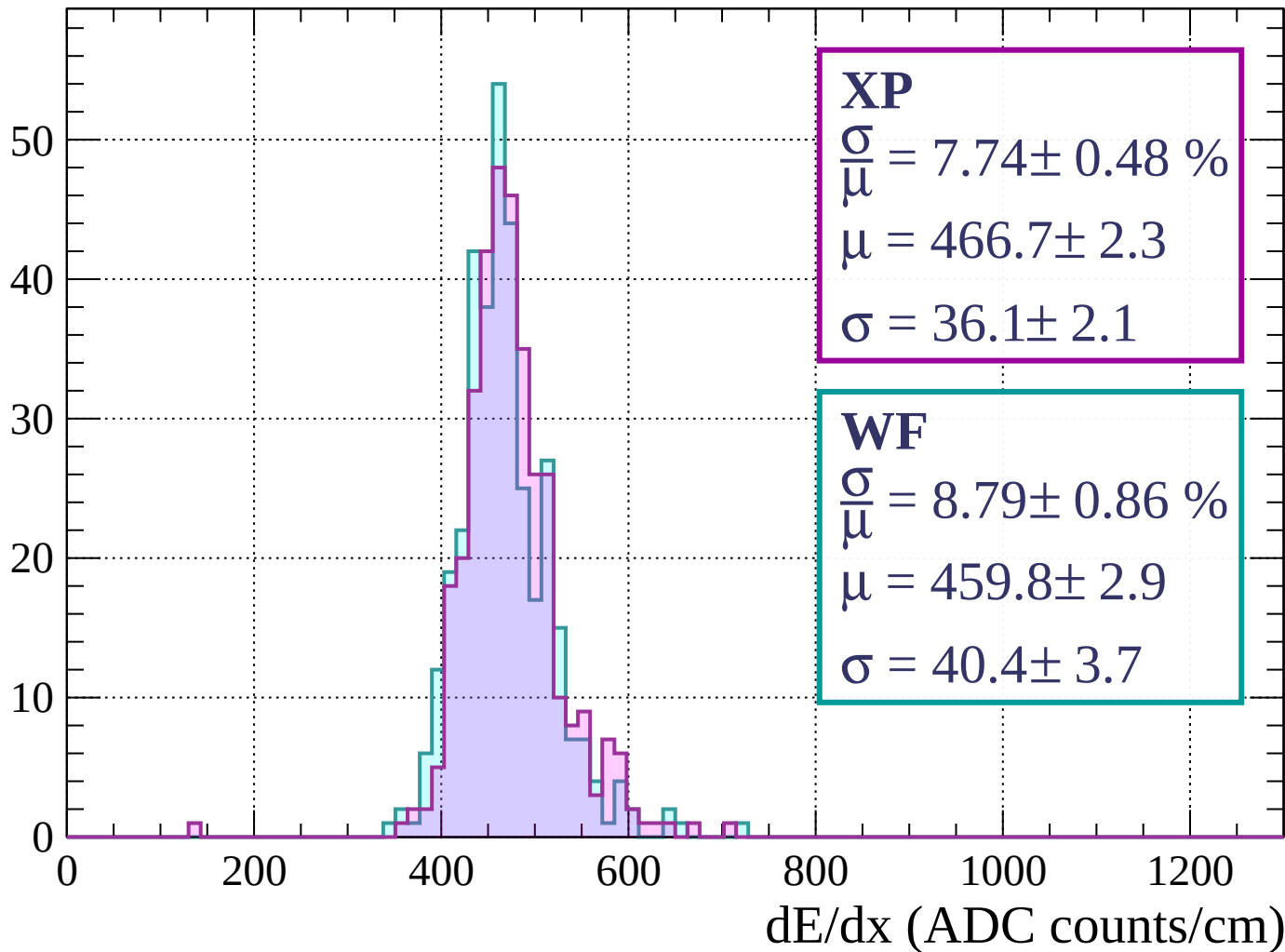
$$\mu = 464.1 \pm 2.9$$

$$\sigma = 40.4 \pm 2.2$$

dE/dx (ADC counts/cm)

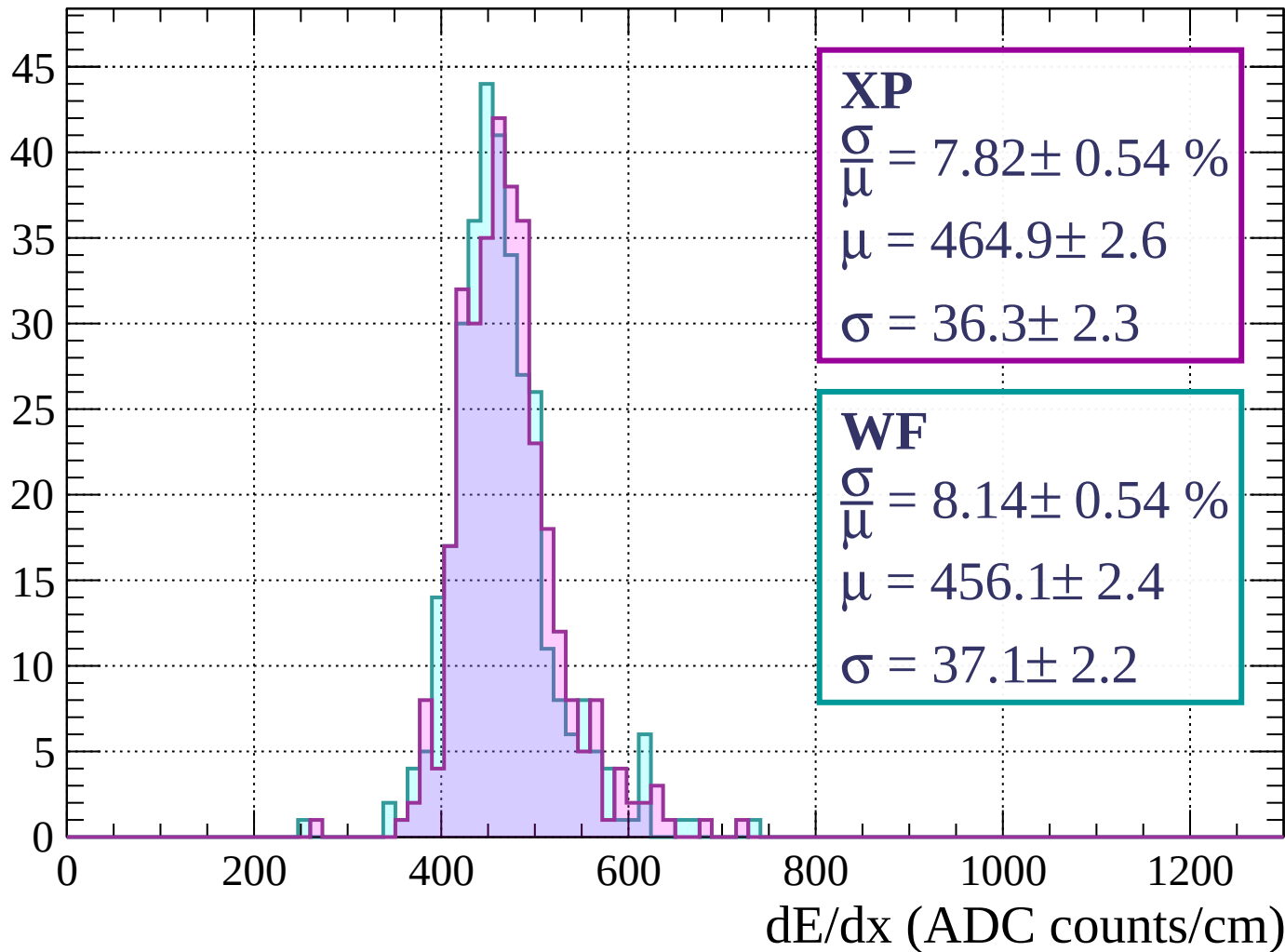
Energy loss | $-1560 < p < -1520$

Count



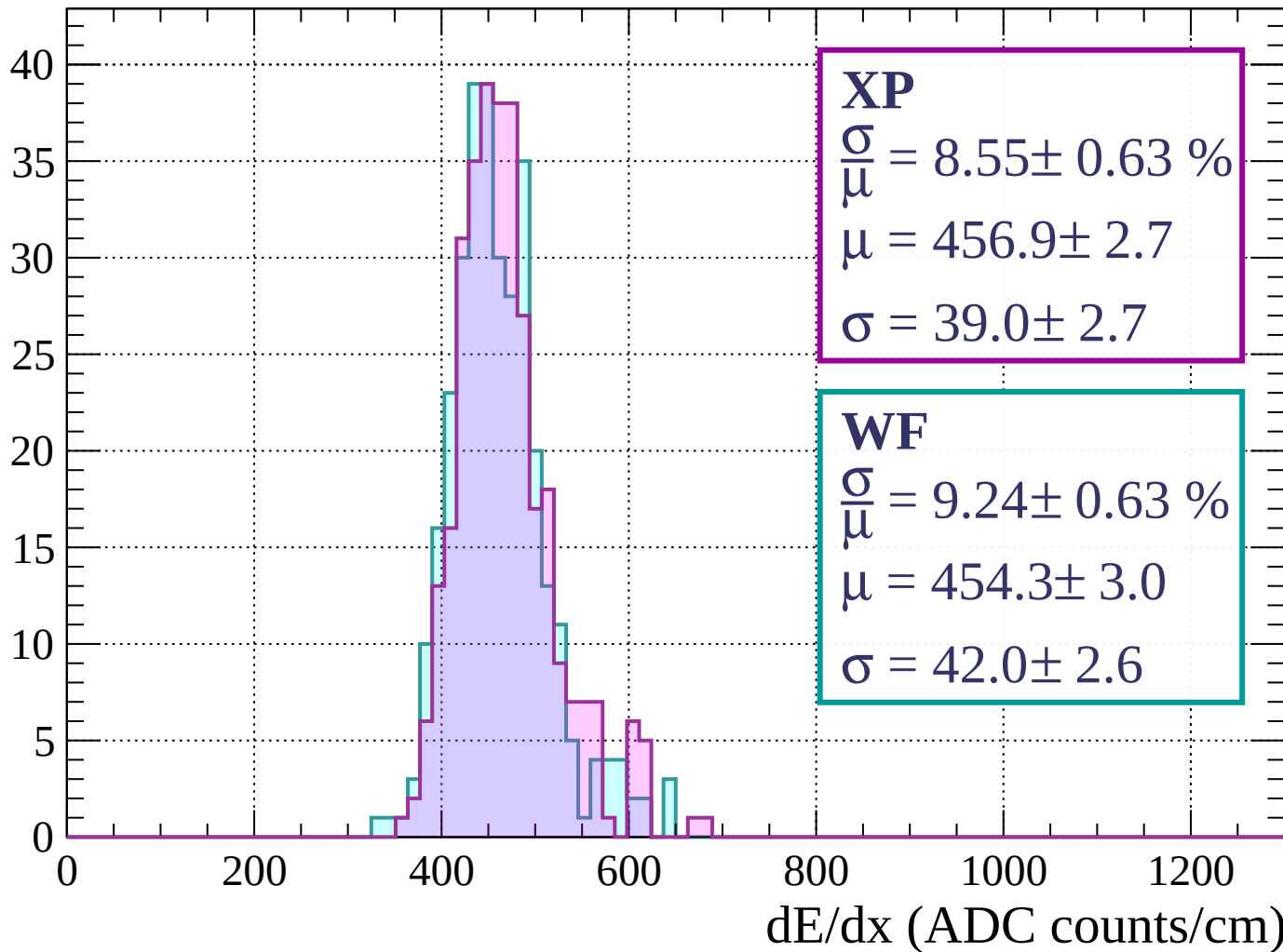
Energy loss | $-1520 < p < -1480$

Count



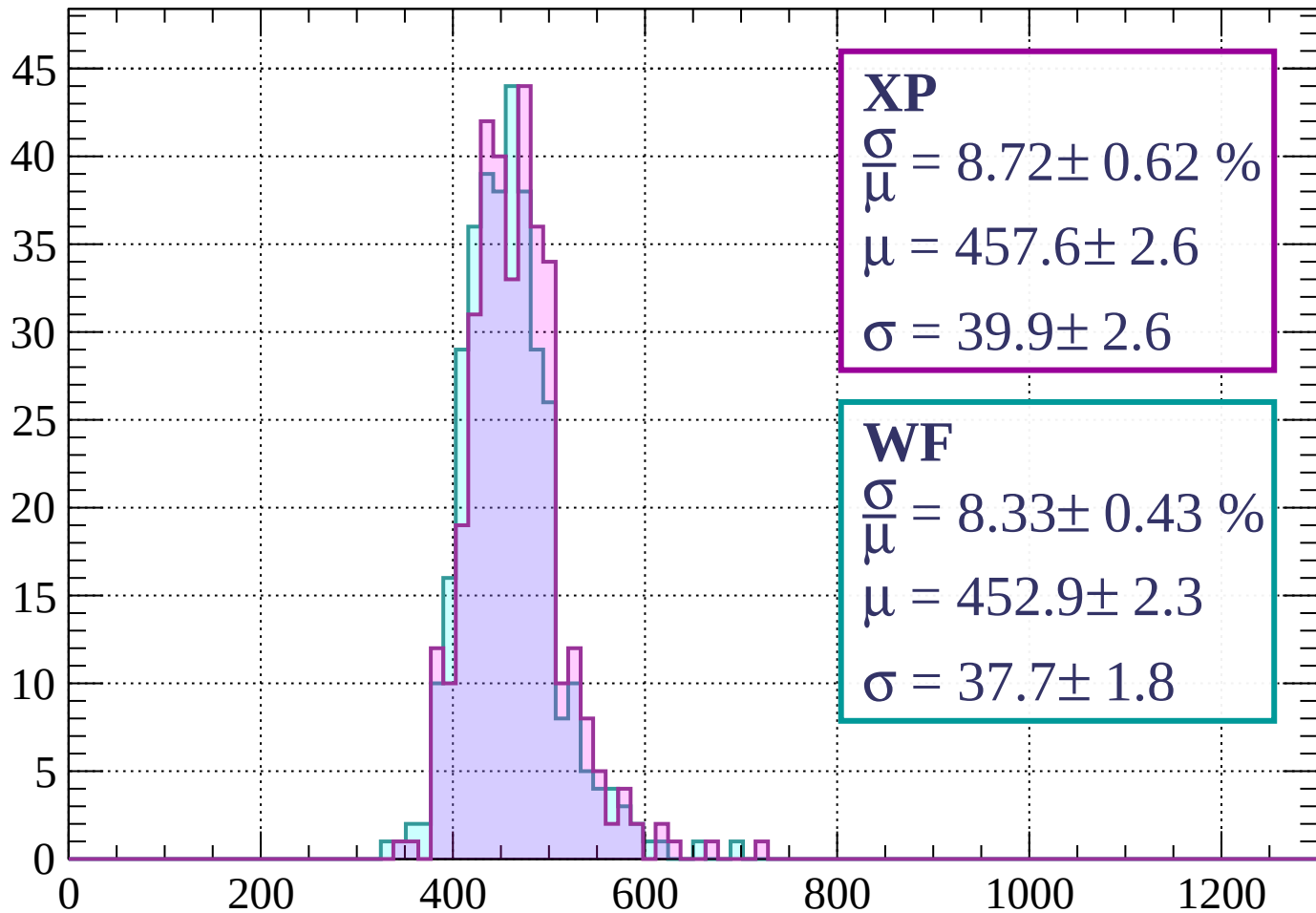
Energy loss | $-1480 < p < -1440$

Count



Energy loss | -1440 < p < -1400

Count



XP

$$\frac{\sigma}{\mu} = 8.72 \pm 0.62 \%$$

$$\mu = 457.6 \pm 2.6$$

$$\sigma = 39.9 \pm 2.6$$

WF

$$\frac{\sigma}{\mu} = 8.33 \pm 0.43 \%$$

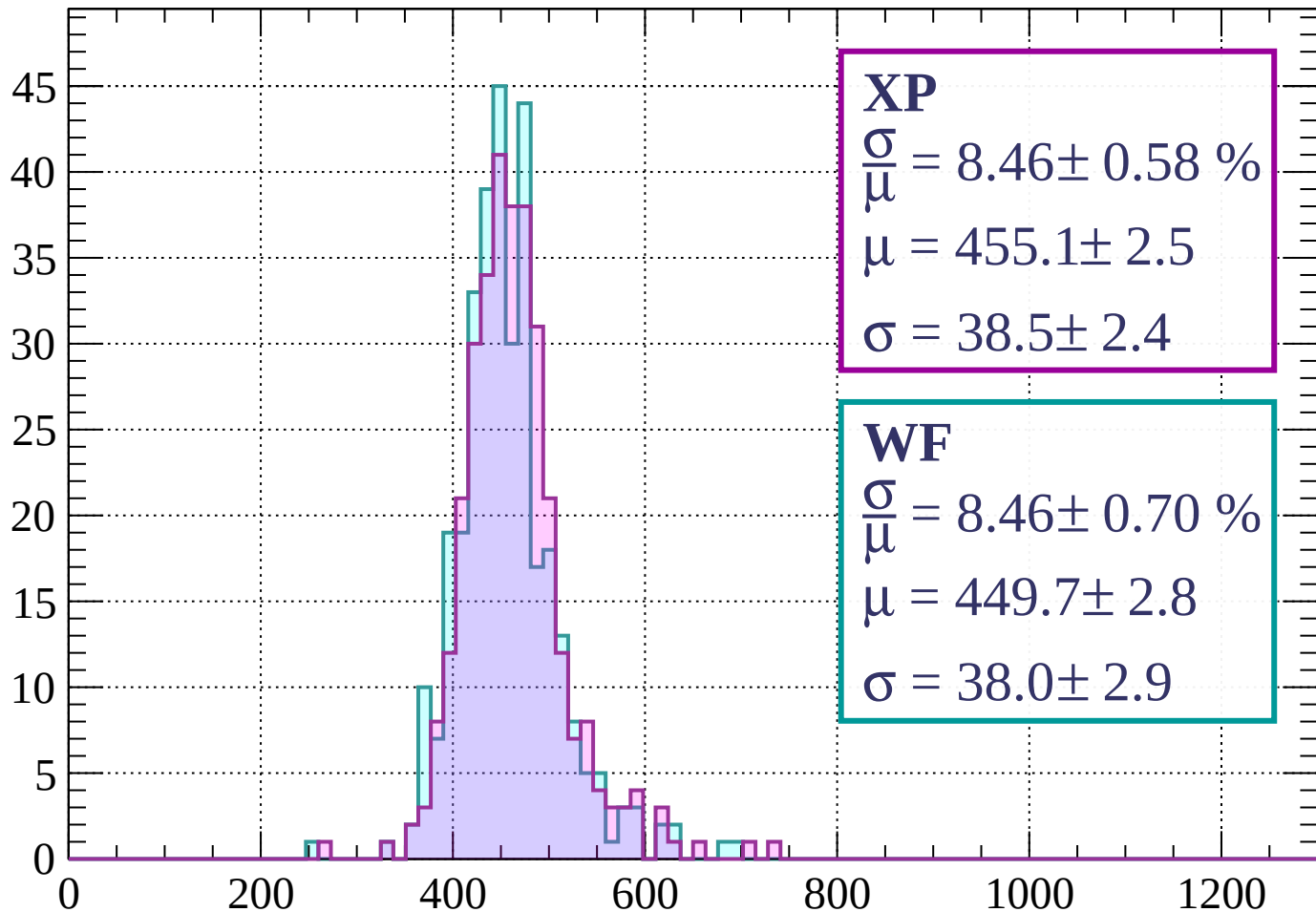
$$\mu = 452.9 \pm 2.3$$

$$\sigma = 37.7 \pm 1.8$$

dE/dx (ADC counts/cm)

Energy loss | $-1400 < p < -1360$

Count



XP

$$\frac{\sigma}{\mu} = 8.46 \pm 0.58 \%$$

$$\mu = 455.1 \pm 2.5$$

$$\sigma = 38.5 \pm 2.4$$

WF

$$\frac{\sigma}{\mu} = 8.46 \pm 0.70 \%$$

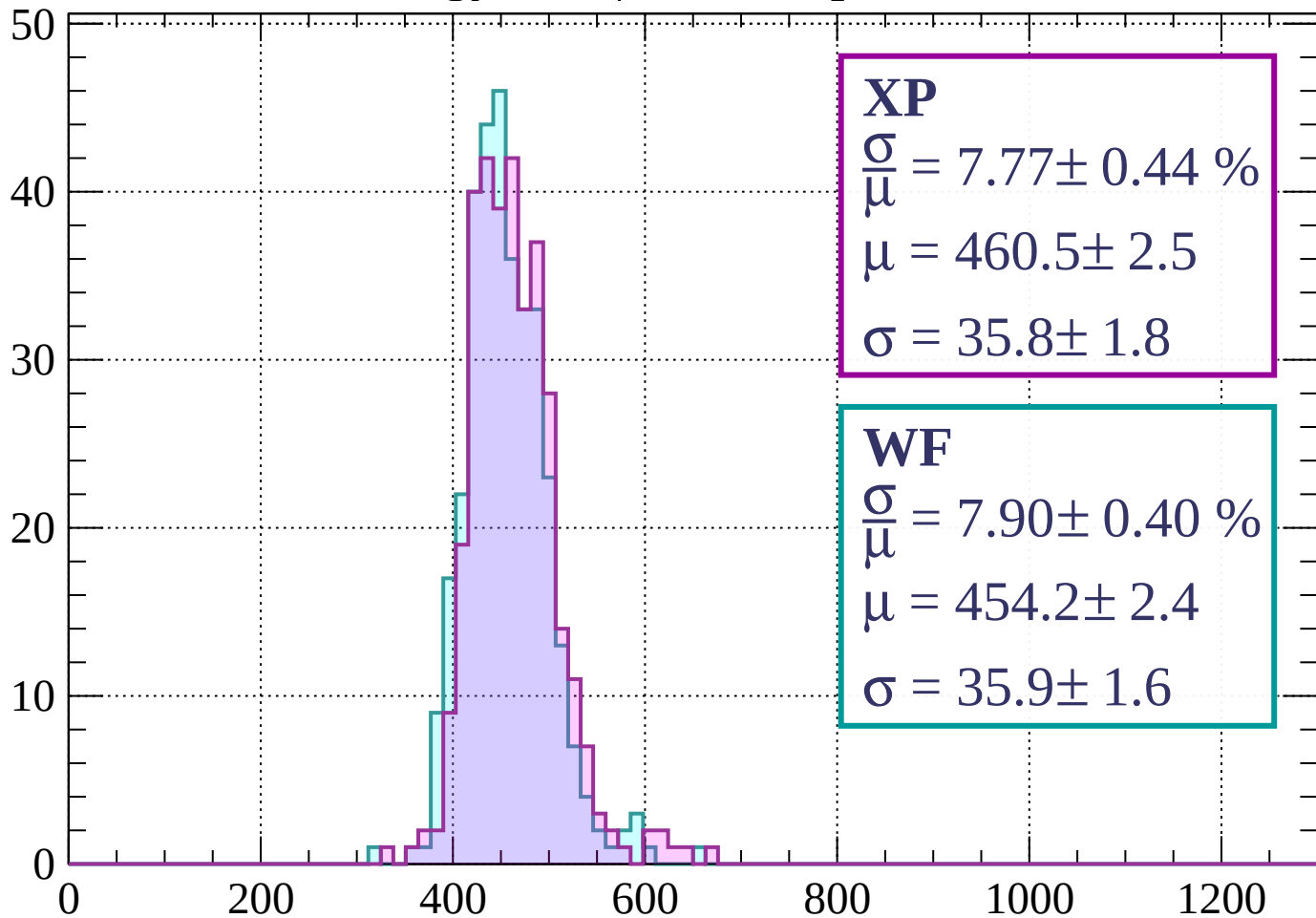
$$\mu = 449.7 \pm 2.8$$

$$\sigma = 38.0 \pm 2.9$$

dE/dx (ADC counts/cm)

Energy loss | $-1360 < p < -1320$

Count



XP

$$\frac{\sigma}{\mu} = 7.77 \pm 0.44 \%$$

$$\mu = 460.5 \pm 2.5$$

$$\sigma = 35.8 \pm 1.8$$

WF

$$\frac{\sigma}{\mu} = 7.90 \pm 0.40 \%$$

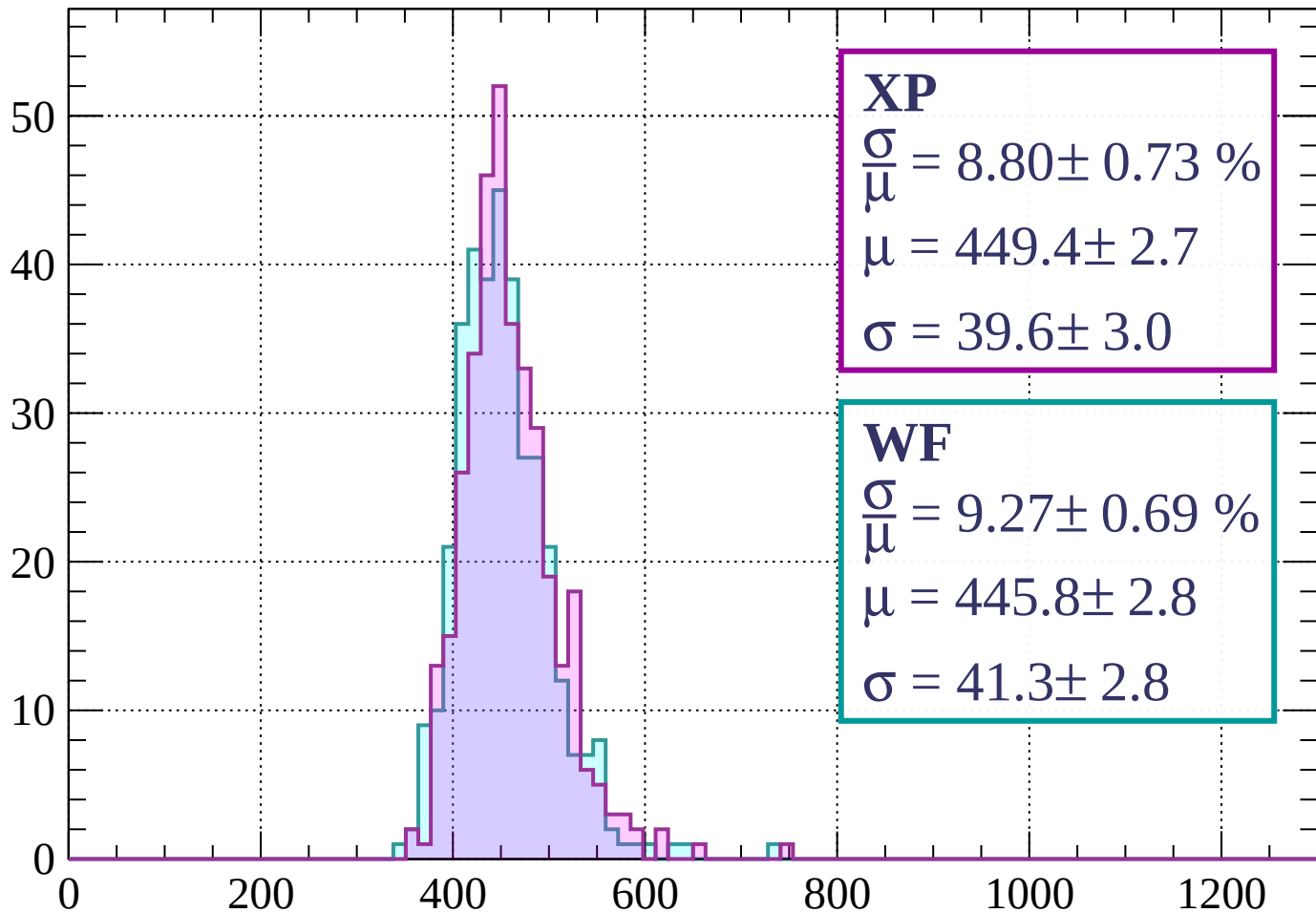
$$\mu = 454.2 \pm 2.4$$

$$\sigma = 35.9 \pm 1.6$$

dE/dx (ADC counts/cm)

Energy loss | $-1320 < p < -1280$

Count



XP

$$\frac{\sigma}{\mu} = 8.80 \pm 0.73 \%$$

$$\mu = 449.4 \pm 2.7$$

$$\sigma = 39.6 \pm 3.0$$

WF

$$\frac{\sigma}{\mu} = 9.27 \pm 0.69 \%$$

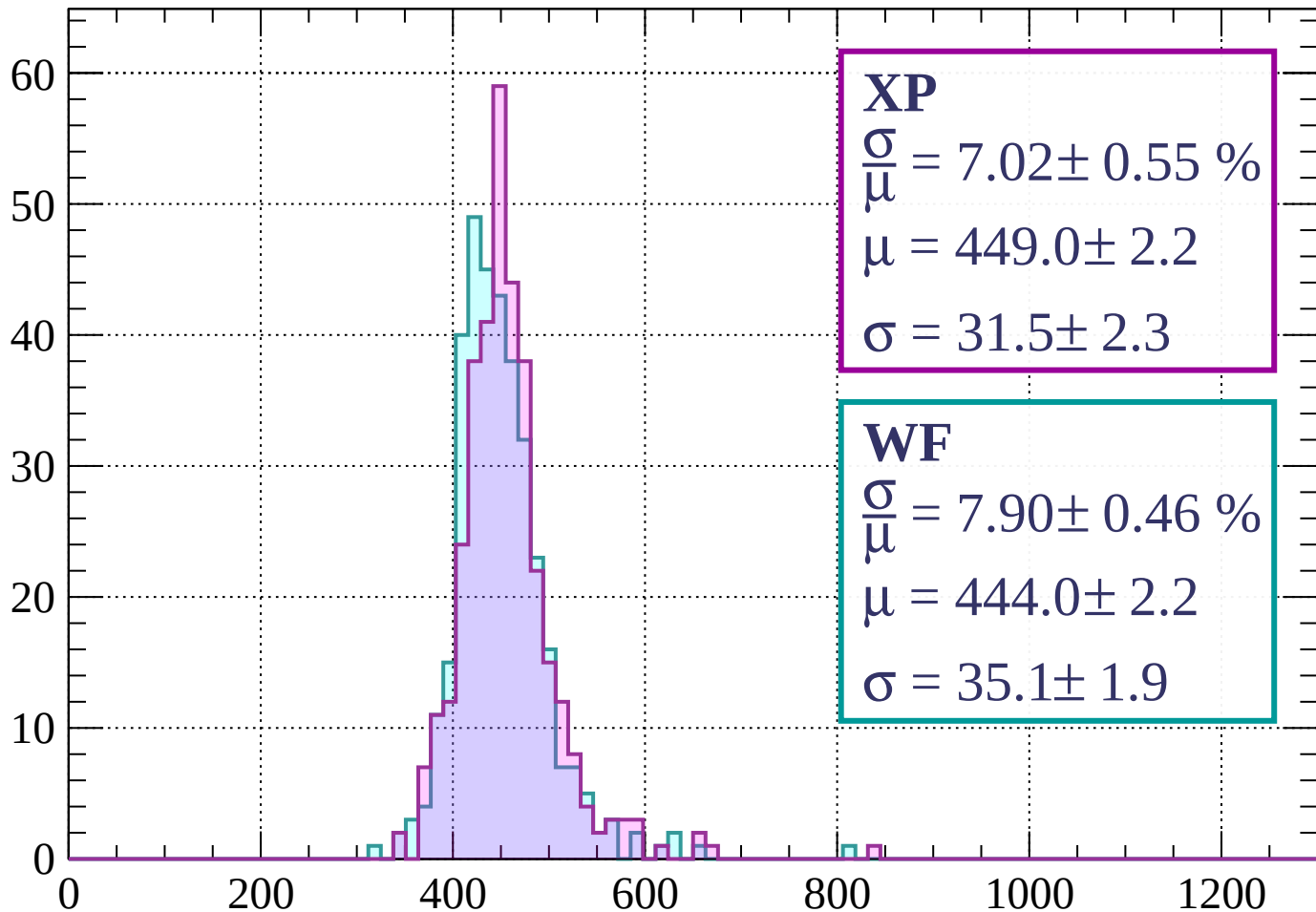
$$\mu = 445.8 \pm 2.8$$

$$\sigma = 41.3 \pm 2.8$$

dE/dx (ADC counts/cm)

Energy loss | $-1280 < p < -1240$

Count



XP

$$\frac{\sigma}{\mu} = 7.02 \pm 0.55 \%$$

$$\mu = 449.0 \pm 2.2$$

$$\sigma = 31.5 \pm 2.3$$

WF

$$\frac{\sigma}{\mu} = 7.90 \pm 0.46 \%$$

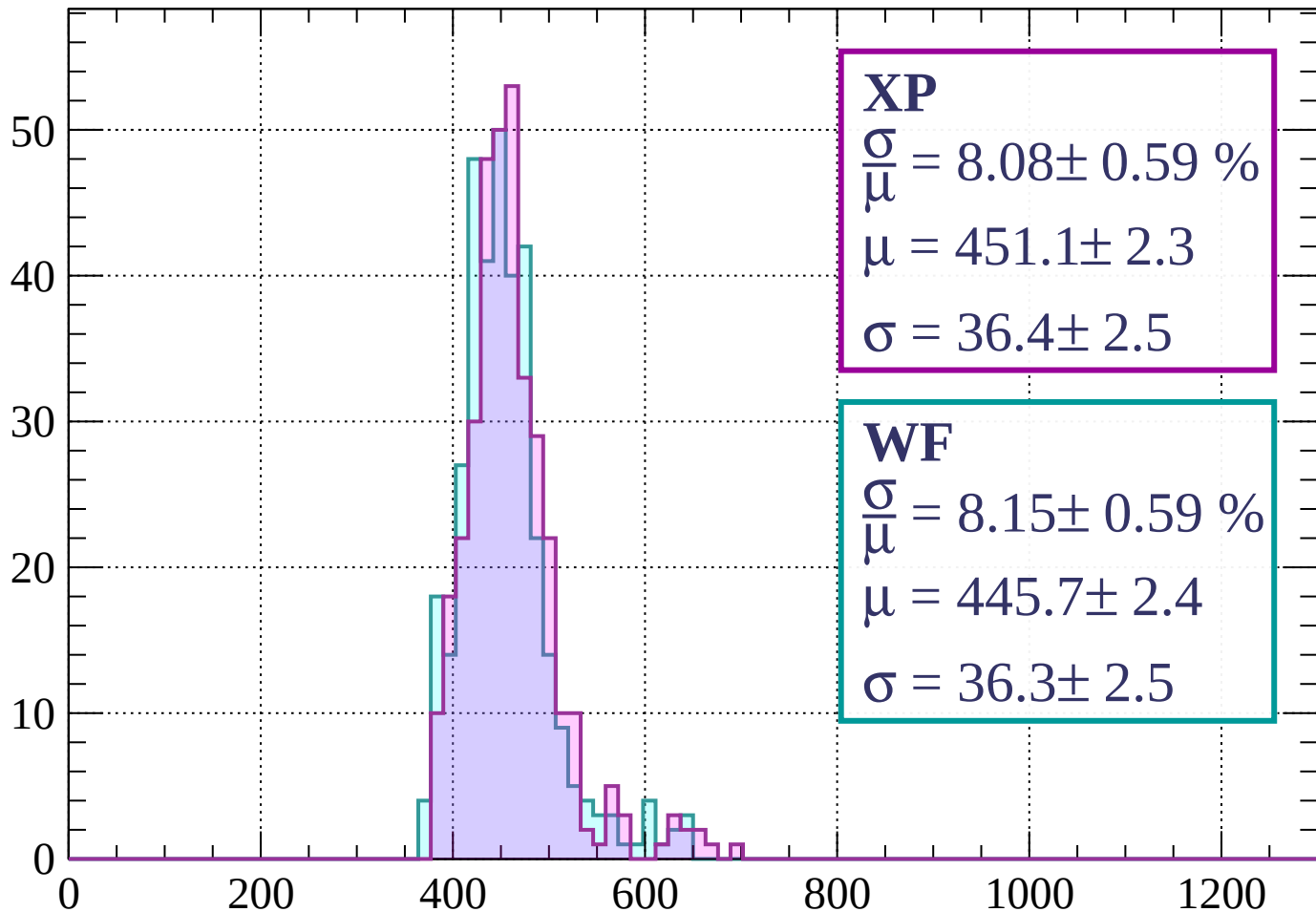
$$\mu = 444.0 \pm 2.2$$

$$\sigma = 35.1 \pm 1.9$$

dE/dx (ADC counts/cm)

Energy loss | $-1240 < p < -1200$

Count



XP

$$\frac{\sigma}{\mu} = 8.08 \pm 0.59 \%$$

$$\mu = 451.1 \pm 2.3$$

$$\sigma = 36.4 \pm 2.5$$

WF

$$\frac{\sigma}{\mu} = 8.15 \pm 0.59 \%$$

$$\mu = 445.7 \pm 2.4$$

$$\sigma = 36.3 \pm 2.5$$

dE/dx (ADC counts/cm)

Energy loss | $-1200 < p < -1160$

Count

XP

$$\frac{\sigma}{\mu} = 8.86 \pm 0.61 \%$$

$$\mu = 443.1 \pm 2.5$$

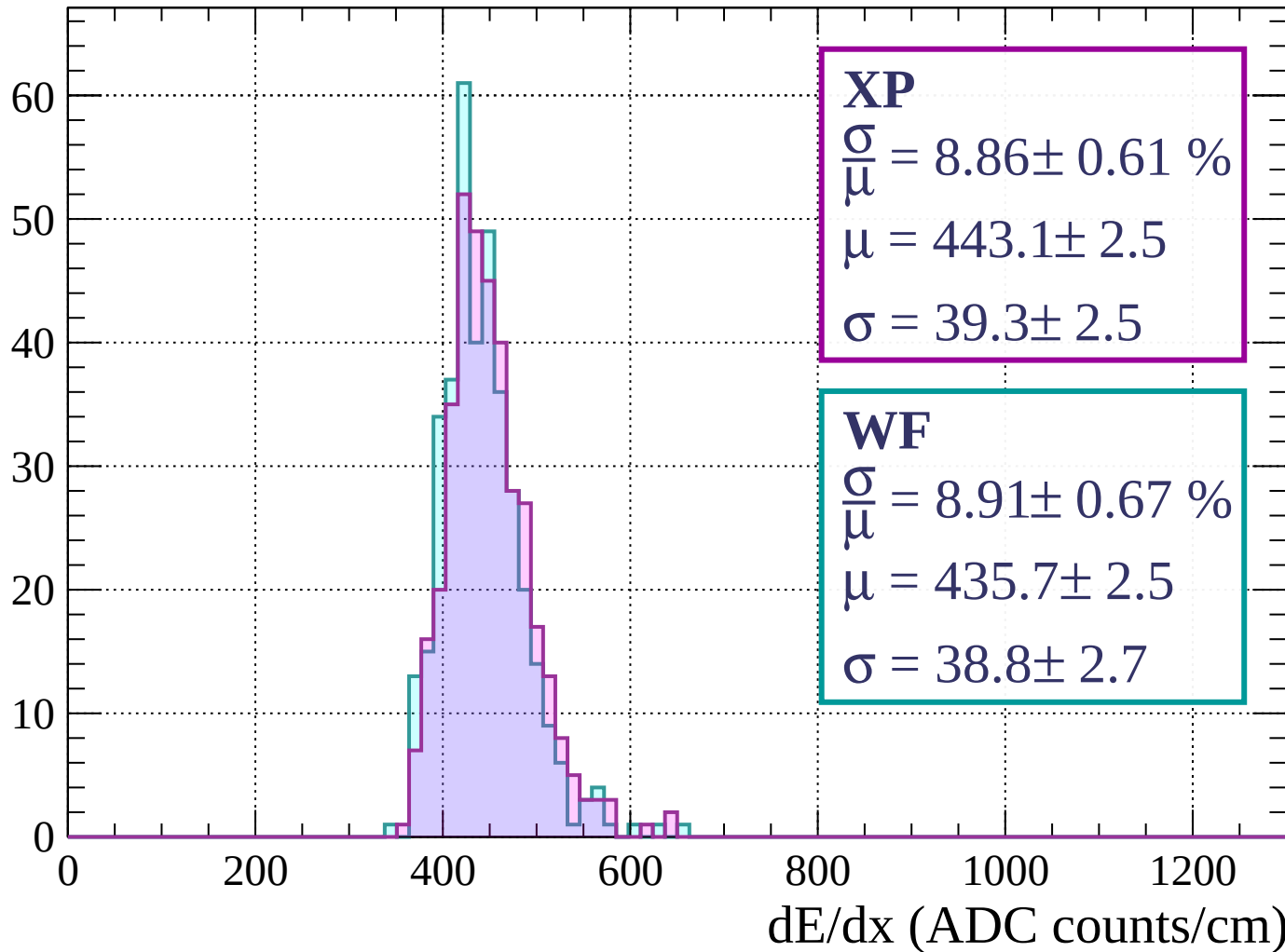
$$\sigma = 39.3 \pm 2.5$$

WF

$$\frac{\sigma}{\mu} = 8.91 \pm 0.67 \%$$

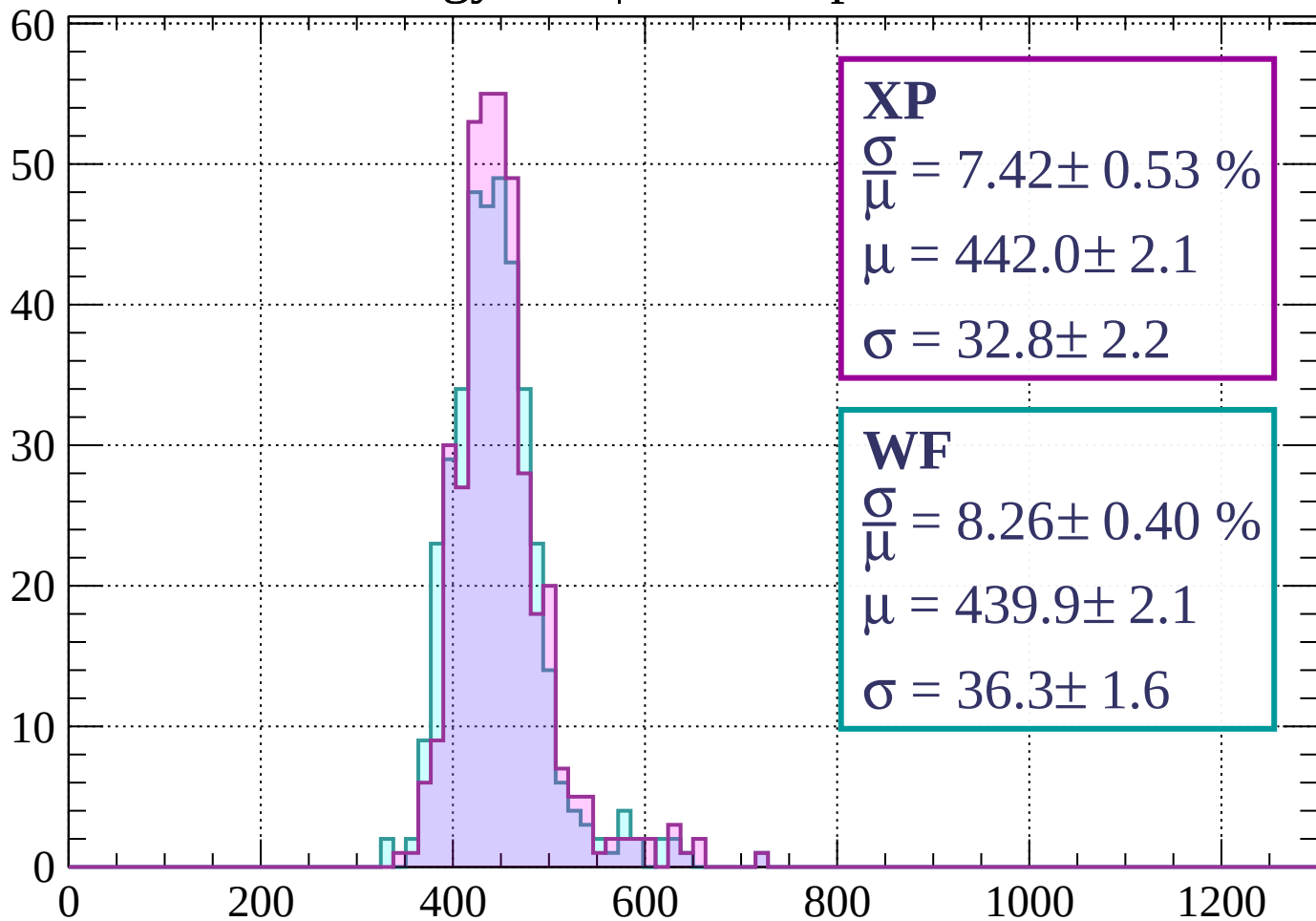
$$\mu = 435.7 \pm 2.5$$

$$\sigma = 38.8 \pm 2.7$$



Energy loss | $-1160 < p < -1120$

Count



XP

$$\frac{\sigma}{\bar{\mu}} = 7.42 \pm 0.53 \%$$

$$\mu = 442.0 \pm 2.1$$

$$\sigma = 32.8 \pm 2.2$$

WF

$$\frac{\sigma}{\bar{\mu}} = 8.26 \pm 0.40 \%$$

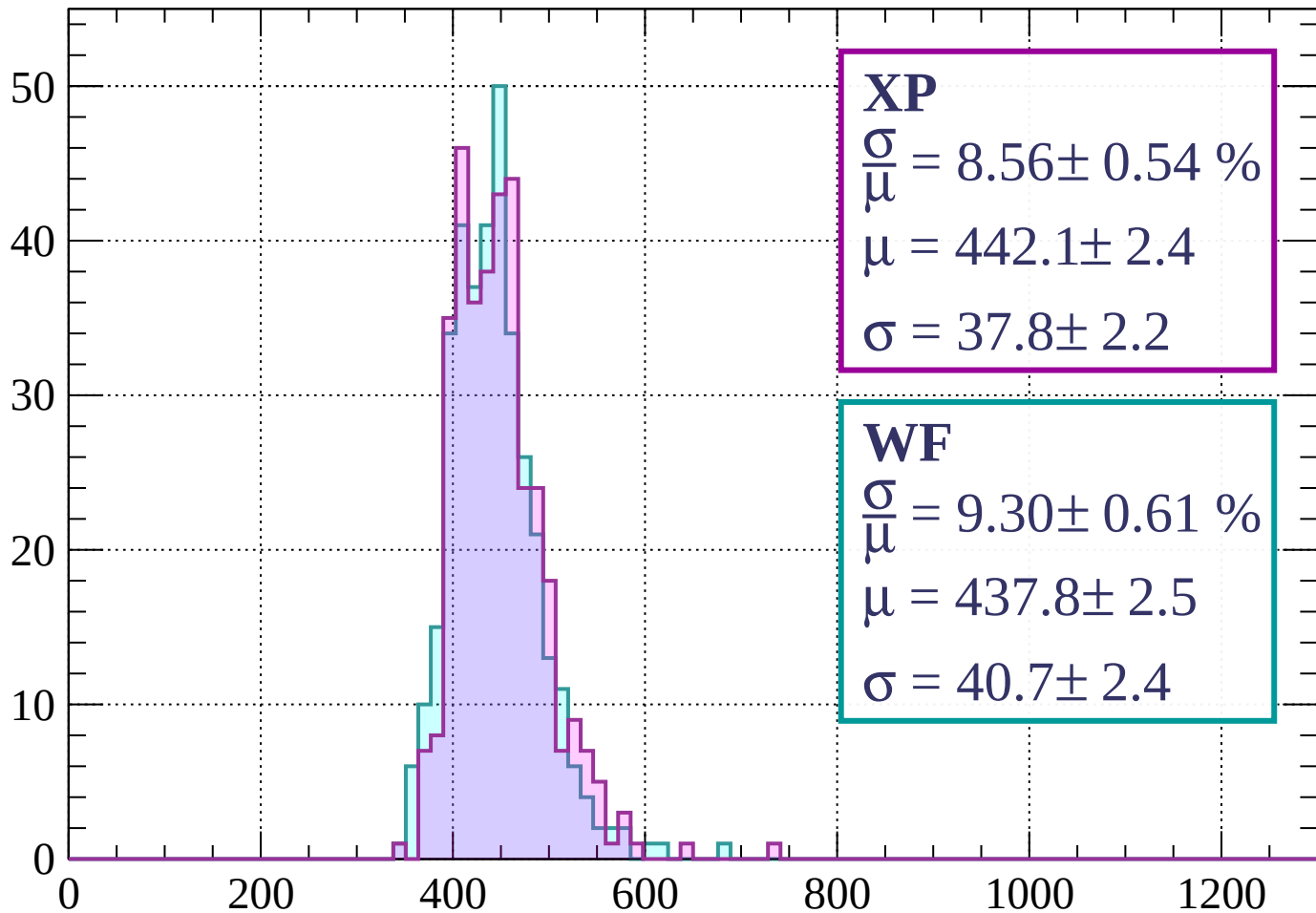
$$\mu = 439.9 \pm 2.1$$

$$\sigma = 36.3 \pm 1.6$$

dE/dx (ADC counts/cm)

Energy loss | $-1120 < p < -1080$

Count



XP

$$\frac{\sigma}{\mu} = 8.56 \pm 0.54 \%$$

$$\mu = 442.1 \pm 2.4$$

$$\sigma = 37.8 \pm 2.2$$

WF

$$\frac{\sigma}{\mu} = 9.30 \pm 0.61 \%$$

$$\mu = 437.8 \pm 2.5$$

$$\sigma = 40.7 \pm 2.4$$

dE/dx (ADC counts/cm)

Energy loss | $-1080 < p < -1040$

Count

XP

$$\frac{\sigma}{\mu} = 8.40 \pm 0.75 \%$$

$$\mu = 438.7 \pm 2.4$$

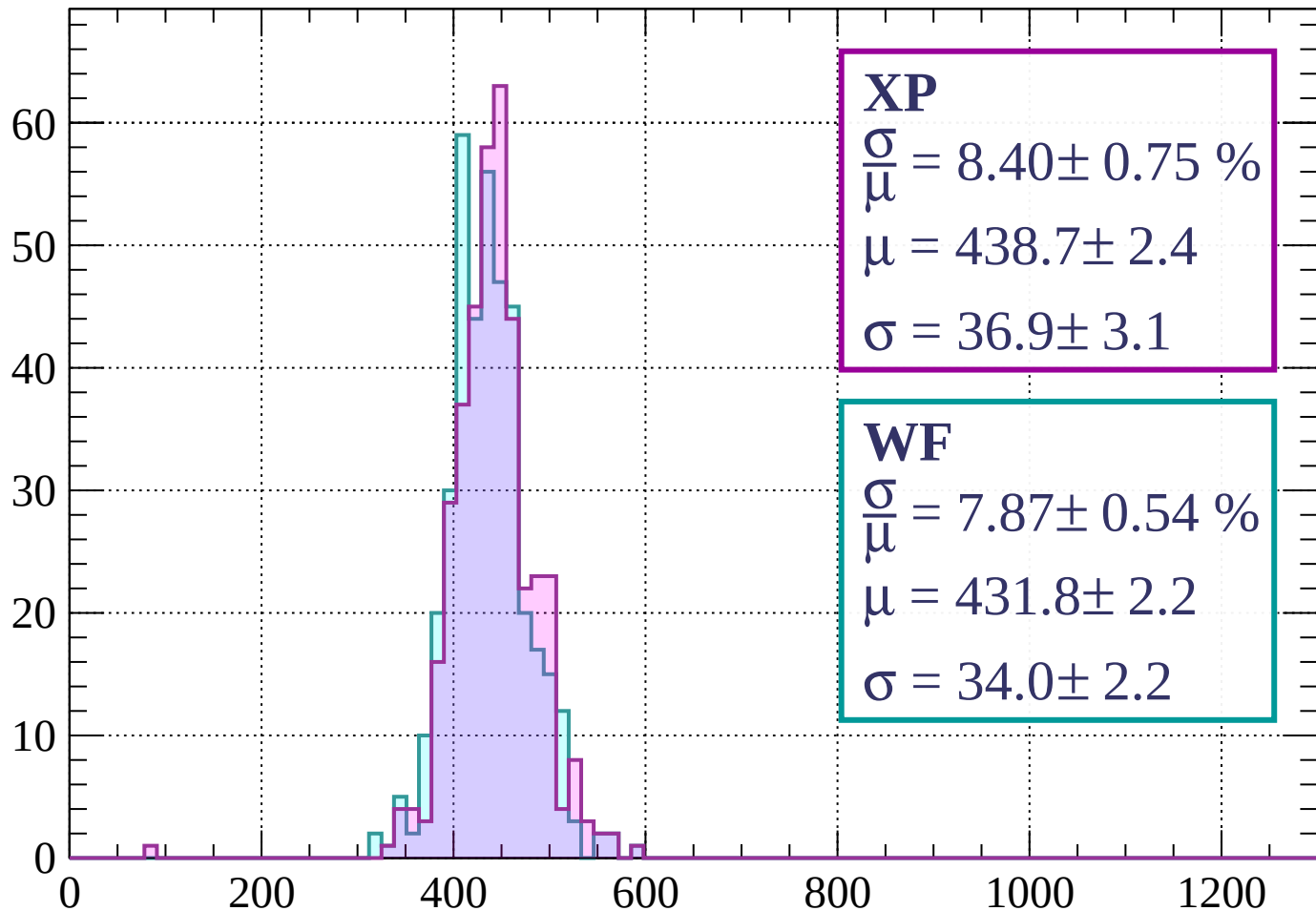
$$\sigma = 36.9 \pm 3.1$$

WF

$$\frac{\sigma}{\mu} = 7.87 \pm 0.54 \%$$

$$\mu = 431.8 \pm 2.2$$

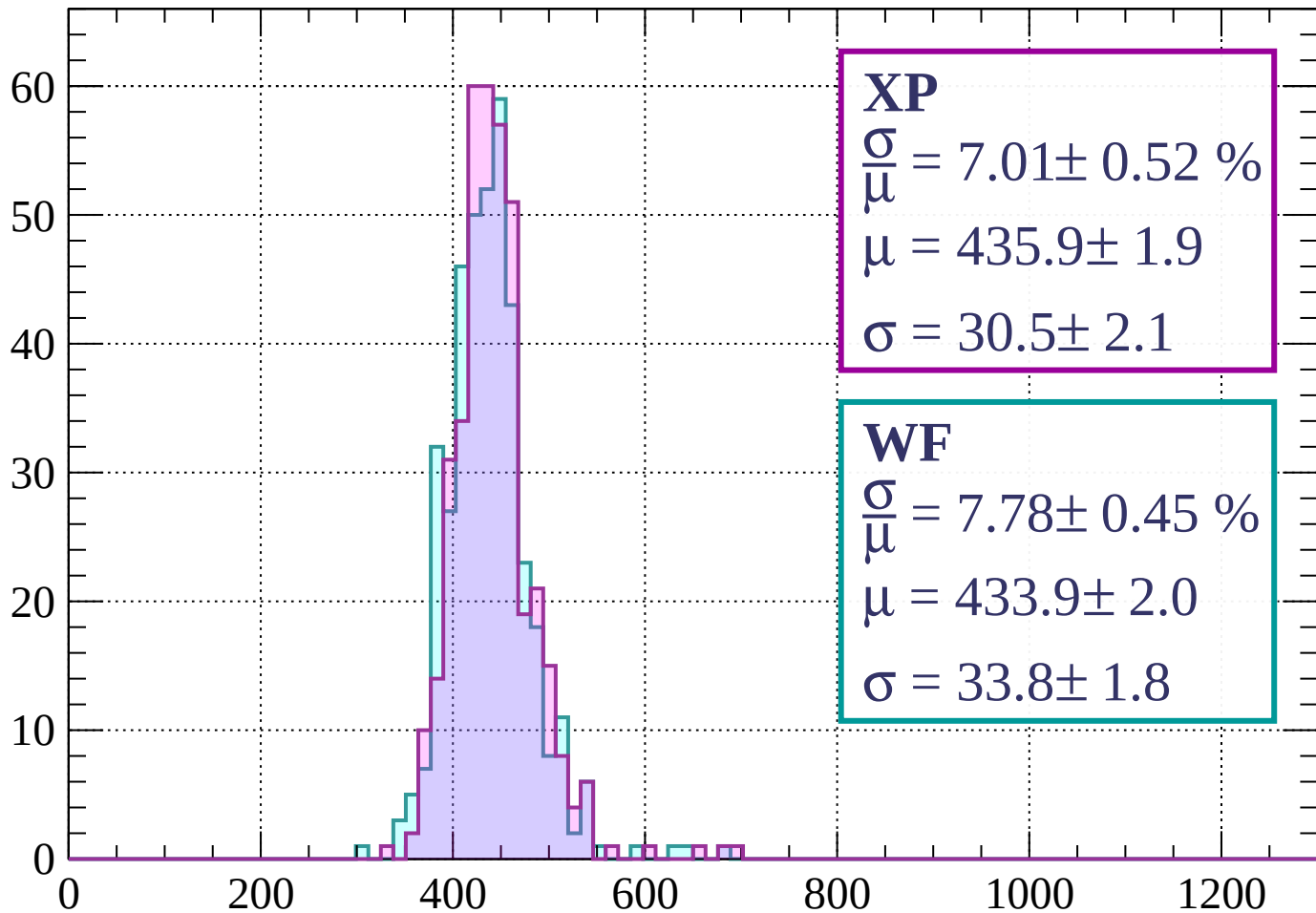
$$\sigma = 34.0 \pm 2.2$$



dE/dx (ADC counts/cm)

Energy loss | $-1040 < p < -1000$

Count



XP

$$\frac{\sigma}{\mu} = 7.01 \pm 0.52 \%$$

$$\mu = 435.9 \pm 1.9$$

$$\sigma = 30.5 \pm 2.1$$

WF

$$\frac{\sigma}{\mu} = 7.78 \pm 0.45 \%$$

$$\mu = 433.9 \pm 2.0$$

$$\sigma = 33.8 \pm 1.8$$

dE/dx (ADC counts/cm)

Energy loss | $-1000 < p < -960$

Count

XP

$$\frac{\sigma}{\mu} = 8.62 \pm 0.59 \%$$

$$\mu = 434.6 \pm 2.3$$

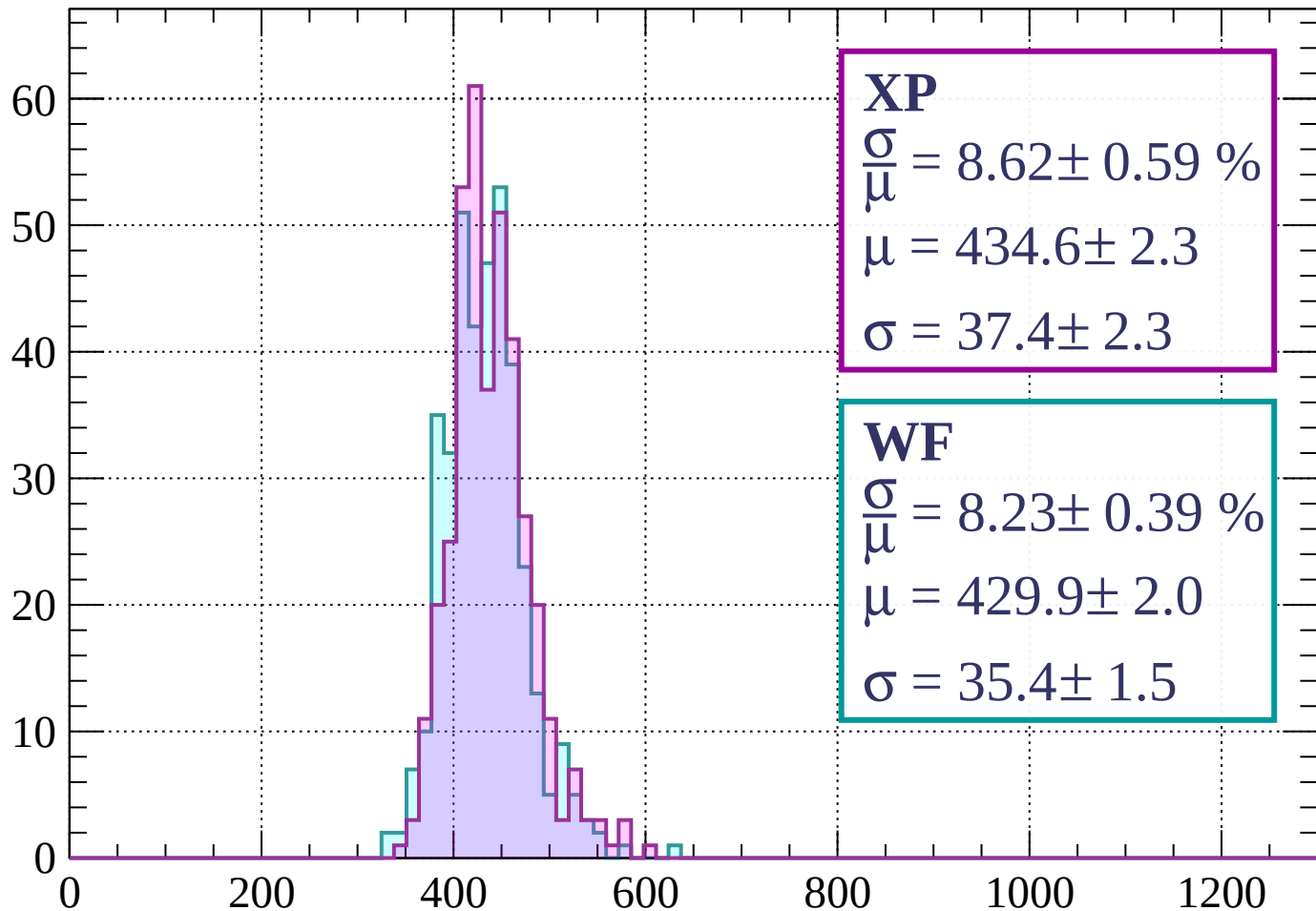
$$\sigma = 37.4 \pm 2.3$$

WF

$$\frac{\sigma}{\mu} = 8.23 \pm 0.39 \%$$

$$\mu = 429.9 \pm 2.0$$

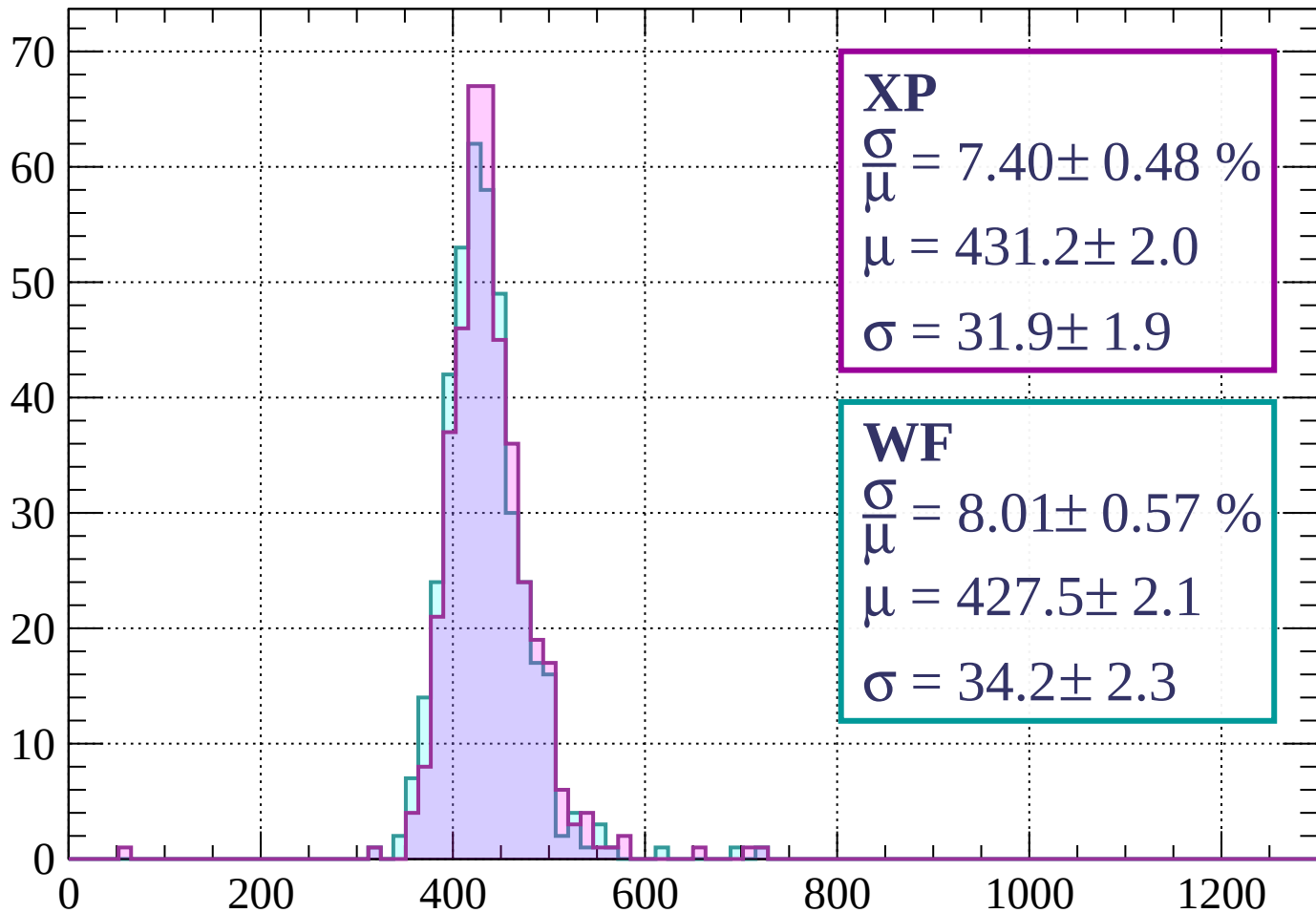
$$\sigma = 35.4 \pm 1.5$$



dE/dx (ADC counts/cm)

Energy loss | $-960 < p < -920$

Count



XP

$$\frac{\sigma}{\mu} = 7.40 \pm 0.48 \%$$

$$\mu = 431.2 \pm 2.0$$

$$\sigma = 31.9 \pm 1.9$$

WF

$$\frac{\sigma}{\mu} = 8.01 \pm 0.57 \%$$

$$\mu = 427.5 \pm 2.1$$

$$\sigma = 34.2 \pm 2.3$$

dE/dx (ADC counts/cm)

Energy loss | $-920 < p < -880$

Count

XP

$$\frac{\sigma}{\mu} = 7.89 \pm 0.47 \%$$

$$\mu = 431.4 \pm 2.0$$

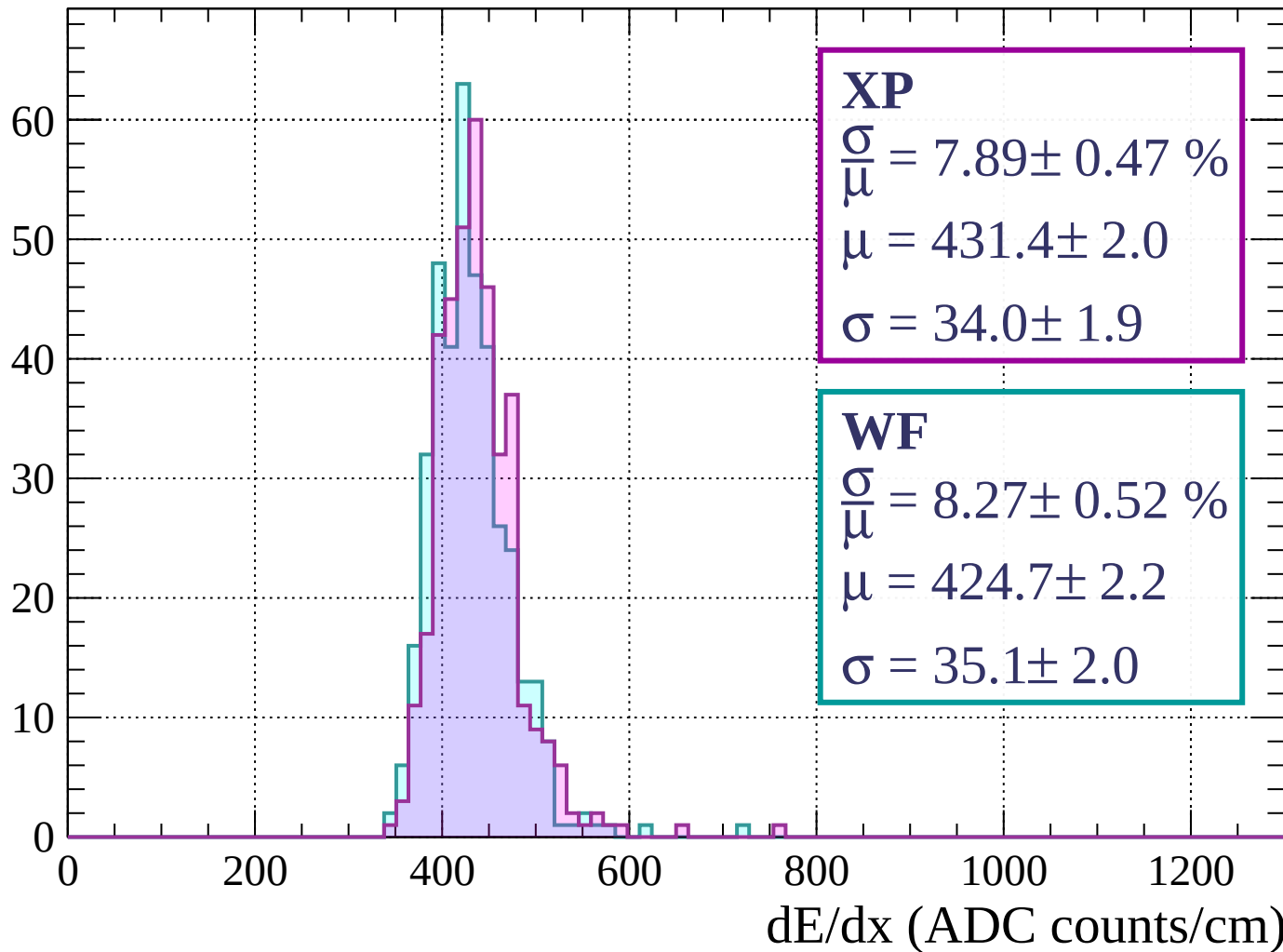
$$\sigma = 34.0 \pm 1.9$$

WF

$$\frac{\sigma}{\mu} = 8.27 \pm 0.52 \%$$

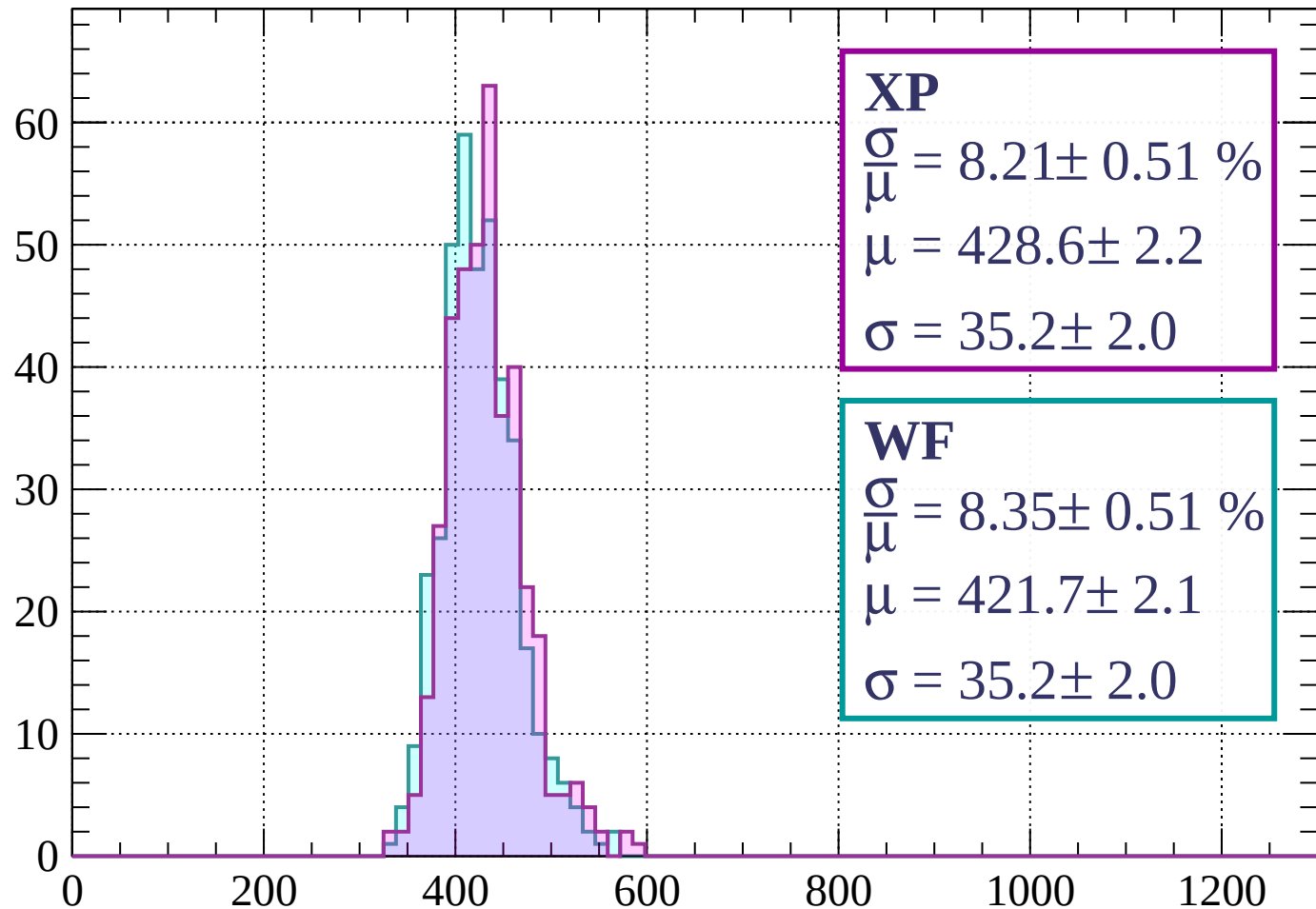
$$\mu = 424.7 \pm 2.2$$

$$\sigma = 35.1 \pm 2.0$$



Energy loss | $-880 < p < -840$

Count



XP

$$\frac{\sigma}{\mu} = 8.21 \pm 0.51 \%$$

$$\mu = 428.6 \pm 2.2$$

$$\sigma = 35.2 \pm 2.0$$

WF

$$\frac{\sigma}{\mu} = 8.35 \pm 0.51 \%$$

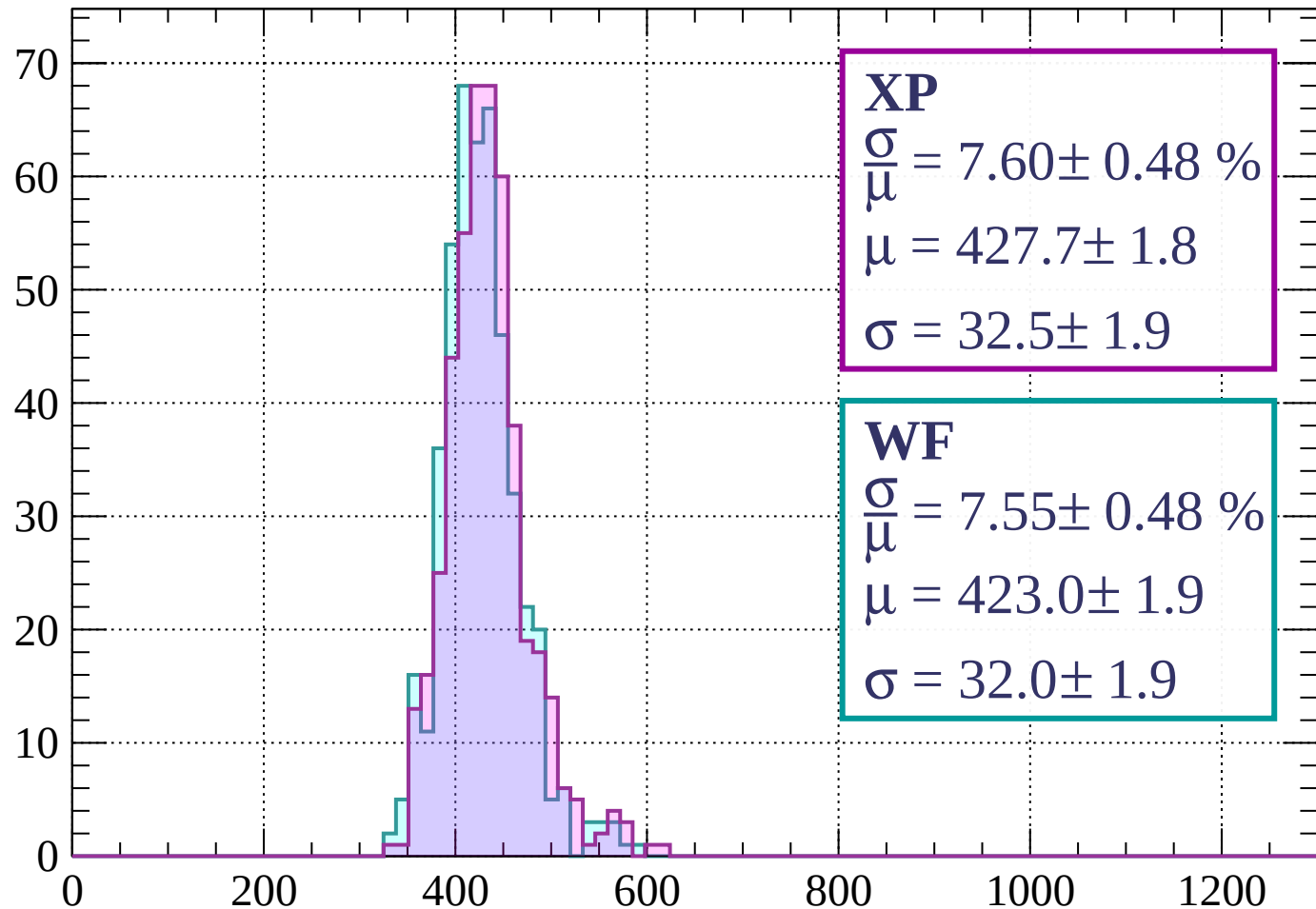
$$\mu = 421.7 \pm 2.1$$

$$\sigma = 35.2 \pm 2.0$$

dE/dx (ADC counts/cm)

Energy loss | $-840 < p < -800$

Count



XP

$$\frac{\sigma}{\mu} = 7.60 \pm 0.48 \%$$

$$\mu = 427.7 \pm 1.8$$

$$\sigma = 32.5 \pm 1.9$$

WF

$$\frac{\sigma}{\mu} = 7.55 \pm 0.48 \%$$

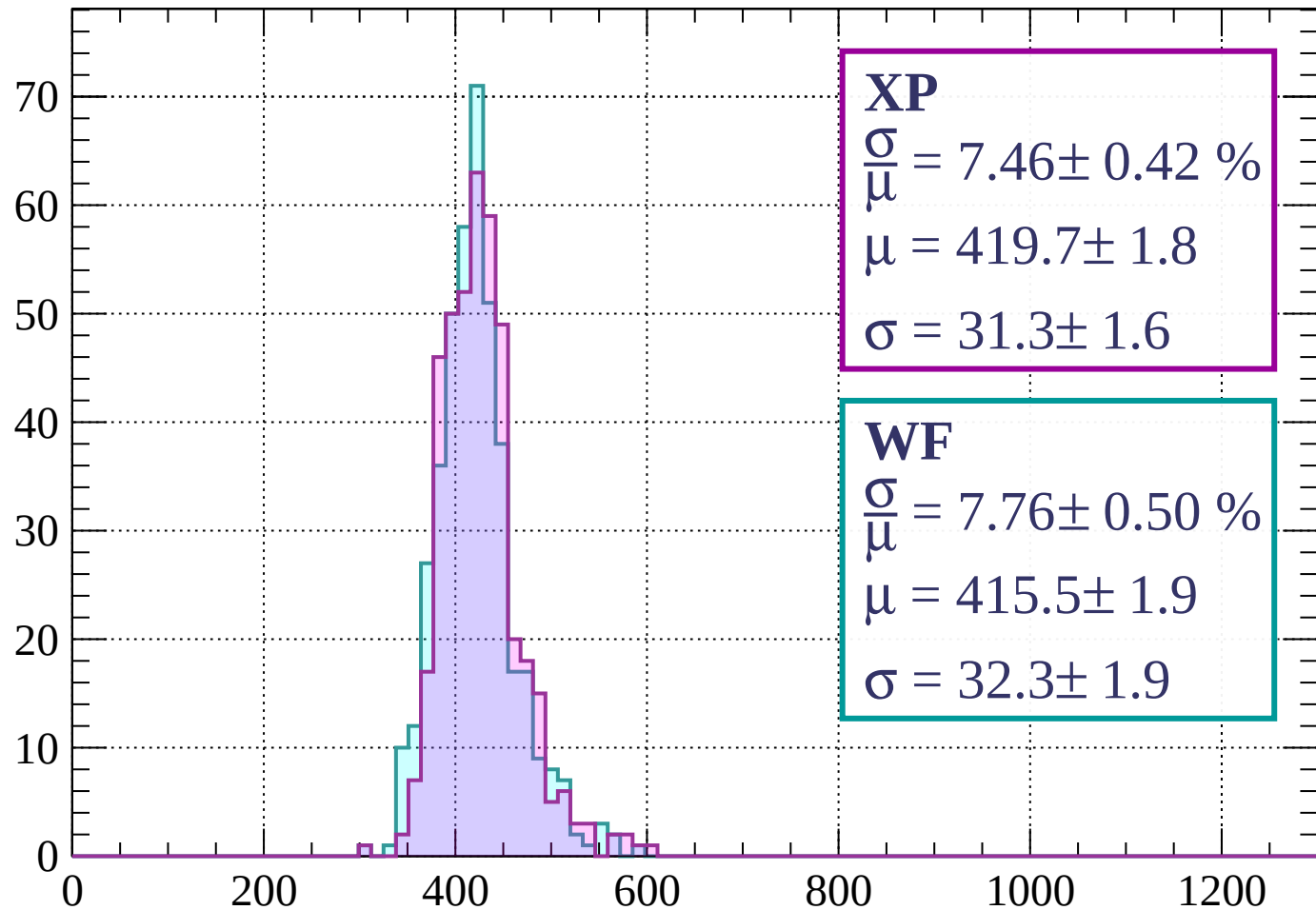
$$\mu = 423.0 \pm 1.9$$

$$\sigma = 32.0 \pm 1.9$$

dE/dx (ADC counts/cm)

Energy loss | $-800 < p < -760$

Count



XP

$$\frac{\sigma}{\mu} = 7.46 \pm 0.42 \%$$

$$\mu = 419.7 \pm 1.8$$

$$\sigma = 31.3 \pm 1.6$$

WF

$$\frac{\sigma}{\mu} = 7.76 \pm 0.50 \%$$

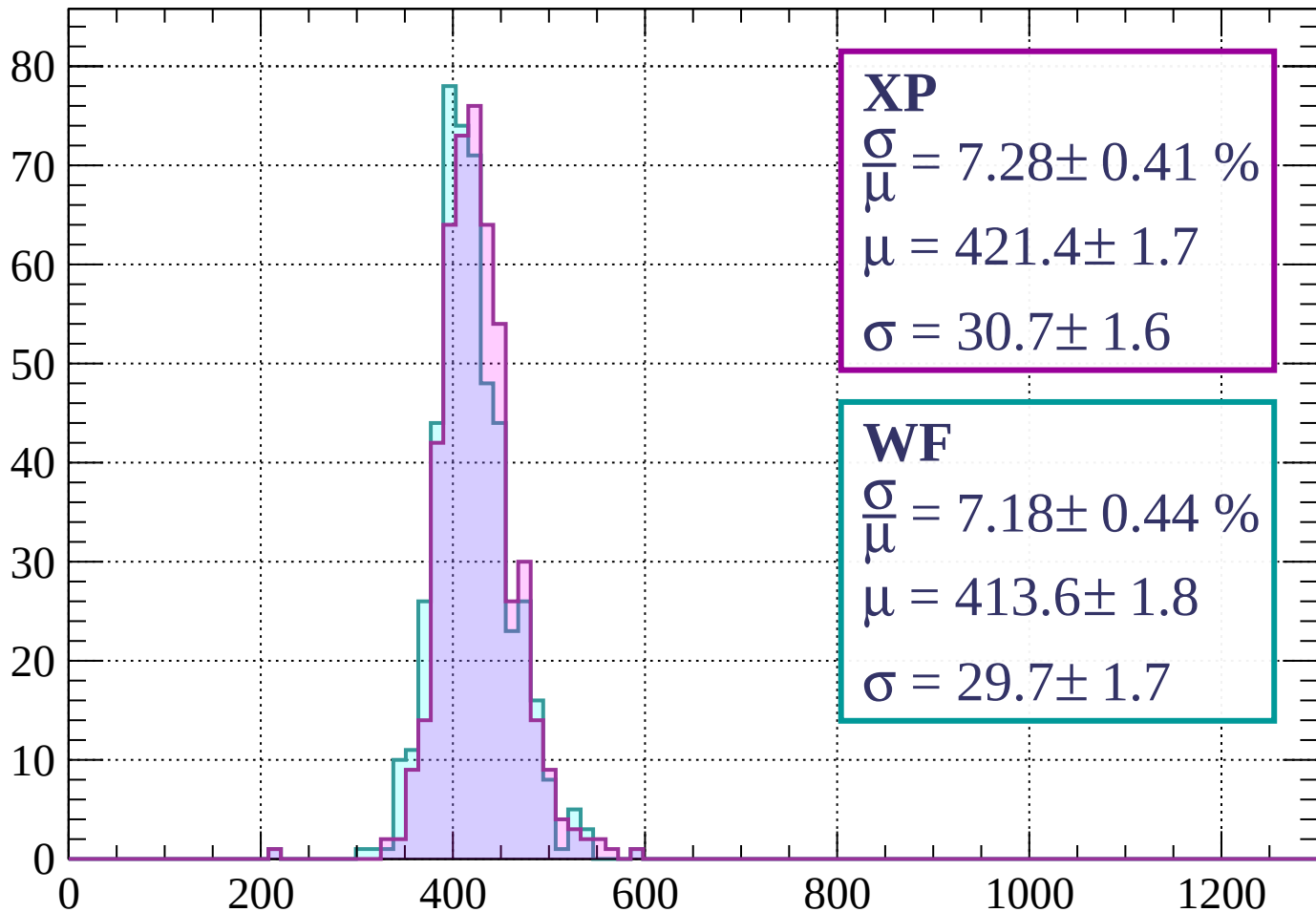
$$\mu = 415.5 \pm 1.9$$

$$\sigma = 32.3 \pm 1.9$$

dE/dx (ADC counts/cm)

Energy loss | $-760 < p < -720$

Count



XP

$$\frac{\sigma}{\mu} = 7.28 \pm 0.41 \%$$

$$\mu = 421.4 \pm 1.7$$

$$\sigma = 30.7 \pm 1.6$$

WF

$$\frac{\sigma}{\mu} = 7.18 \pm 0.44 \%$$

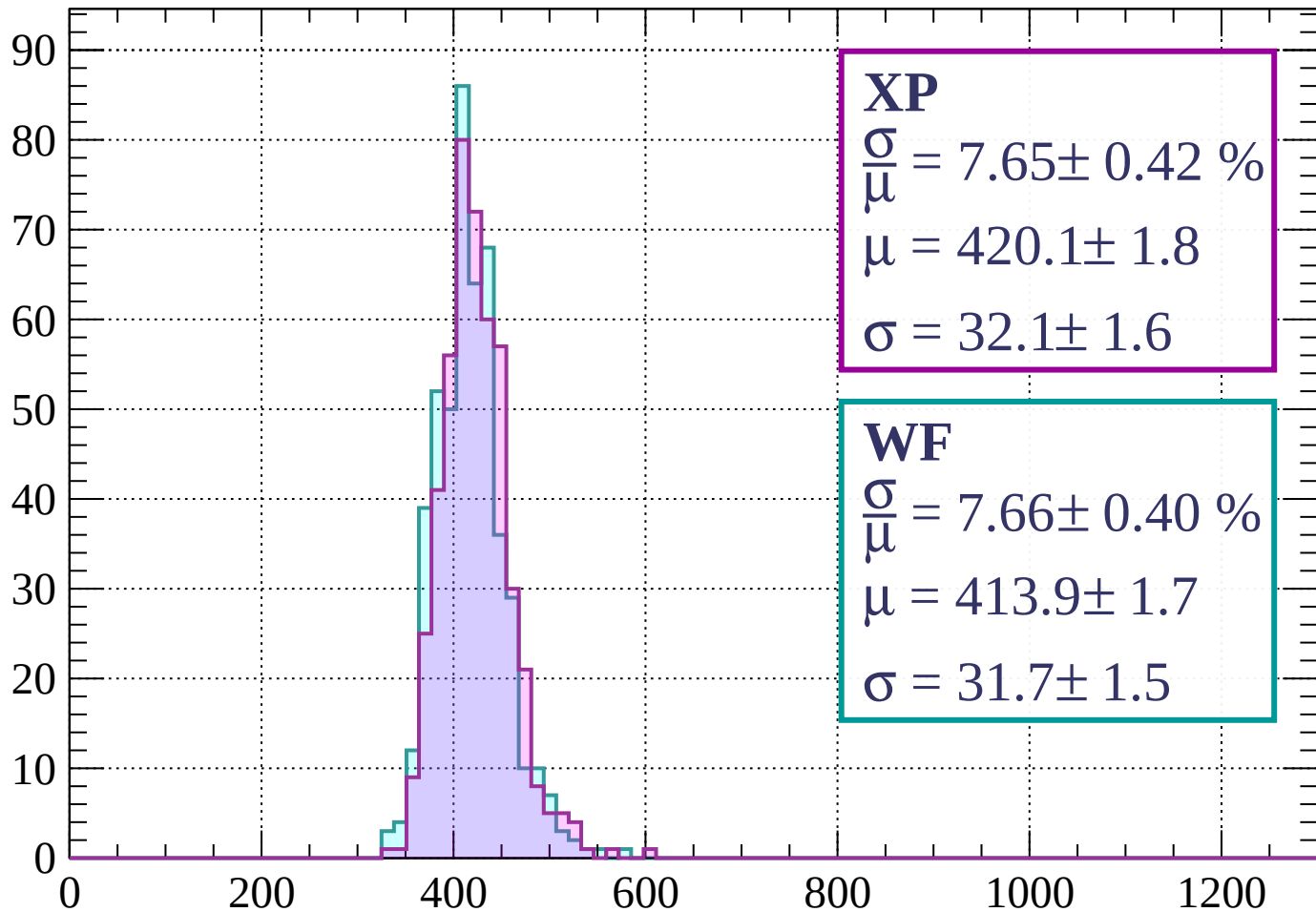
$$\mu = 413.6 \pm 1.8$$

$$\sigma = 29.7 \pm 1.7$$

dE/dx (ADC counts/cm)

Energy loss | $-720 < p < -680$

Count



XP

$$\frac{\sigma}{\mu} = 7.65 \pm 0.42 \%$$

$$\mu = 420.1 \pm 1.8$$

$$\sigma = 32.1 \pm 1.6$$

WF

$$\frac{\sigma}{\mu} = 7.66 \pm 0.40 \%$$

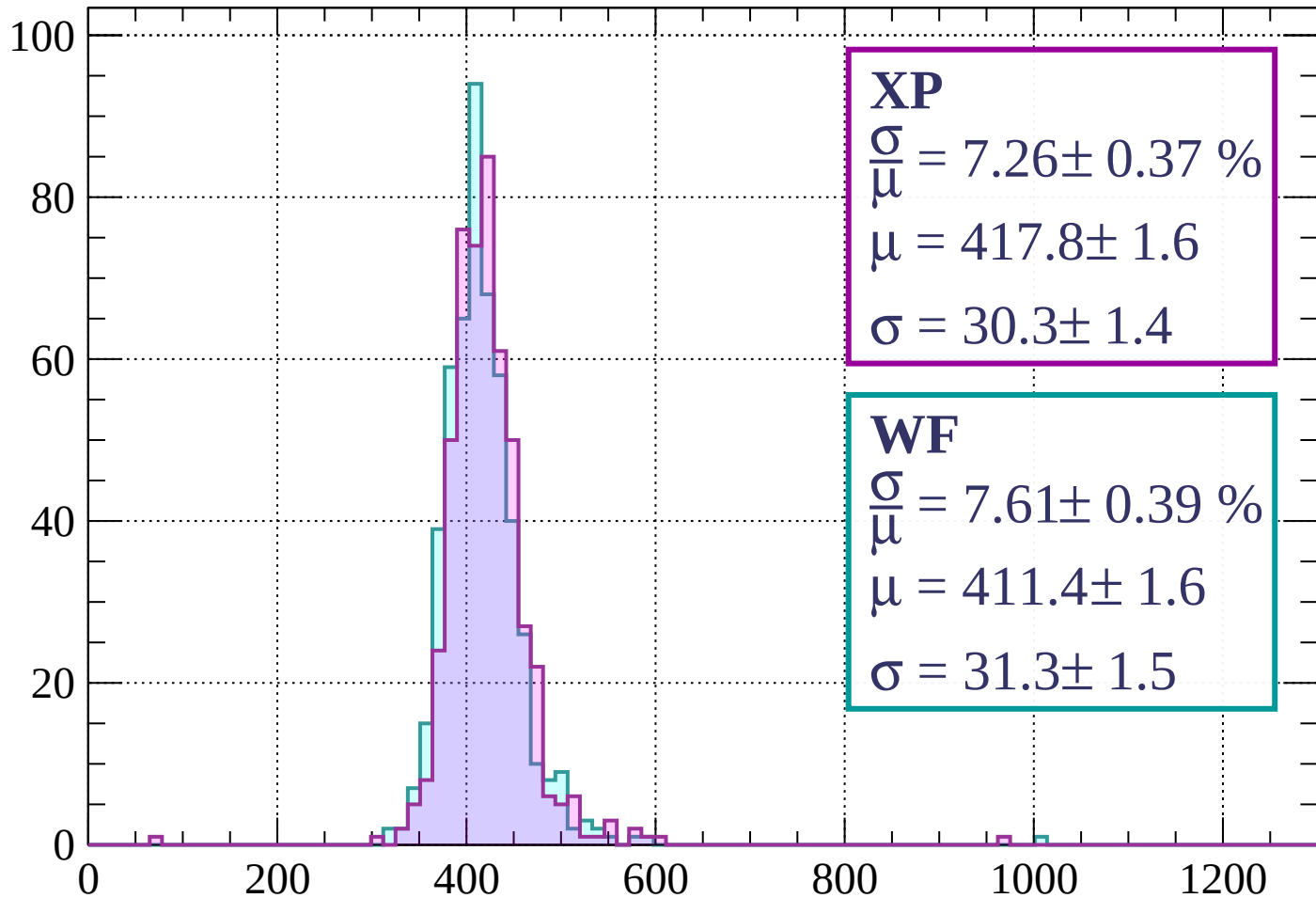
$$\mu = 413.9 \pm 1.7$$

$$\sigma = 31.7 \pm 1.5$$

dE/dx (ADC counts/cm)

Energy loss | $-680 < p < -640$

Count



XP

$$\frac{\sigma}{\mu} = 7.26 \pm 0.37 \%$$

$$\mu = 417.8 \pm 1.6$$

$$\sigma = 30.3 \pm 1.4$$

WF

$$\frac{\sigma}{\mu} = 7.61 \pm 0.39 \%$$

$$\mu = 411.4 \pm 1.6$$

$$\sigma = 31.3 \pm 1.5$$

dE/dx (ADC counts/cm)

Energy loss | $-640 < p < -600$

Count

XP

$$\frac{\sigma}{\mu} = 7.98 \pm 0.39 \%$$

$$\mu = 411.1 \pm 1.7$$

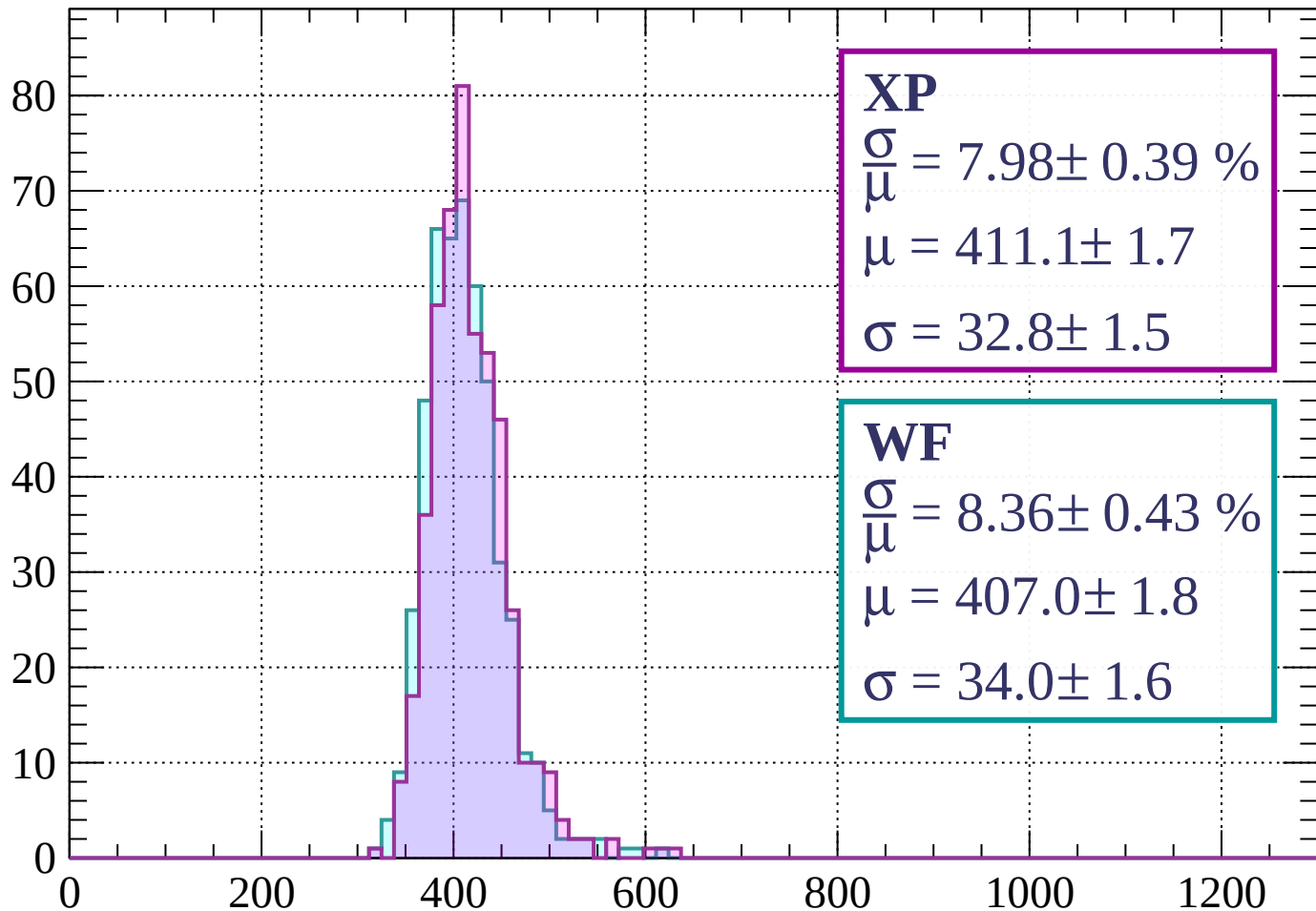
$$\sigma = 32.8 \pm 1.5$$

WF

$$\frac{\sigma}{\mu} = 8.36 \pm 0.43 \%$$

$$\mu = 407.0 \pm 1.8$$

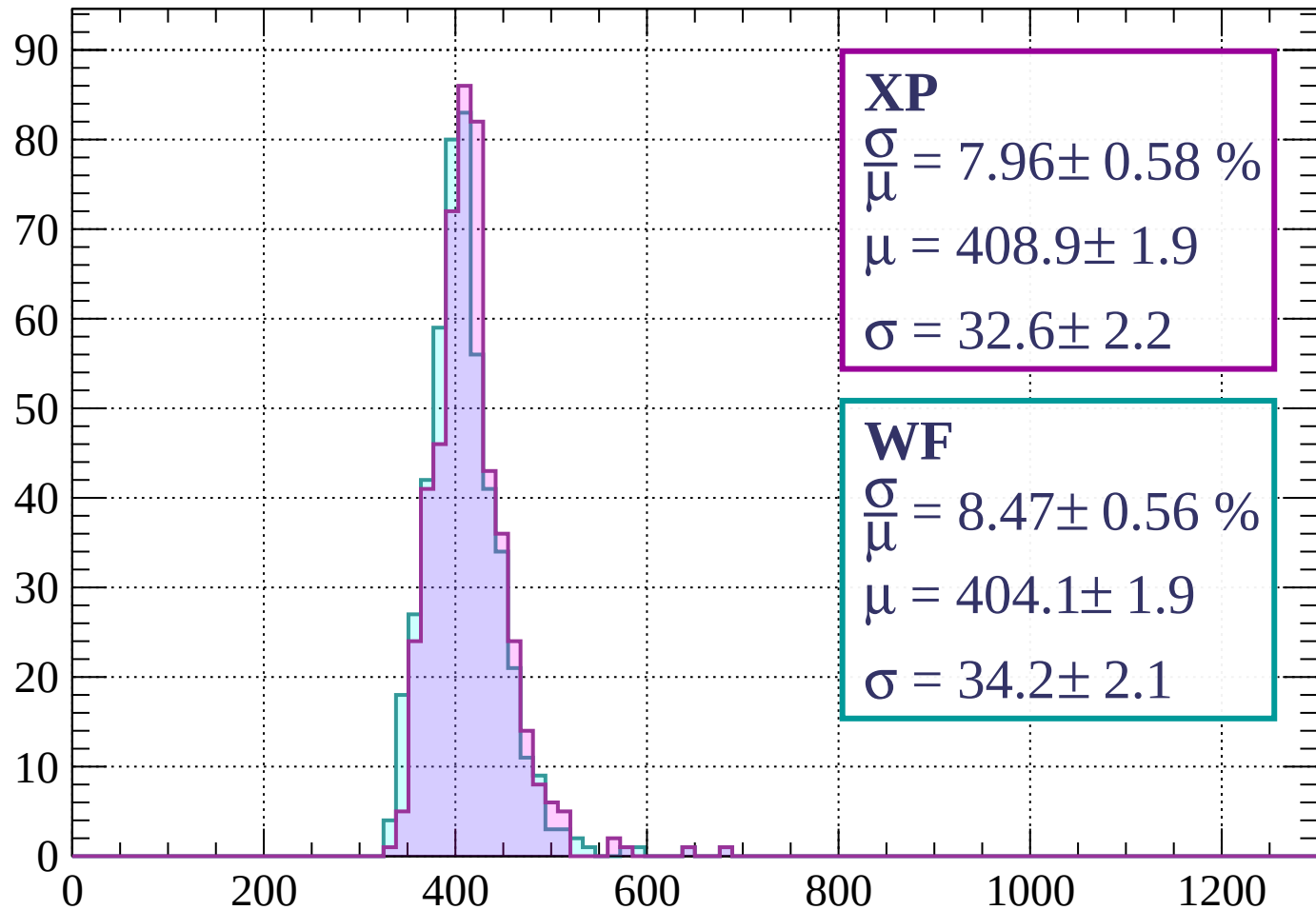
$$\sigma = 34.0 \pm 1.6$$



dE/dx (ADC counts/cm)

Energy loss | $-600 < p < -560$

Count



XP

$$\frac{\sigma}{\mu} = 7.96 \pm 0.58 \%$$

$$\mu = 408.9 \pm 1.9$$

$$\sigma = 32.6 \pm 2.2$$

WF

$$\frac{\sigma}{\mu} = 8.47 \pm 0.56 \%$$

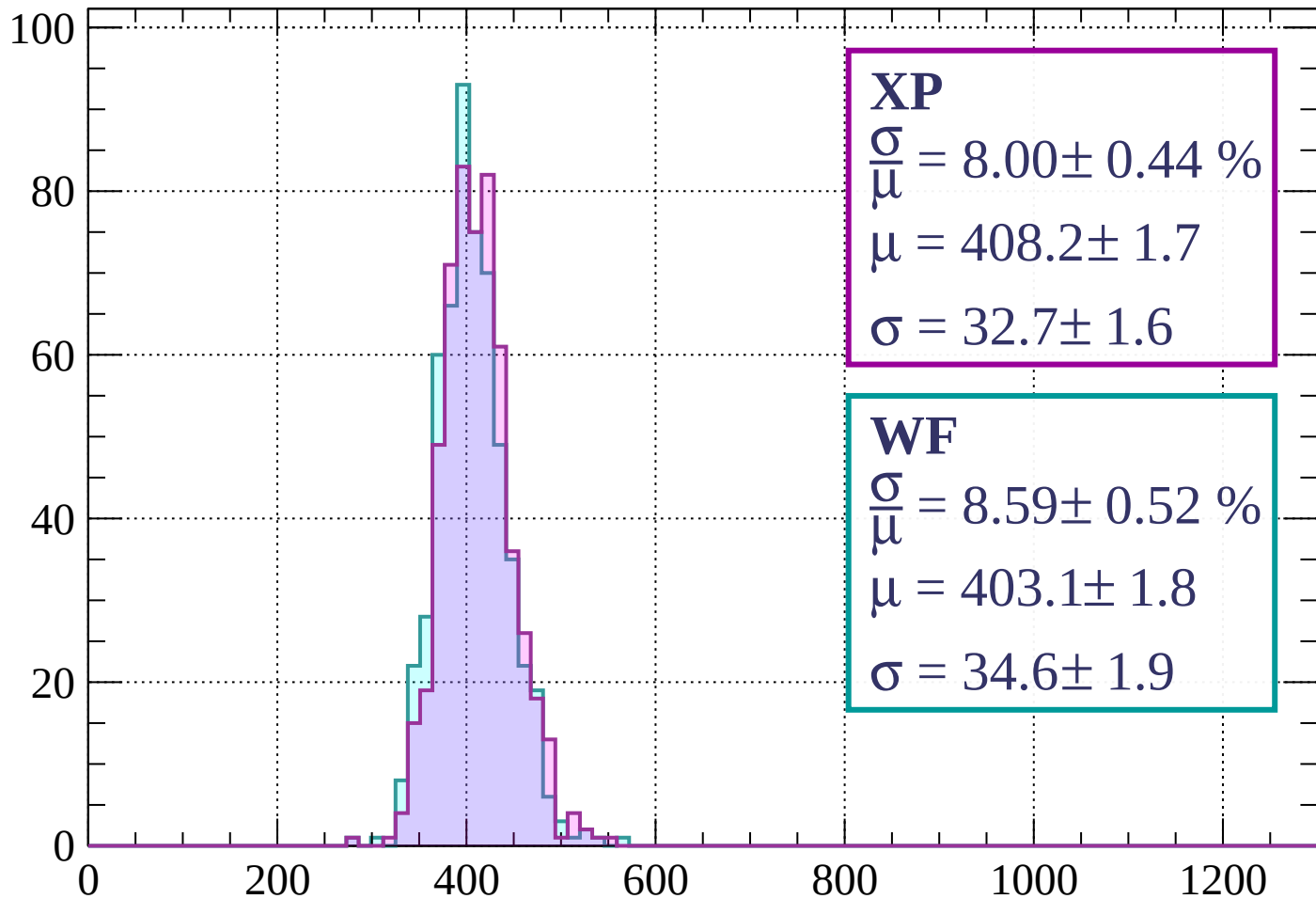
$$\mu = 404.1 \pm 1.9$$

$$\sigma = 34.2 \pm 2.1$$

dE/dx (ADC counts/cm)

Energy loss | $-560 < p < -520$

Count



XP

$$\frac{\sigma}{\mu} = 8.00 \pm 0.44 \%$$

$$\mu = 408.2 \pm 1.7$$

$$\sigma = 32.7 \pm 1.6$$

WF

$$\frac{\sigma}{\mu} = 8.59 \pm 0.52 \%$$

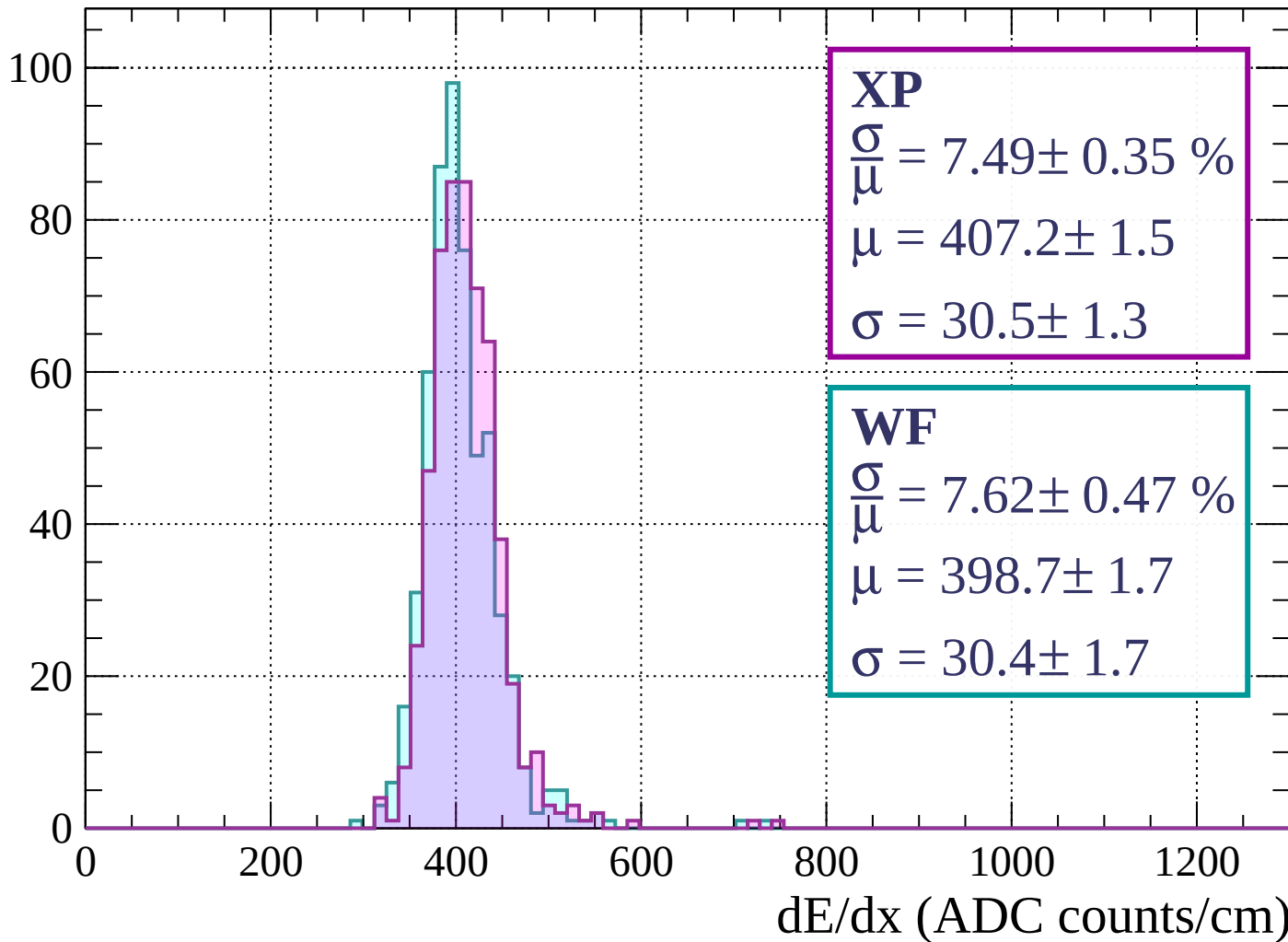
$$\mu = 403.1 \pm 1.8$$

$$\sigma = 34.6 \pm 1.9$$

dE/dx (ADC counts/cm)

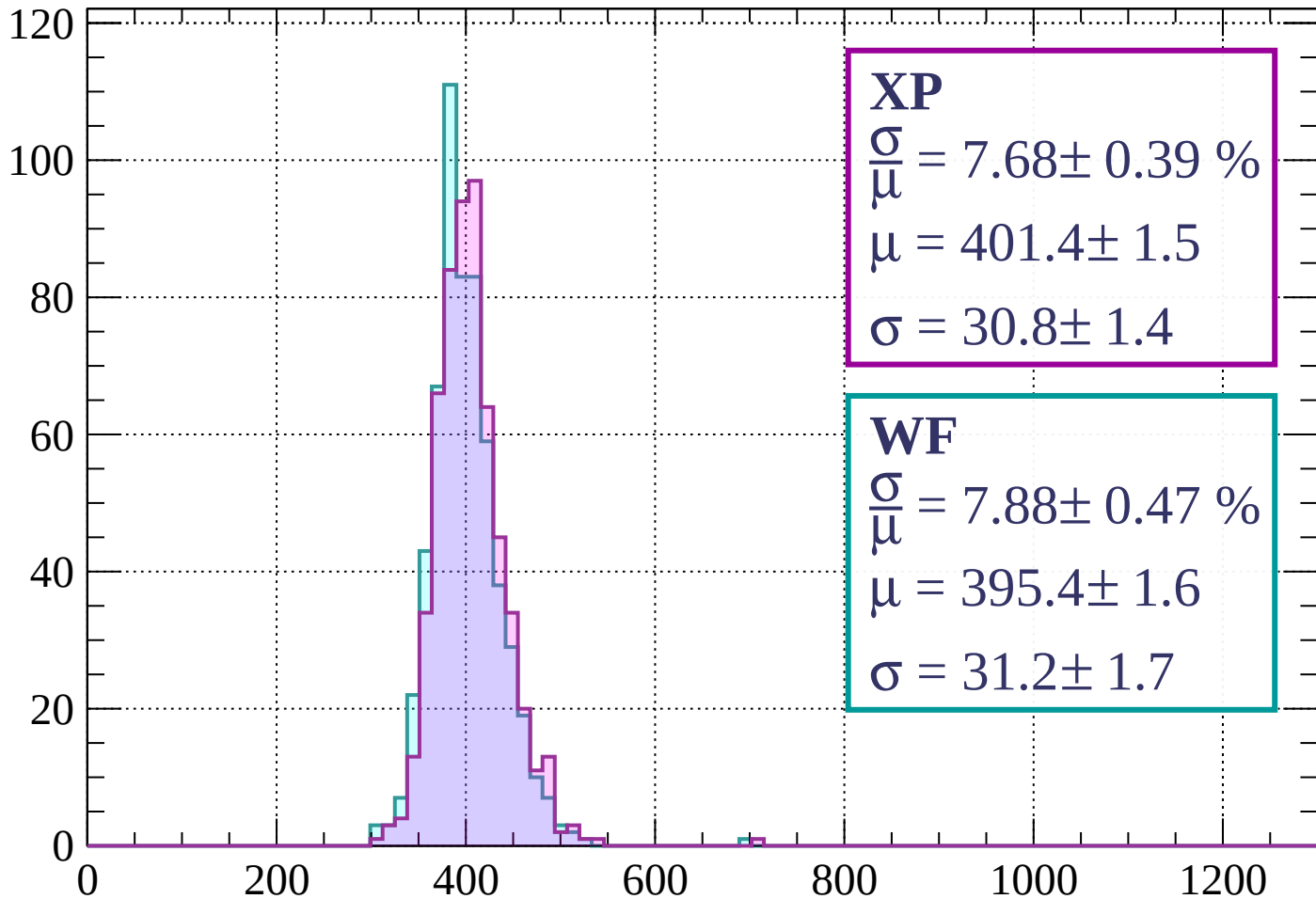
Energy loss | $-520 < p < -480$

Count



Energy loss | $-480 < p < -440$

Count



XP

$$\frac{\sigma}{\mu} = 7.68 \pm 0.39 \%$$

$$\mu = 401.4 \pm 1.5$$

$$\sigma = 30.8 \pm 1.4$$

WF

$$\frac{\sigma}{\mu} = 7.88 \pm 0.47 \%$$

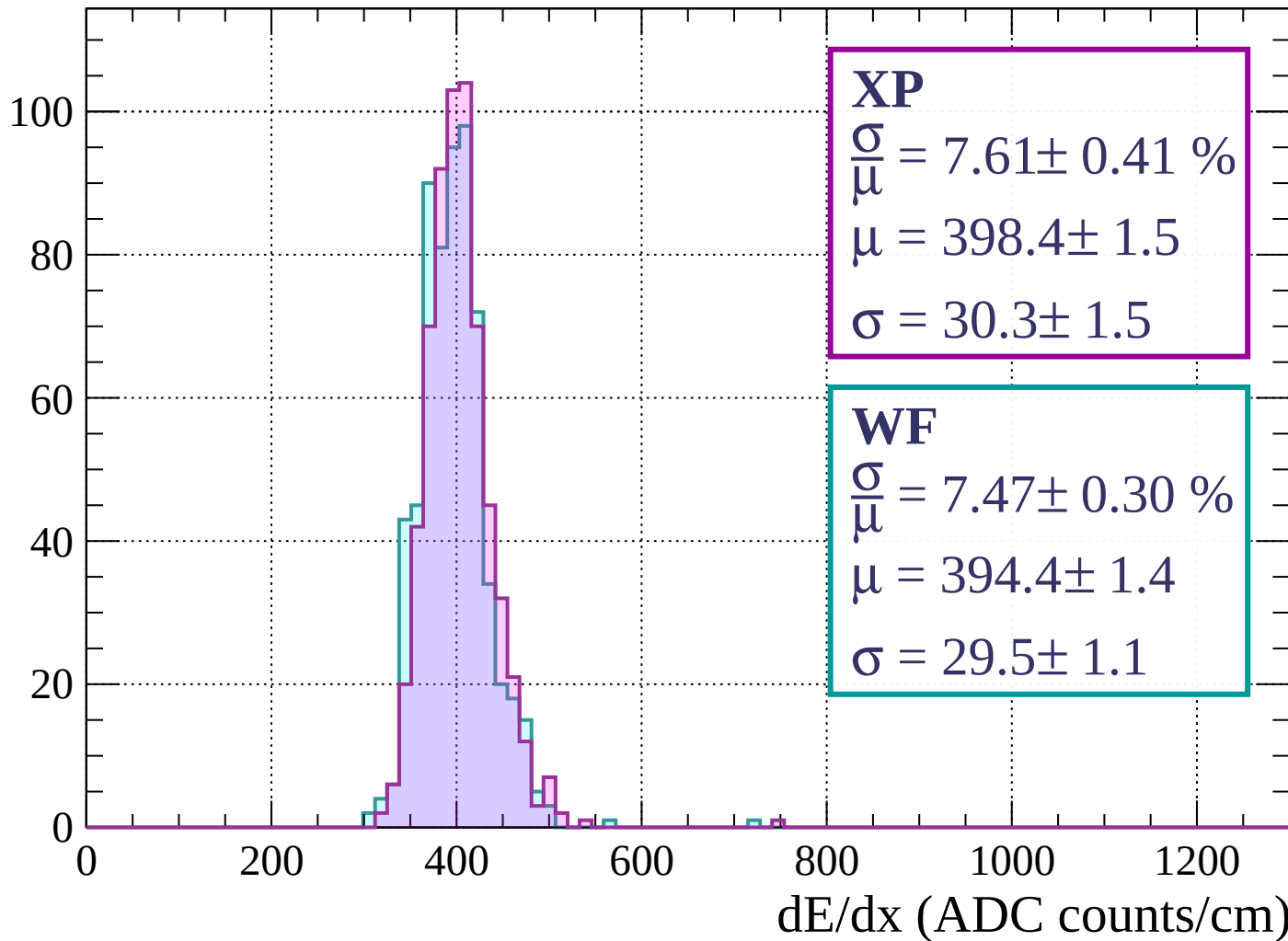
$$\mu = 395.4 \pm 1.6$$

$$\sigma = 31.2 \pm 1.7$$

dE/dx (ADC counts/cm)

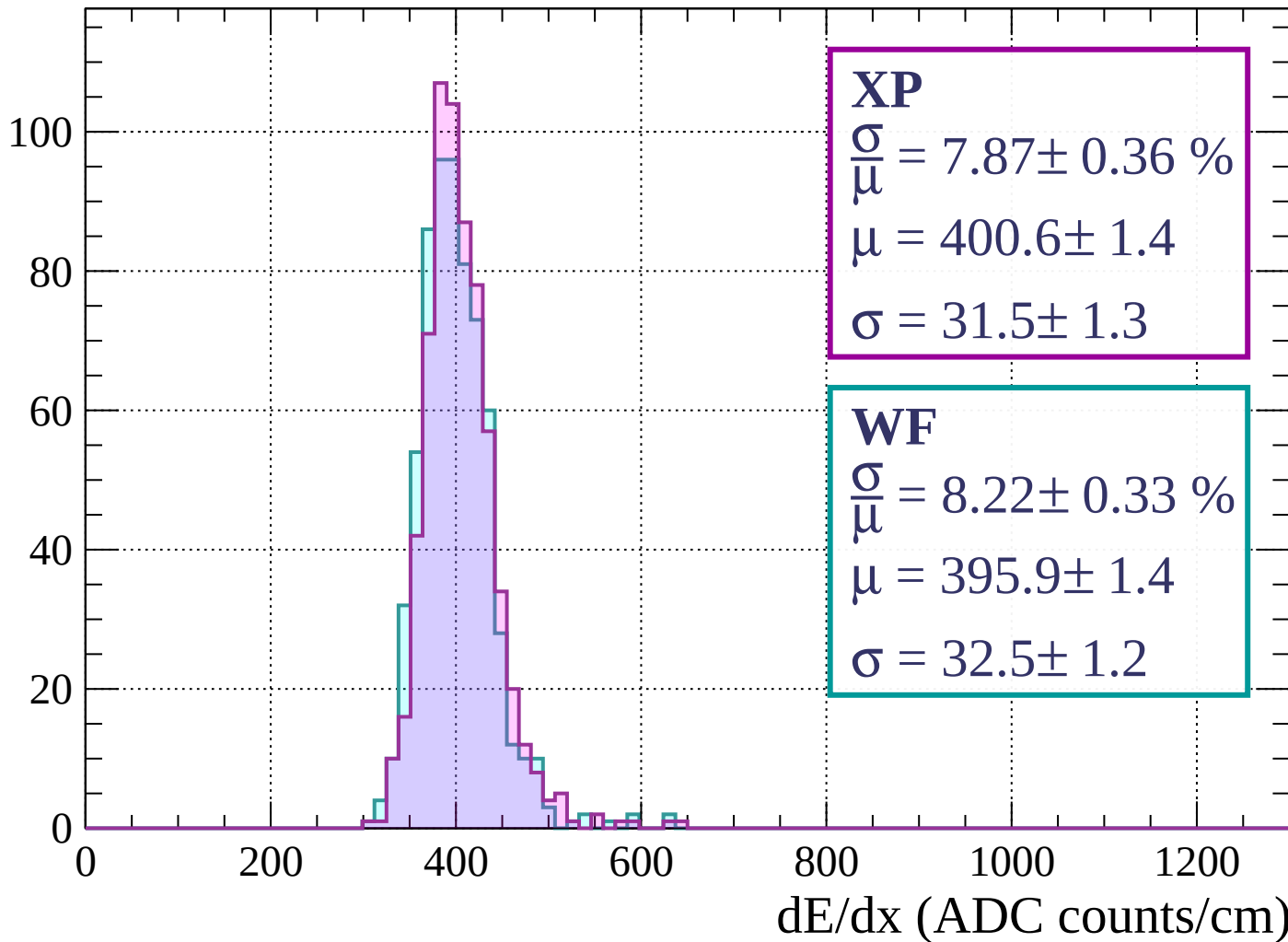
Energy loss | $-440 < p < -400$

Count



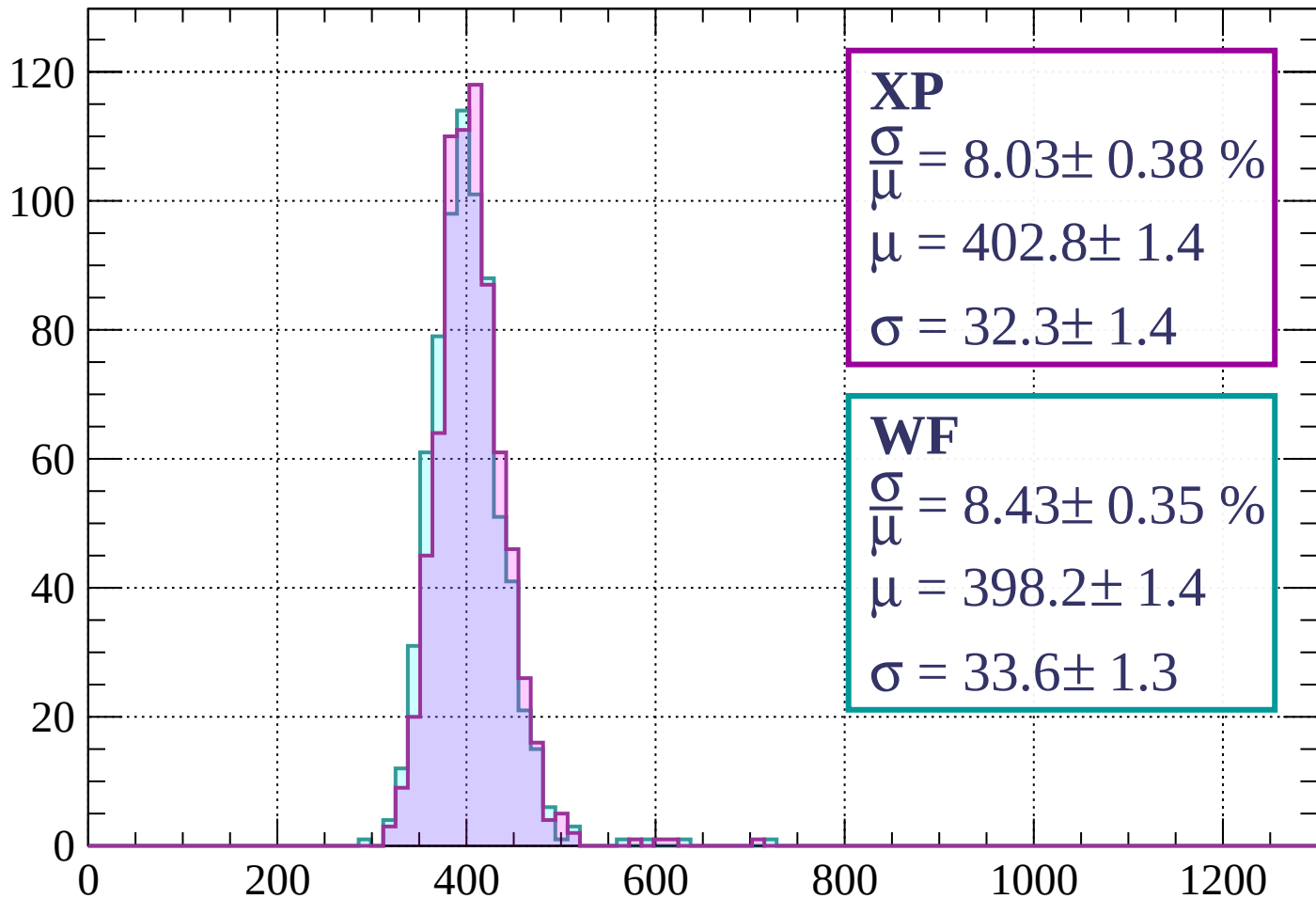
Energy loss | $-400 < p < -360$

Count



Energy loss | $-360 < p < -320$

Count



XP

$$\frac{\sigma}{\mu} = 8.03 \pm 0.38 \%$$

$$\mu = 402.8 \pm 1.4$$

$$\sigma = 32.3 \pm 1.4$$

WF

$$\frac{\sigma}{\mu} = 8.43 \pm 0.35 \%$$

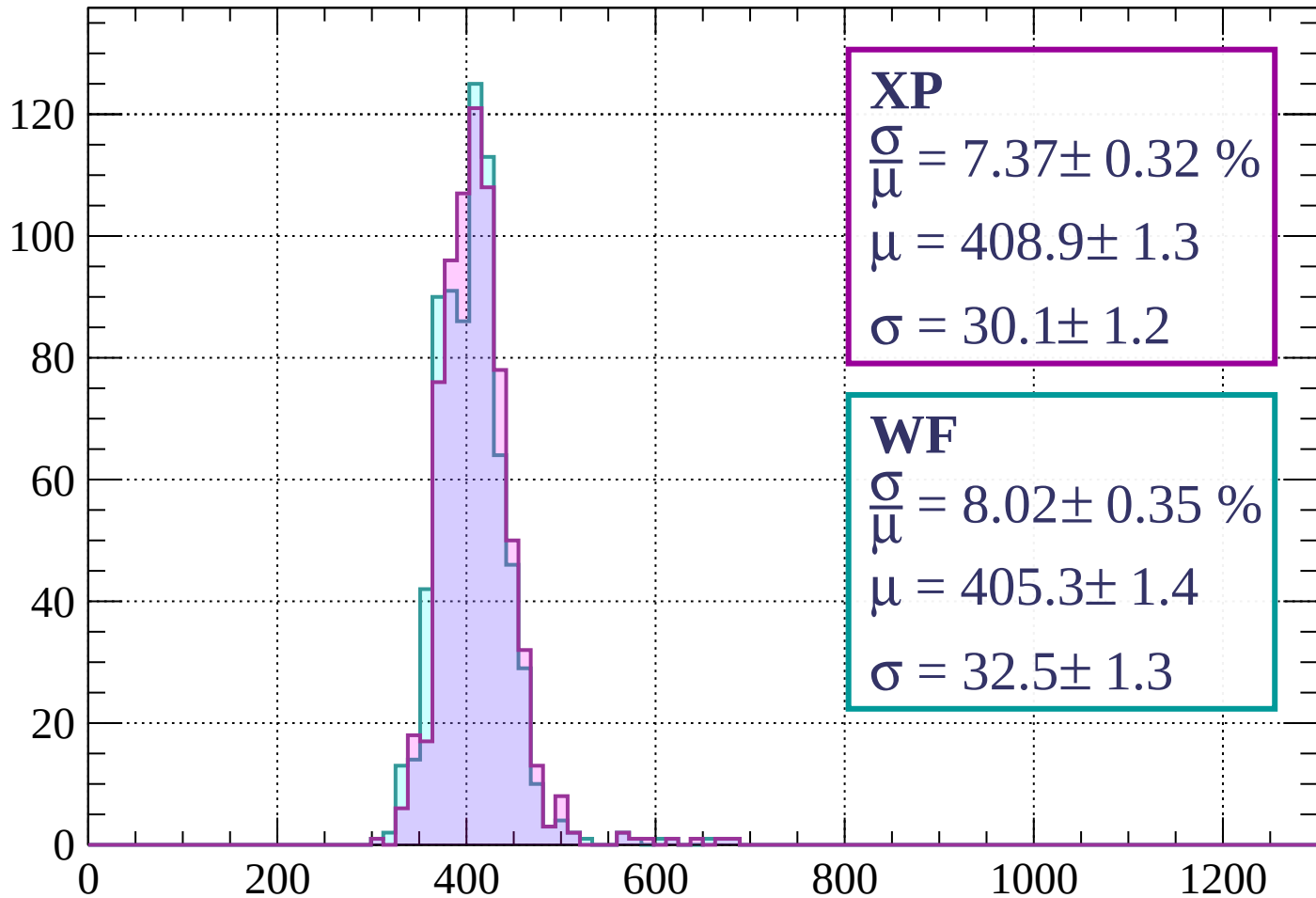
$$\mu = 398.2 \pm 1.4$$

$$\sigma = 33.6 \pm 1.3$$

dE/dx (ADC counts/cm)

Energy loss | $-320 < p < -280$

Count



XP

$$\frac{\sigma}{\mu} = 7.37 \pm 0.32 \%$$

$$\mu = 408.9 \pm 1.3$$

$$\sigma = 30.1 \pm 1.2$$

WF

$$\frac{\sigma}{\mu} = 8.02 \pm 0.35 \%$$

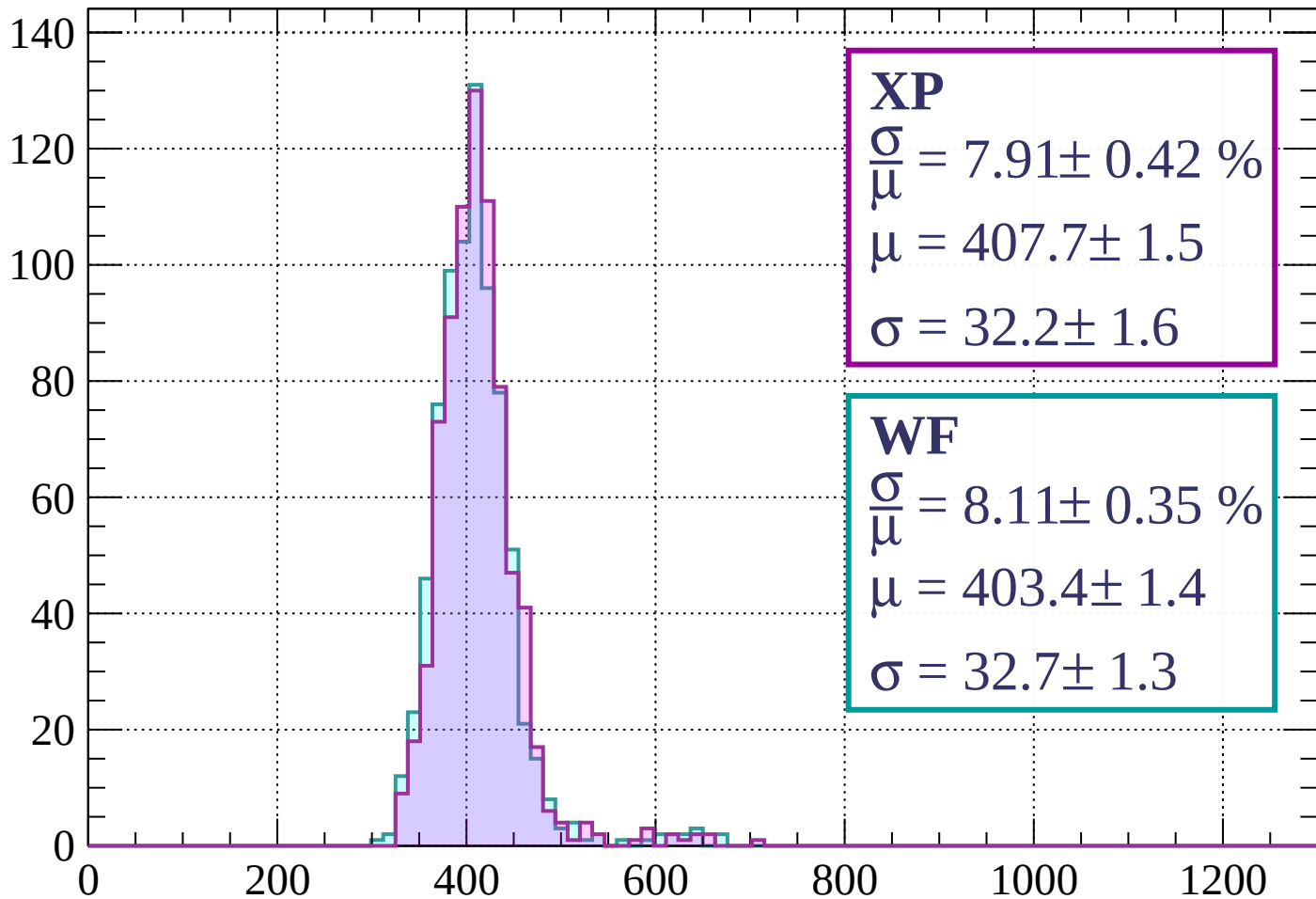
$$\mu = 405.3 \pm 1.4$$

$$\sigma = 32.5 \pm 1.3$$

dE/dx (ADC counts/cm)

Energy loss | $-280 < p < -240$

Count



XP

$$\frac{\sigma}{\mu} = 7.91 \pm 0.42 \%$$

$$\mu = 407.7 \pm 1.5$$

$$\sigma = 32.2 \pm 1.6$$

WF

$$\frac{\sigma}{\mu} = 8.11 \pm 0.35 \%$$

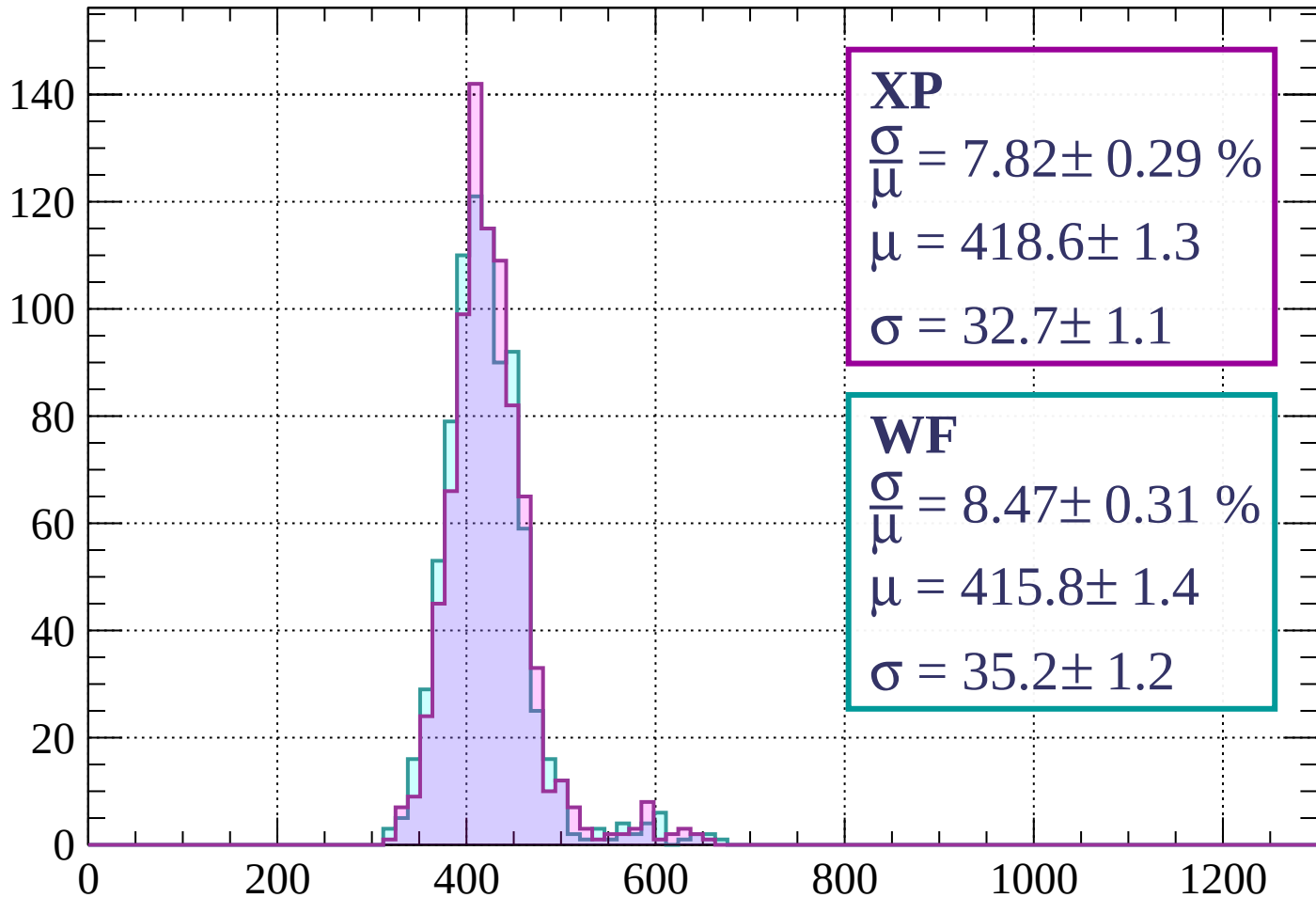
$$\mu = 403.4 \pm 1.4$$

$$\sigma = 32.7 \pm 1.3$$

dE/dx (ADC counts/cm)

Energy loss | $-240 < p < -200$

Count



XP

$$\frac{\sigma}{\mu} = 7.82 \pm 0.29 \%$$

$$\mu = 418.6 \pm 1.3$$

$$\sigma = 32.7 \pm 1.1$$

WF

$$\frac{\sigma}{\mu} = 8.47 \pm 0.31 \%$$

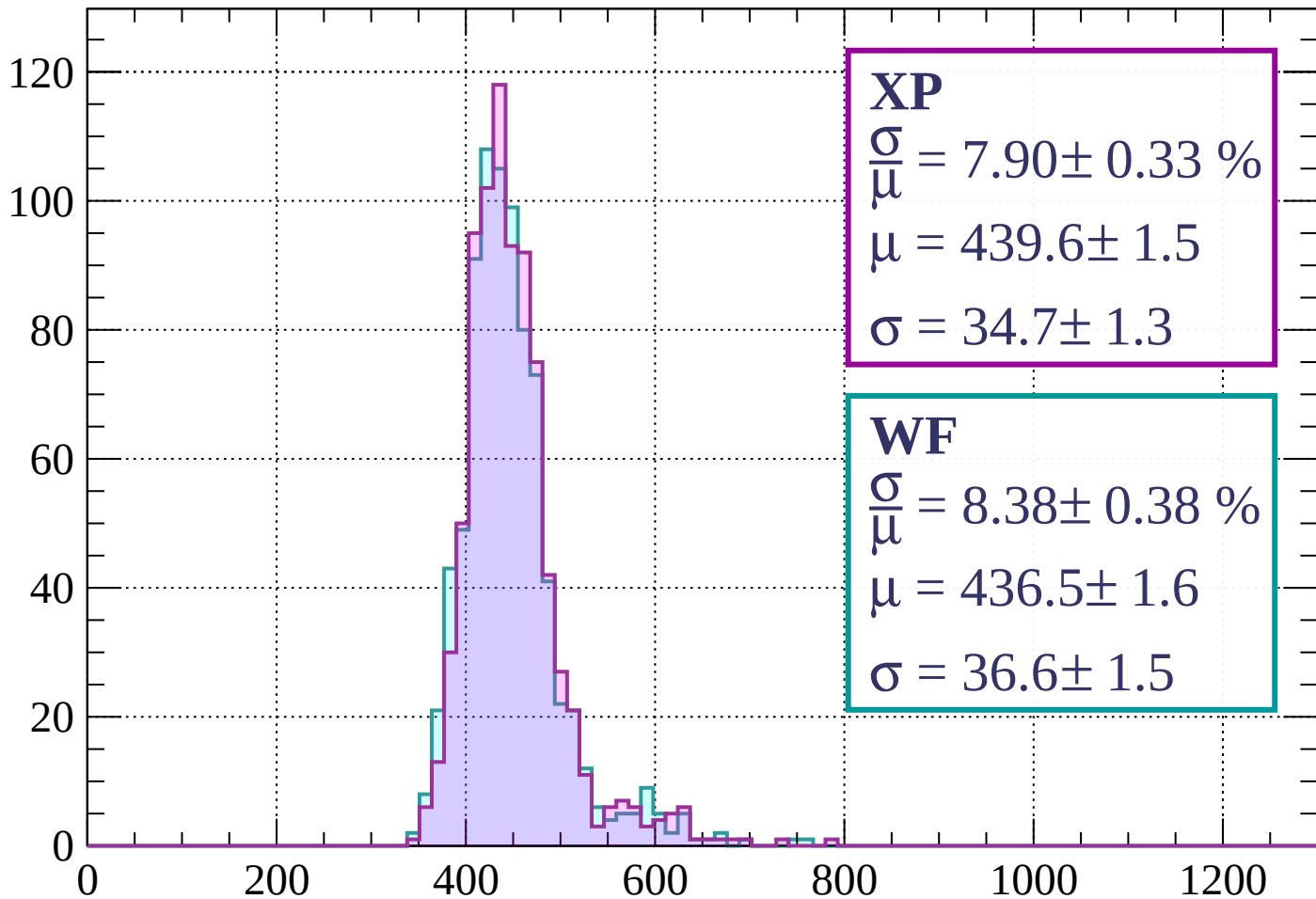
$$\mu = 415.8 \pm 1.4$$

$$\sigma = 35.2 \pm 1.2$$

dE/dx (ADC counts/cm)

Energy loss | $-200 < p < -160$

Count



XP

$$\frac{\sigma}{\mu} = 7.90 \pm 0.33 \%$$

$$\mu = 439.6 \pm 1.5$$

$$\sigma = 34.7 \pm 1.3$$

WF

$$\frac{\sigma}{\mu} = 8.38 \pm 0.38 \%$$

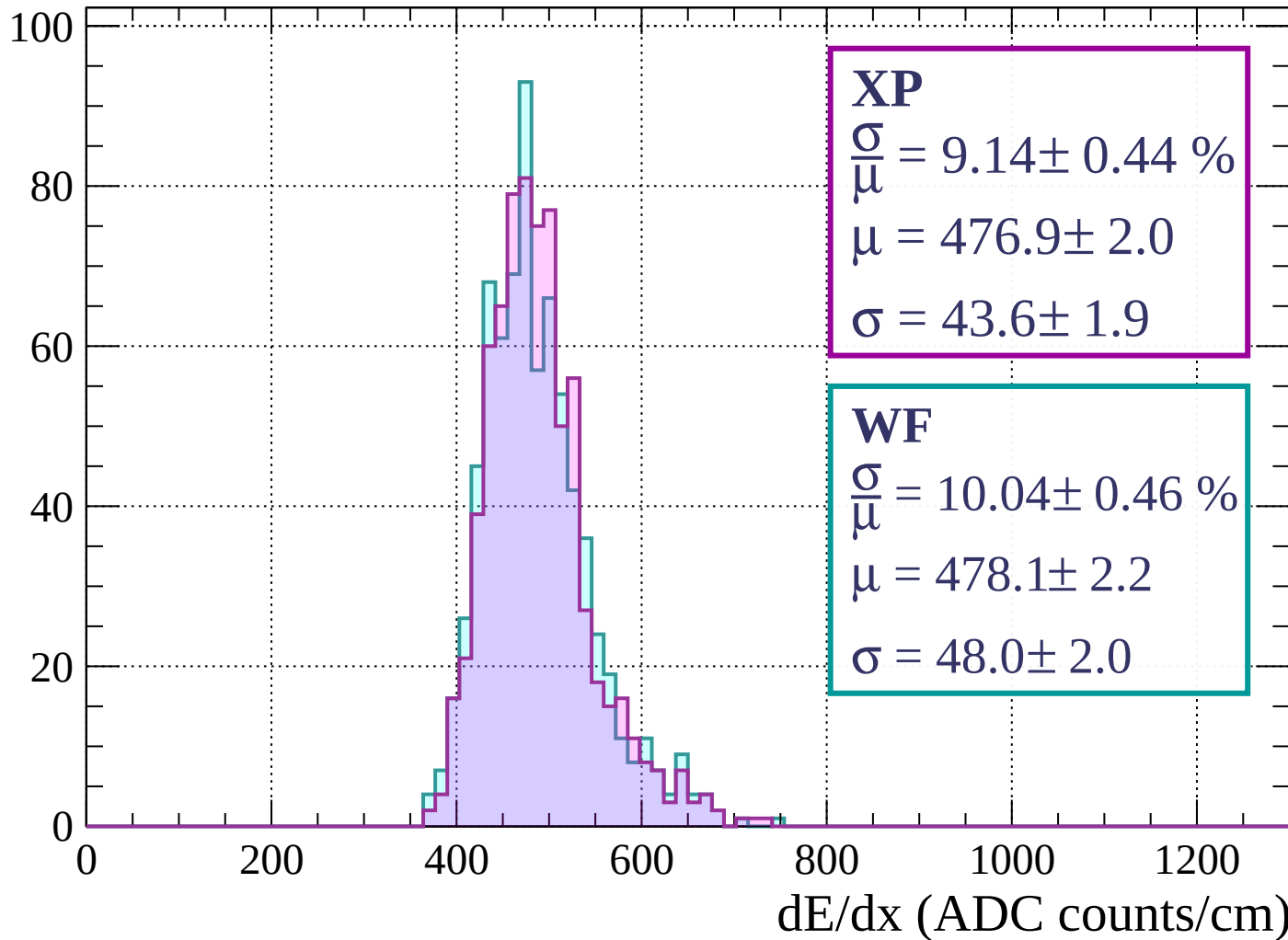
$$\mu = 436.5 \pm 1.6$$

$$\sigma = 36.6 \pm 1.5$$

dE/dx (ADC counts/cm)

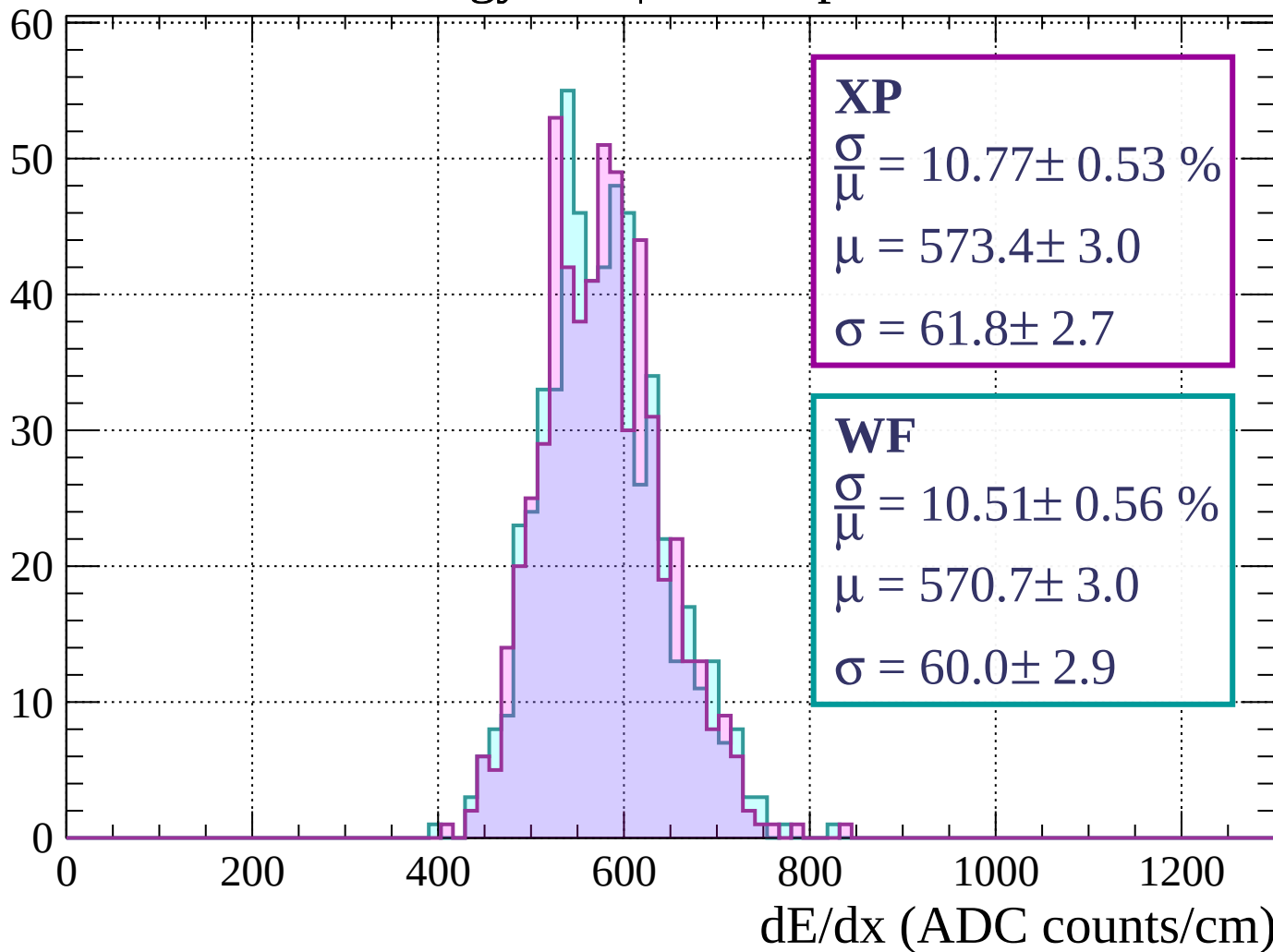
Energy loss | $-160 < p < -120$

Count



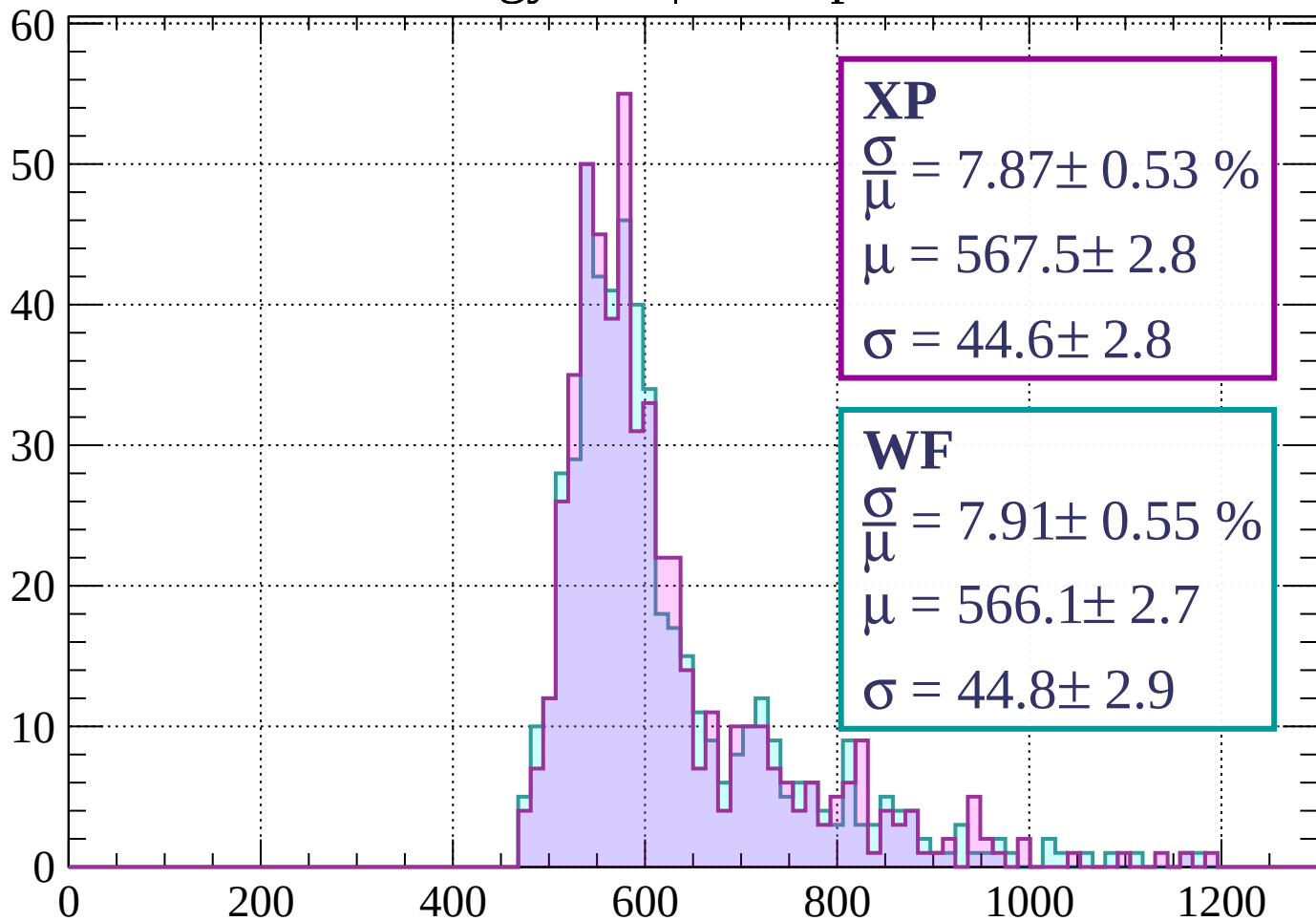
Energy loss | $-120 < p < -80$

Count



Energy loss | $-80 < p < -40$

Count



XP

$$\frac{\sigma}{\mu} = 7.87 \pm 0.53 \%$$

$$\mu = 567.5 \pm 2.8$$

$$\sigma = 44.6 \pm 2.8$$

WF

$$\frac{\sigma}{\mu} = 7.91 \pm 0.55 \%$$

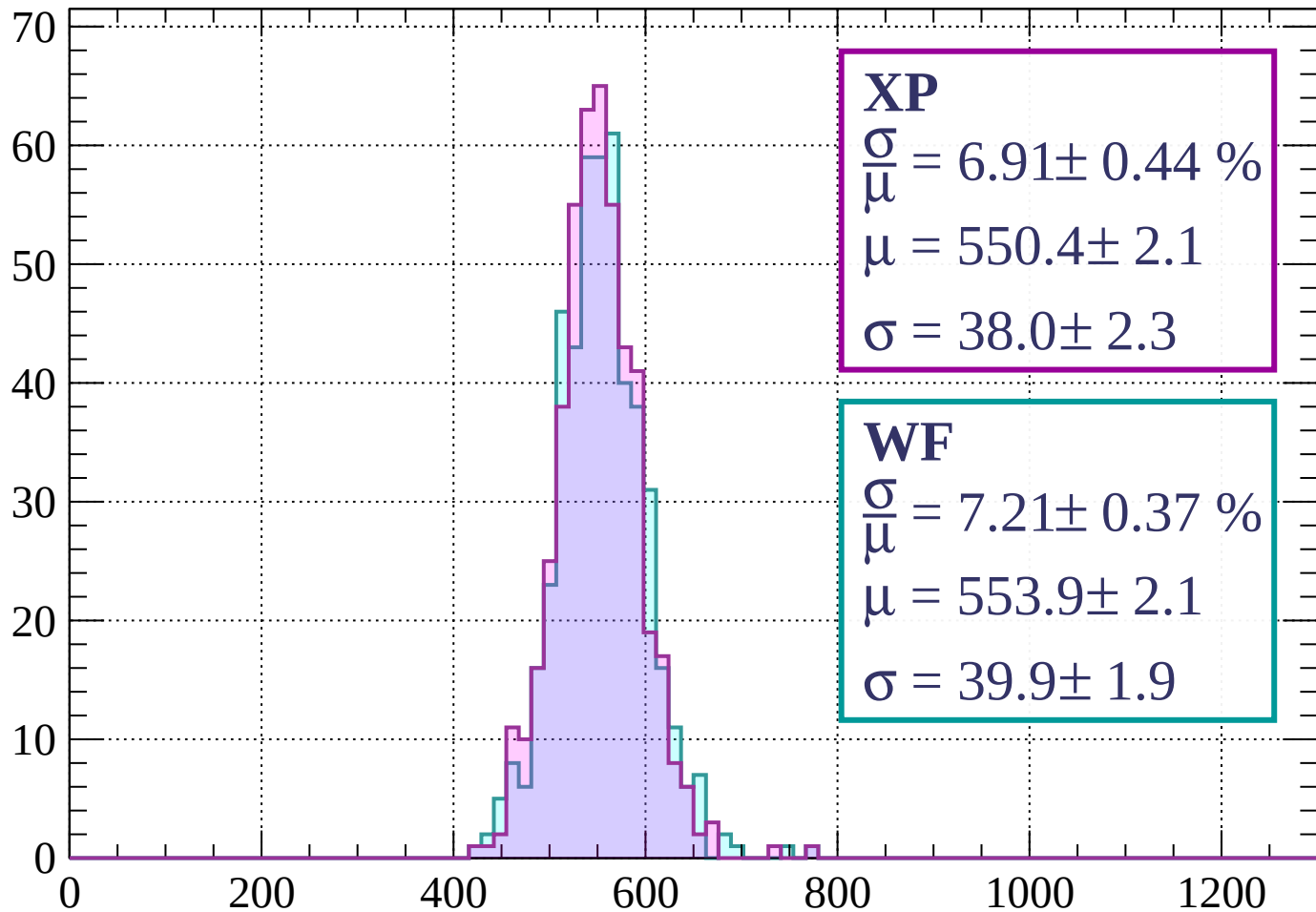
$$\mu = 566.1 \pm 2.7$$

$$\sigma = 44.8 \pm 2.9$$

dE/dx (ADC counts/cm)

Energy loss | $-40 < p < 0$

Count



XP

$$\frac{\sigma}{\mu} = 6.91 \pm 0.44 \%$$

$$\mu = 550.4 \pm 2.1$$

$$\sigma = 38.0 \pm 2.3$$

WF

$$\frac{\sigma}{\mu} = 7.21 \pm 0.37 \%$$

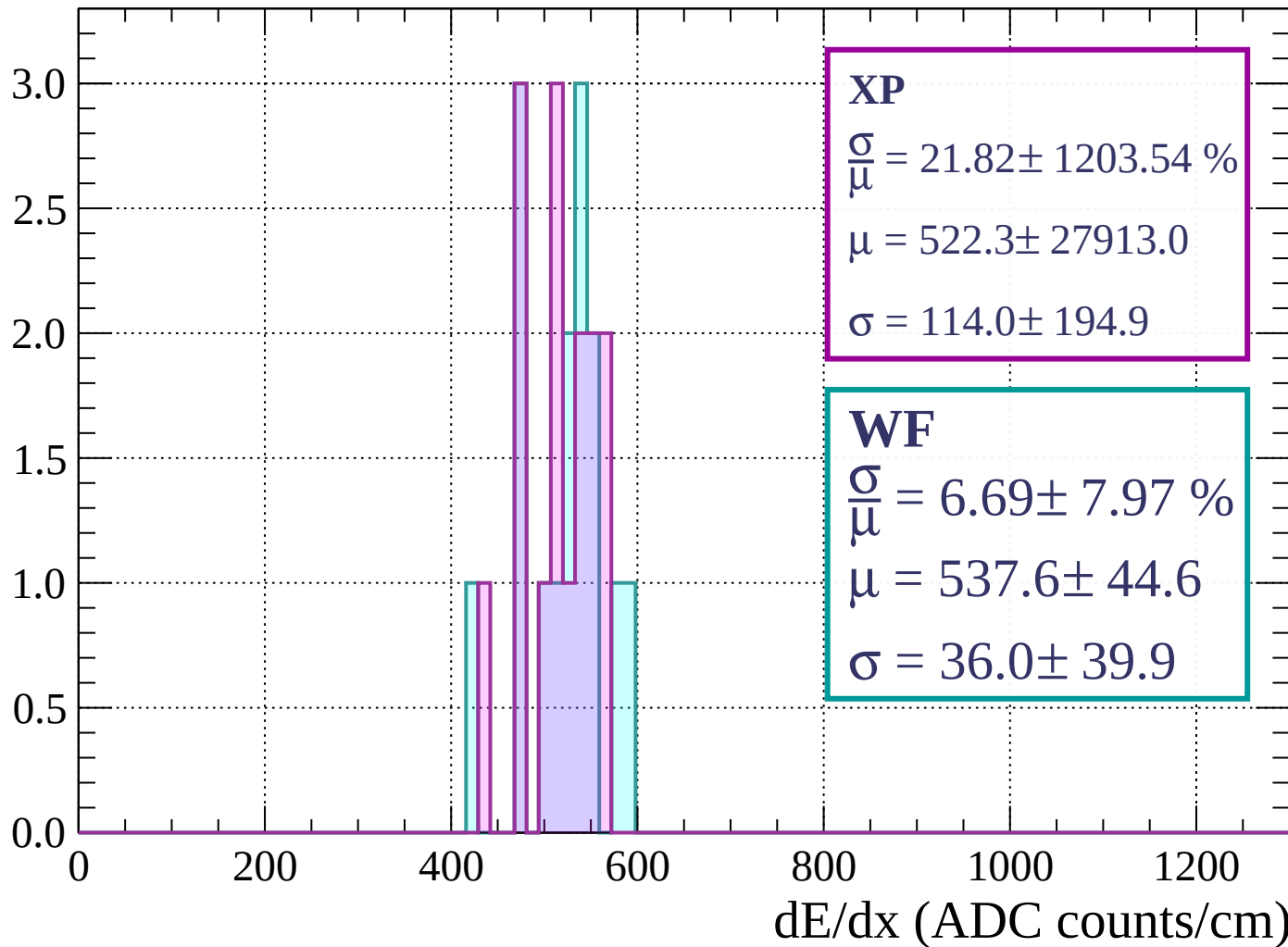
$$\mu = 553.9 \pm 2.1$$

$$\sigma = 39.9 \pm 1.9$$

dE/dx (ADC counts/cm)

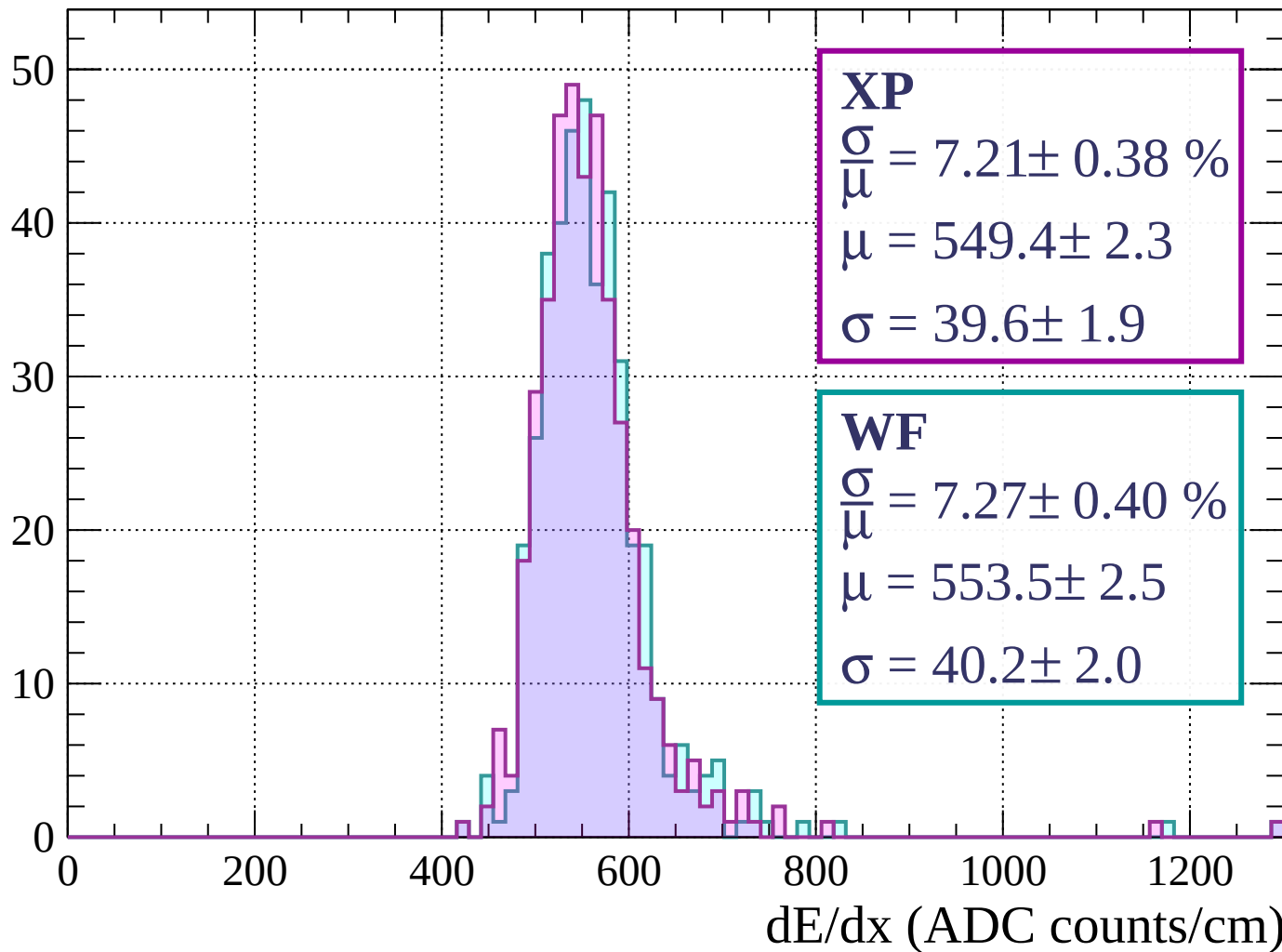
Energy loss | $0 < p < 40$

Count



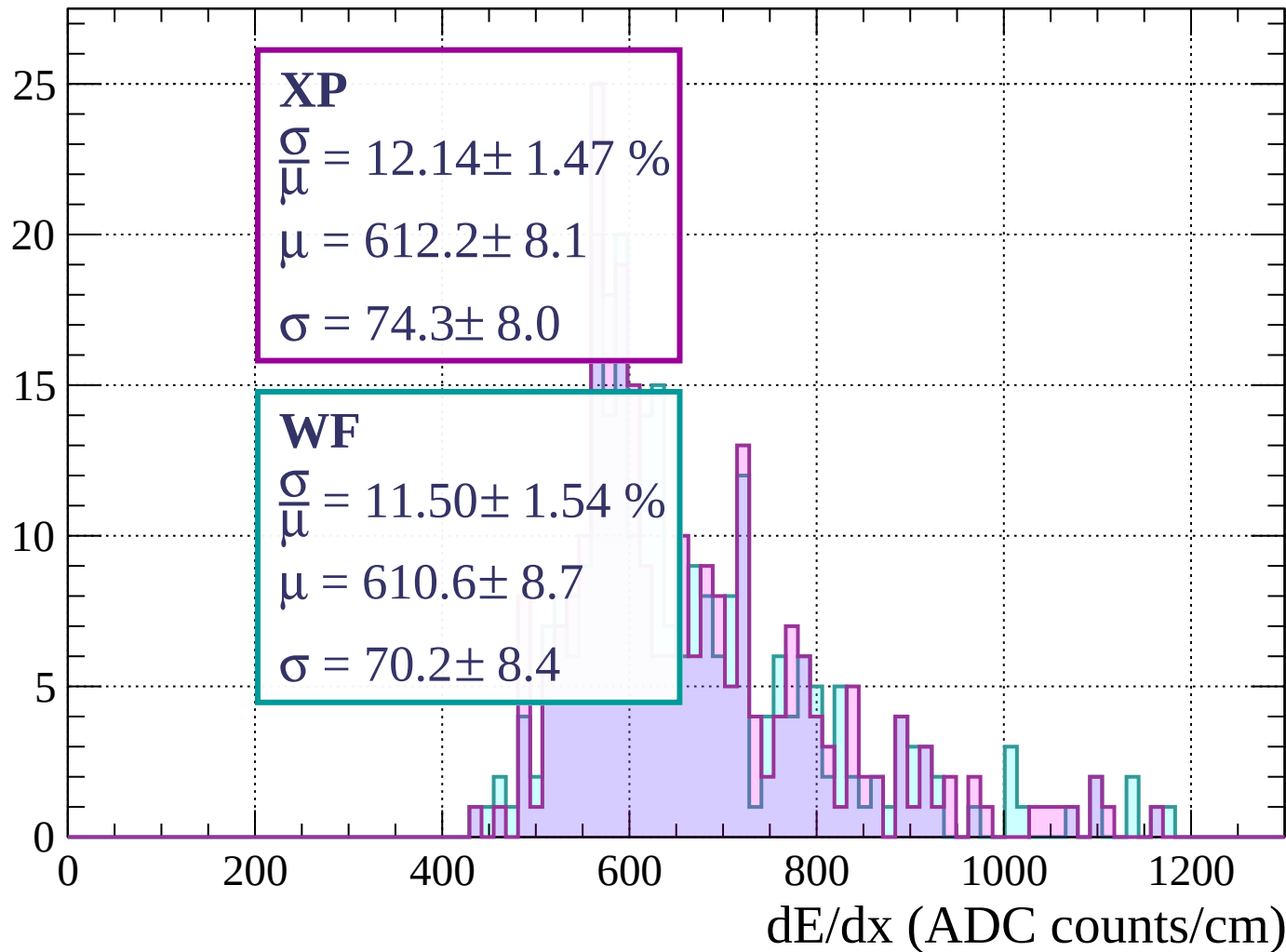
Energy loss | $40 < p < 80$

Count



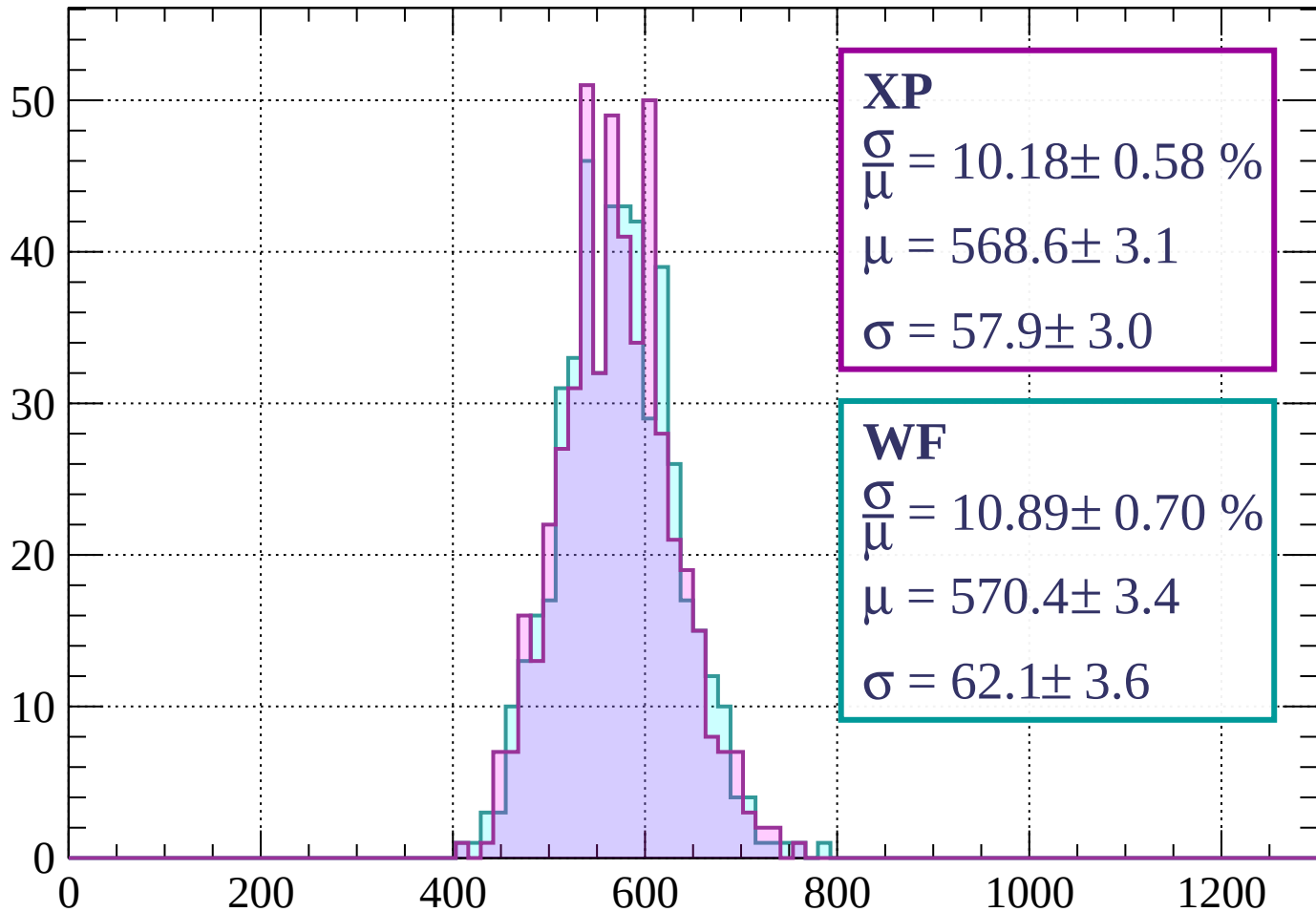
Energy loss | $80 < p < 120$

Count



Energy loss | $120 < p < 160$

Count



XP

$$\frac{\sigma}{\mu} = 10.18 \pm 0.58 \%$$

$$\mu = 568.6 \pm 3.1$$

$$\sigma = 57.9 \pm 3.0$$

WF

$$\frac{\sigma}{\mu} = 10.89 \pm 0.70 \%$$

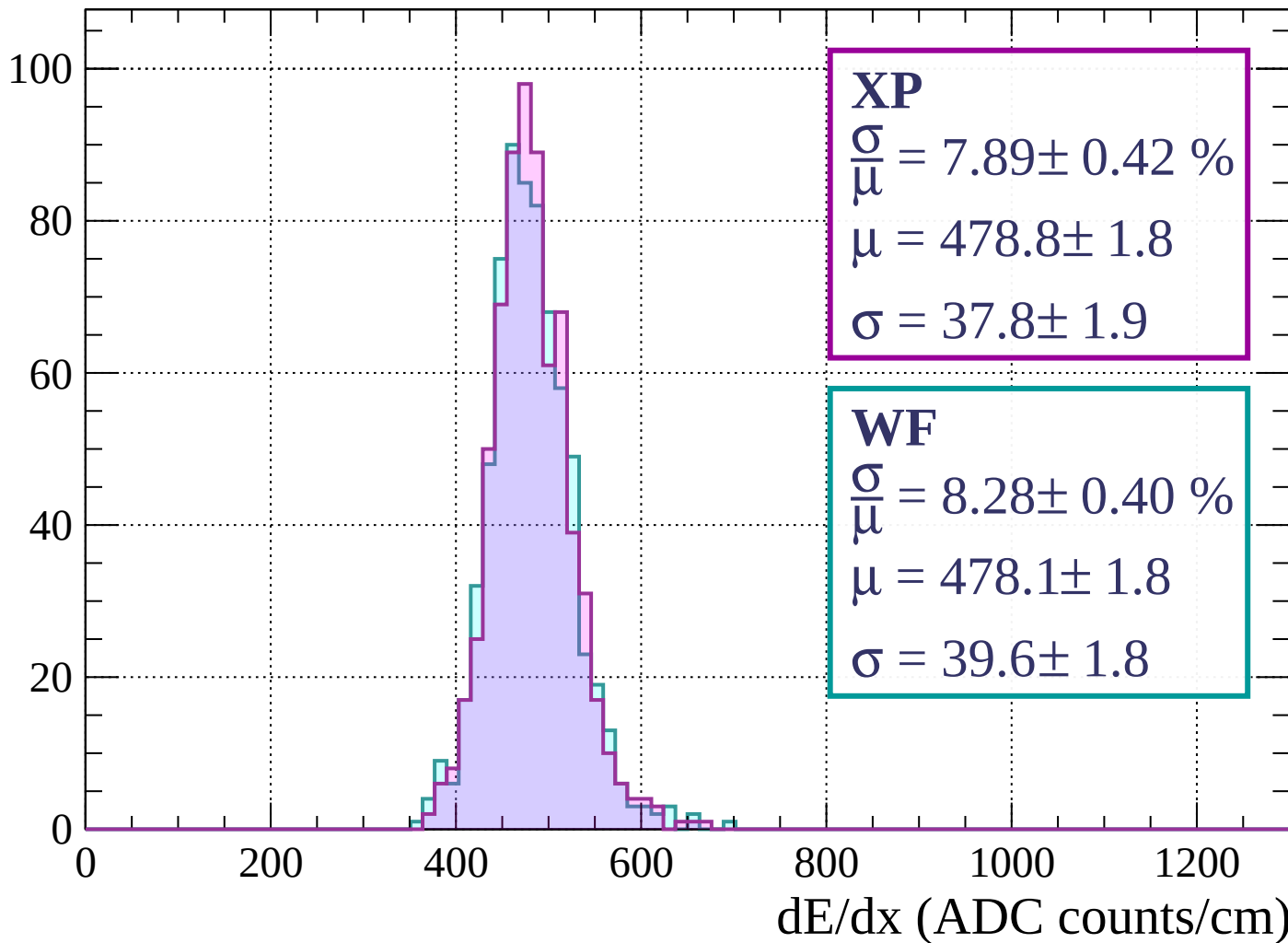
$$\mu = 570.4 \pm 3.4$$

$$\sigma = 62.1 \pm 3.6$$

dE/dx (ADC counts/cm)

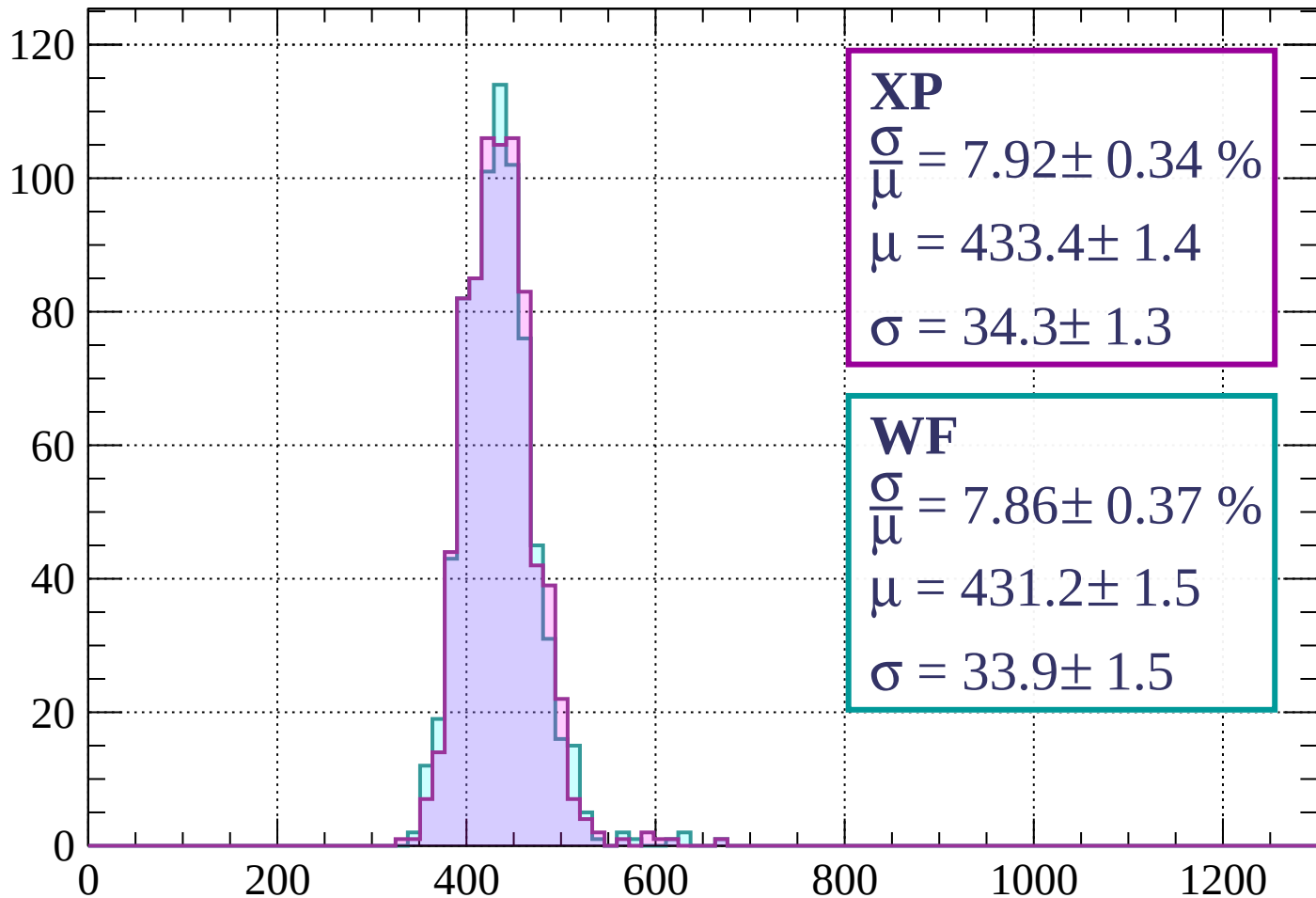
Energy loss | $160 < p < 200$

Count



Energy loss | $200 < p < 240$

Count



XP

$$\frac{\sigma}{\mu} = 7.92 \pm 0.34 \%$$

$$\mu = 433.4 \pm 1.4$$

$$\sigma = 34.3 \pm 1.3$$

WF

$$\frac{\sigma}{\mu} = 7.86 \pm 0.37 \%$$

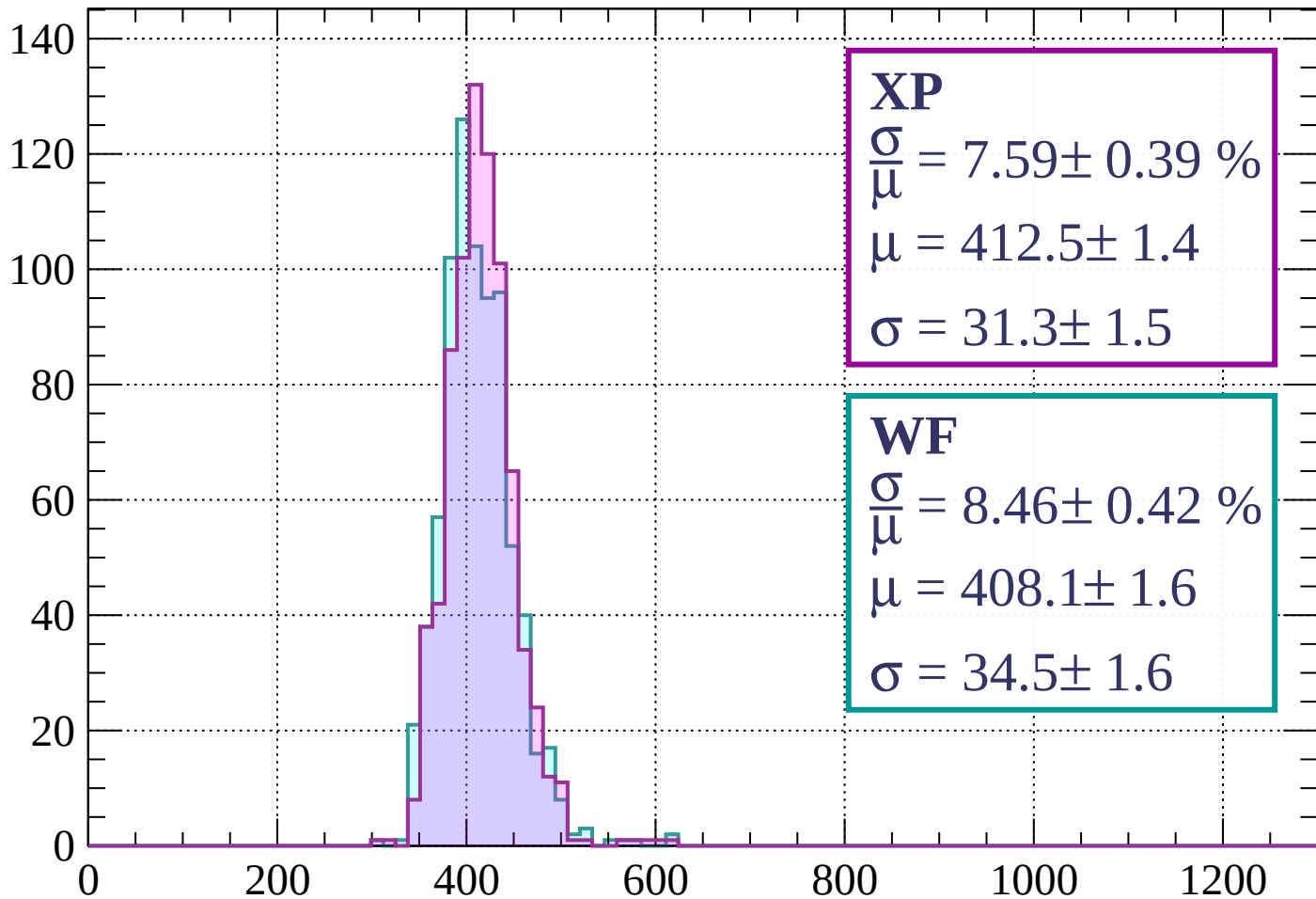
$$\mu = 431.2 \pm 1.5$$

$$\sigma = 33.9 \pm 1.5$$

dE/dx (ADC counts/cm)

Energy loss | $240 < p < 280$

Count



XP

$$\frac{\sigma}{\mu} = 7.59 \pm 0.39 \%$$

$$\mu = 412.5 \pm 1.4$$

$$\sigma = 31.3 \pm 1.5$$

WF

$$\frac{\sigma}{\mu} = 8.46 \pm 0.42 \%$$

$$\mu = 408.1 \pm 1.6$$

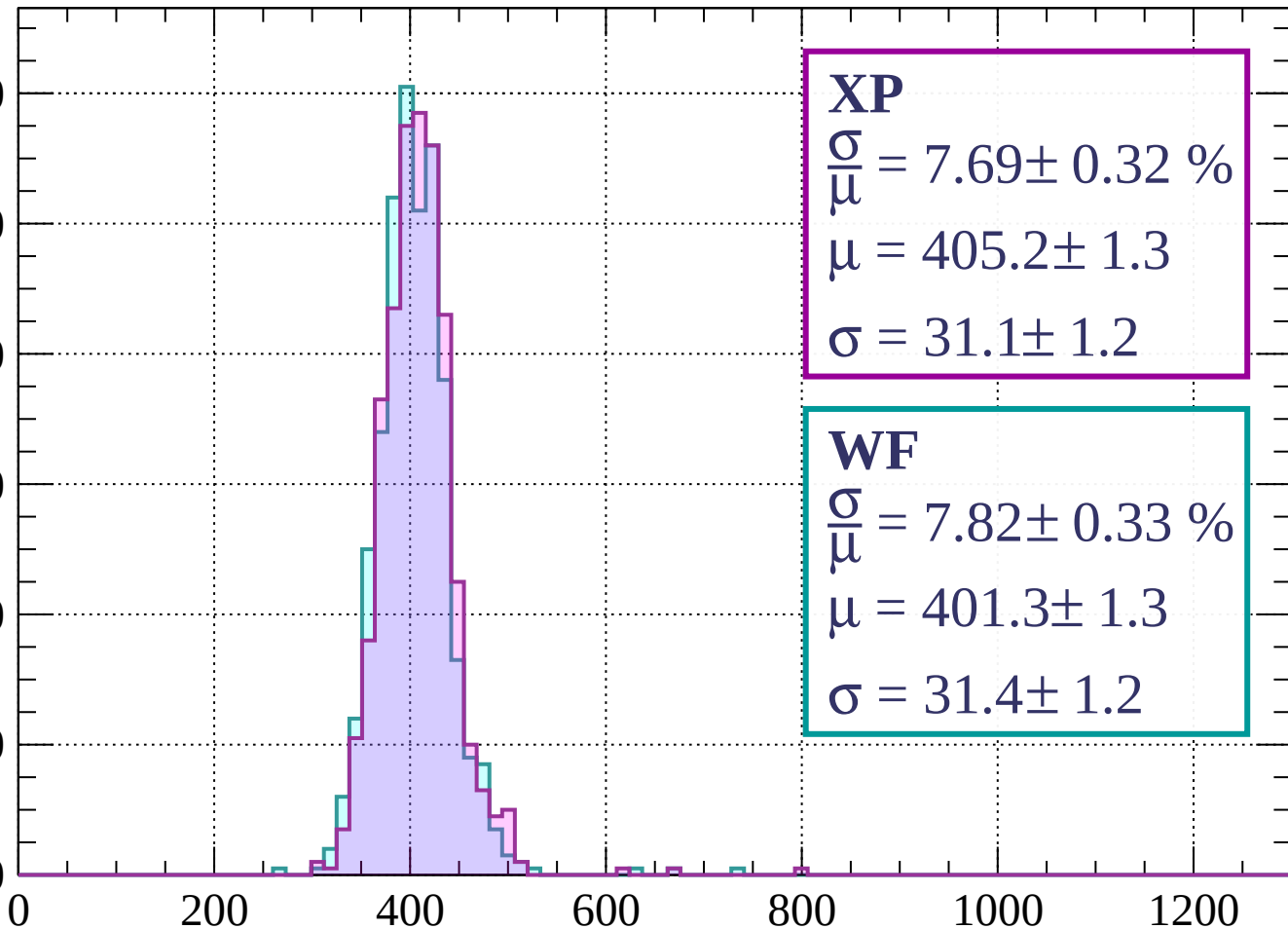
$$\sigma = 34.5 \pm 1.6$$

dE/dx (ADC counts/cm)

Energy loss | $280 < p < 320$

Count

120
100
80
60
40
20
0



XP

$$\frac{\sigma}{\mu} = 7.69 \pm 0.32 \%$$

$$\mu = 405.2 \pm 1.3$$

$$\sigma = 31.1 \pm 1.2$$

WF

$$\frac{\sigma}{\mu} = 7.82 \pm 0.33 \%$$

$$\mu = 401.3 \pm 1.3$$

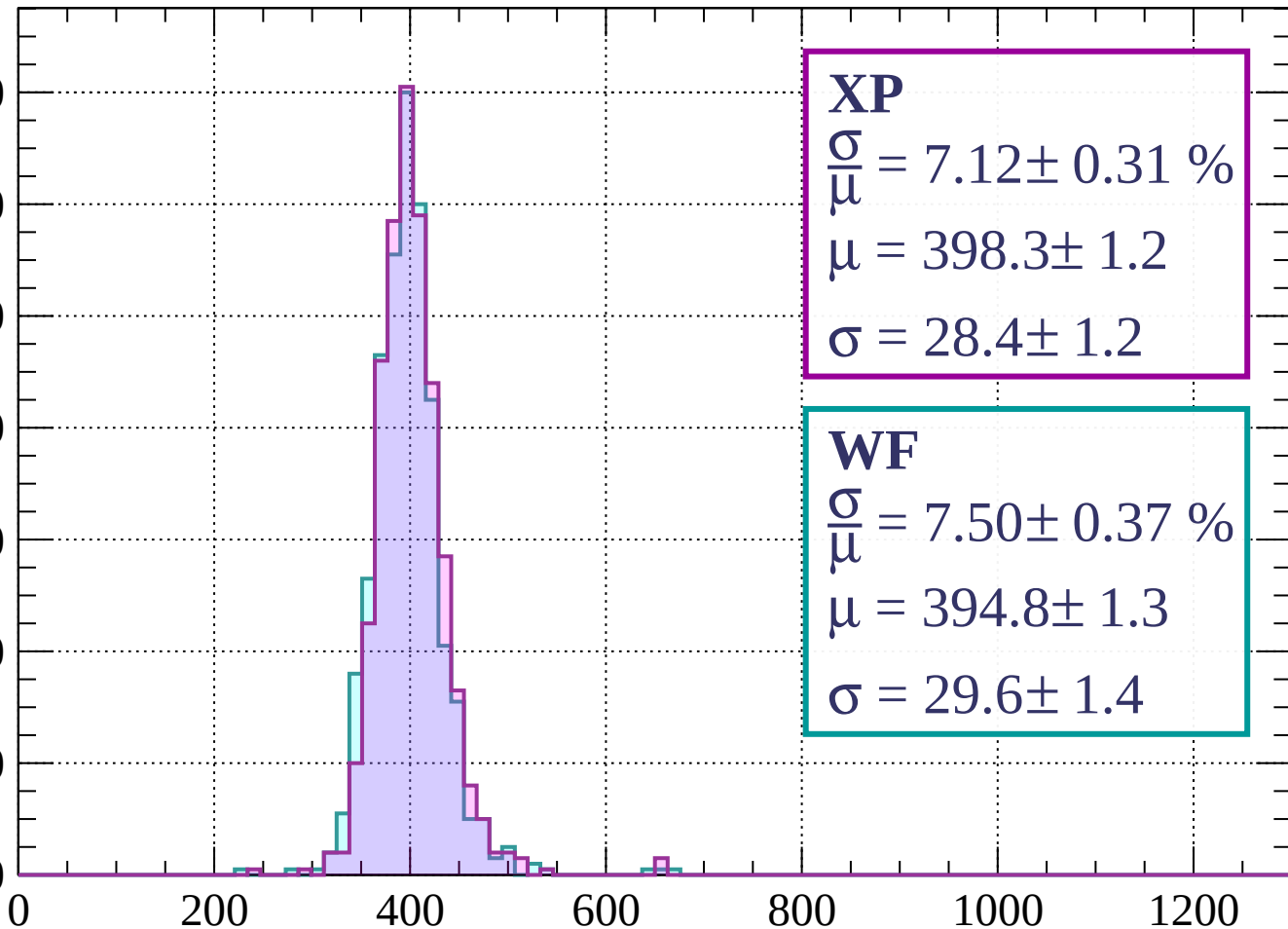
$$\sigma = 31.4 \pm 1.2$$

dE/dx (ADC counts/cm)

Energy loss | $320 < p < 360$

Count

140
120
100
80
60
40
20
0



XP

$$\frac{\sigma}{\mu} = 7.12 \pm 0.31 \%$$

$$\mu = 398.3 \pm 1.2$$

$$\sigma = 28.4 \pm 1.2$$

WF

$$\frac{\sigma}{\mu} = 7.50 \pm 0.37 \%$$

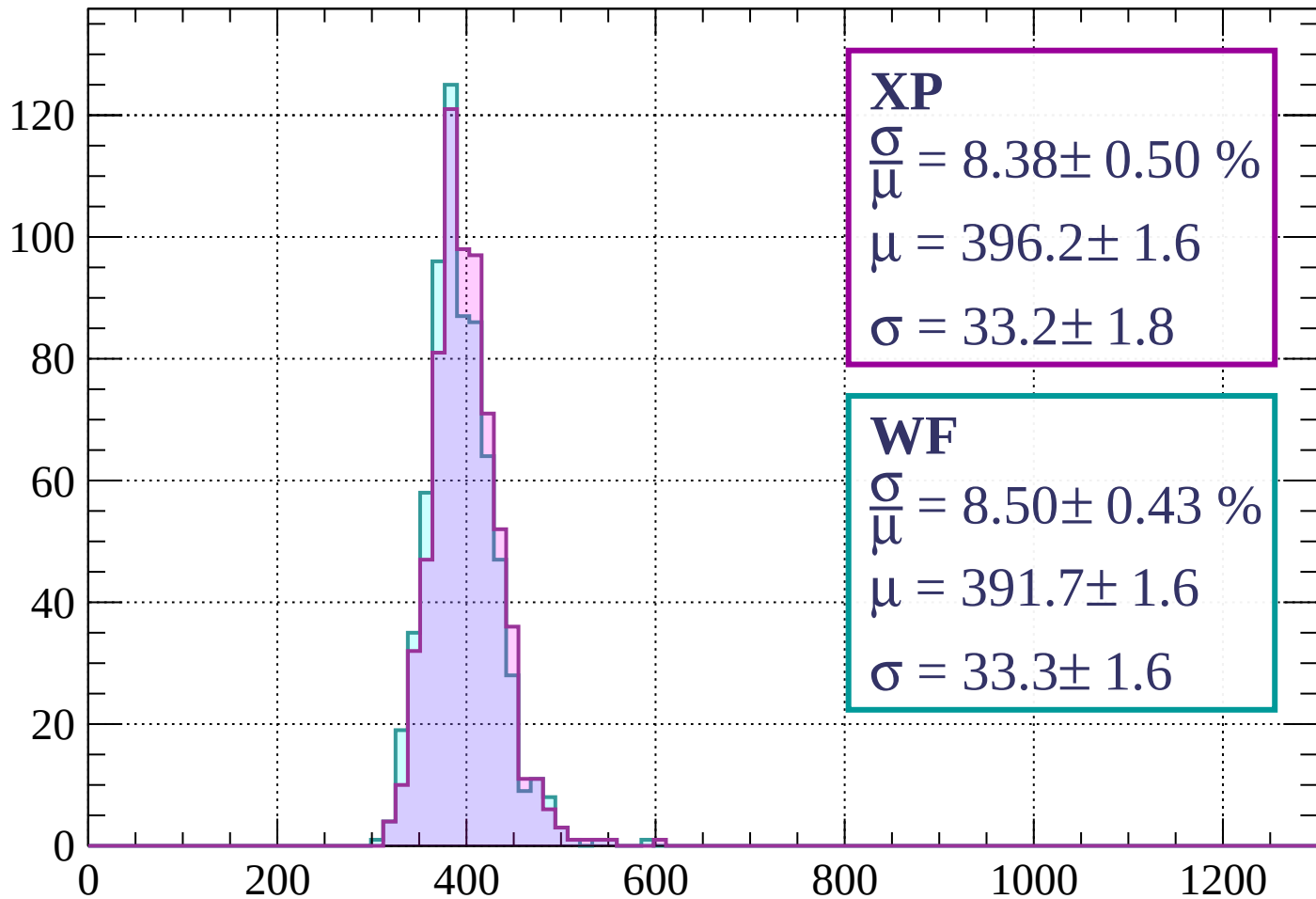
$$\mu = 394.8 \pm 1.3$$

$$\sigma = 29.6 \pm 1.4$$

dE/dx (ADC counts/cm)

Energy loss | $360 < p < 400$

Count



XP

$$\frac{\sigma}{\mu} = 8.38 \pm 0.50 \%$$

$$\mu = 396.2 \pm 1.6$$

$$\sigma = 33.2 \pm 1.8$$

WF

$$\frac{\sigma}{\mu} = 8.50 \pm 0.43 \%$$

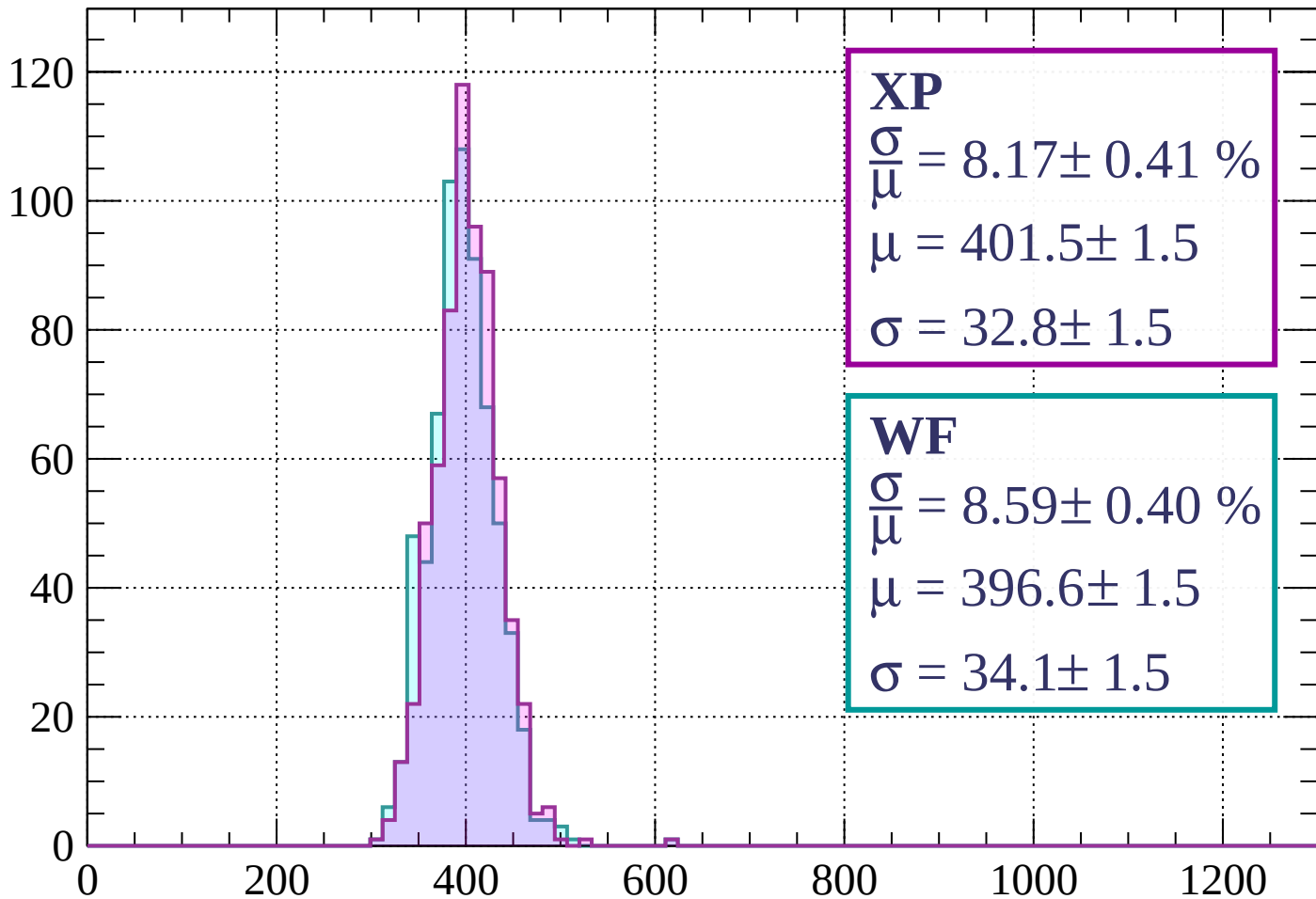
$$\mu = 391.7 \pm 1.6$$

$$\sigma = 33.3 \pm 1.6$$

dE/dx (ADC counts/cm)

Energy loss | $400 < p < 440$

Count



XP

$$\frac{\sigma}{\mu} = 8.17 \pm 0.41 \%$$

$$\mu = 401.5 \pm 1.5$$

$$\sigma = 32.8 \pm 1.5$$

WF

$$\frac{\sigma}{\mu} = 8.59 \pm 0.40 \%$$

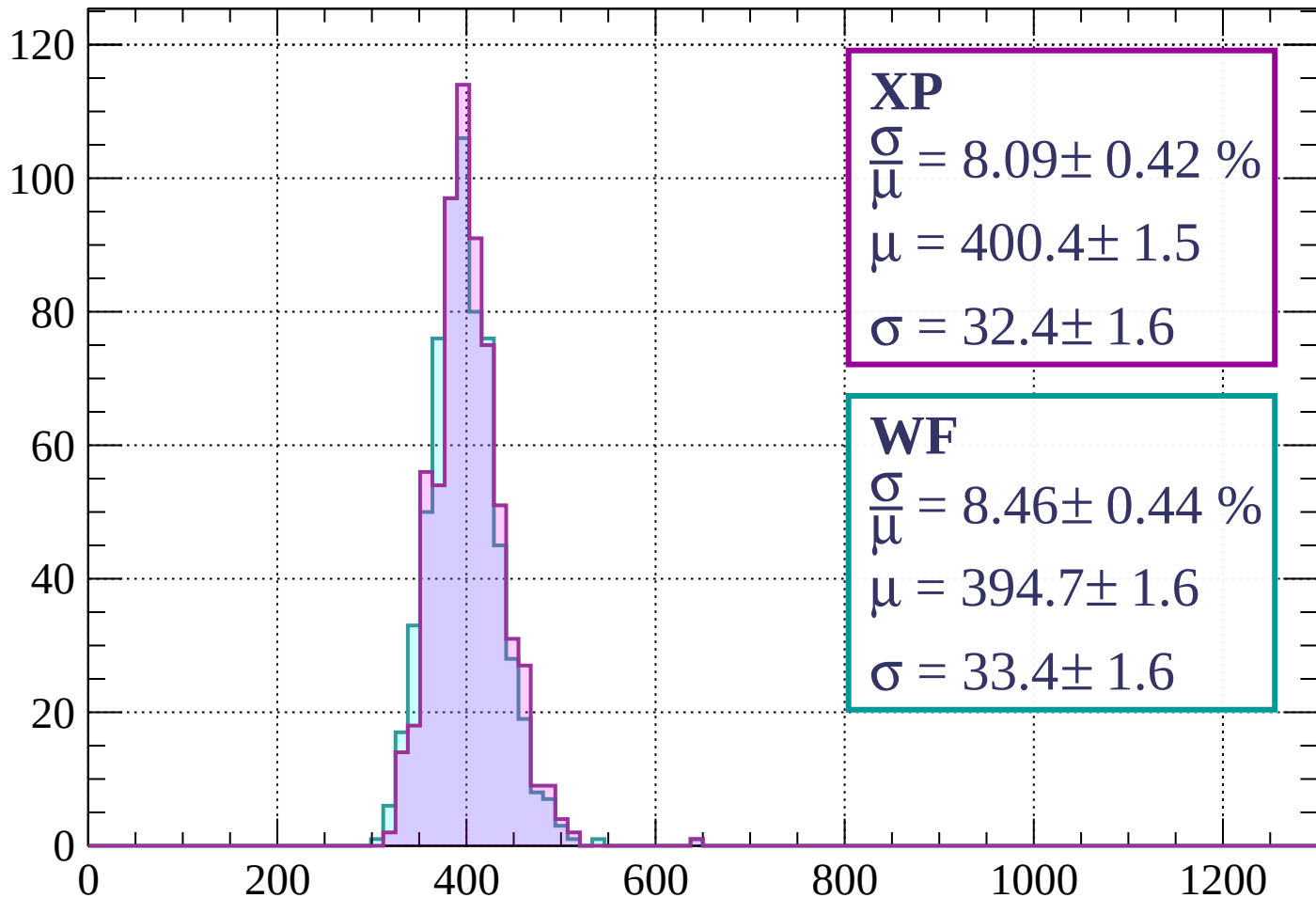
$$\mu = 396.6 \pm 1.5$$

$$\sigma = 34.1 \pm 1.5$$

dE/dx (ADC counts/cm)

Energy loss | $440 < p < 480$

Count



XP

$$\frac{\sigma}{\mu} = 8.09 \pm 0.42 \%$$

$$\mu = 400.4 \pm 1.5$$

$$\sigma = 32.4 \pm 1.6$$

WF

$$\frac{\sigma}{\mu} = 8.46 \pm 0.44 \%$$

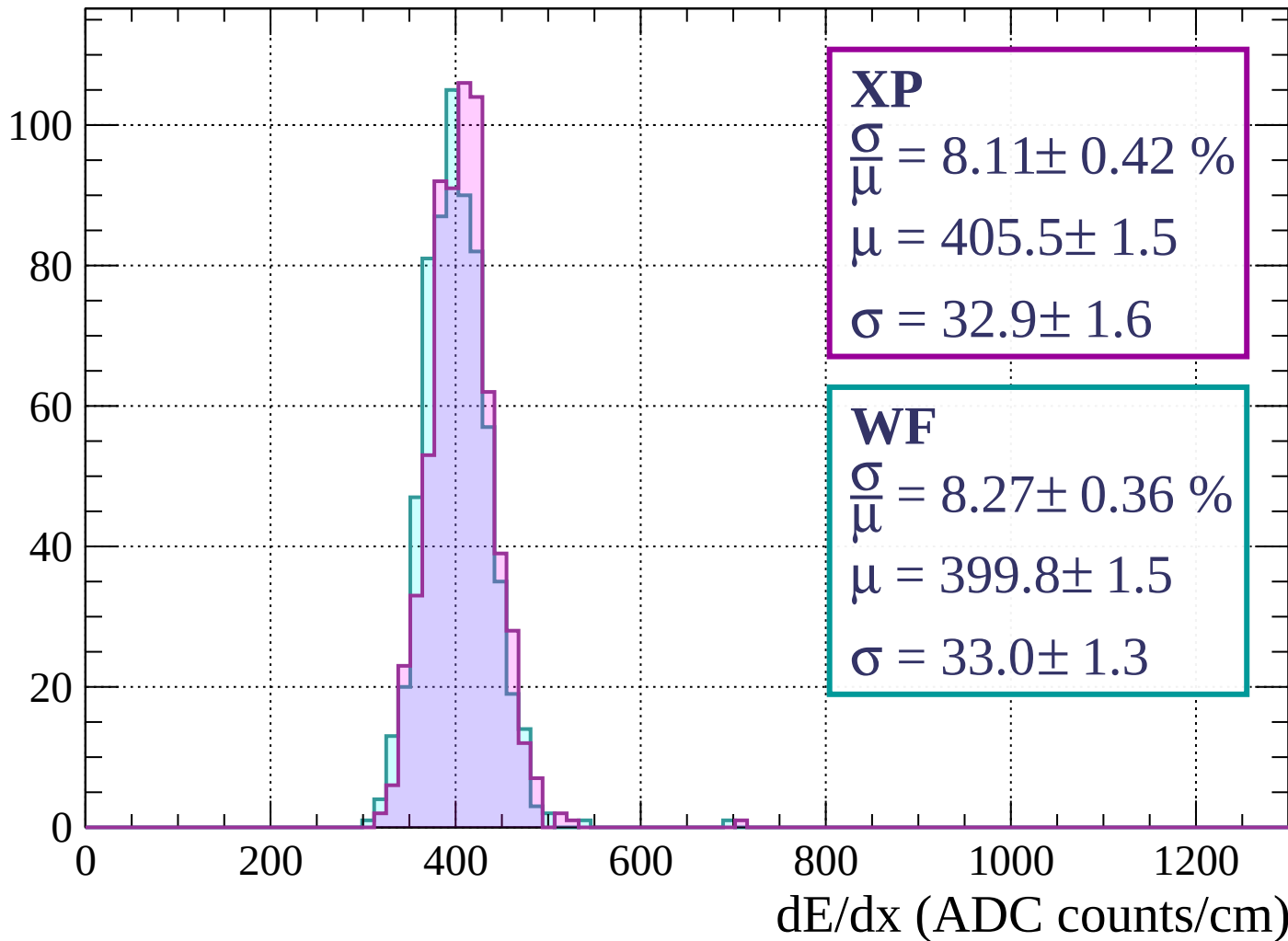
$$\mu = 394.7 \pm 1.6$$

$$\sigma = 33.4 \pm 1.6$$

dE/dx (ADC counts/cm)

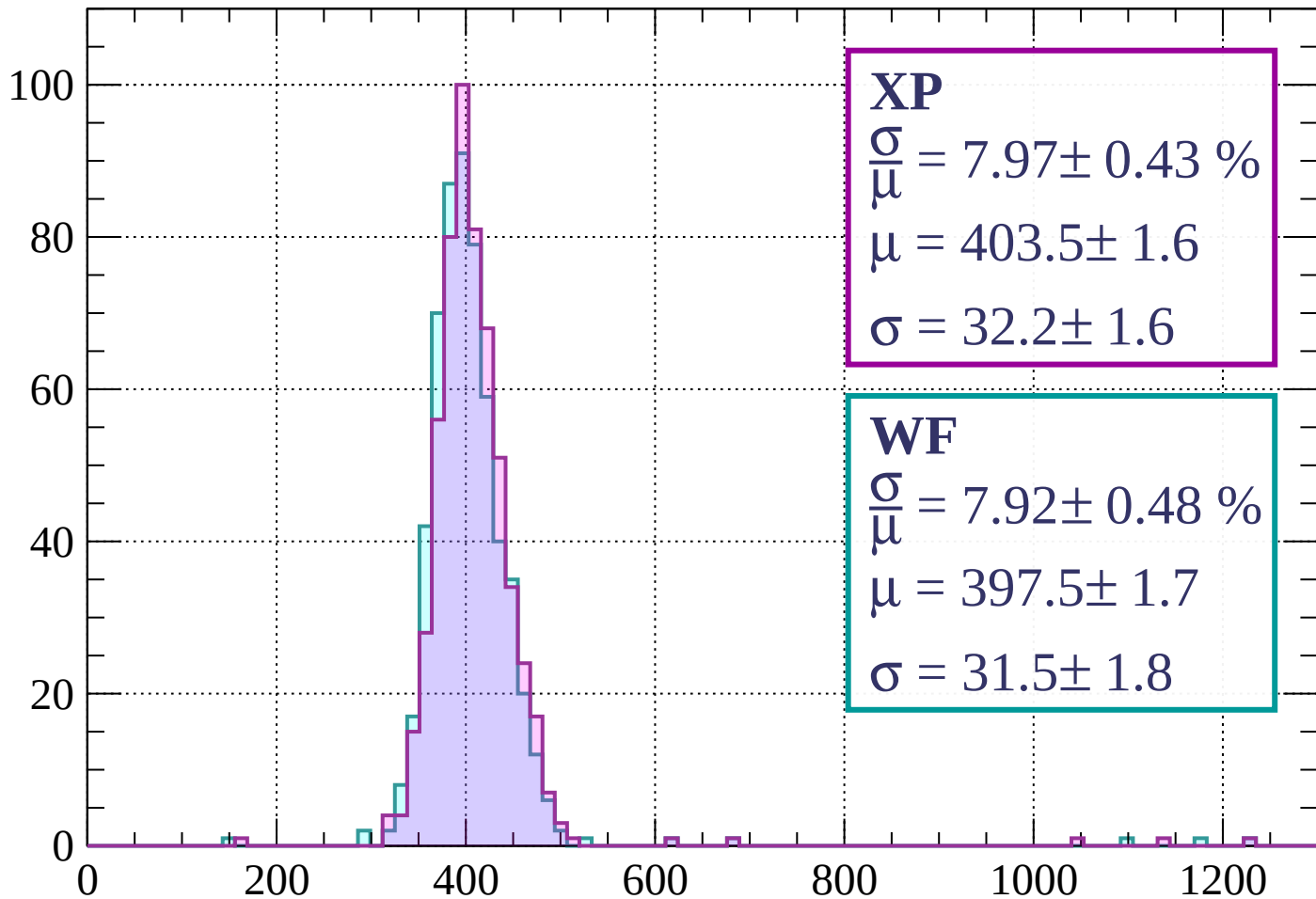
Energy loss | $480 < p < 520$

Count



Energy loss | $520 < p < 560$

Count



XP

$$\frac{\sigma}{\mu} = 7.97 \pm 0.43 \%$$

$$\mu = 403.5 \pm 1.6$$

$$\sigma = 32.2 \pm 1.6$$

WF

$$\frac{\sigma}{\mu} = 7.92 \pm 0.48 \%$$

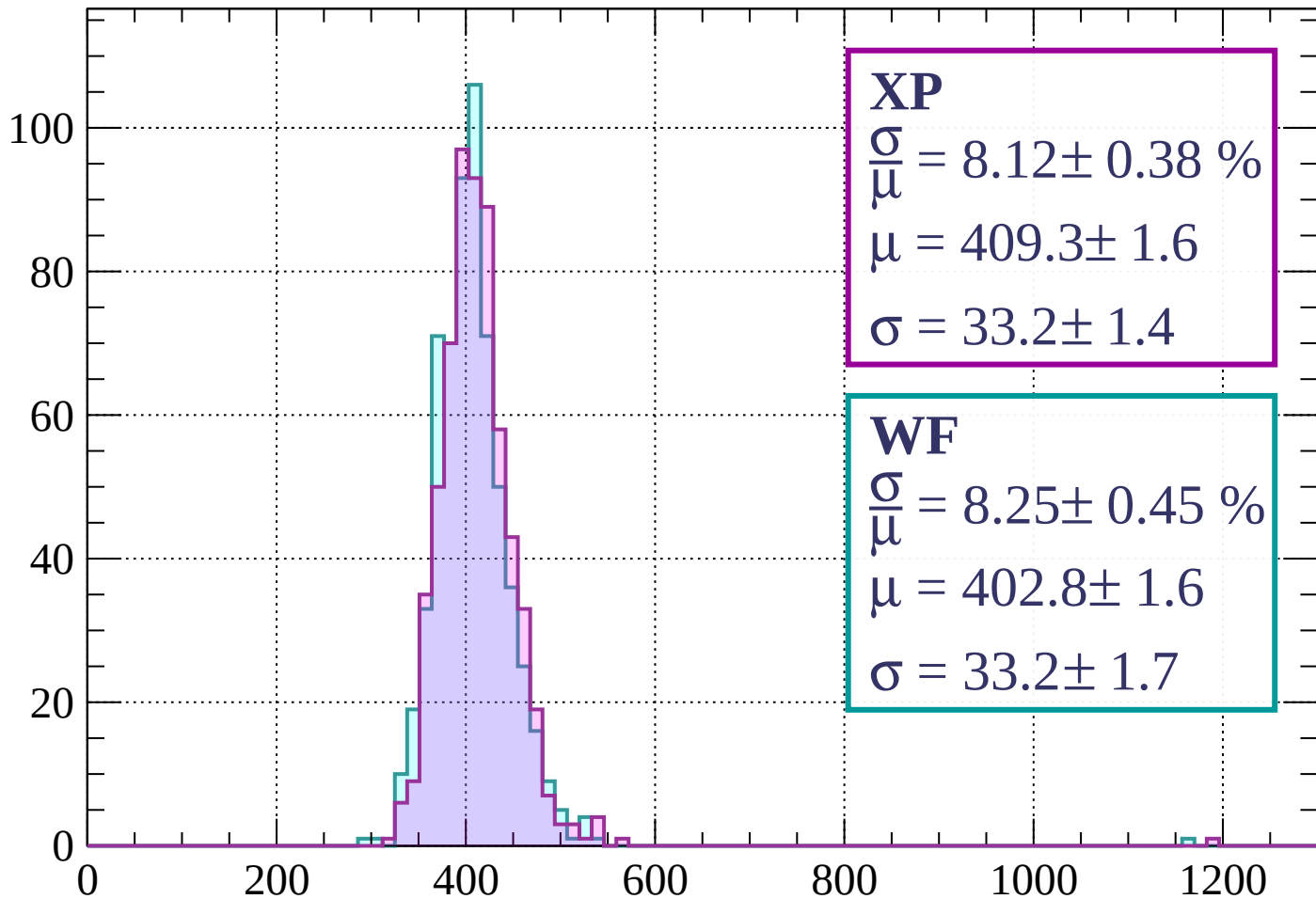
$$\mu = 397.5 \pm 1.7$$

$$\sigma = 31.5 \pm 1.8$$

dE/dx (ADC counts/cm)

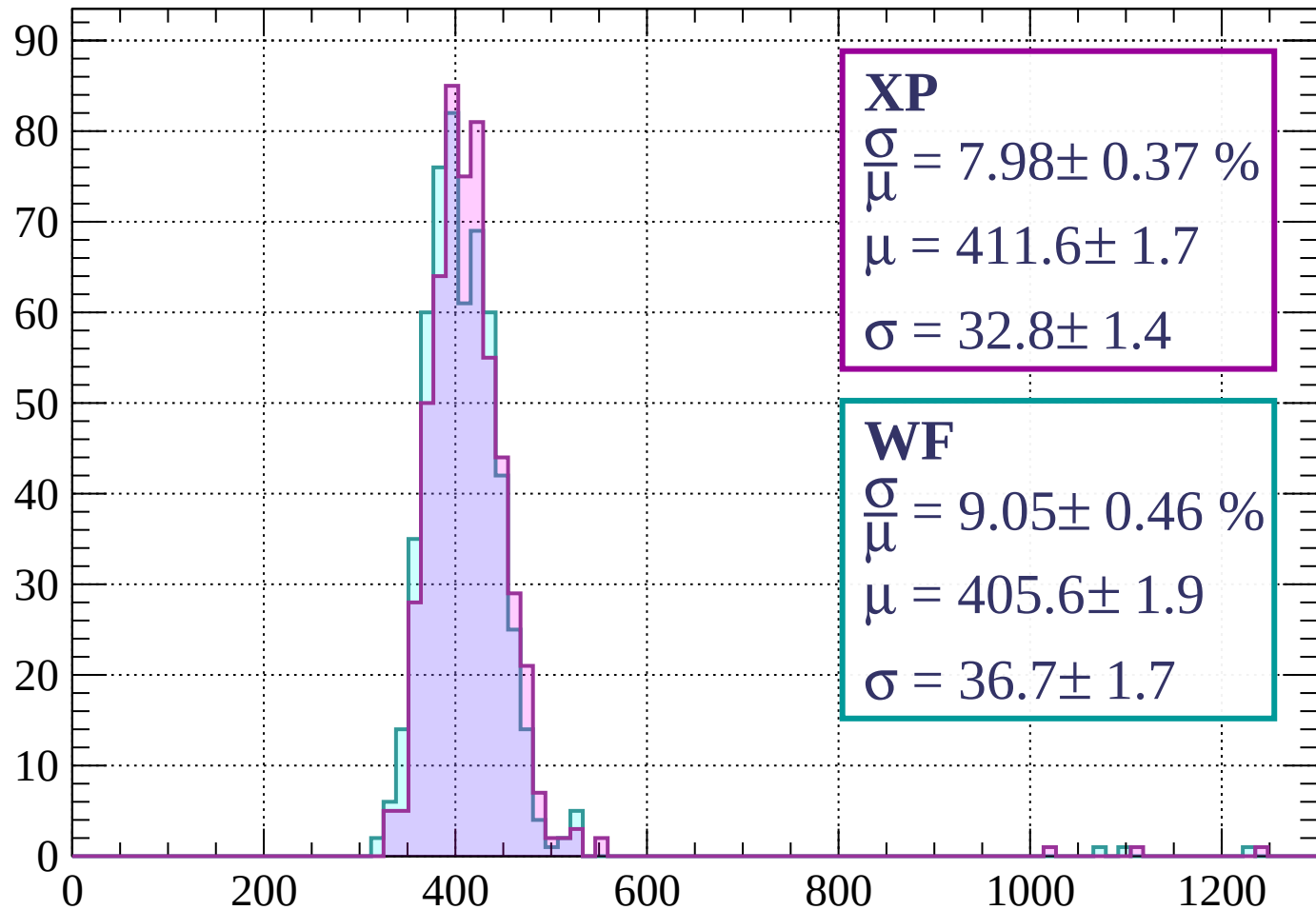
Energy loss | $560 < p < 600$

Count



Energy loss | $600 < p < 640$

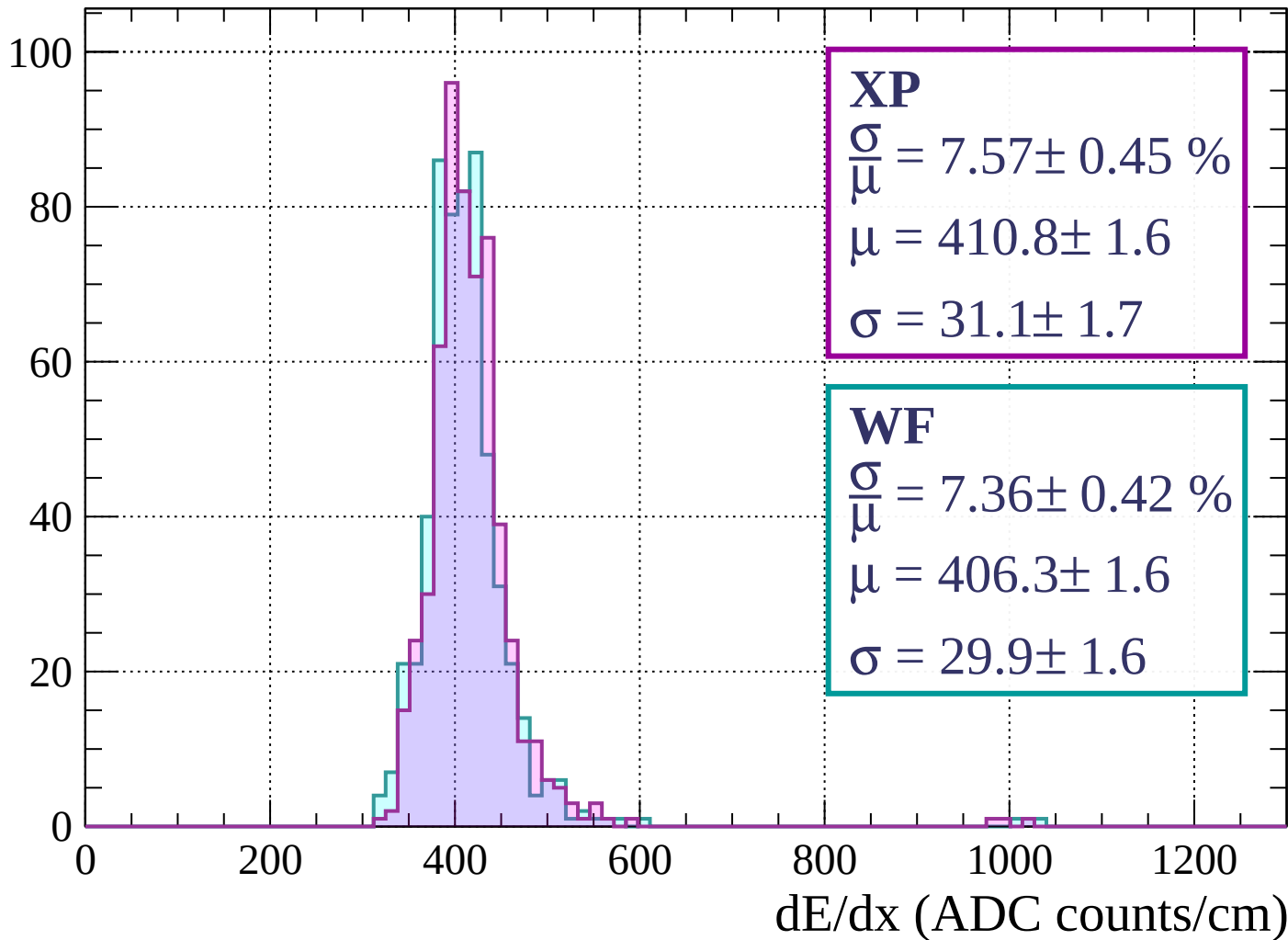
Count



dE/dx (ADC counts/cm)

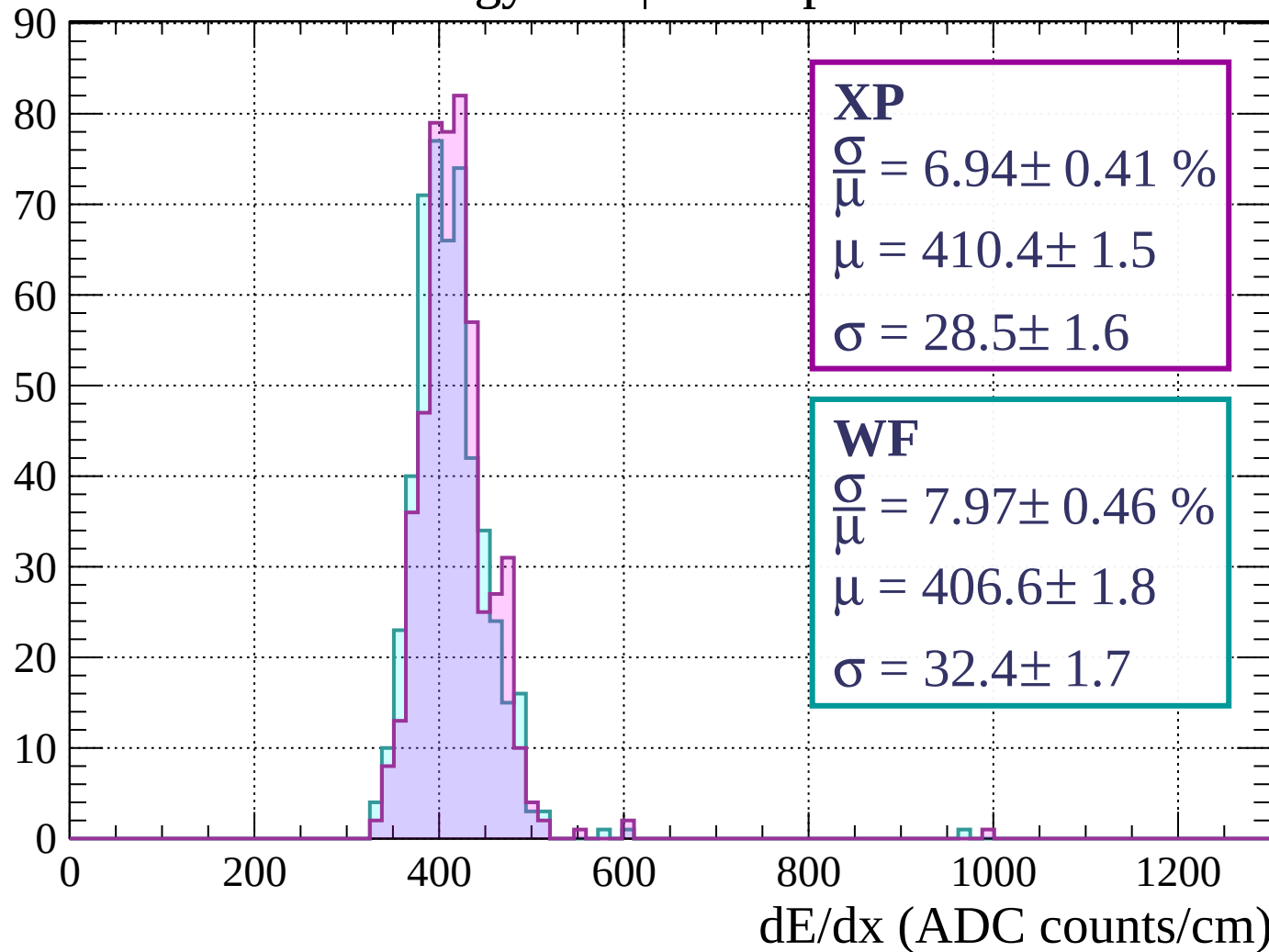
Energy loss | $640 < p < 680$

Count



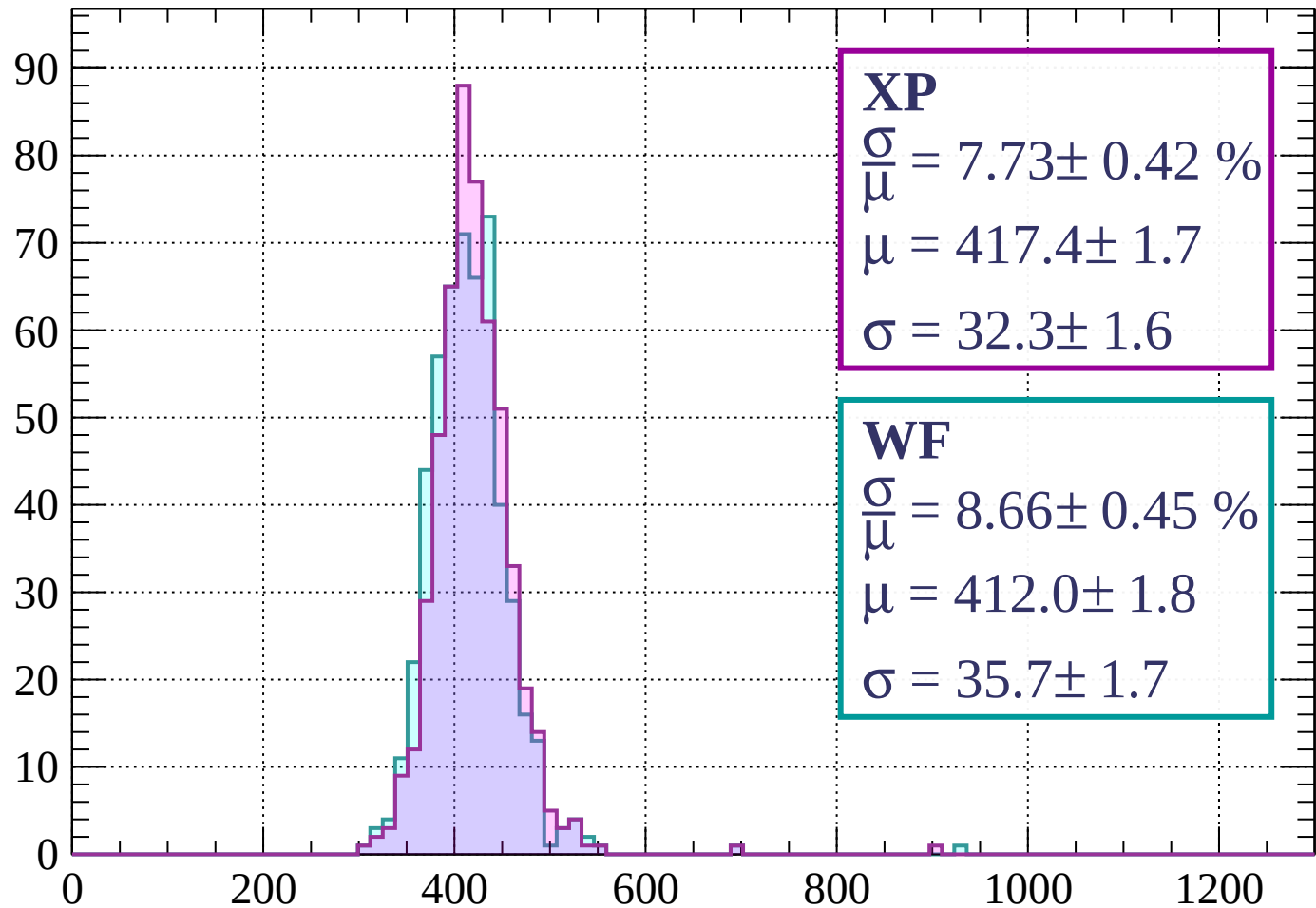
Energy loss | $680 < p < 720$

Count



Energy loss | $720 < p < 760$

Count



XP

$$\frac{\sigma}{\mu} = 7.73 \pm 0.42 \%$$

$$\mu = 417.4 \pm 1.7$$

$$\sigma = 32.3 \pm 1.6$$

WF

$$\frac{\sigma}{\mu} = 8.66 \pm 0.45 \%$$

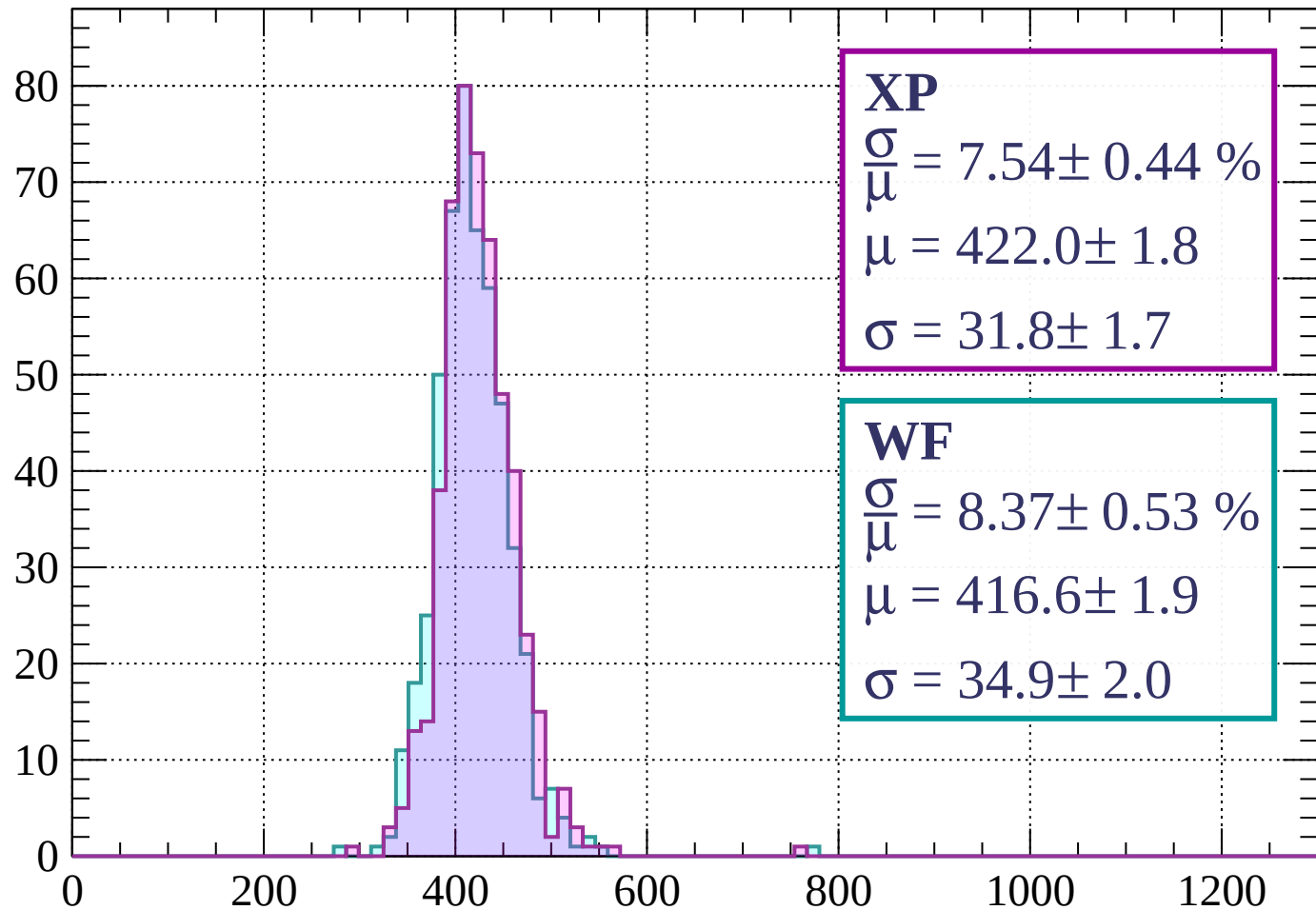
$$\mu = 412.0 \pm 1.8$$

$$\sigma = 35.7 \pm 1.7$$

dE/dx (ADC counts/cm)

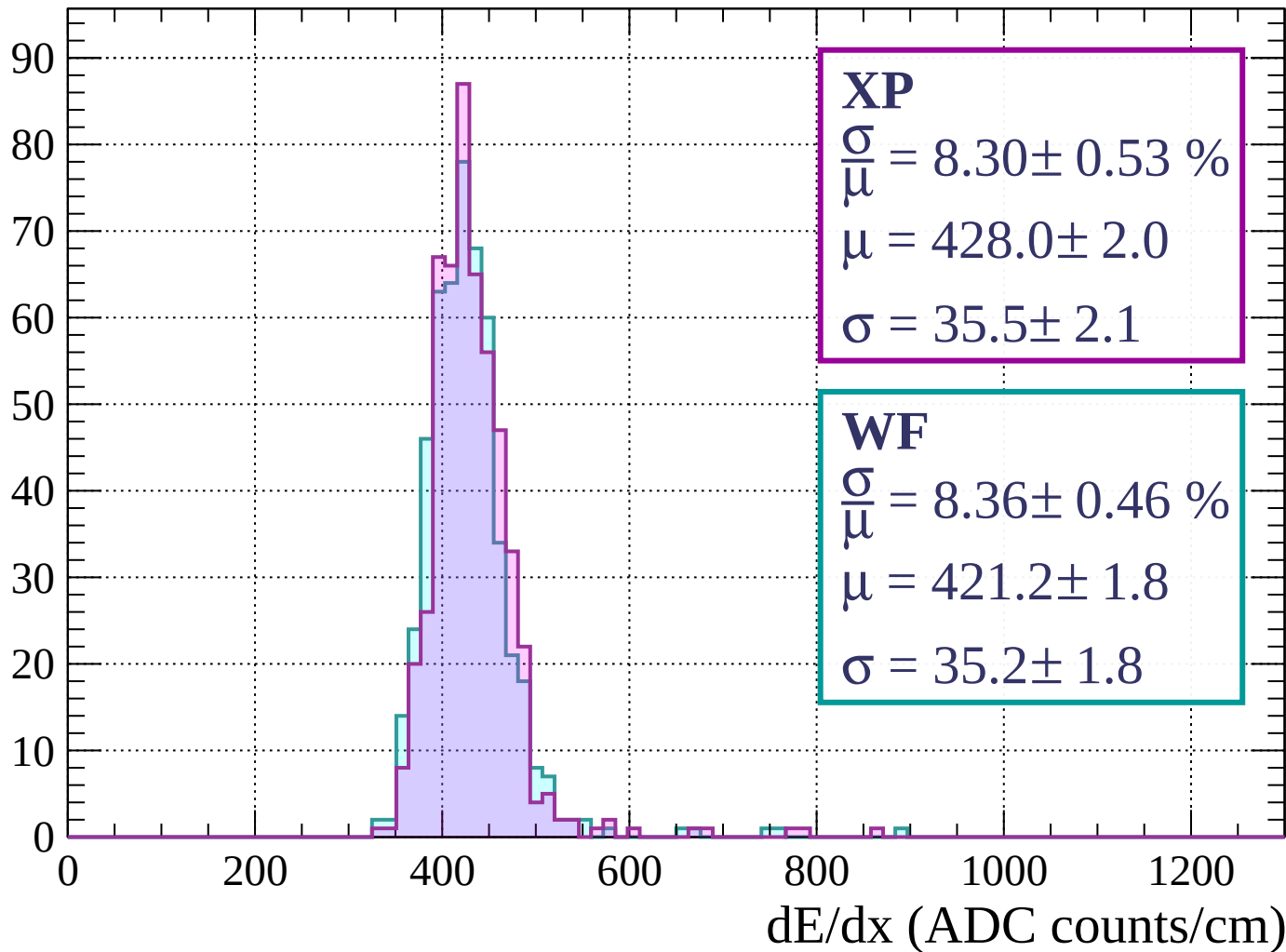
Energy loss | $760 < p < 800$

Count



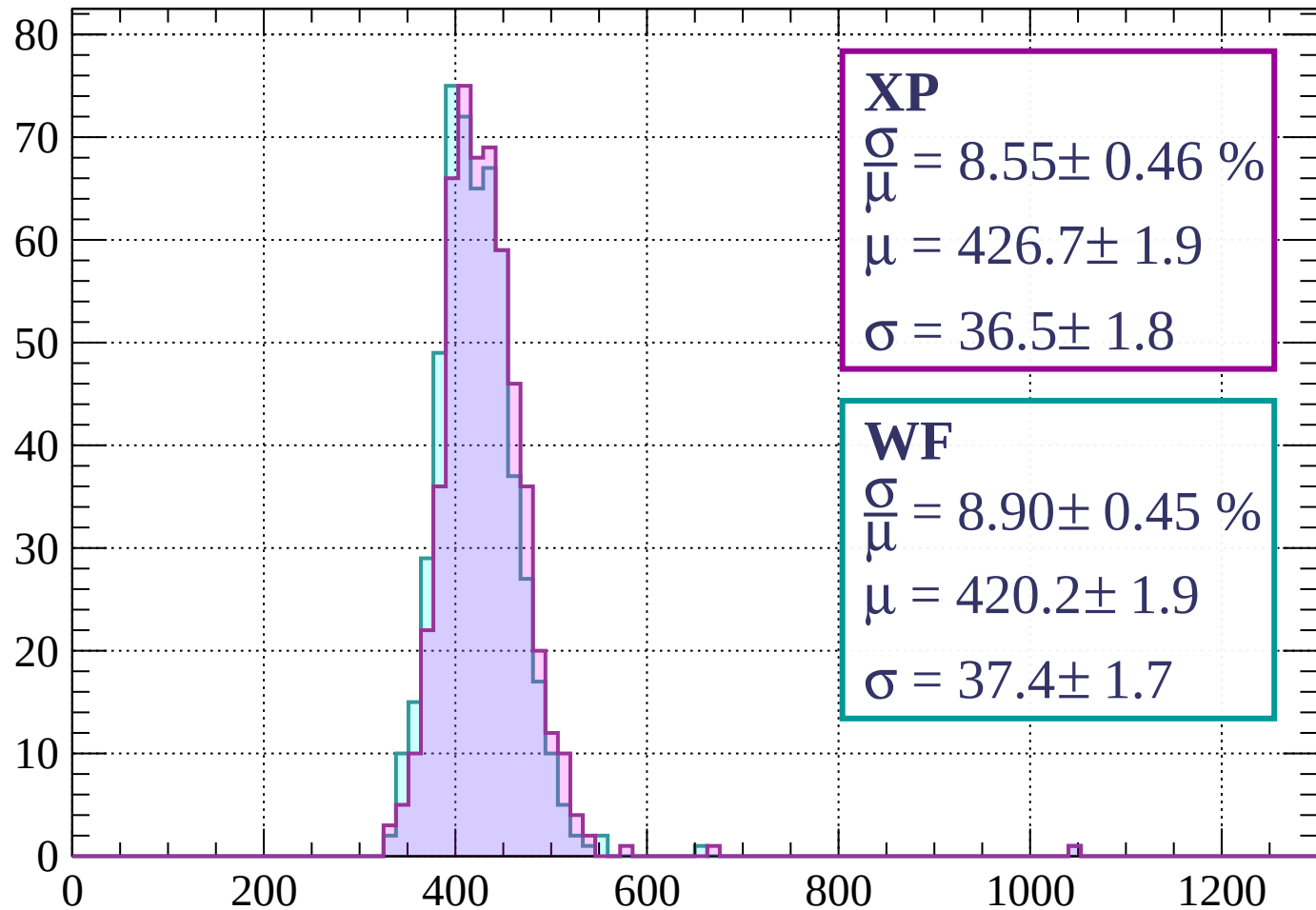
Energy loss | $800 < p < 840$

Count



Energy loss | $840 < p < 880$

Count



XP

$$\frac{\sigma}{\mu} = 8.55 \pm 0.46 \%$$

$$\mu = 426.7 \pm 1.9$$

$$\sigma = 36.5 \pm 1.8$$

WF

$$\frac{\sigma}{\mu} = 8.90 \pm 0.45 \%$$

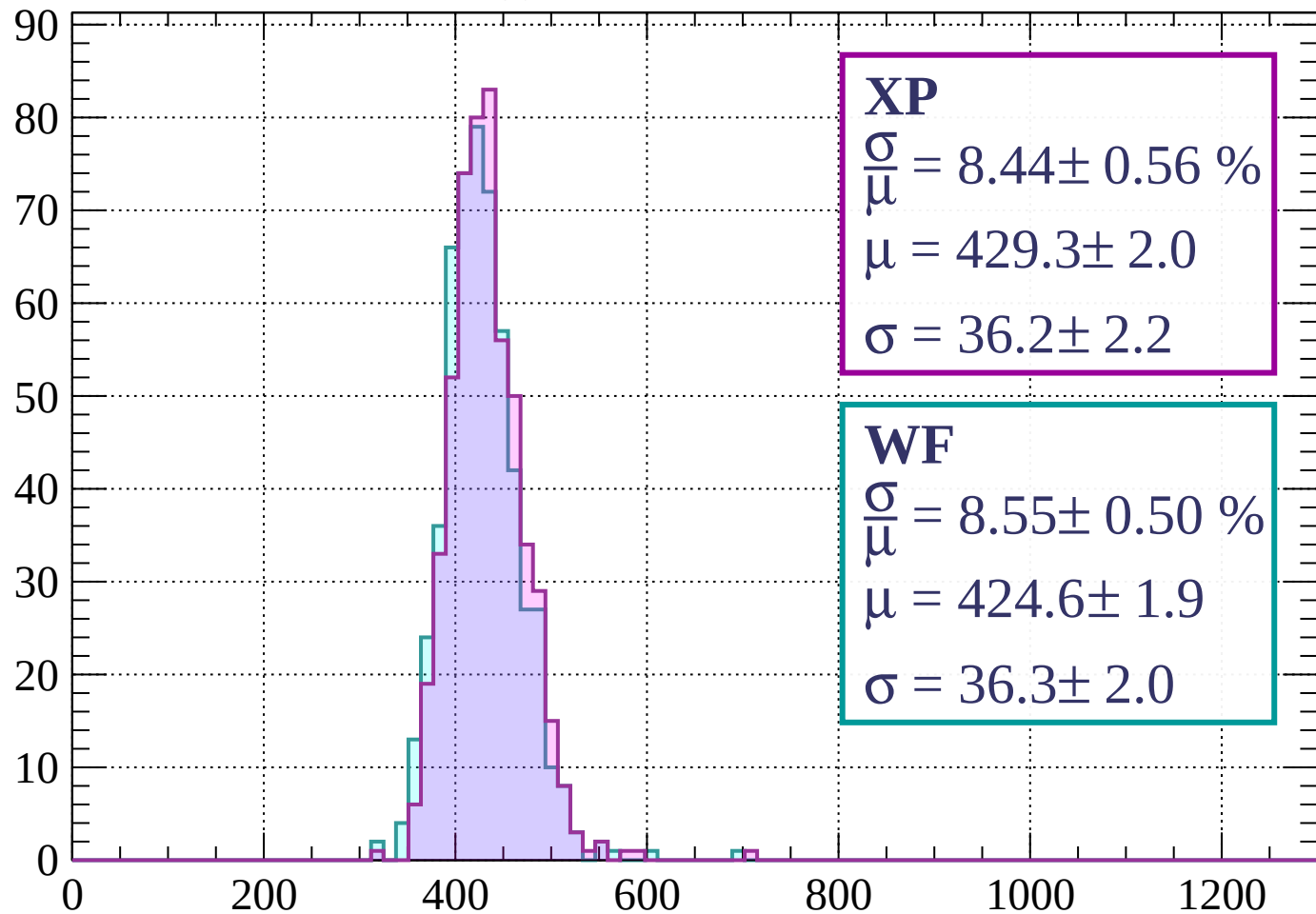
$$\mu = 420.2 \pm 1.9$$

$$\sigma = 37.4 \pm 1.7$$

dE/dx (ADC counts/cm)

Energy loss | $880 < p < 920$

Count



XP

$$\frac{\sigma}{\mu} = 8.44 \pm 0.56 \%$$

$$\mu = 429.3 \pm 2.0$$

$$\sigma = 36.2 \pm 2.2$$

WF

$$\frac{\sigma}{\mu} = 8.55 \pm 0.50 \%$$

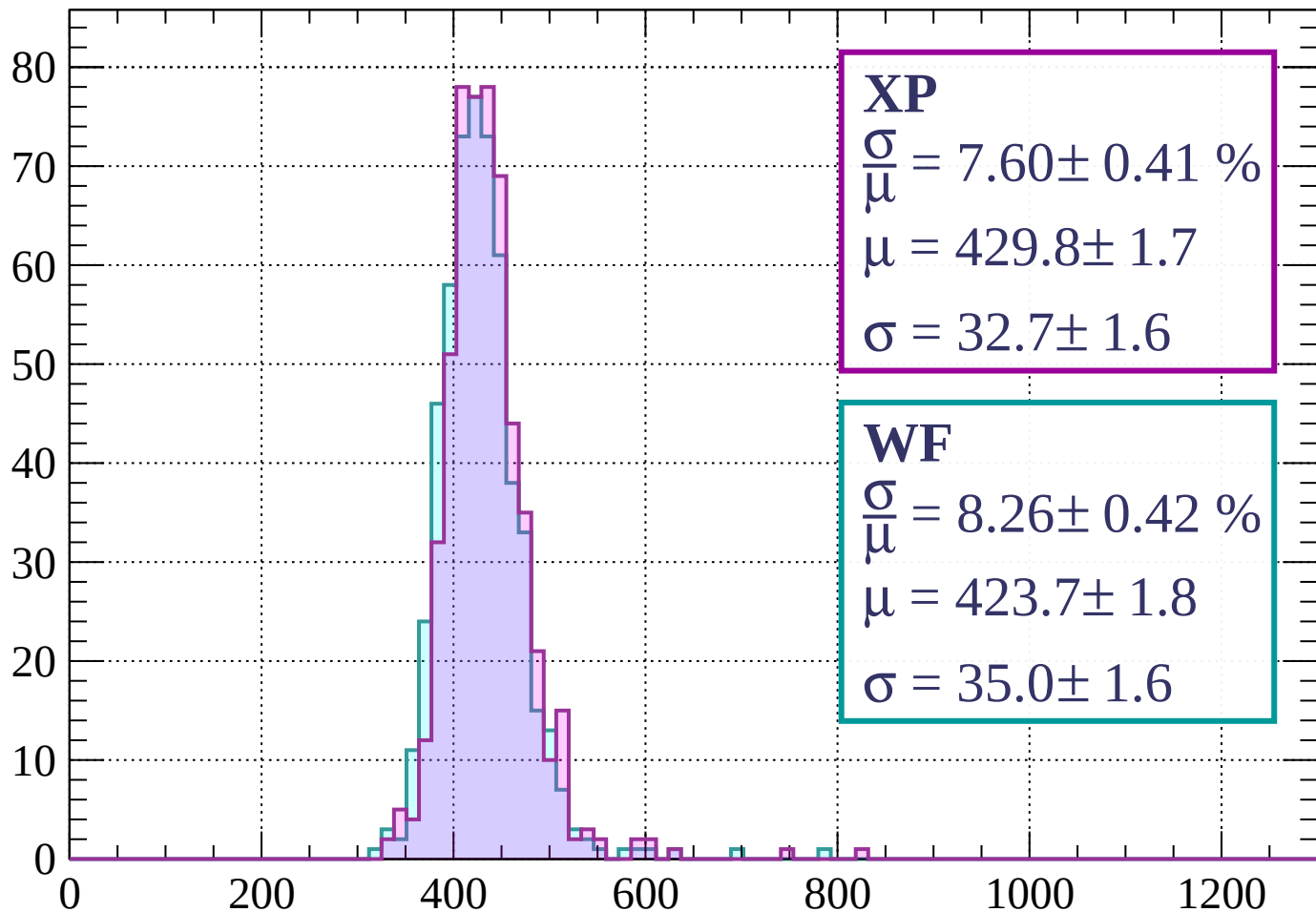
$$\mu = 424.6 \pm 1.9$$

$$\sigma = 36.3 \pm 2.0$$

dE/dx (ADC counts/cm)

Energy loss | $920 < p < 960$

Count



XP

$$\frac{\sigma}{\bar{\mu}} = 7.60 \pm 0.41 \%$$

$$\mu = 429.8 \pm 1.7$$

$$\sigma = 32.7 \pm 1.6$$

WF

$$\frac{\sigma}{\bar{\mu}} = 8.26 \pm 0.42 \%$$

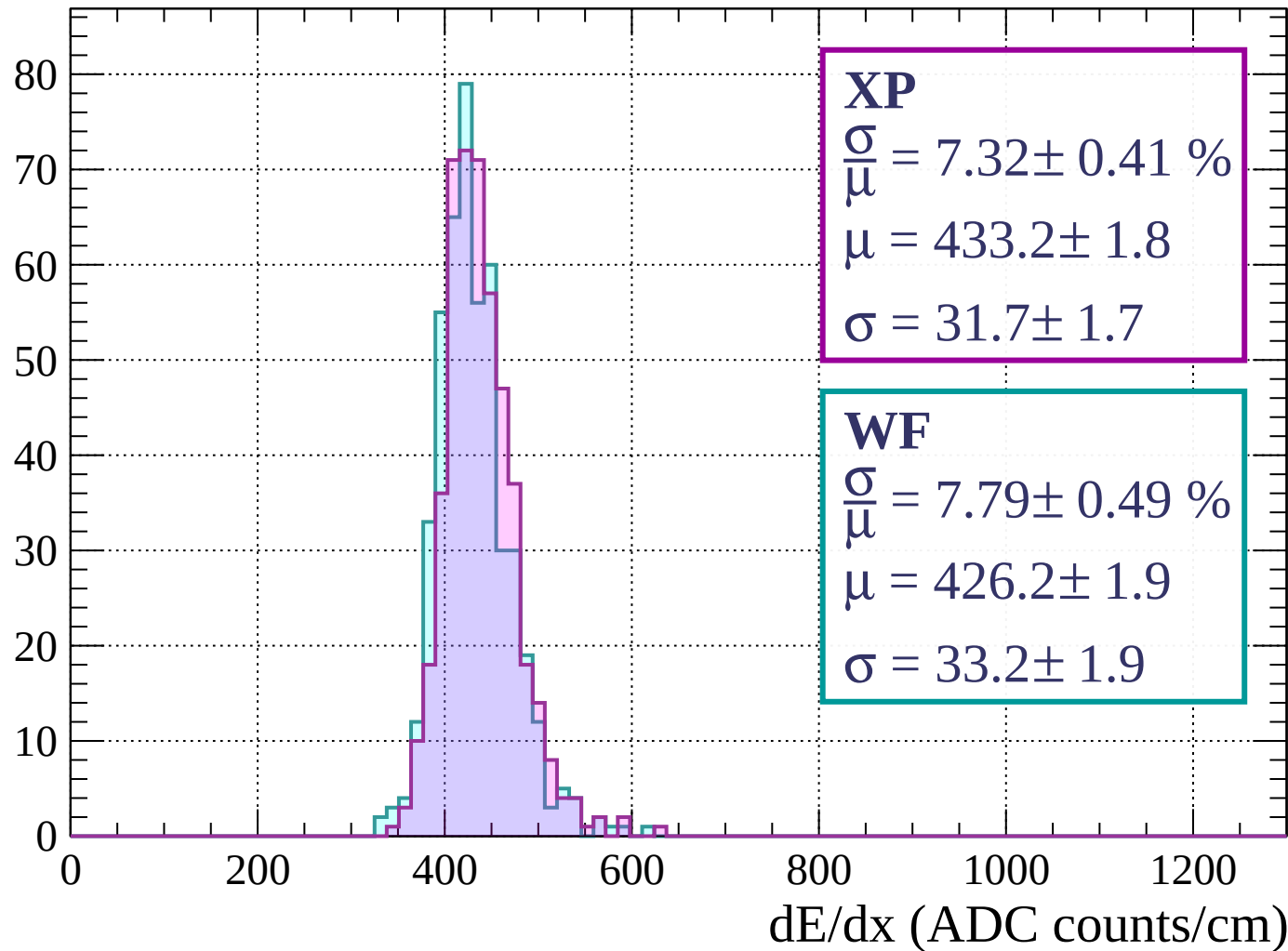
$$\mu = 423.7 \pm 1.8$$

$$\sigma = 35.0 \pm 1.6$$

dE/dx (ADC counts/cm)

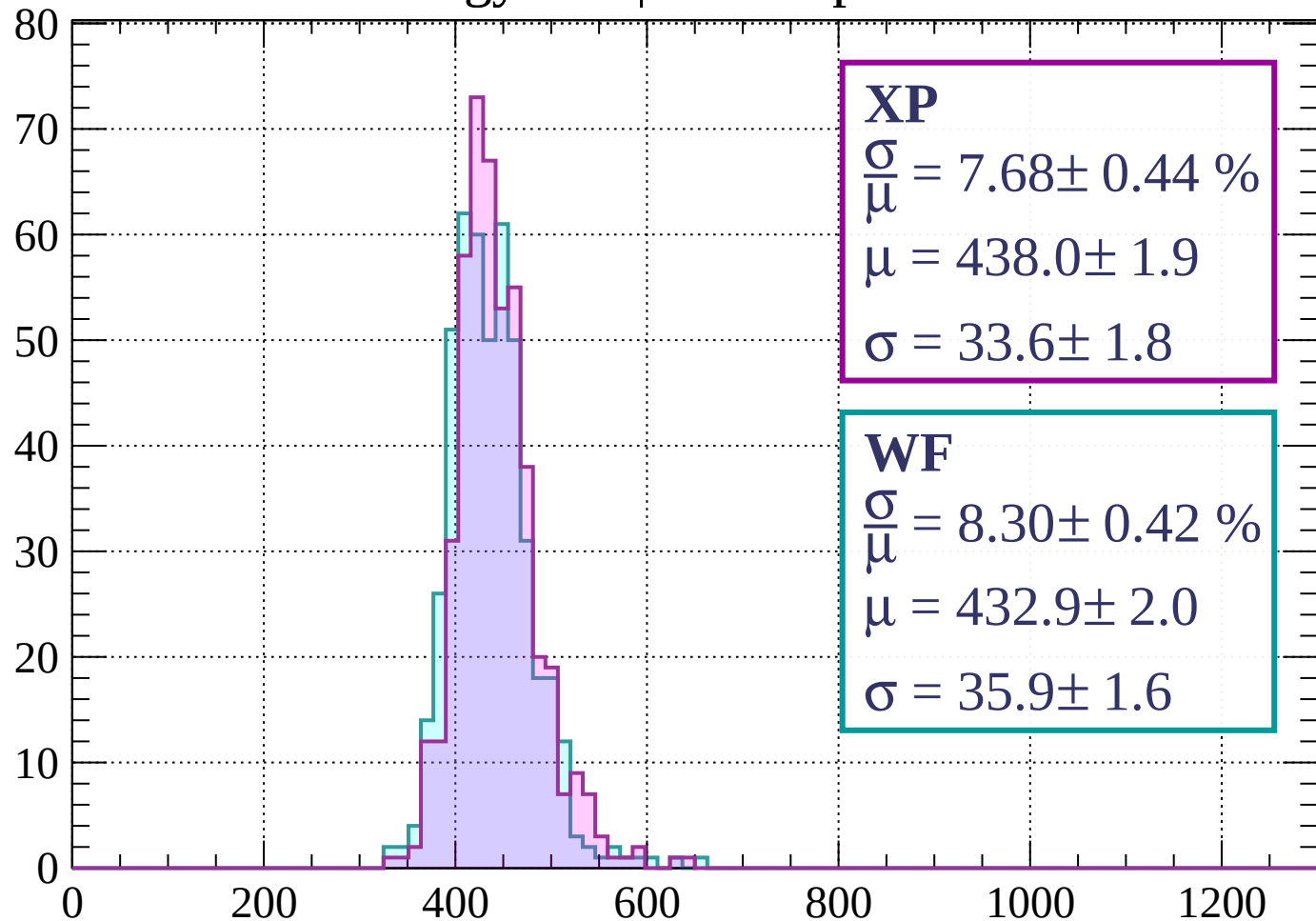
Energy loss | $960 < p < 1000$

Count



Energy loss | $1000 < p < 1040$

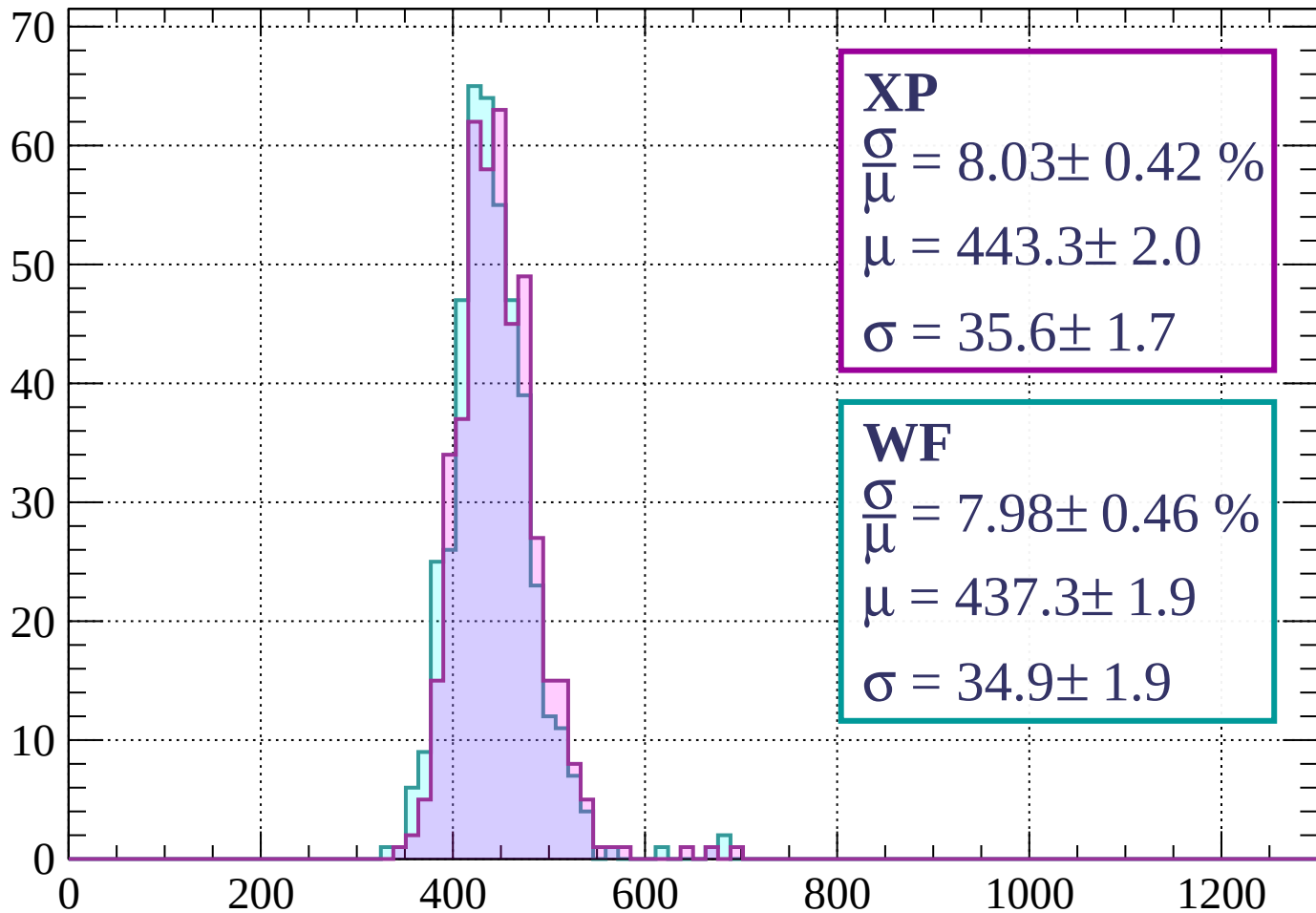
Count



dE/dx (ADC counts/cm)

Energy loss | $1040 < p < 1080$

Count



XP

$$\frac{\sigma}{\mu} = 8.03 \pm 0.42 \%$$

$$\mu = 443.3 \pm 2.0$$

$$\sigma = 35.6 \pm 1.7$$

WF

$$\frac{\sigma}{\mu} = 7.98 \pm 0.46 \%$$

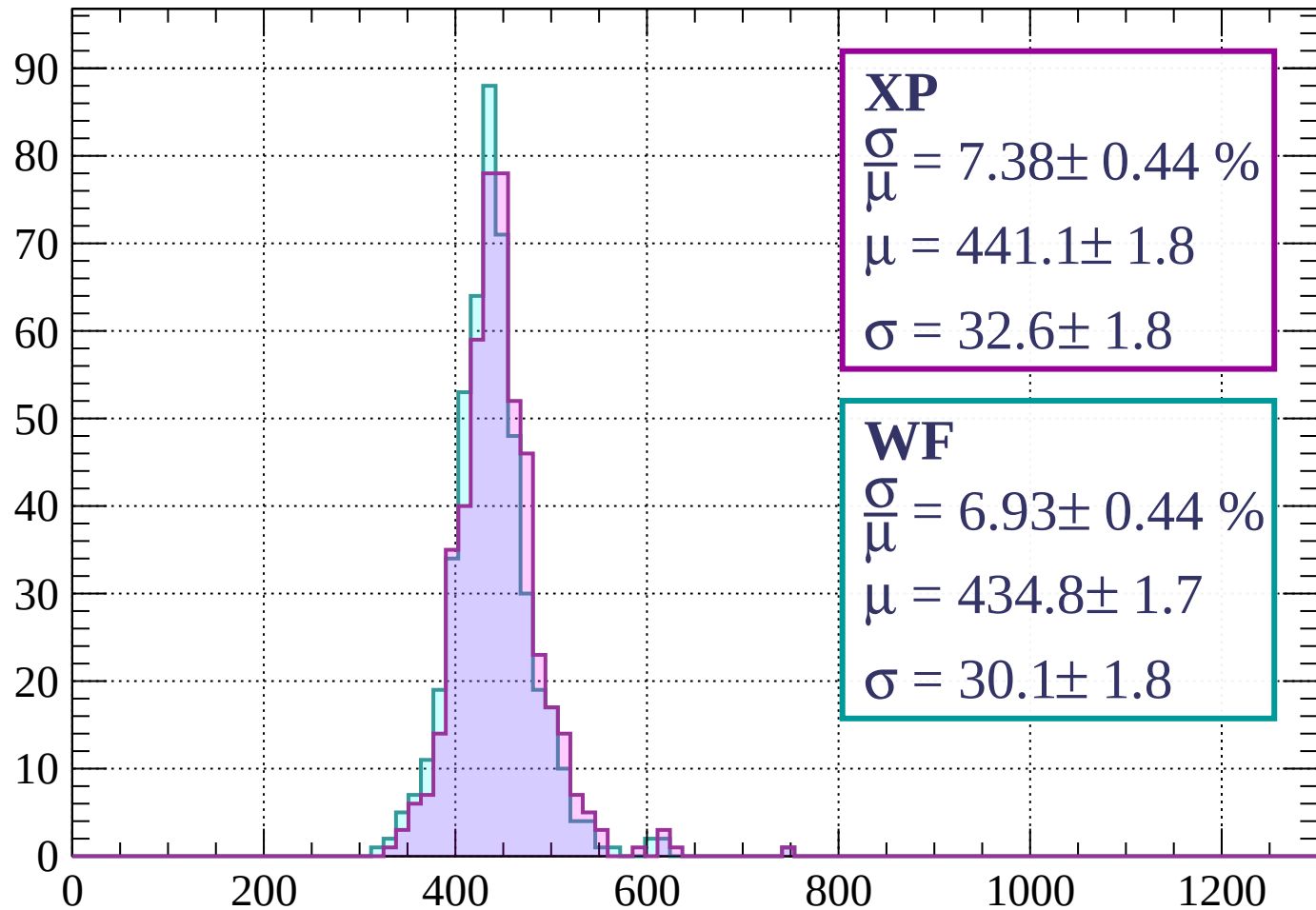
$$\mu = 437.3 \pm 1.9$$

$$\sigma = 34.9 \pm 1.9$$

dE/dx (ADC counts/cm)

Energy loss | $1080 < p < 1120$

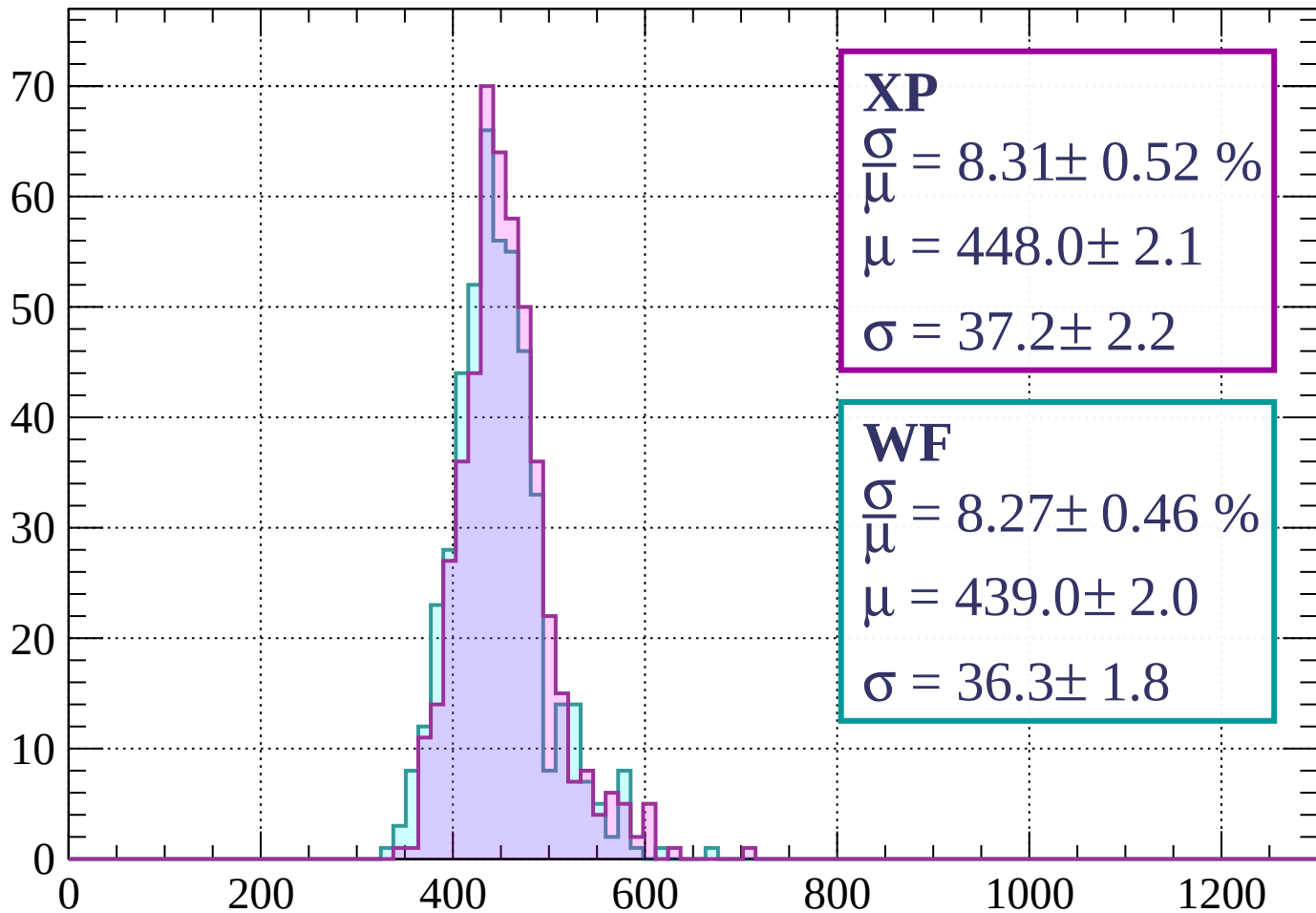
Count



dE/dx (ADC counts/cm)

Energy loss | $1120 < p < 1160$

Count



XP

$$\frac{\sigma}{\mu} = 8.31 \pm 0.52 \%$$

$$\mu = 448.0 \pm 2.1$$

$$\sigma = 37.2 \pm 2.2$$

WF

$$\frac{\sigma}{\mu} = 8.27 \pm 0.46 \%$$

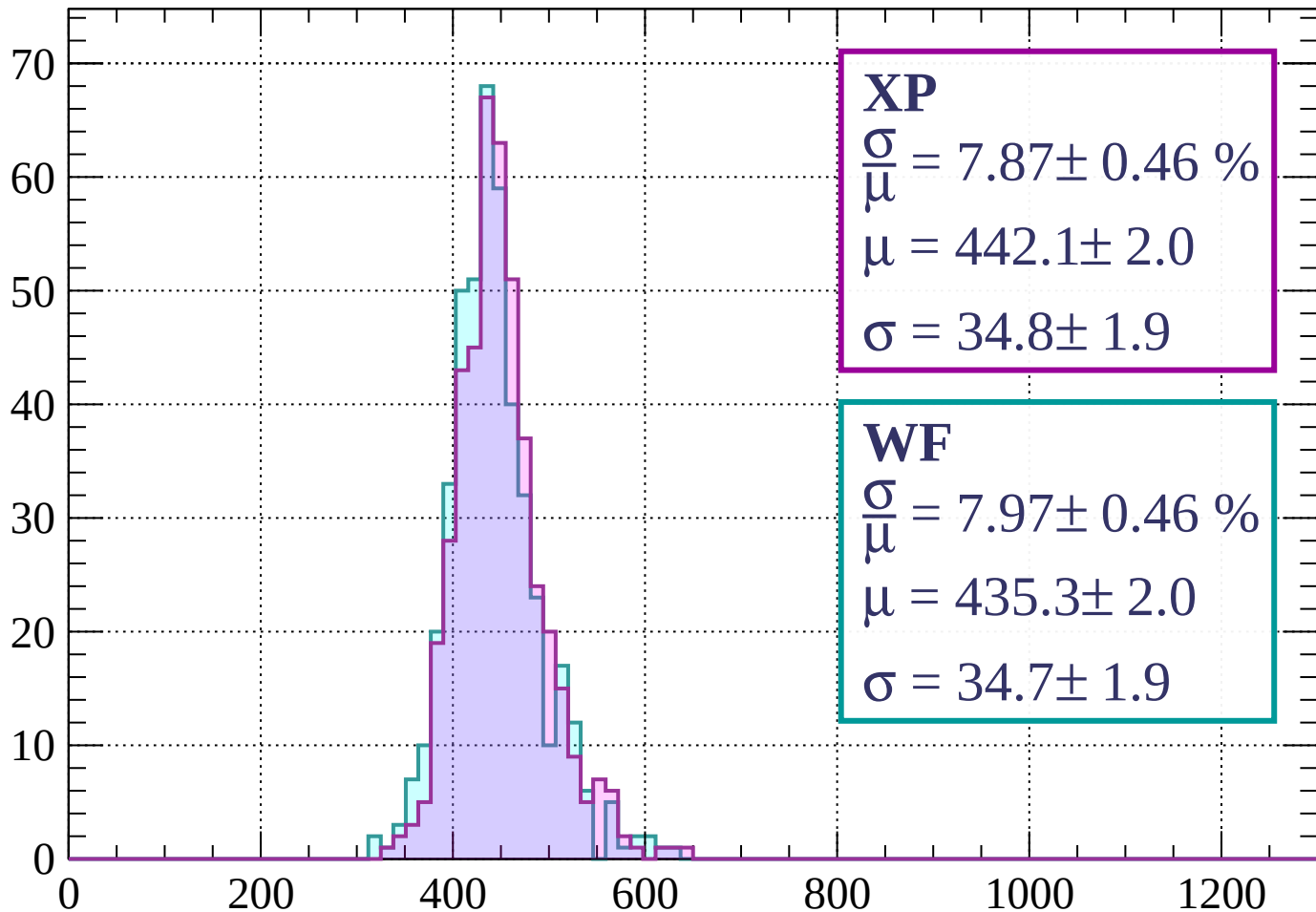
$$\mu = 439.0 \pm 2.0$$

$$\sigma = 36.3 \pm 1.8$$

dE/dx (ADC counts/cm)

Energy loss | $1160 < p < 1200$

Count



XP

$$\frac{\sigma}{\mu} = 7.87 \pm 0.46 \%$$

$$\mu = 442.1 \pm 2.0$$

$$\sigma = 34.8 \pm 1.9$$

WF

$$\frac{\sigma}{\mu} = 7.97 \pm 0.46 \%$$

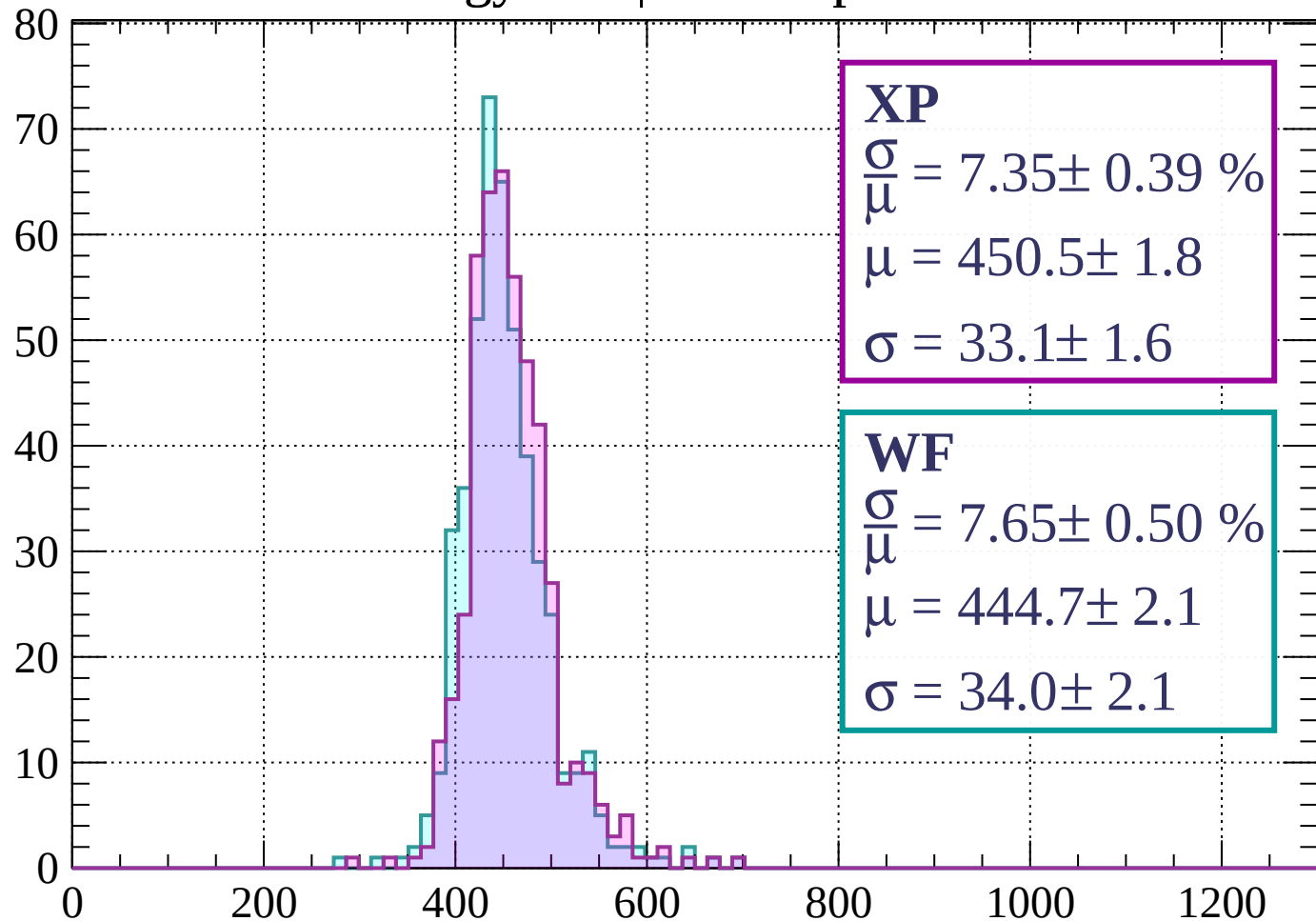
$$\mu = 435.3 \pm 2.0$$

$$\sigma = 34.7 \pm 1.9$$

dE/dx (ADC counts/cm)

Energy loss | $1200 < p < 1240$

Count



XP

$$\frac{\sigma}{\mu} = 7.35 \pm 0.39 \%$$

$$\mu = 450.5 \pm 1.8$$

$$\sigma = 33.1 \pm 1.6$$

WF

$$\frac{\sigma}{\mu} = 7.65 \pm 0.50 \%$$

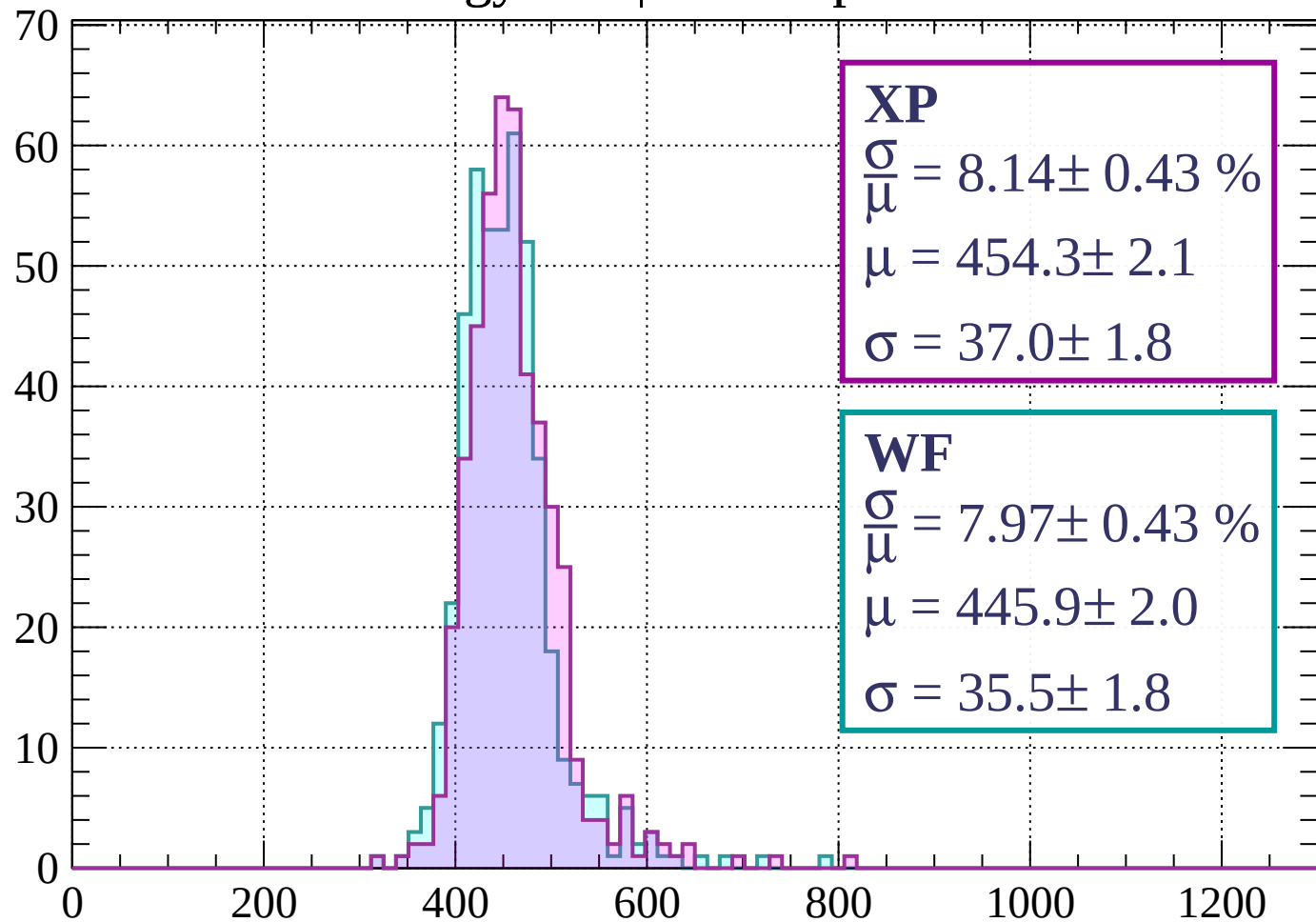
$$\mu = 444.7 \pm 2.1$$

$$\sigma = 34.0 \pm 2.1$$

dE/dx (ADC counts/cm)

Energy loss | $1240 < p < 1280$

Count



XP

$$\frac{\sigma}{\mu} = 8.14 \pm 0.43 \%$$

$$\mu = 454.3 \pm 2.1$$

$$\sigma = 37.0 \pm 1.8$$

WF

$$\frac{\sigma}{\mu} = 7.97 \pm 0.43 \%$$

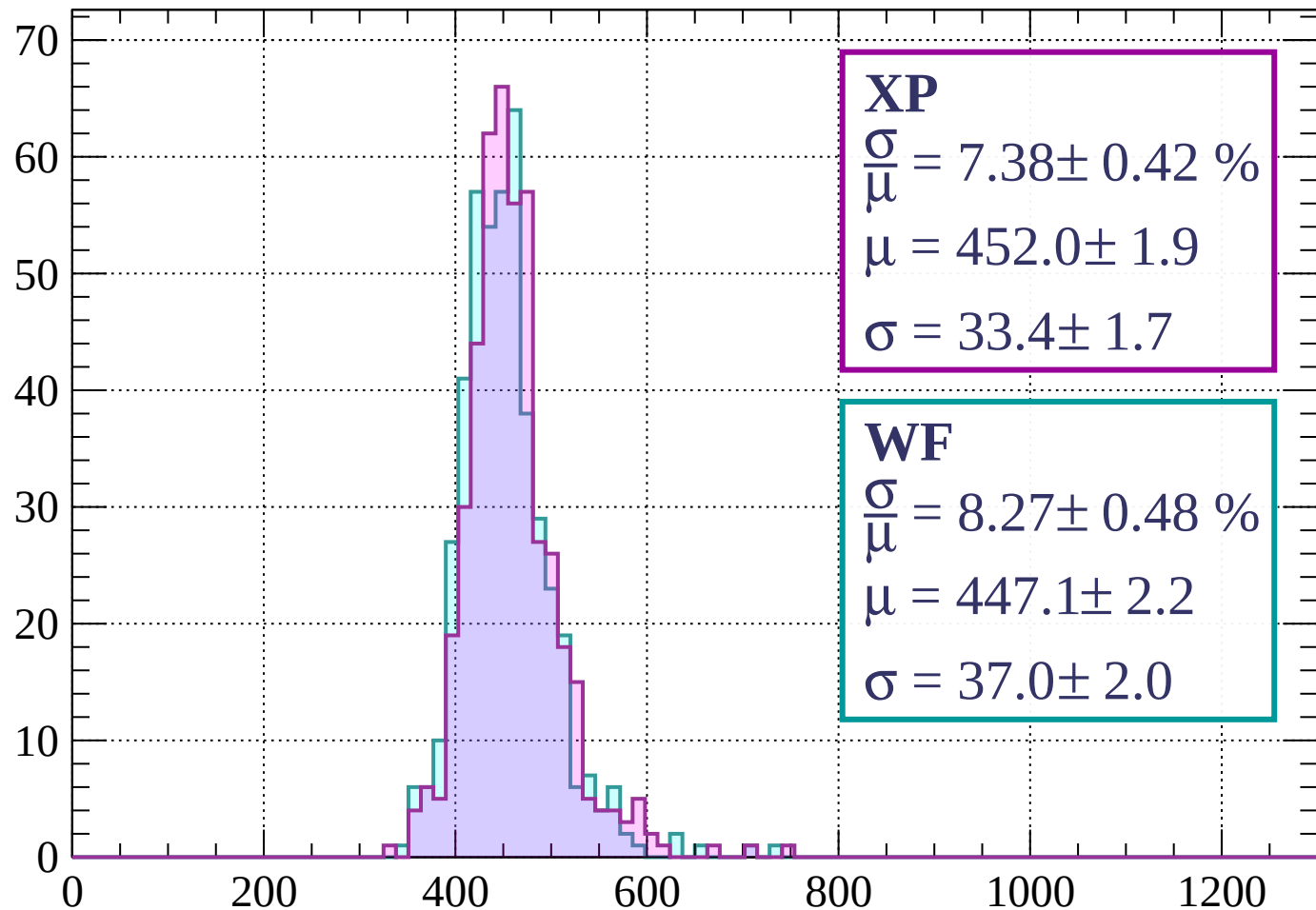
$$\mu = 445.9 \pm 2.0$$

$$\sigma = 35.5 \pm 1.8$$

dE/dx (ADC counts/cm)

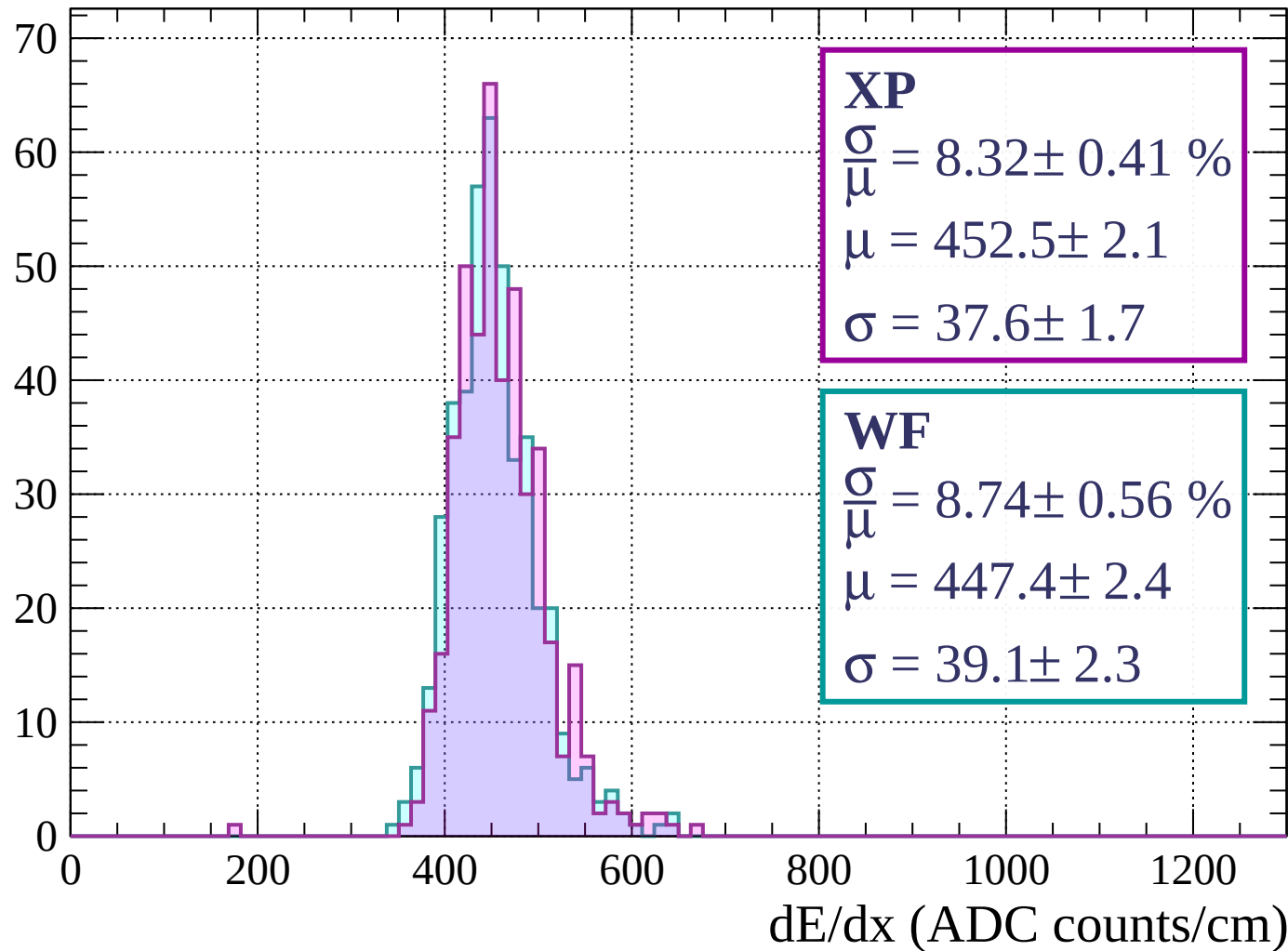
Energy loss | $1280 < p < 1320$

Count



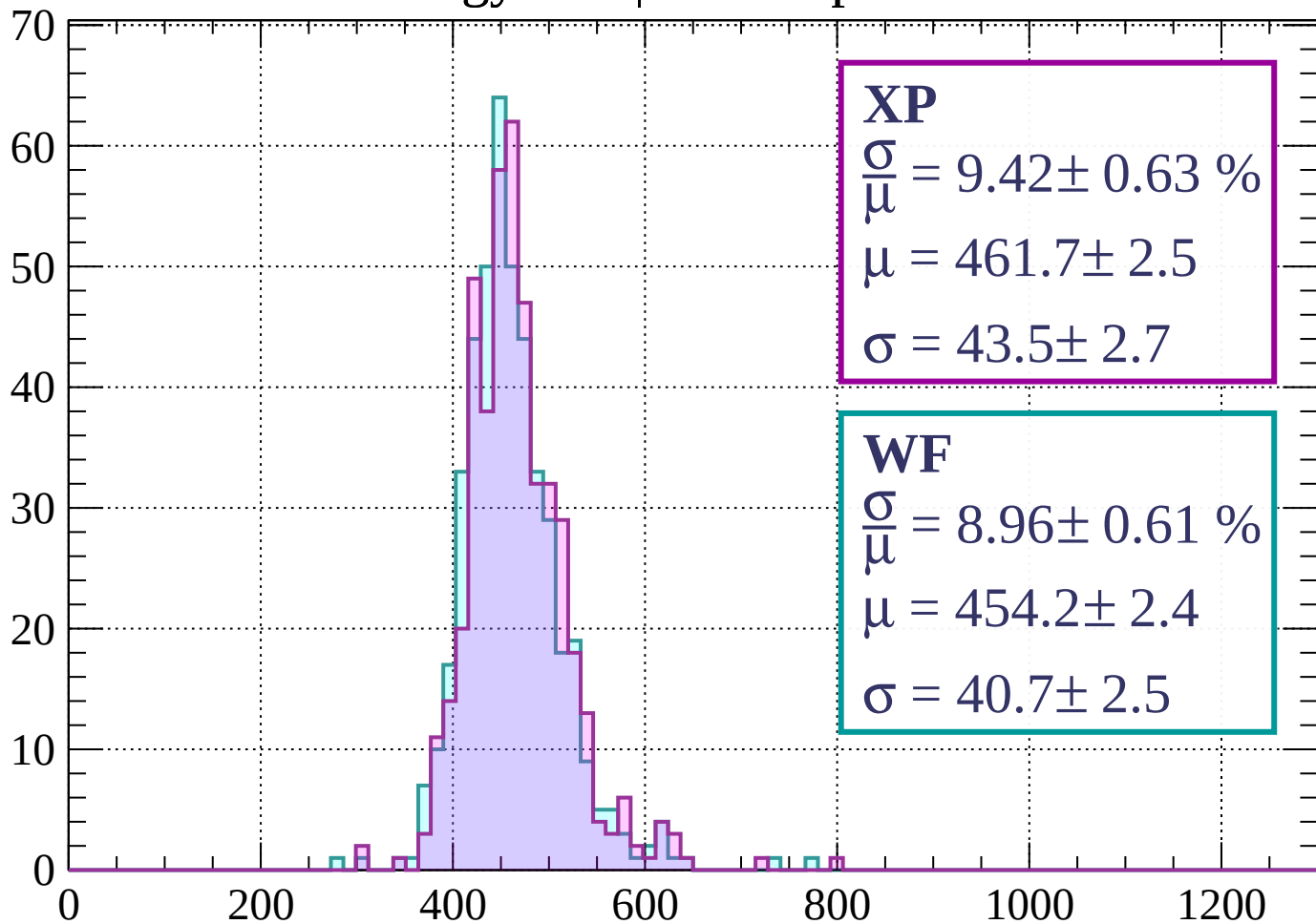
Energy loss | $1320 < p < 1360$

Count



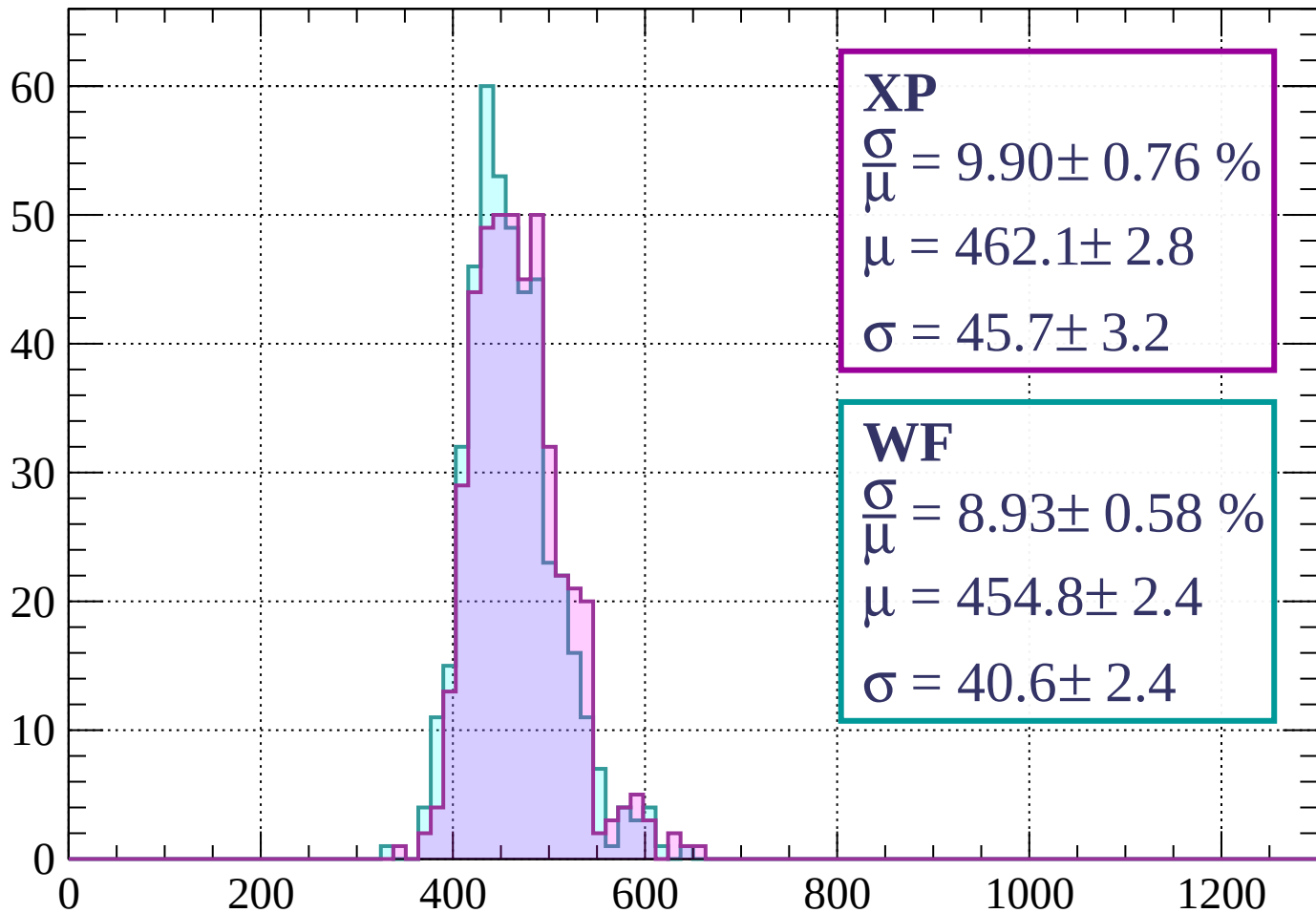
Energy loss | $1360 < p < 1400$

Count



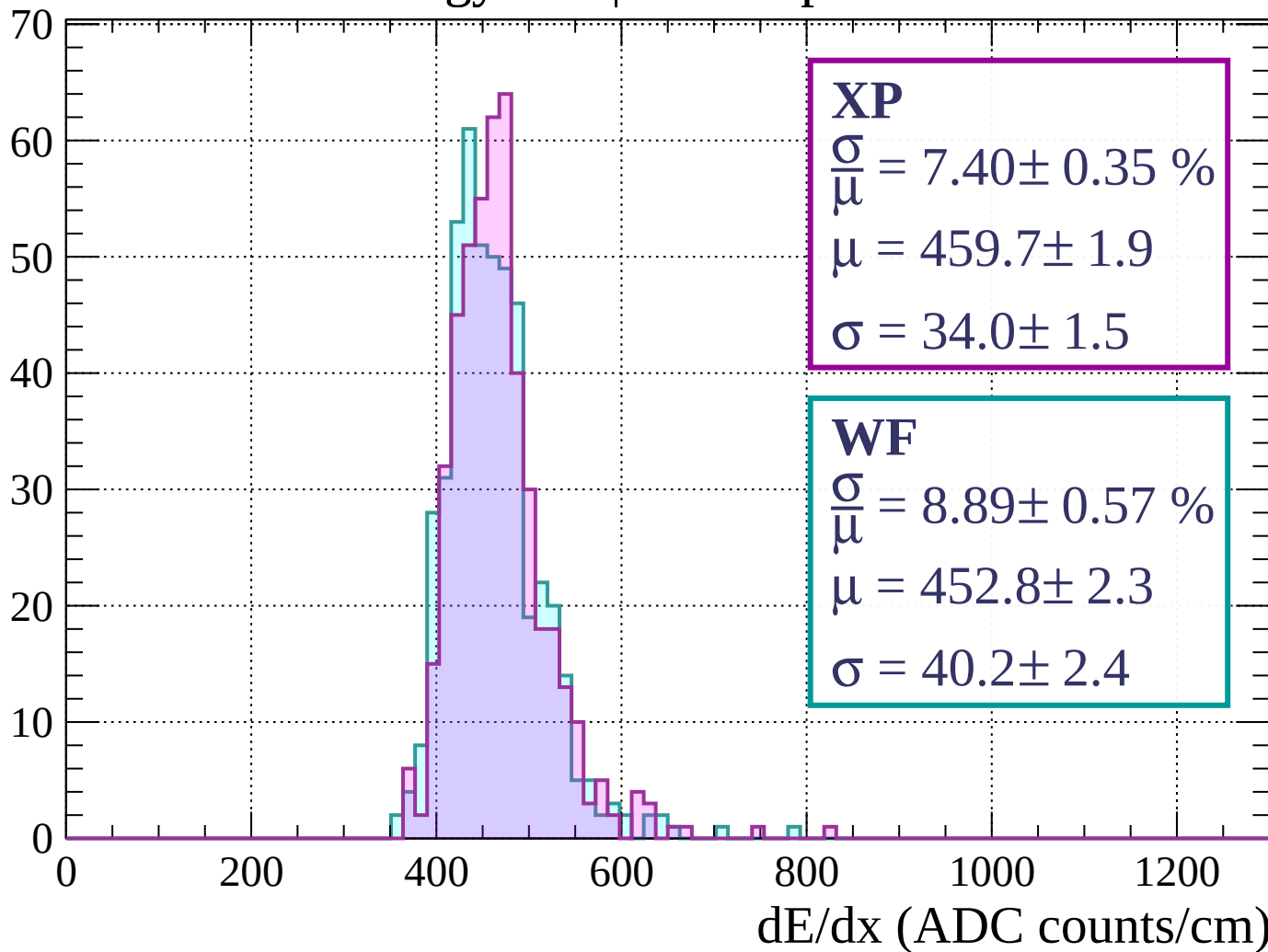
Energy loss | $1400 < p < 1440$

Count



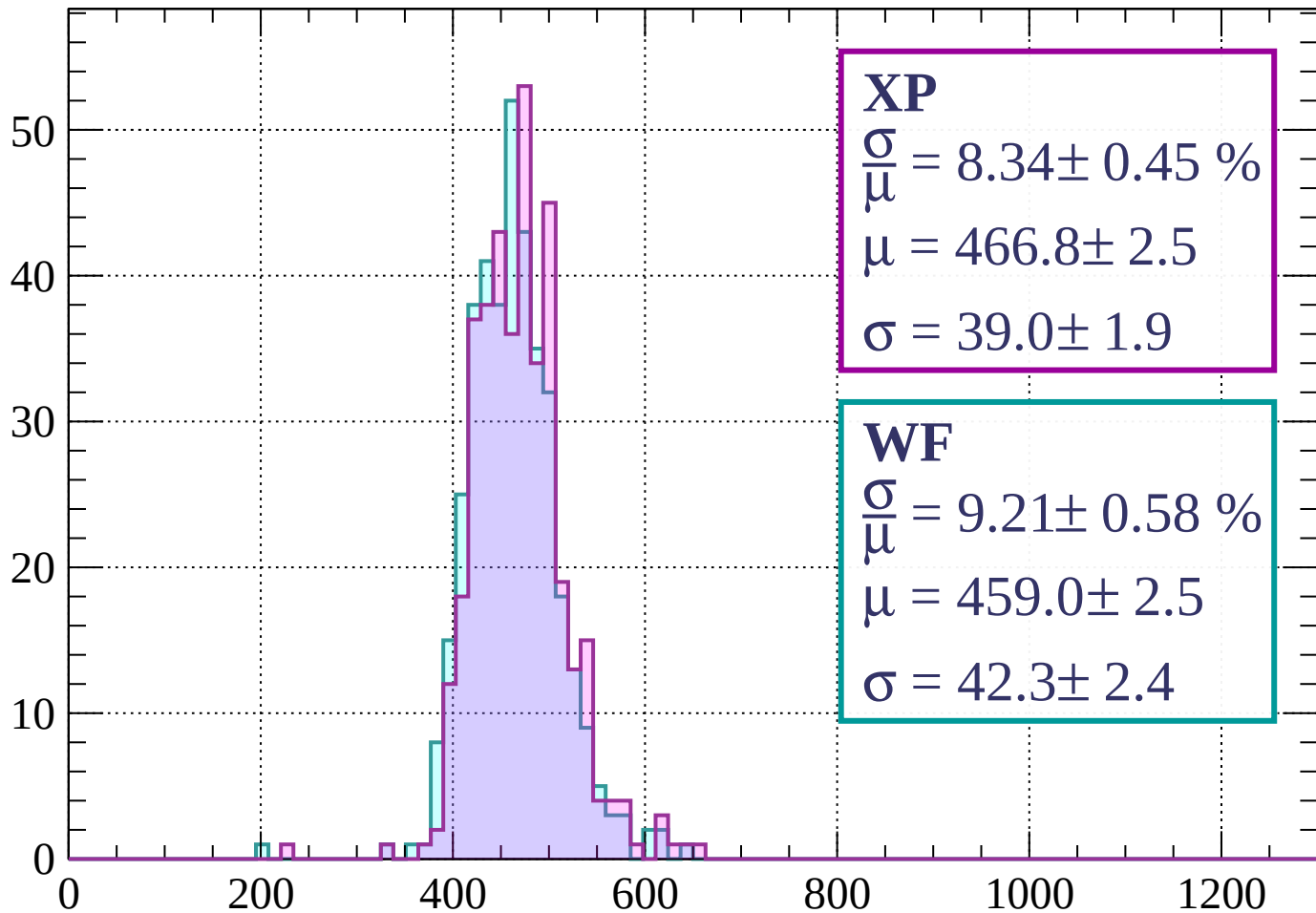
Energy loss | $1440 < p < 1480$

Count



Energy loss | $1480 < p < 1520$

Count



XP

$$\frac{\sigma}{\mu} = 8.34 \pm 0.45 \%$$

$$\mu = 466.8 \pm 2.5$$

$$\sigma = 39.0 \pm 1.9$$

WF

$$\frac{\sigma}{\mu} = 9.21 \pm 0.58 \%$$

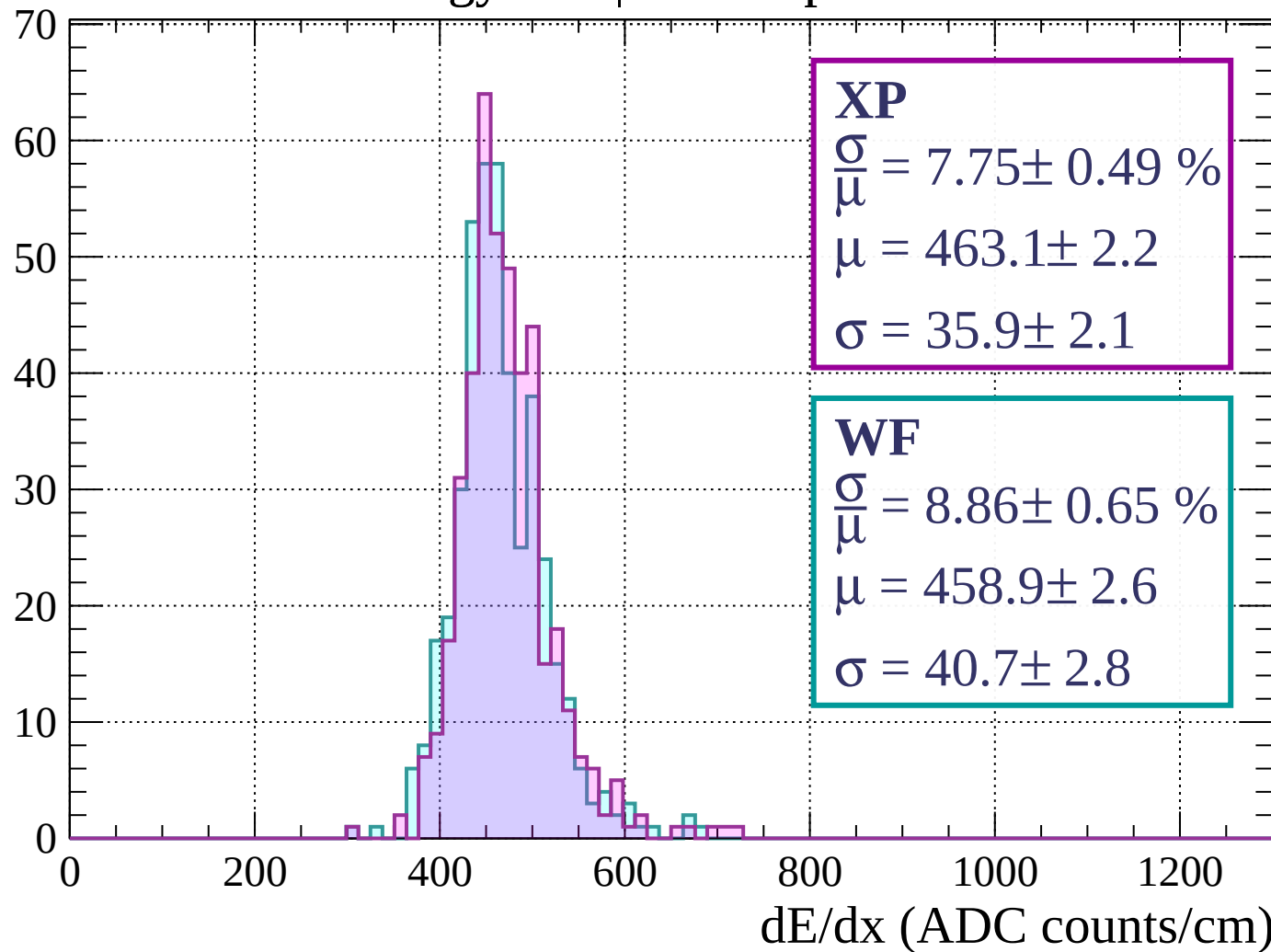
$$\mu = 459.0 \pm 2.5$$

$$\sigma = 42.3 \pm 2.4$$

dE/dx (ADC counts/cm)

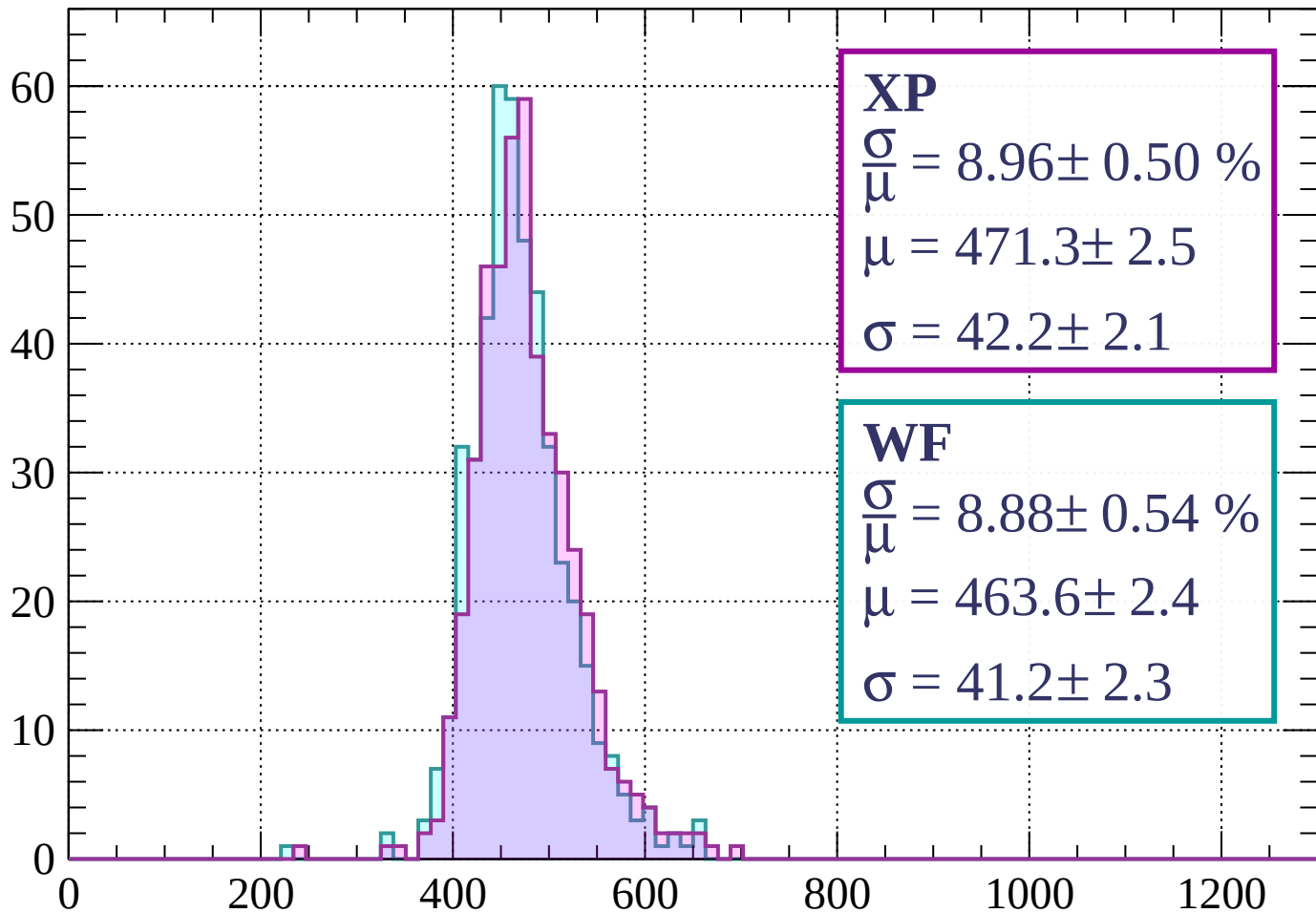
Energy loss | $1520 < p < 1560$

Count



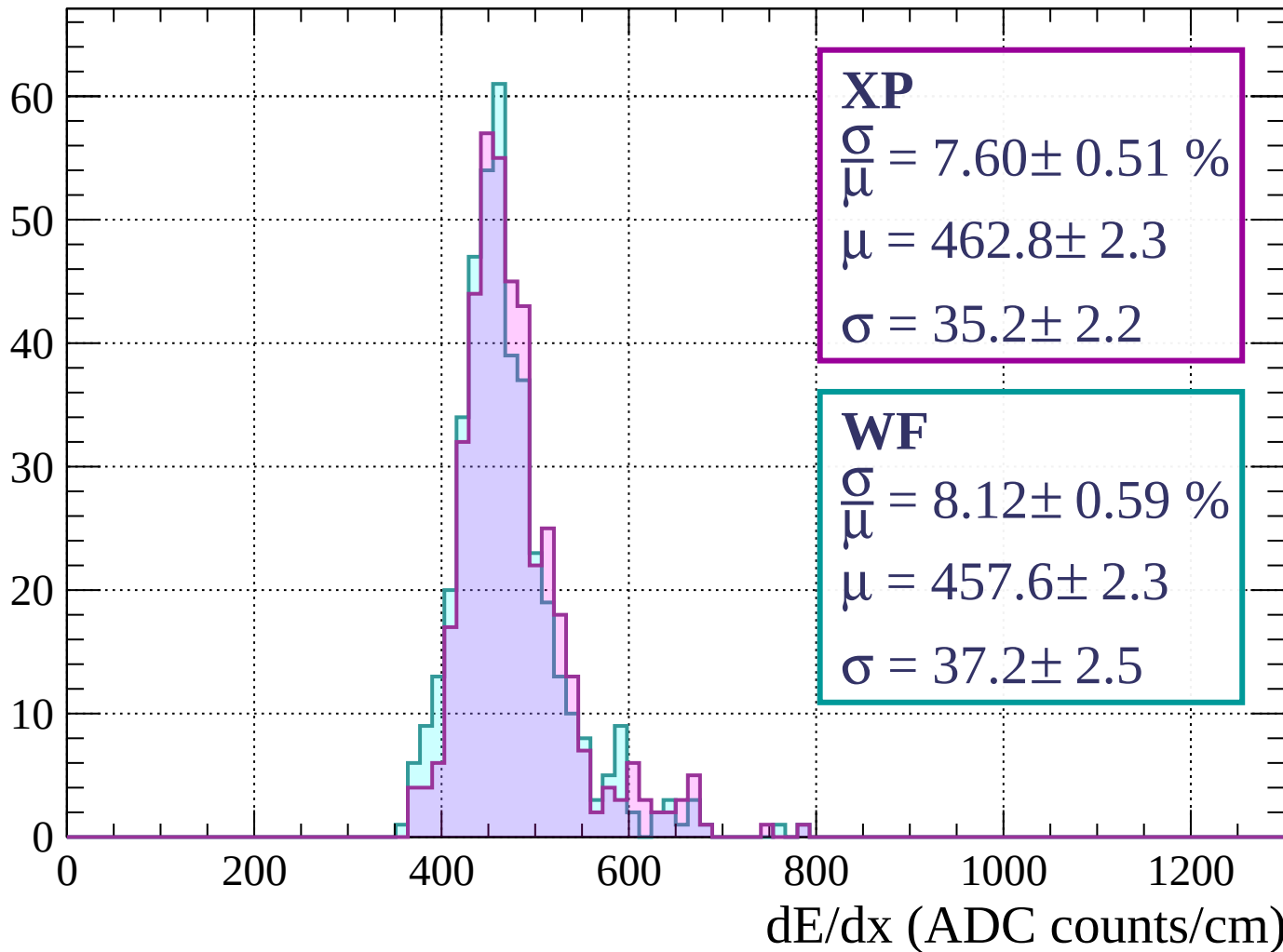
Energy loss | $1560 < p < 1600$

Count



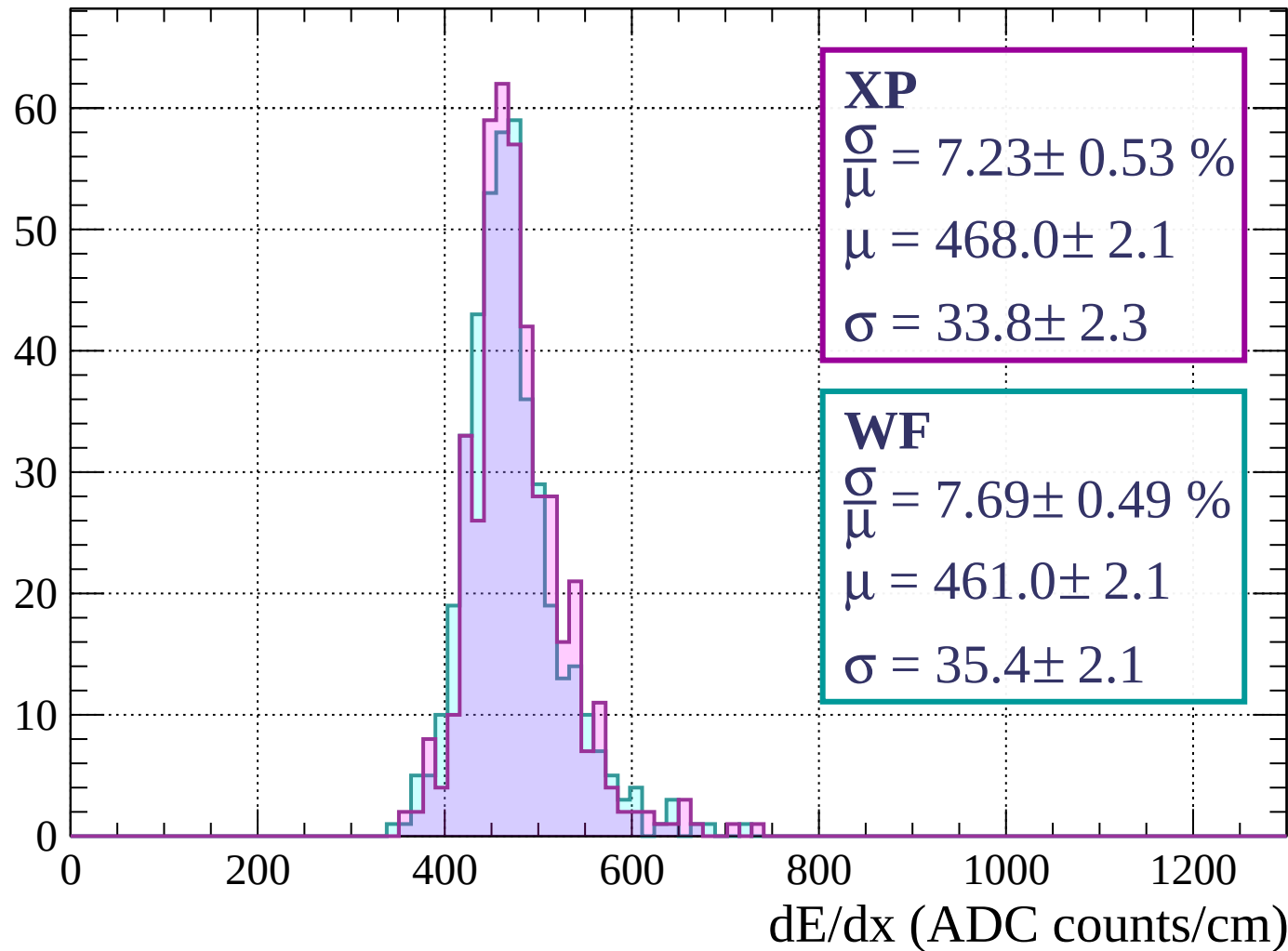
Energy loss | $1600 < p < 1640$

Count



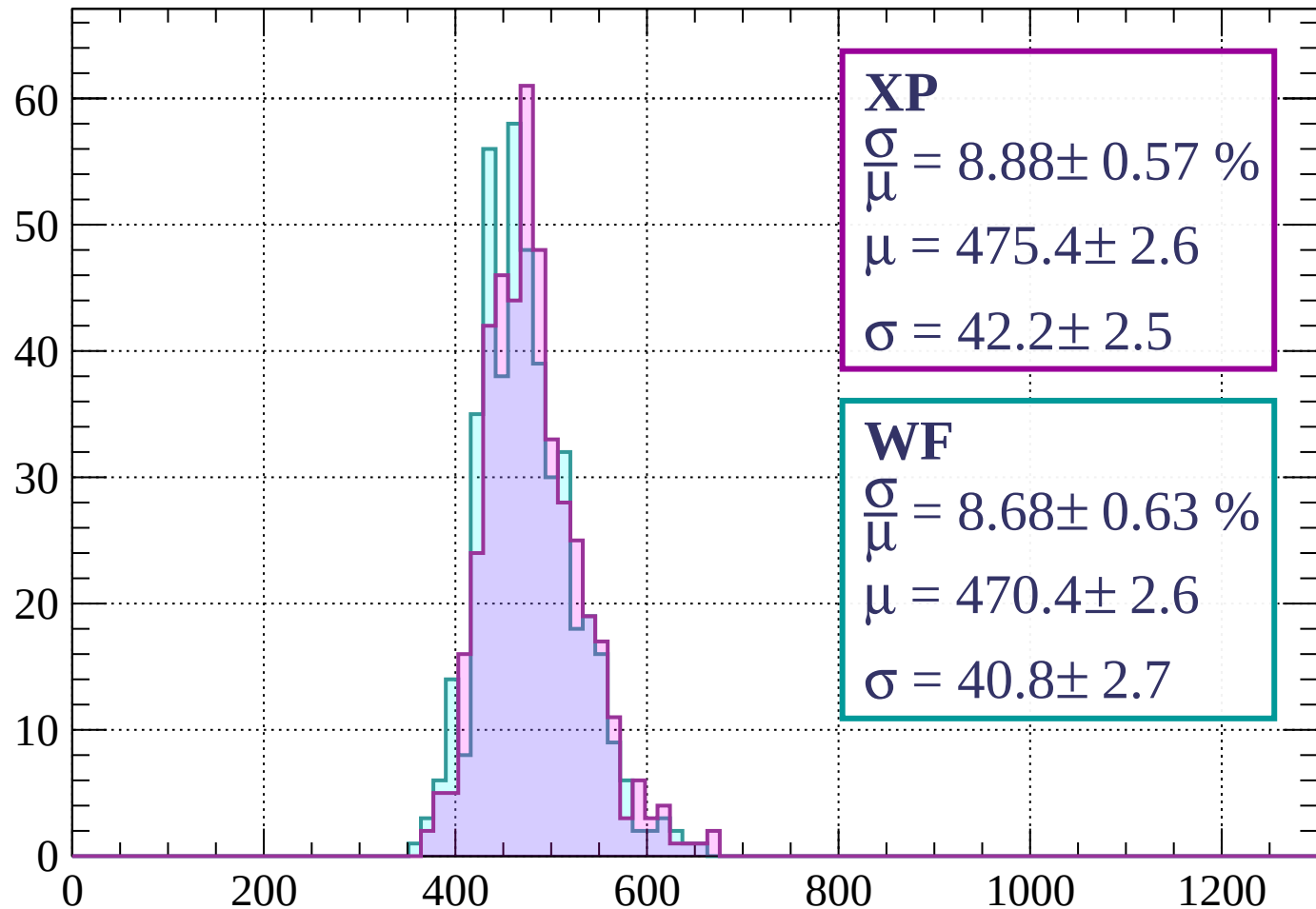
Energy loss | $1640 < p < 1680$

Count



Energy loss | $1680 < p < 1720$

Count



XP

$$\frac{\sigma}{\mu} = 8.88 \pm 0.57 \%$$

$$\mu = 475.4 \pm 2.6$$

$$\sigma = 42.2 \pm 2.5$$

WF

$$\frac{\sigma}{\mu} = 8.68 \pm 0.63 \%$$

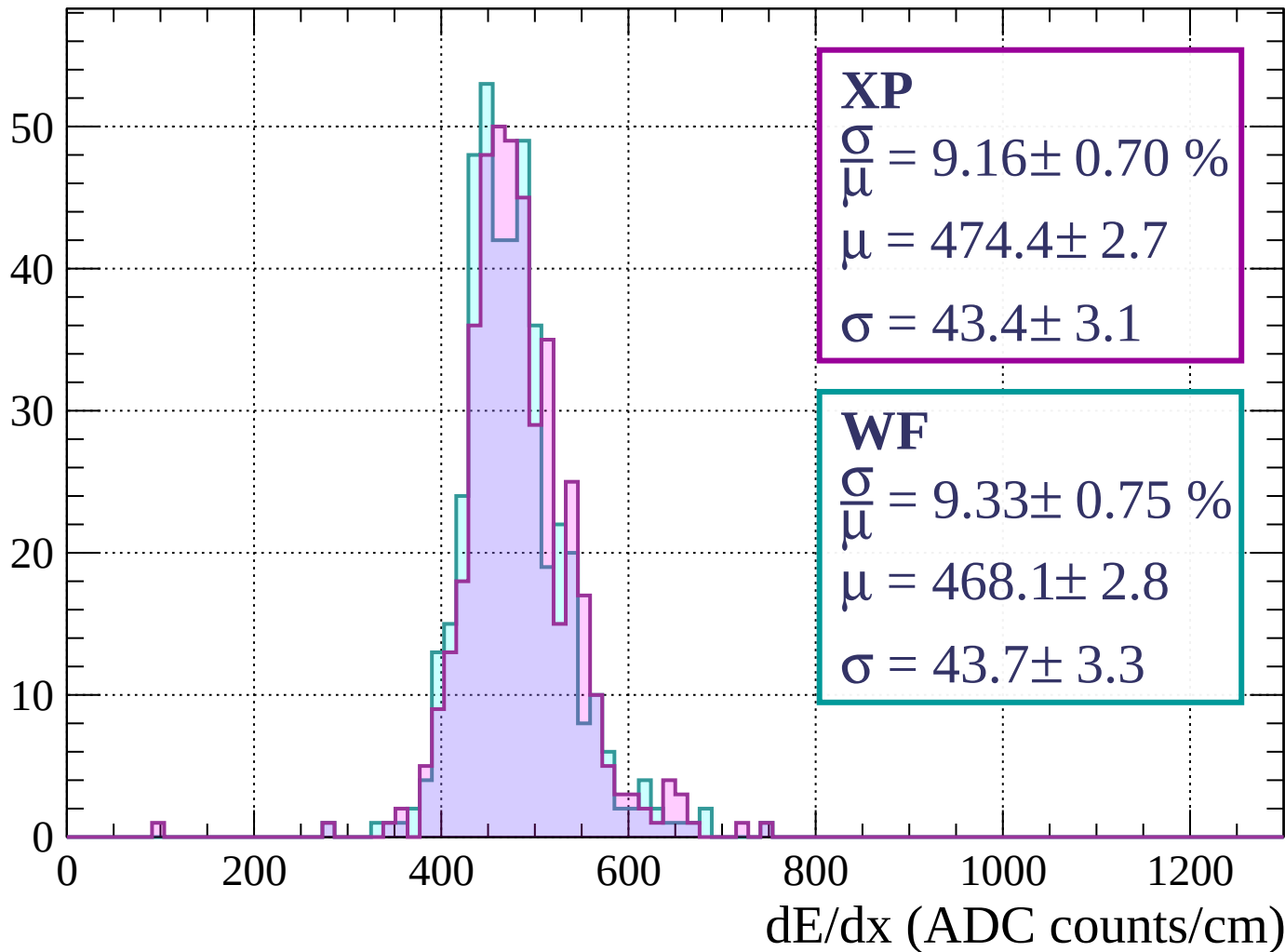
$$\mu = 470.4 \pm 2.6$$

$$\sigma = 40.8 \pm 2.7$$

dE/dx (ADC counts/cm)

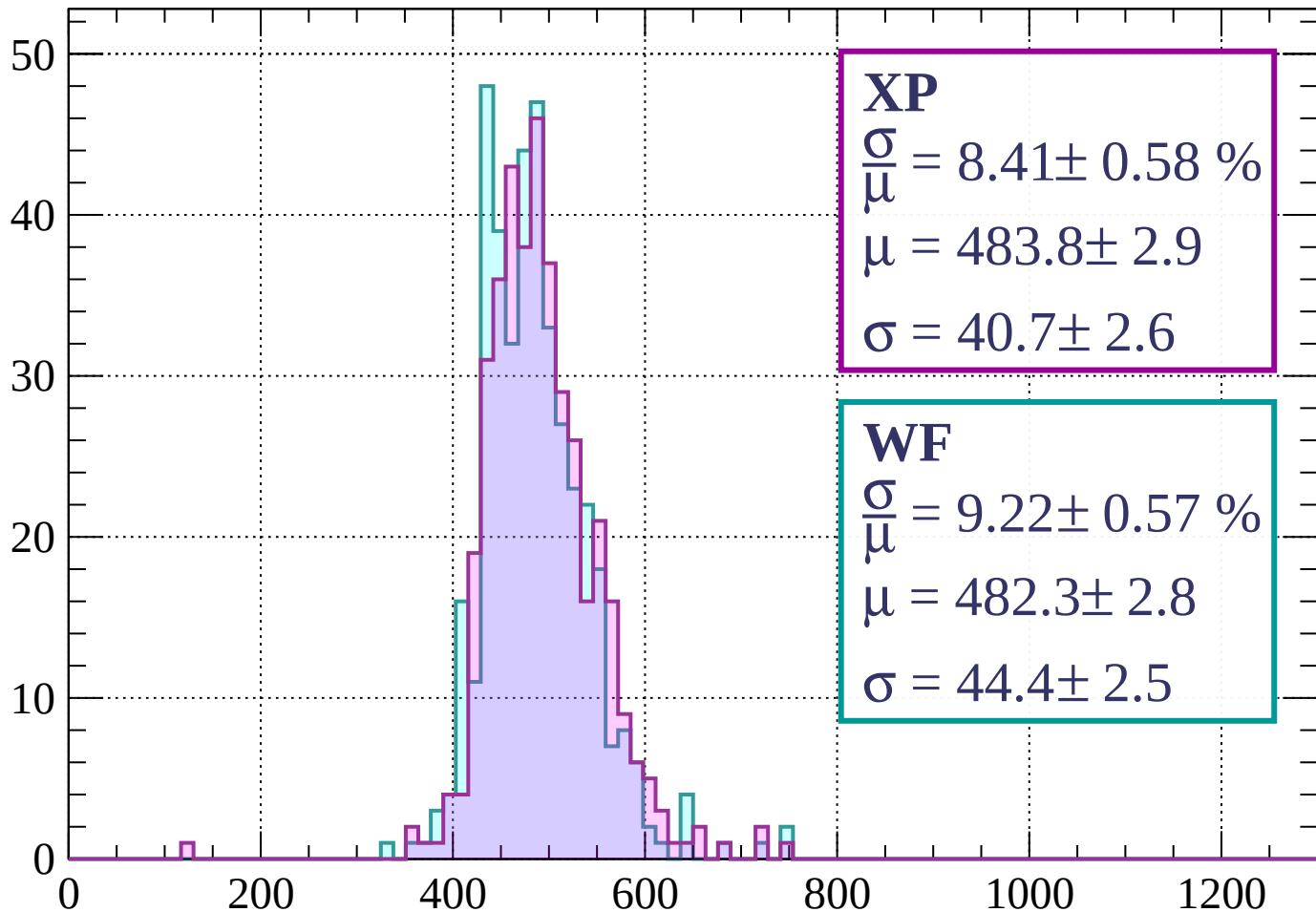
Energy loss | $1720 < p < 1760$

Count



Energy loss | $1760 < p < 1800$

Count



XP

$$\frac{\sigma}{\mu} = 8.41 \pm 0.58 \%$$

$$\mu = 483.8 \pm 2.9$$

$$\sigma = 40.7 \pm 2.6$$

WF

$$\frac{\sigma}{\mu} = 9.22 \pm 0.57 \%$$

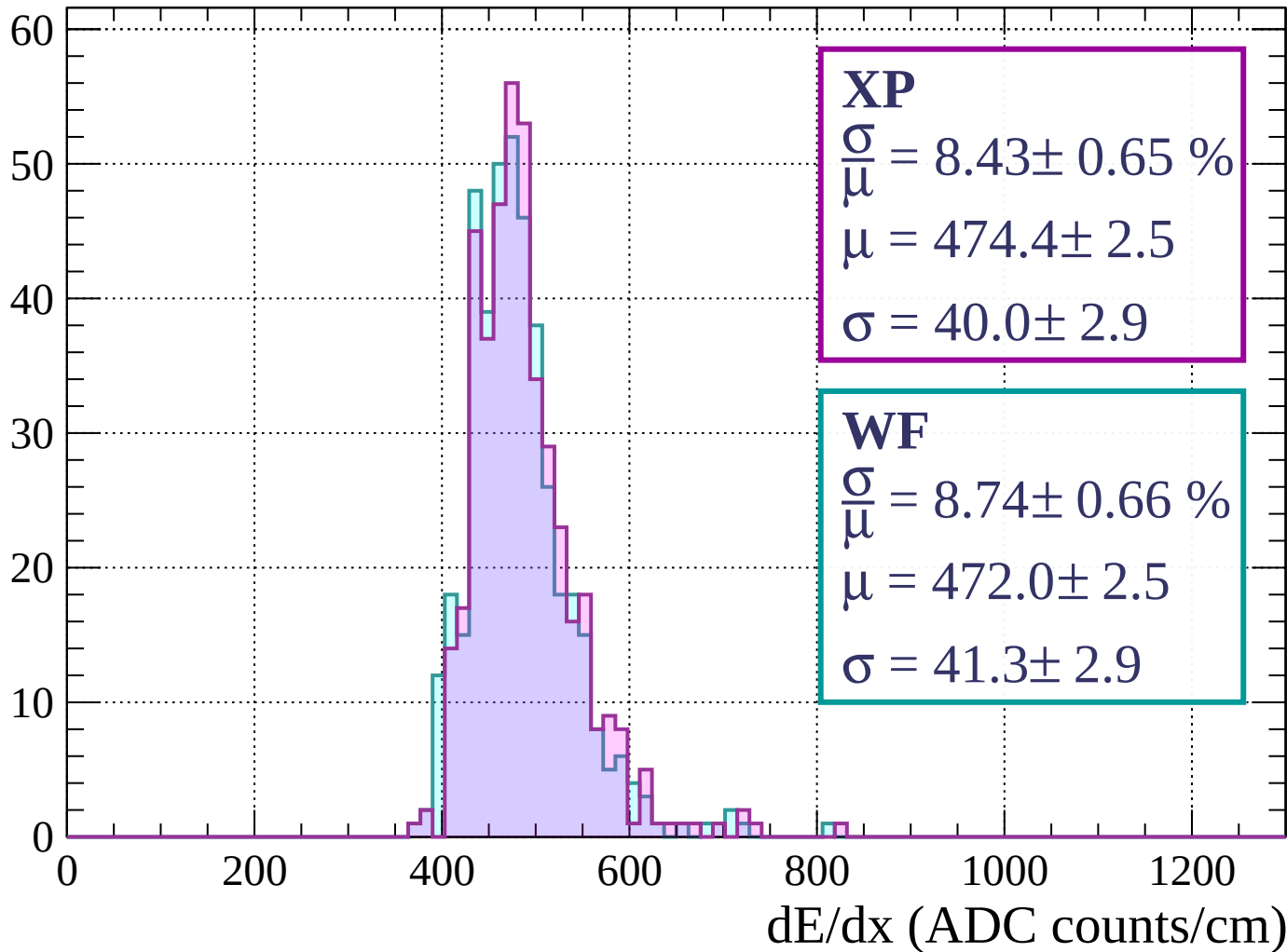
$$\mu = 482.3 \pm 2.8$$

$$\sigma = 44.4 \pm 2.5$$

dE/dx (ADC counts/cm)

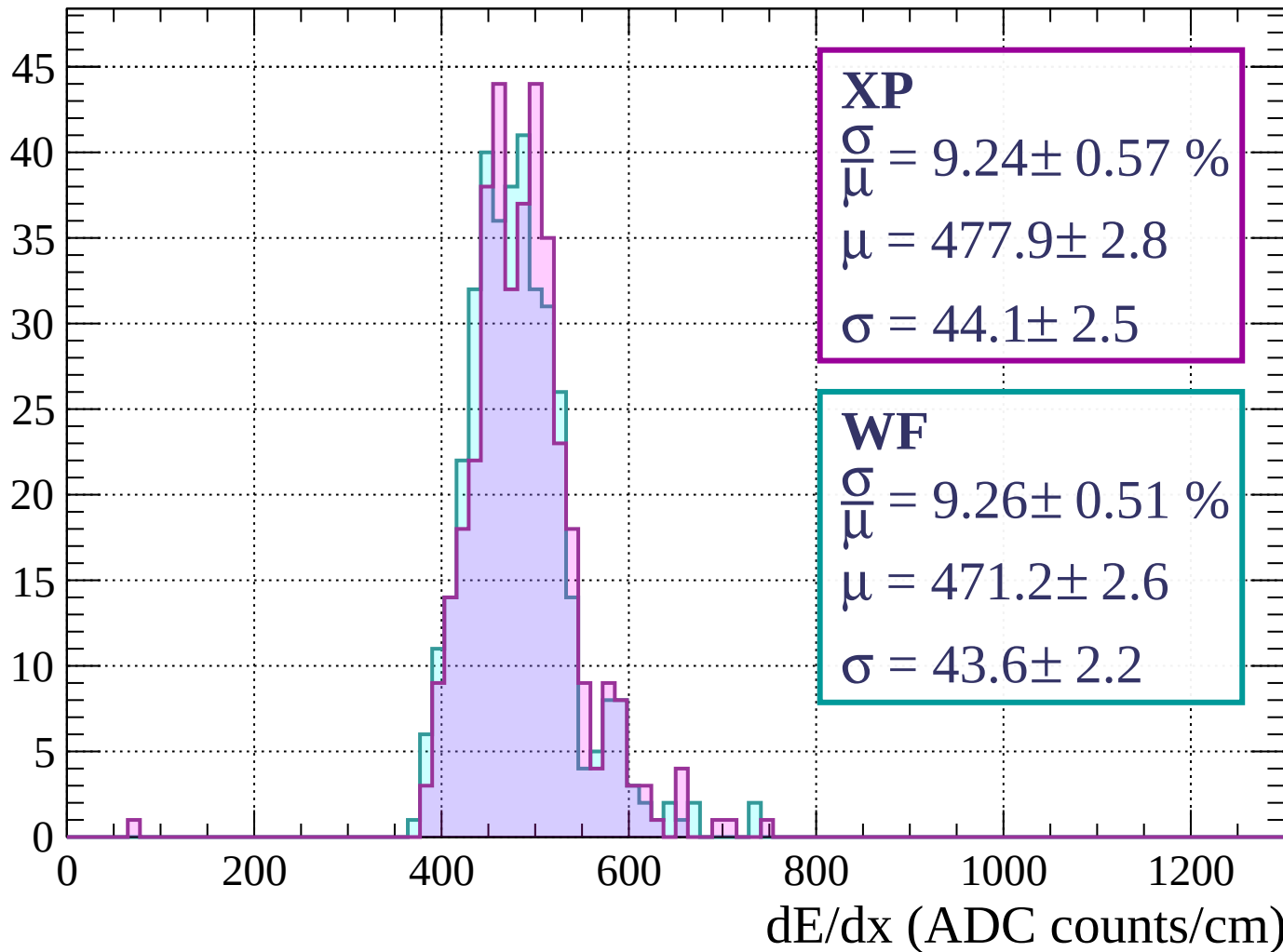
Energy loss | $1800 < p < 1840$

Count



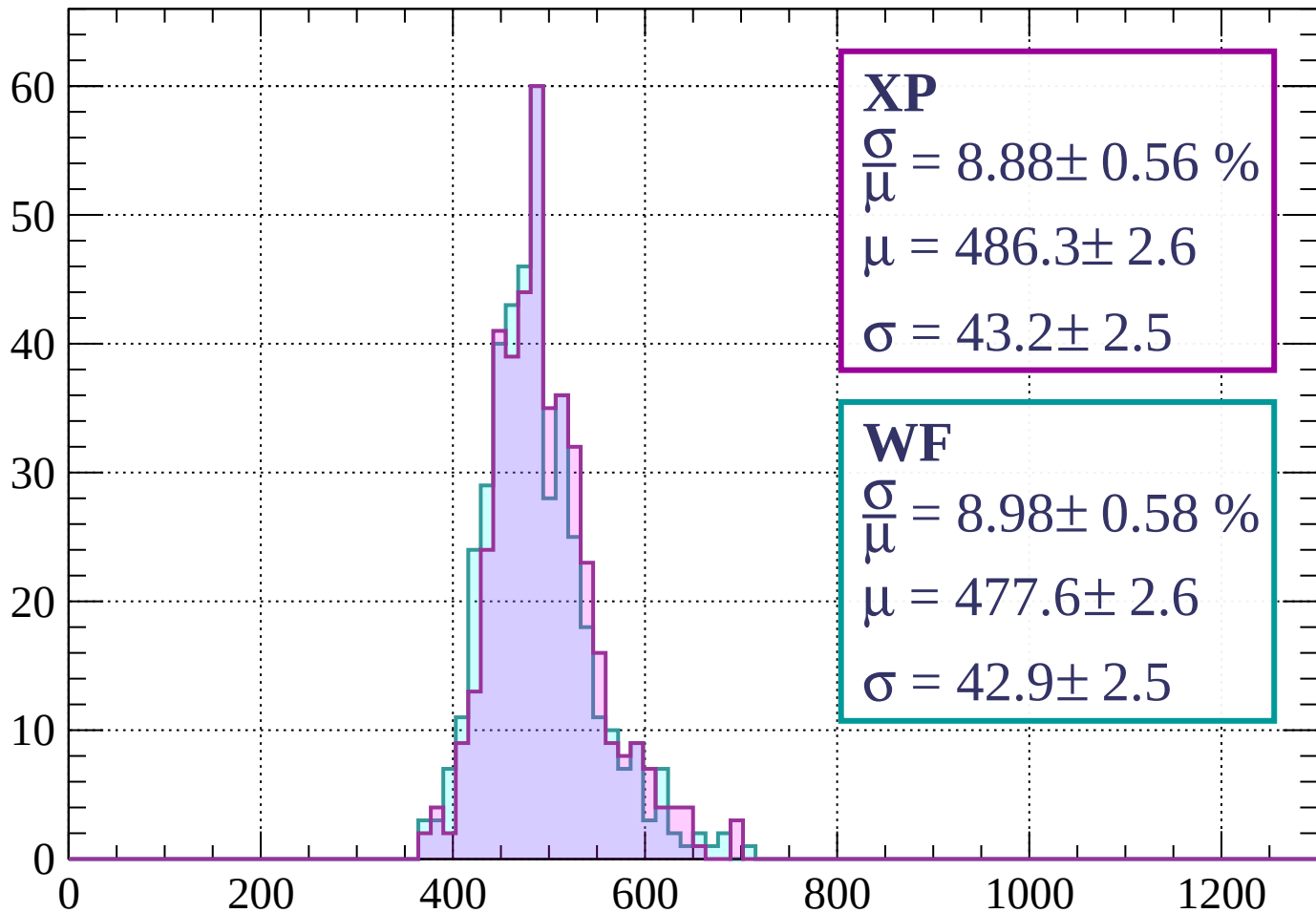
Energy loss | $1840 < p < 1880$

Count



Energy loss | $1880 < p < 1920$

Count



XP

$$\frac{\sigma}{\mu} = 8.88 \pm 0.56 \%$$

$$\mu = 486.3 \pm 2.6$$

$$\sigma = 43.2 \pm 2.5$$

WF

$$\frac{\sigma}{\mu} = 8.98 \pm 0.58 \%$$

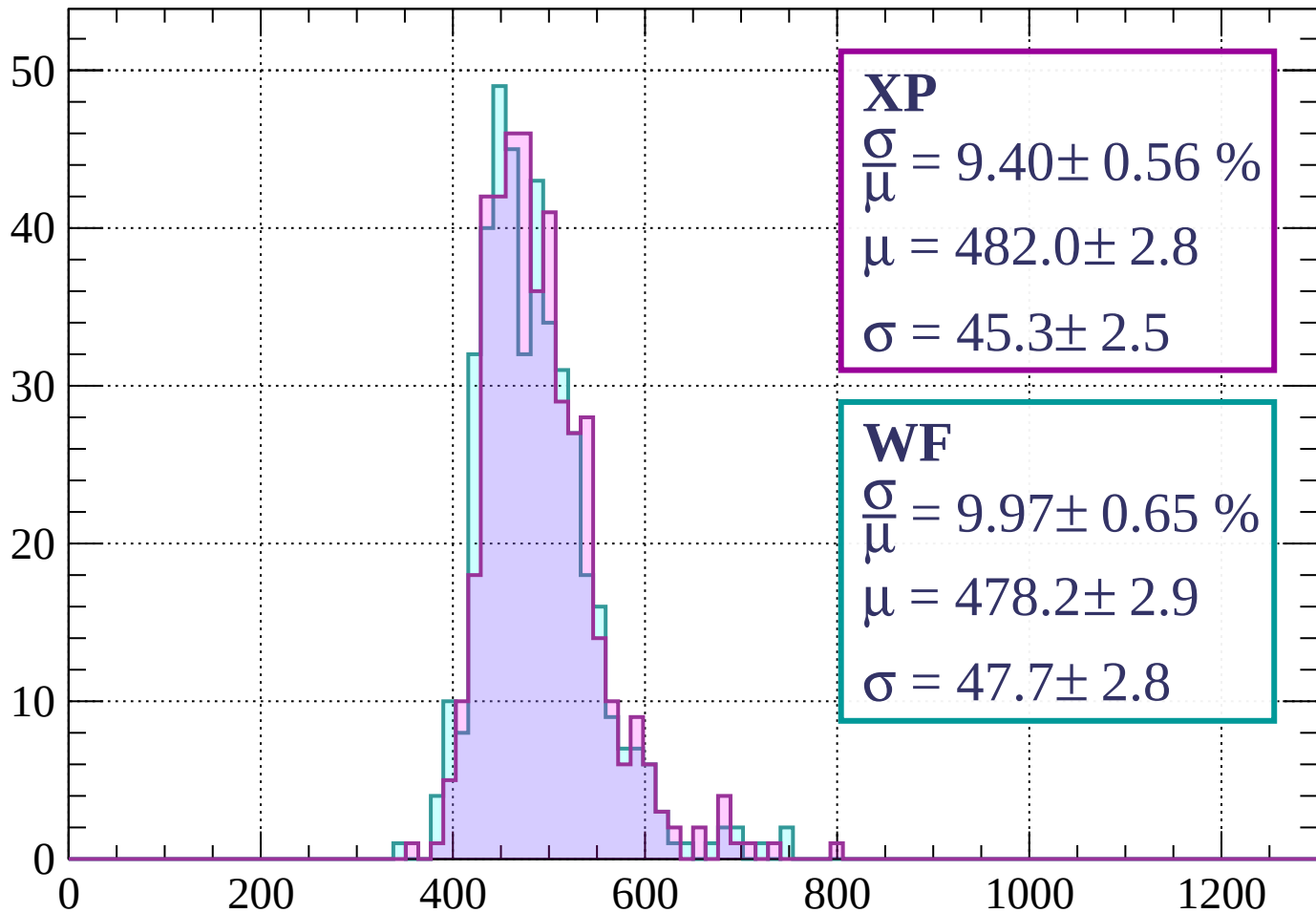
$$\mu = 477.6 \pm 2.6$$

$$\sigma = 42.9 \pm 2.5$$

dE/dx (ADC counts/cm)

Energy loss | $1920 < p < 1960$

Count



XP

$$\frac{\sigma}{\mu} = 9.40 \pm 0.56 \%$$

$$\mu = 482.0 \pm 2.8$$

$$\sigma = 45.3 \pm 2.5$$

WF

$$\frac{\sigma}{\mu} = 9.97 \pm 0.65 \%$$

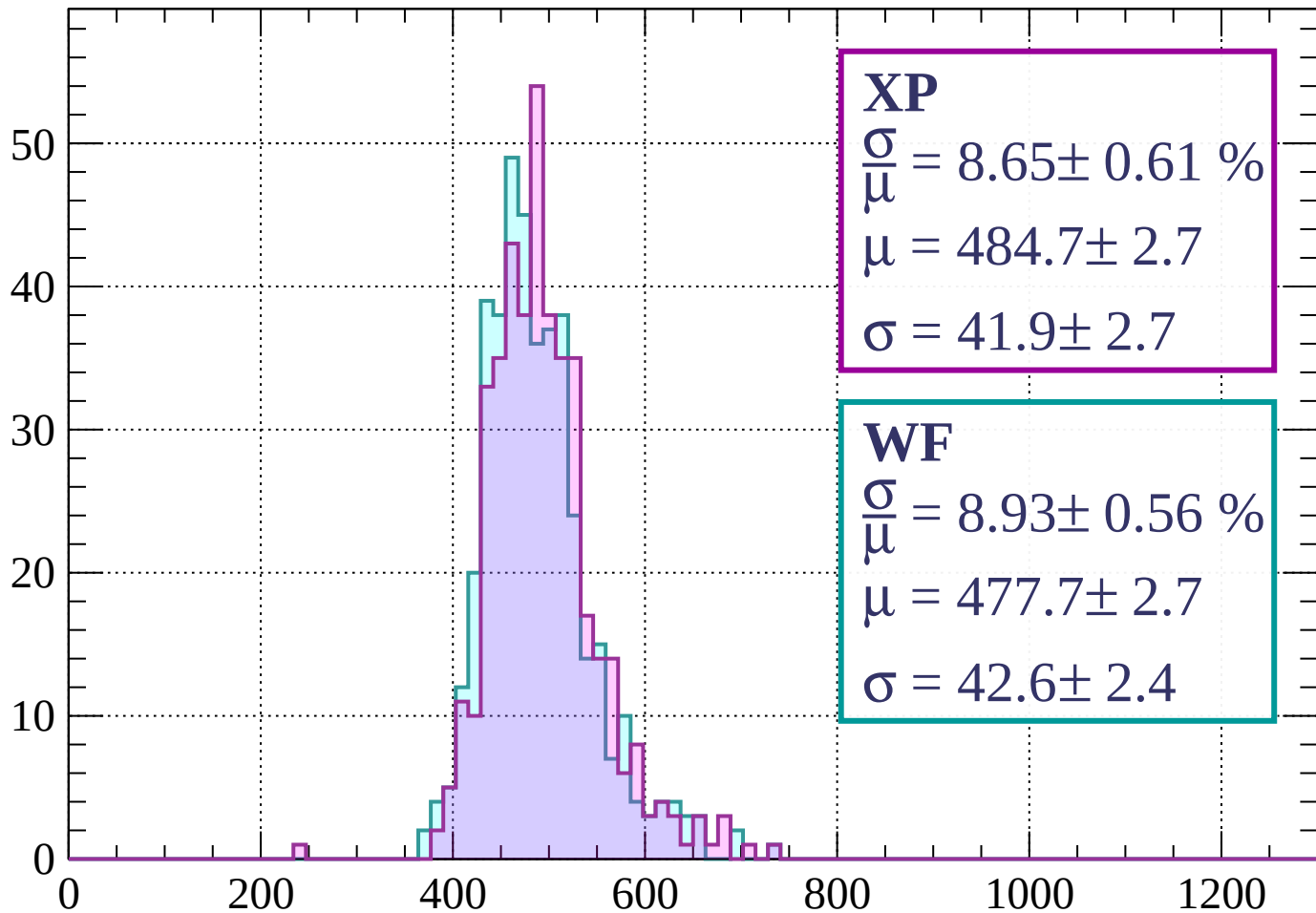
$$\mu = 478.2 \pm 2.9$$

$$\sigma = 47.7 \pm 2.8$$

dE/dx (ADC counts/cm)

Energy loss | 1960 < p < 2000

Count



XP

$$\frac{\sigma}{\mu} = 8.65 \pm 0.61 \%$$

$$\mu = 484.7 \pm 2.7$$

$$\sigma = 41.9 \pm 2.7$$

WF

$$\frac{\sigma}{\mu} = 8.93 \pm 0.56 \%$$

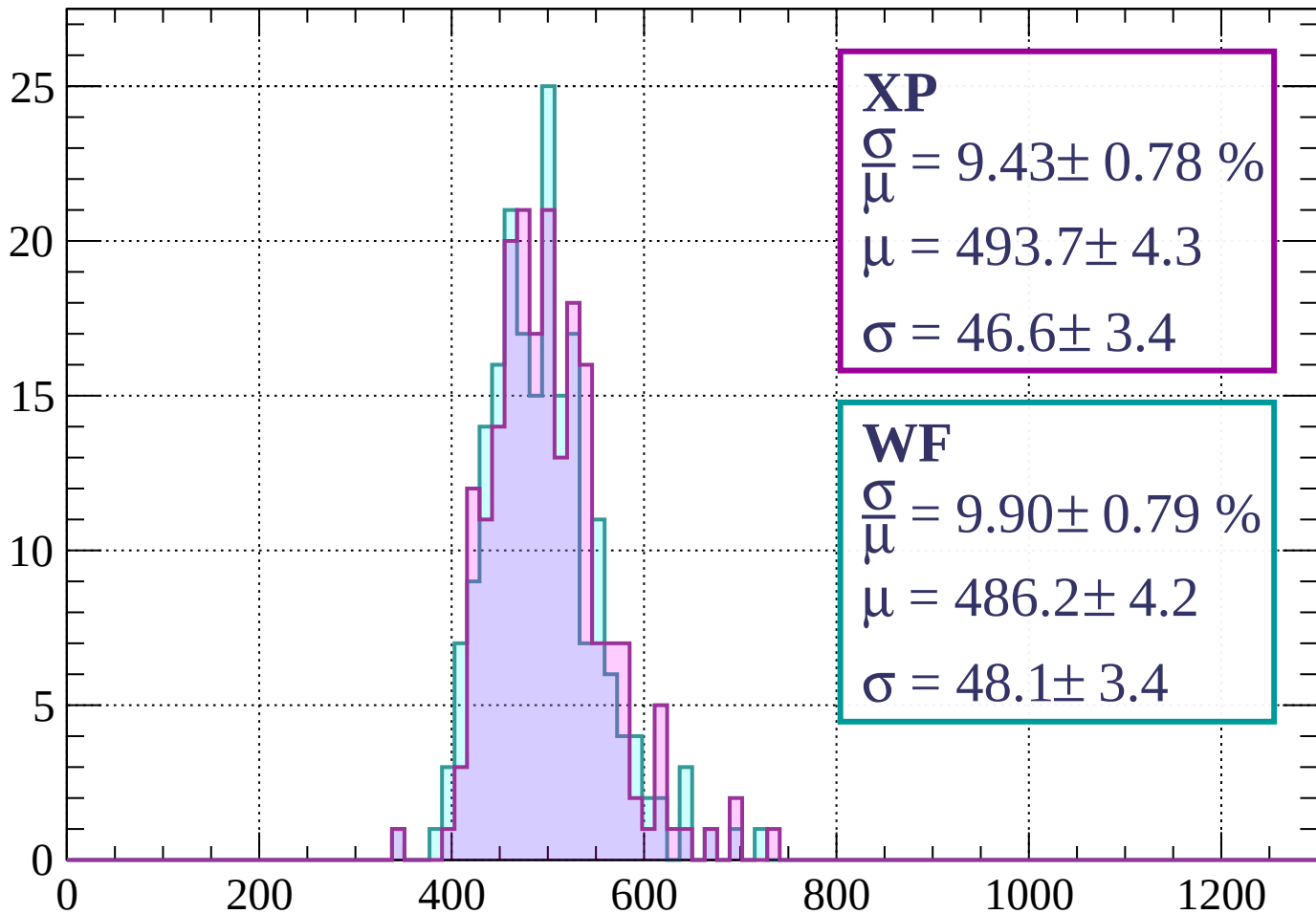
$$\mu = 477.7 \pm 2.7$$

$$\sigma = 42.6 \pm 2.4$$

dE/dx (ADC counts/cm)

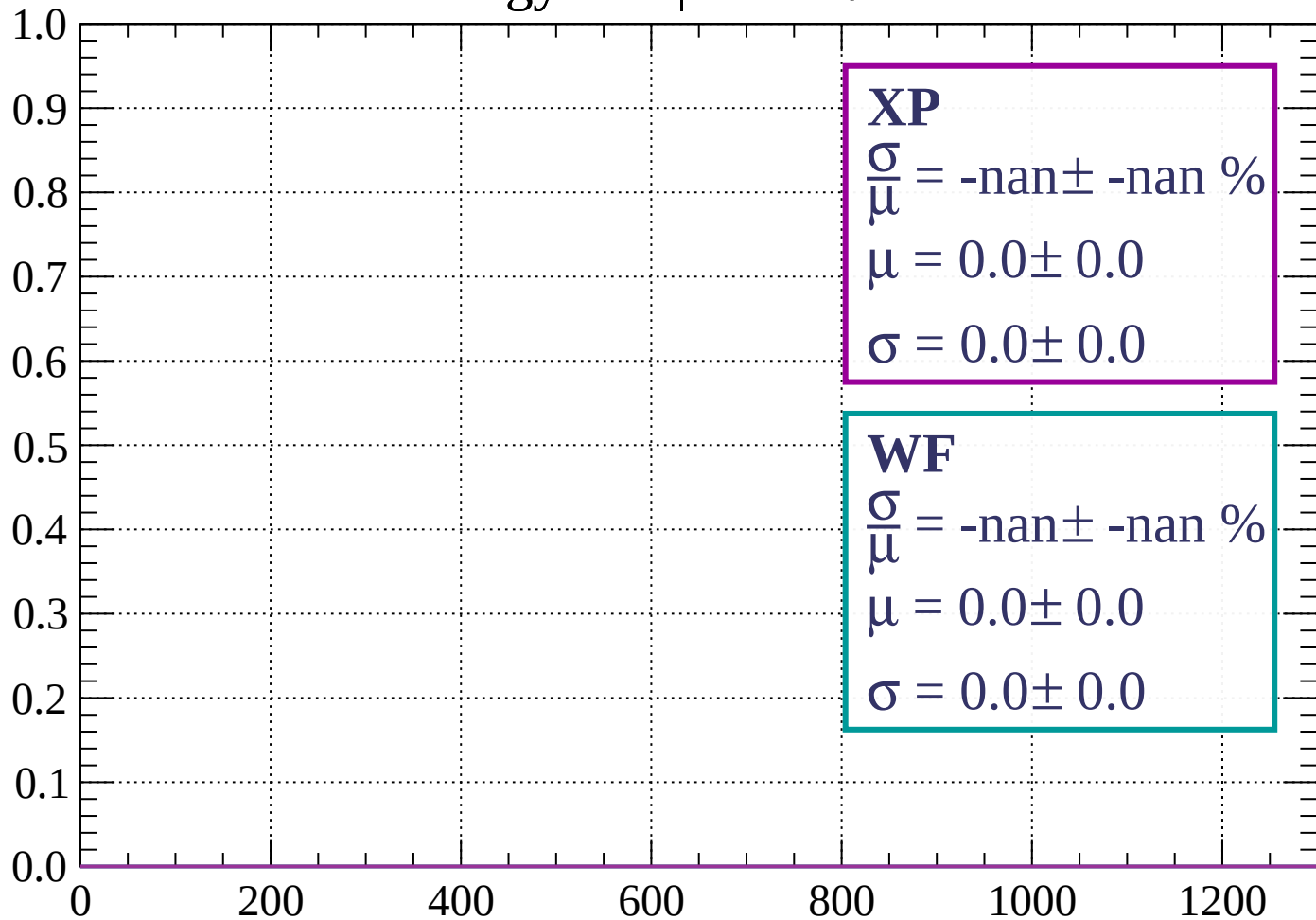
Energy loss | $2000 < p < 2040$

Count



Energy loss | $-90 < \theta < -84$

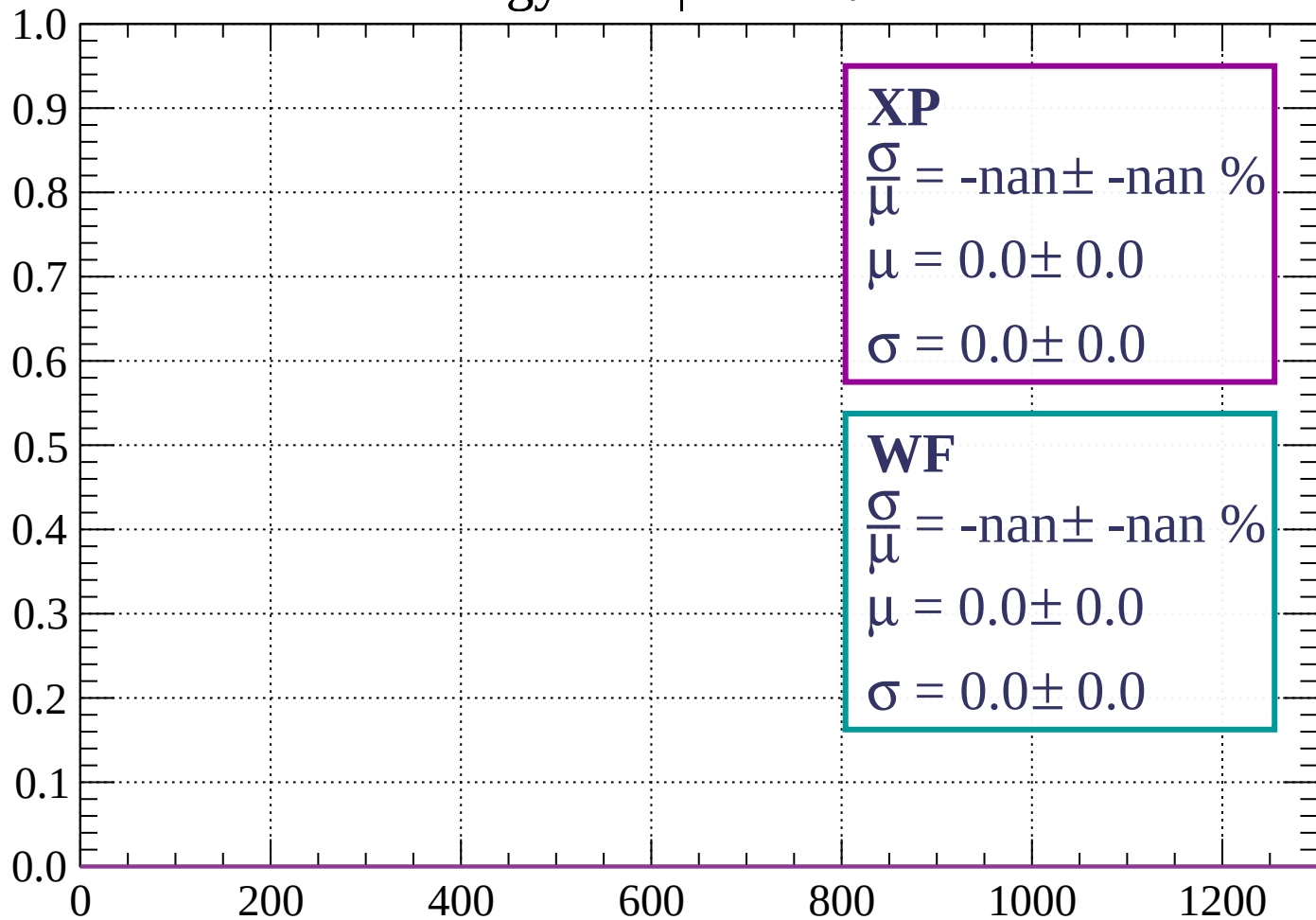
Count



dE/dx (ADC counts/cm)

Energy loss | $-84 < \theta < -78$

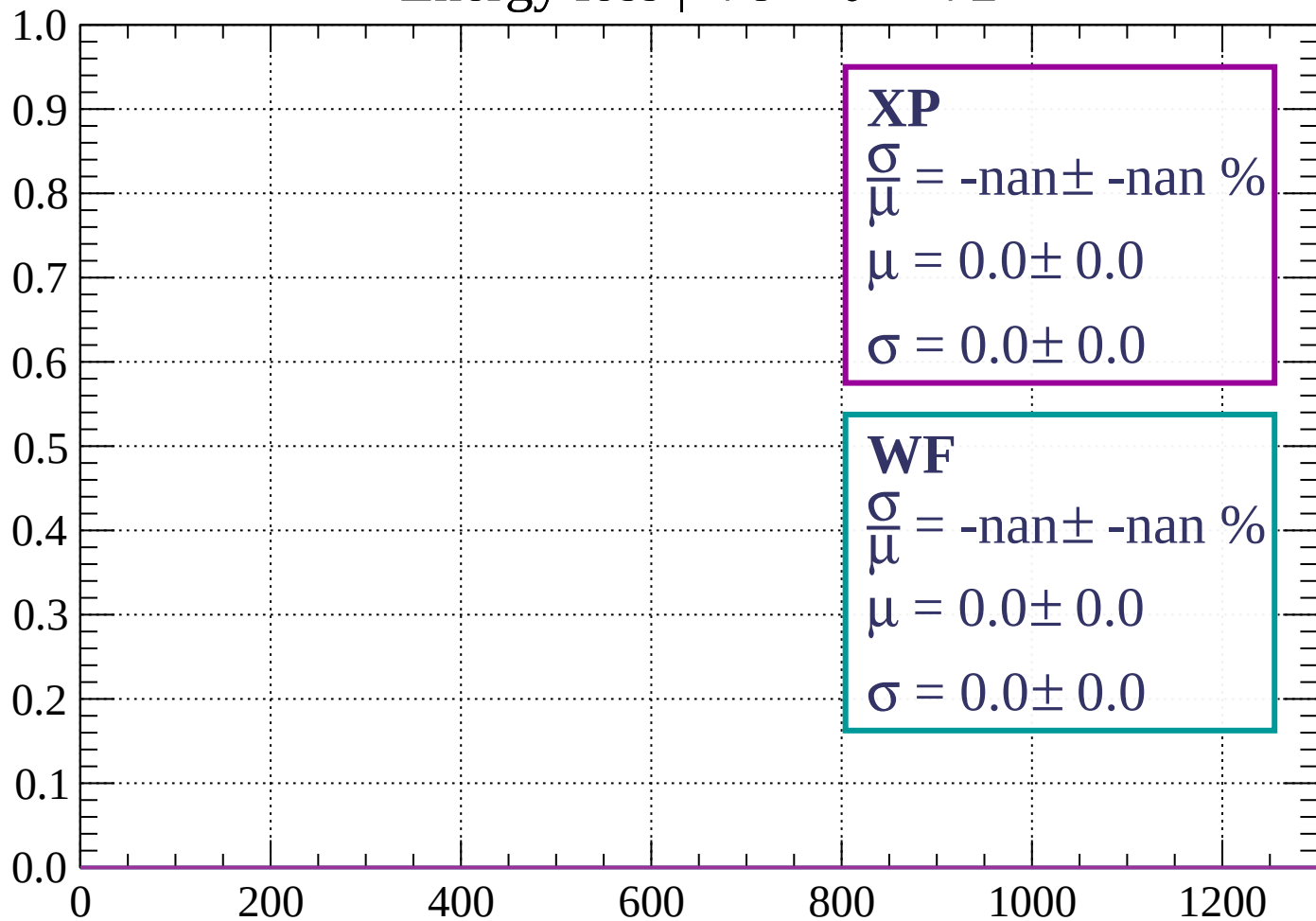
Count



dE/dx (ADC counts/cm)

Energy loss | $-78 < \theta < -72$

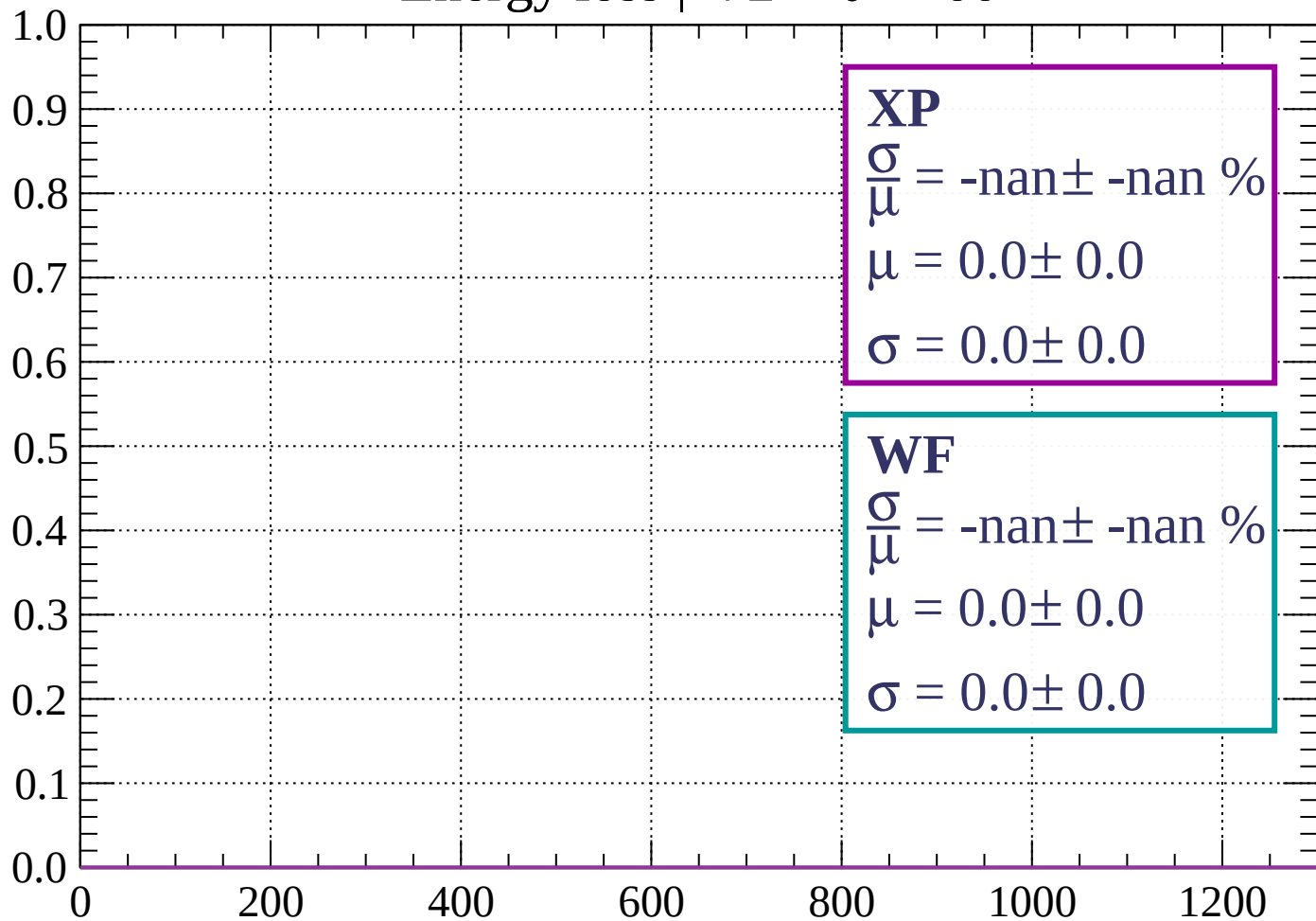
Count



dE/dx (ADC counts/cm)

Energy loss | $-72 < \theta < -66$

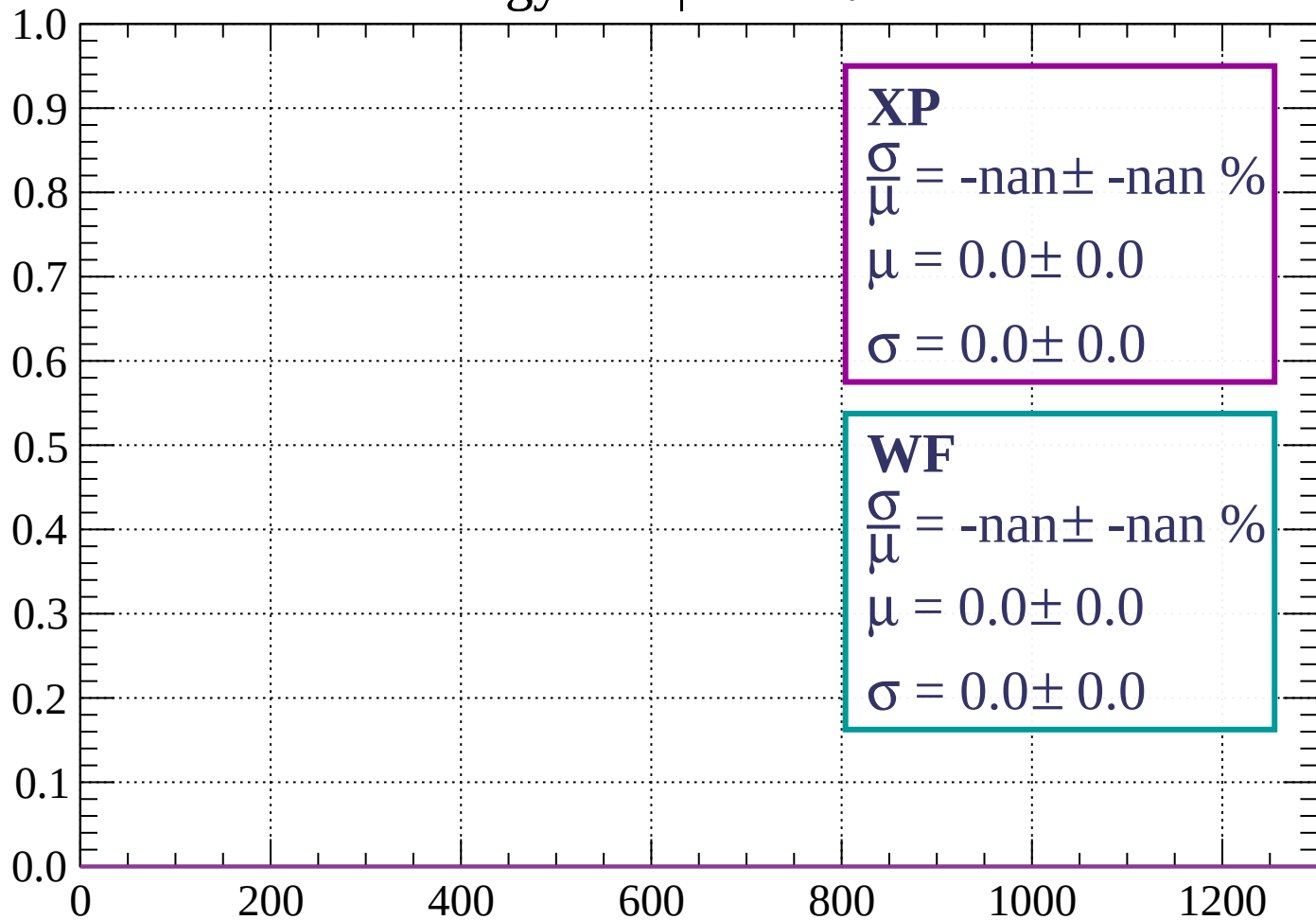
Count



dE/dx (ADC counts/cm)

Energy loss | $-66 < \theta < -60$

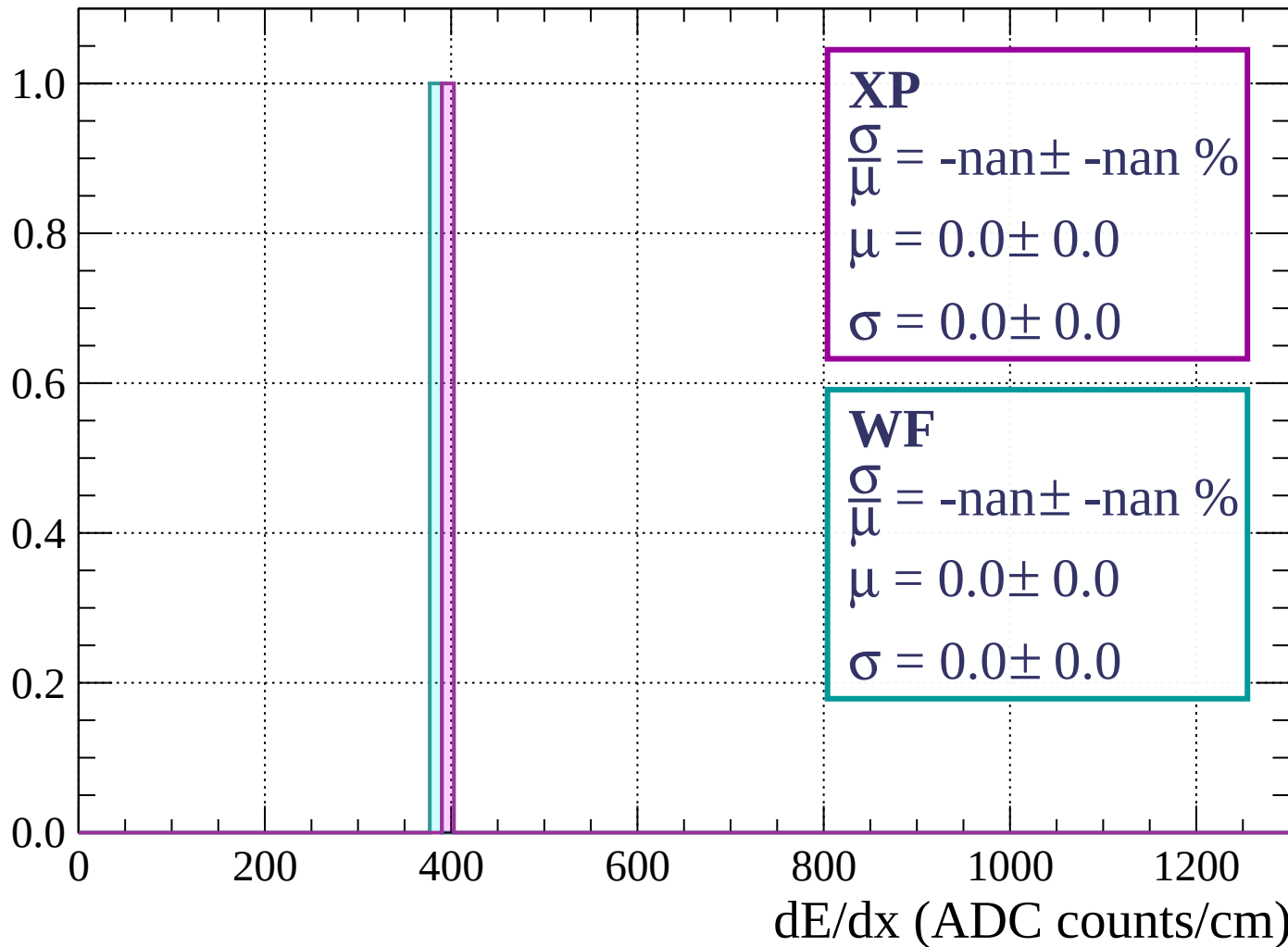
Count



dE/dx (ADC counts/cm)

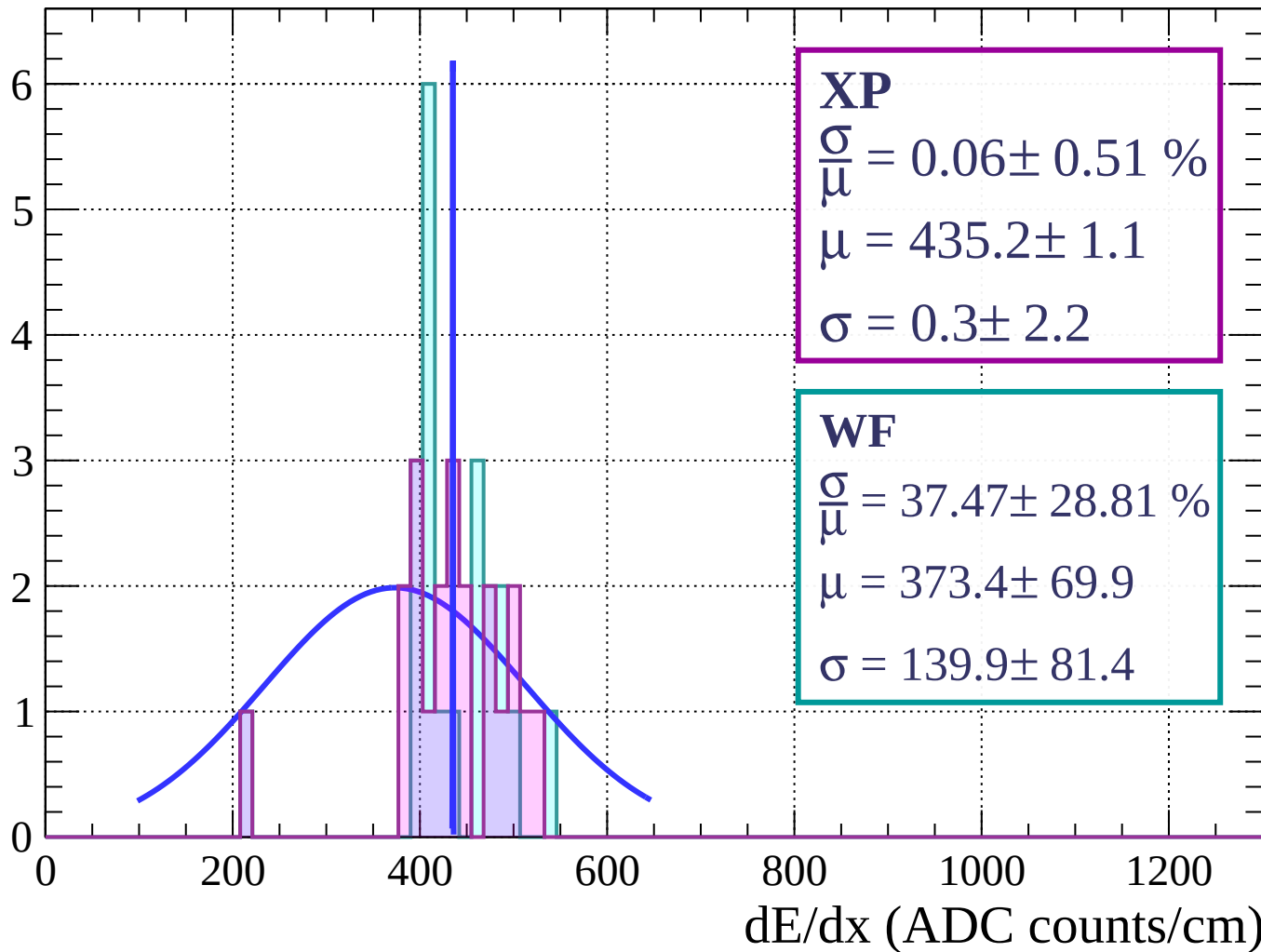
Energy loss | $-60 < \theta < -54$

Count



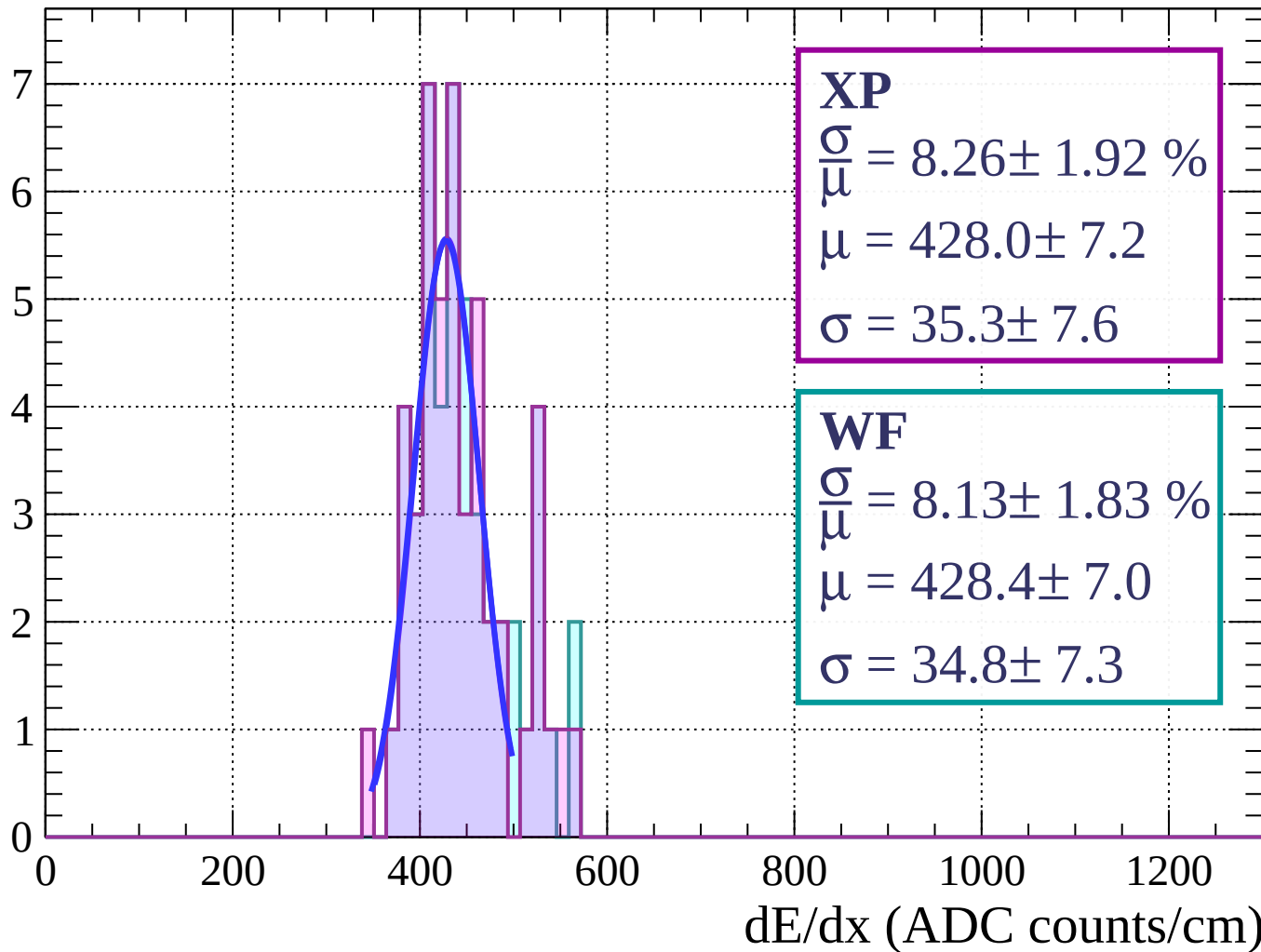
Energy loss | $-54 < \theta < -48$

Count



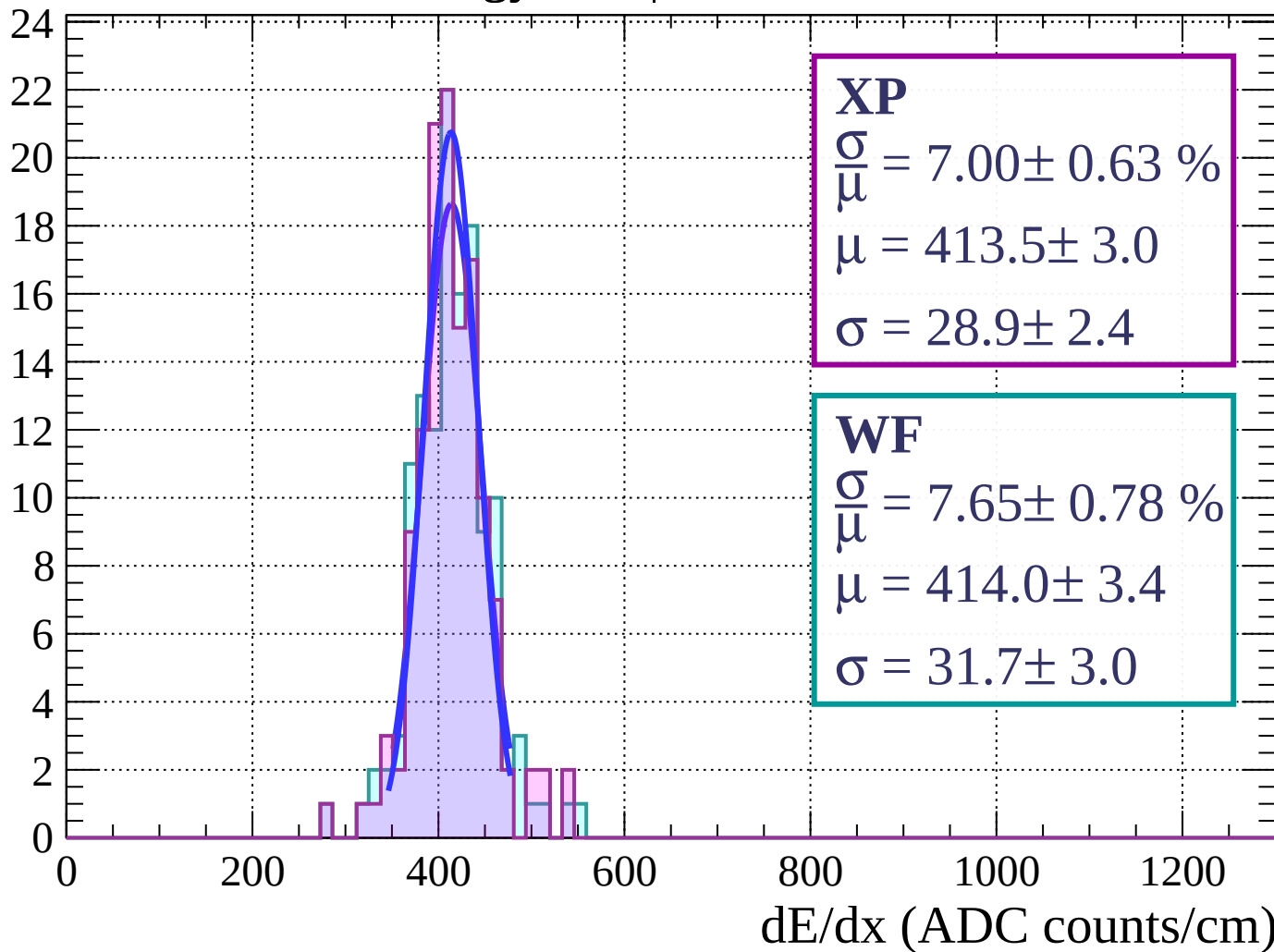
Energy loss | $-48 < \theta < -42$

Count



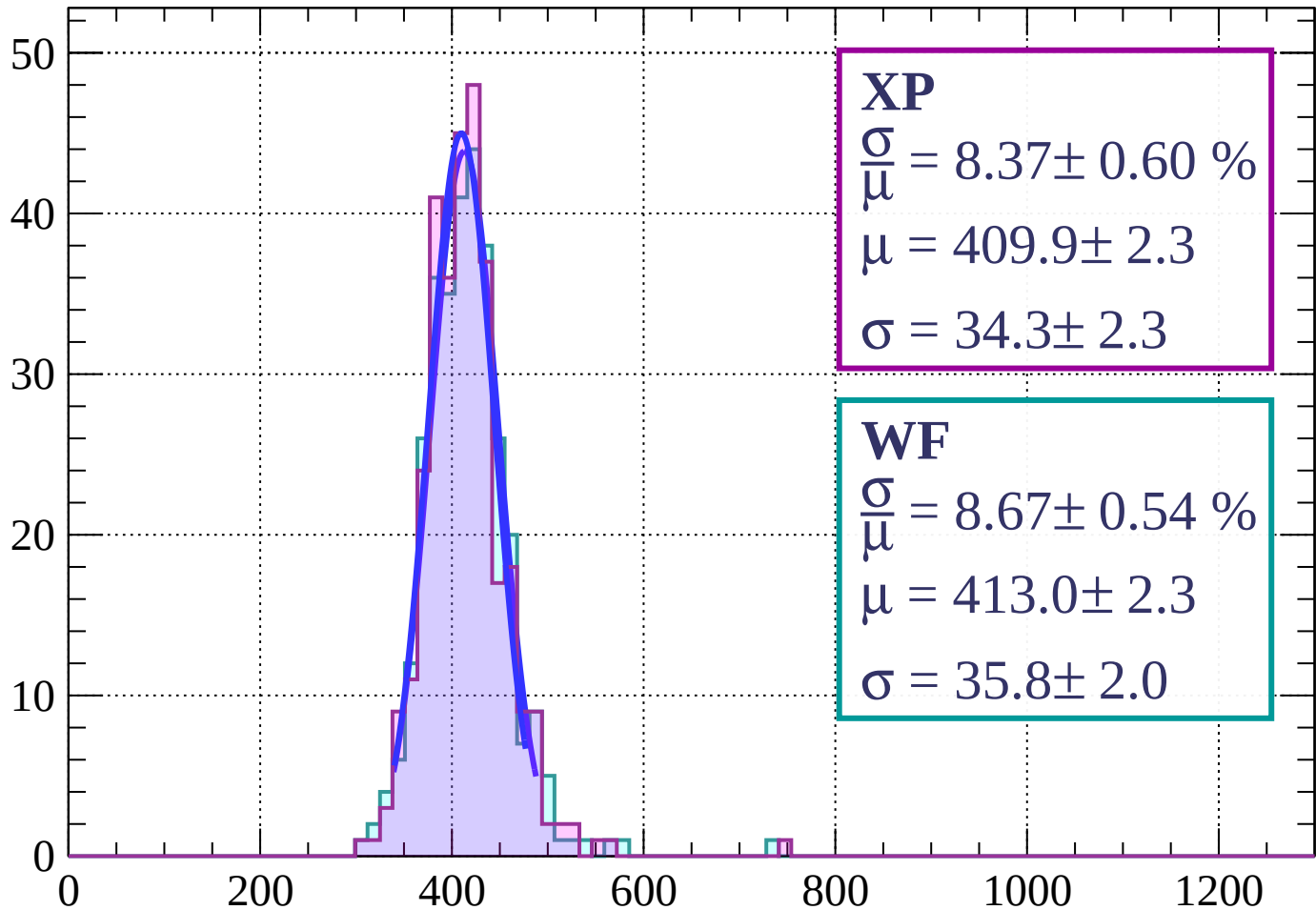
Energy loss $|-42 < \theta < -36$

Count



Energy loss | $-36 < \theta < -30$

Count



XP

$$\frac{\sigma}{\mu} = 8.37 \pm 0.60 \%$$

$$\mu = 409.9 \pm 2.3$$

$$\sigma = 34.3 \pm 2.3$$

WF

$$\frac{\sigma}{\mu} = 8.67 \pm 0.54 \%$$

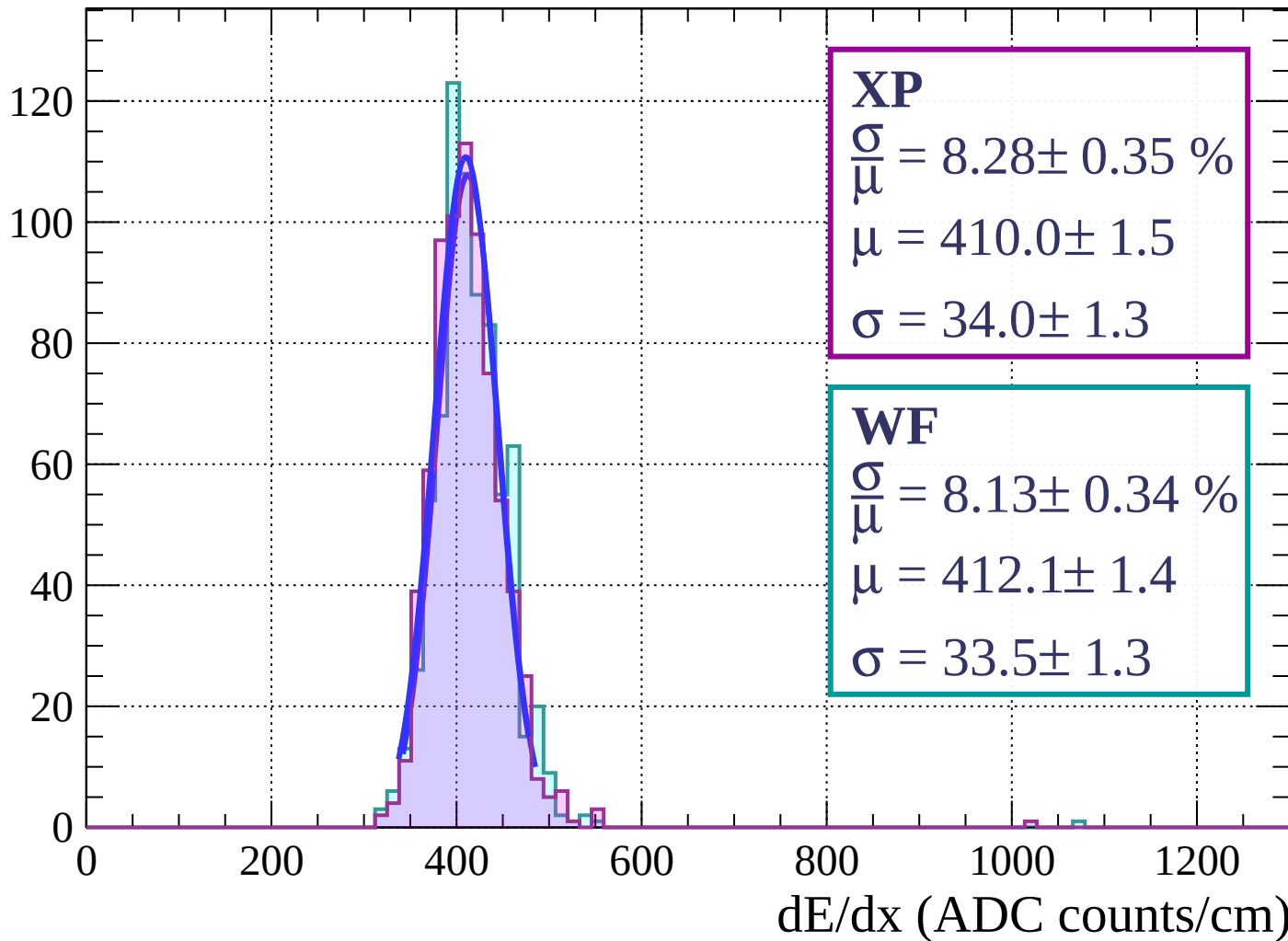
$$\mu = 413.0 \pm 2.3$$

$$\sigma = 35.8 \pm 2.0$$

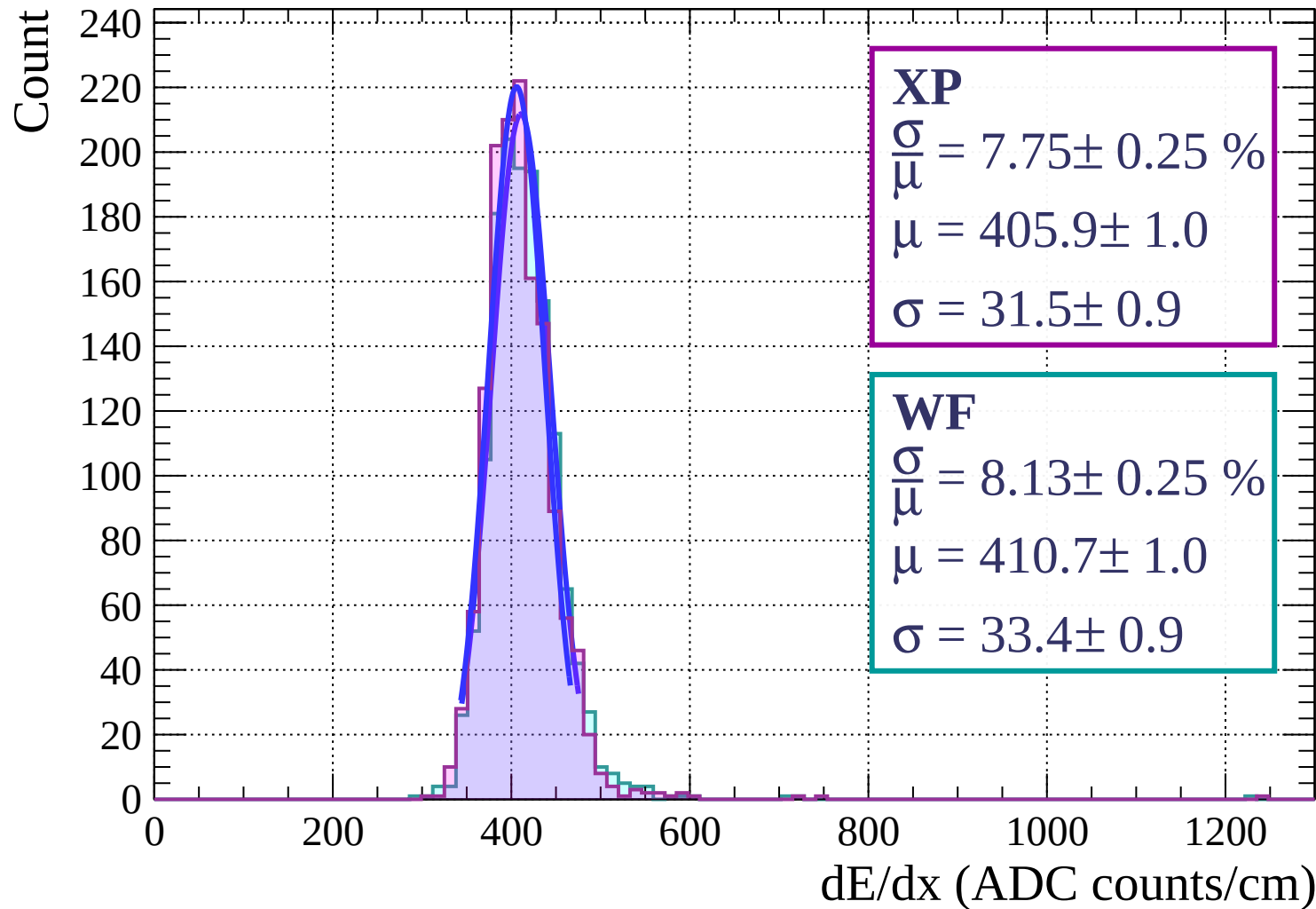
dE/dx (ADC counts/cm)

Energy loss | $-30 < \theta < -24$

Count

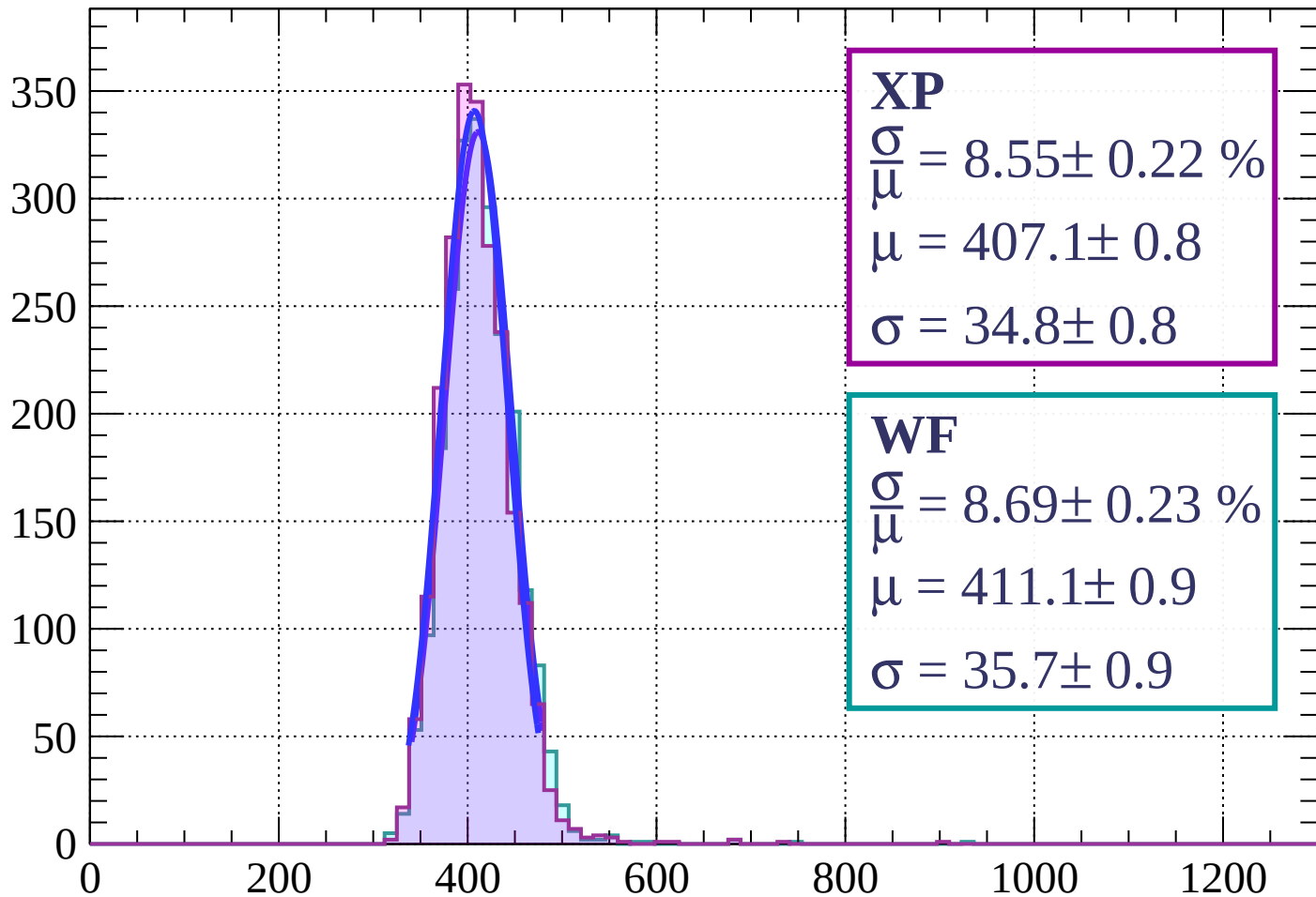


Energy loss | $-24 < \theta < -18$



Energy loss | $-18 < \theta < -12$

Count



XP

$$\frac{\sigma}{\mu} = 8.55 \pm 0.22 \%$$

$$\mu = 407.1 \pm 0.8$$

$$\sigma = 34.8 \pm 0.8$$

WF

$$\frac{\sigma}{\mu} = 8.69 \pm 0.23 \%$$

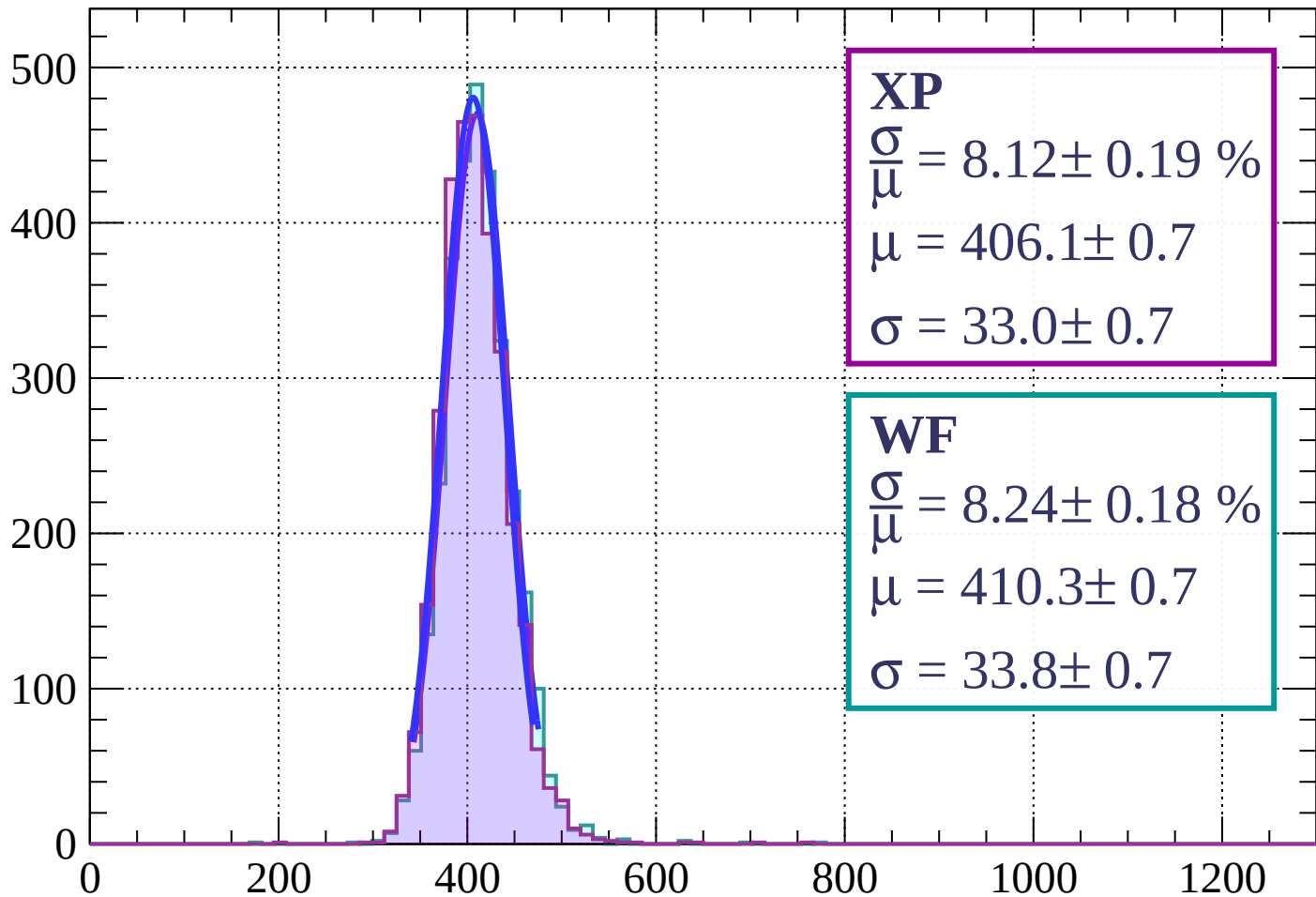
$$\mu = 411.1 \pm 0.9$$

$$\sigma = 35.7 \pm 0.9$$

dE/dx (ADC counts/cm)

Energy loss | $-12 < \theta < -6$

Count



XP

$$\frac{\sigma}{\mu} = 8.12 \pm 0.19 \%$$

$$\mu = 406.1 \pm 0.7$$

$$\sigma = 33.0 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 8.24 \pm 0.18 \%$$

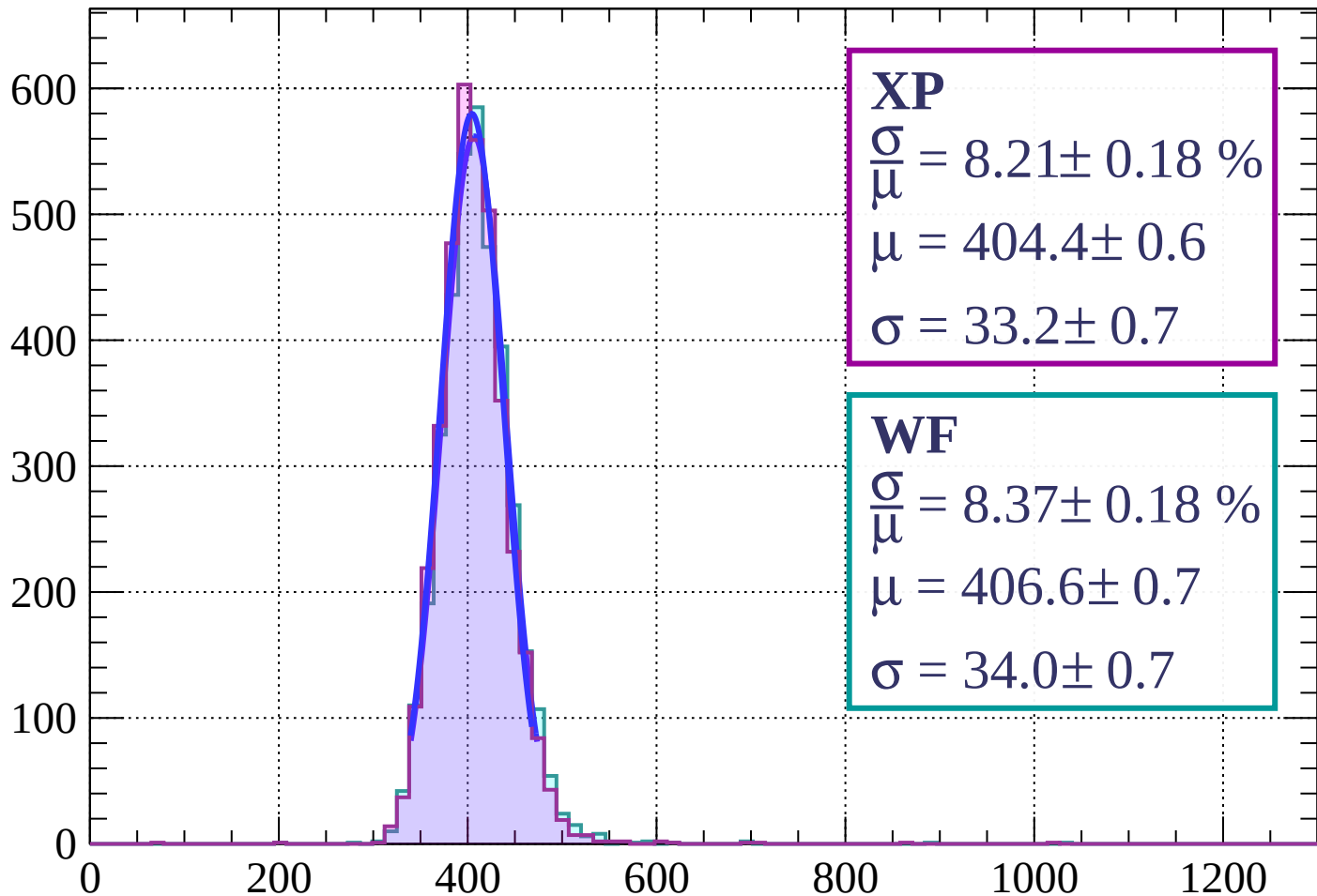
$$\mu = 410.3 \pm 0.7$$

$$\sigma = 33.8 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $-6 < \theta < 0$

Count



XP

$$\frac{\sigma}{\mu} = 8.21 \pm 0.18 \%$$

$$\mu = 404.4 \pm 0.6$$

$$\sigma = 33.2 \pm 0.7$$

WF

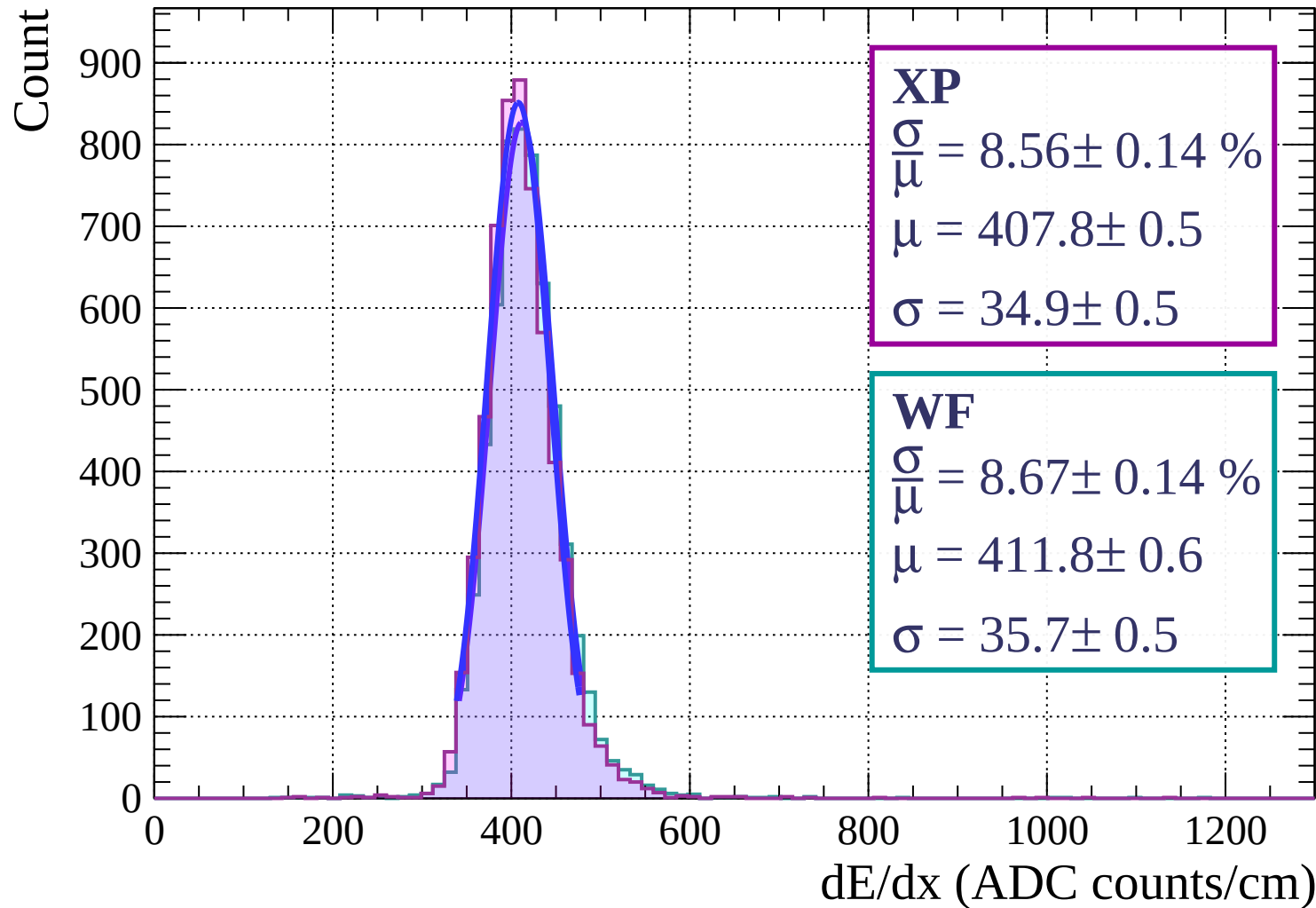
$$\frac{\sigma}{\mu} = 8.37 \pm 0.18 \%$$

$$\mu = 406.6 \pm 0.7$$

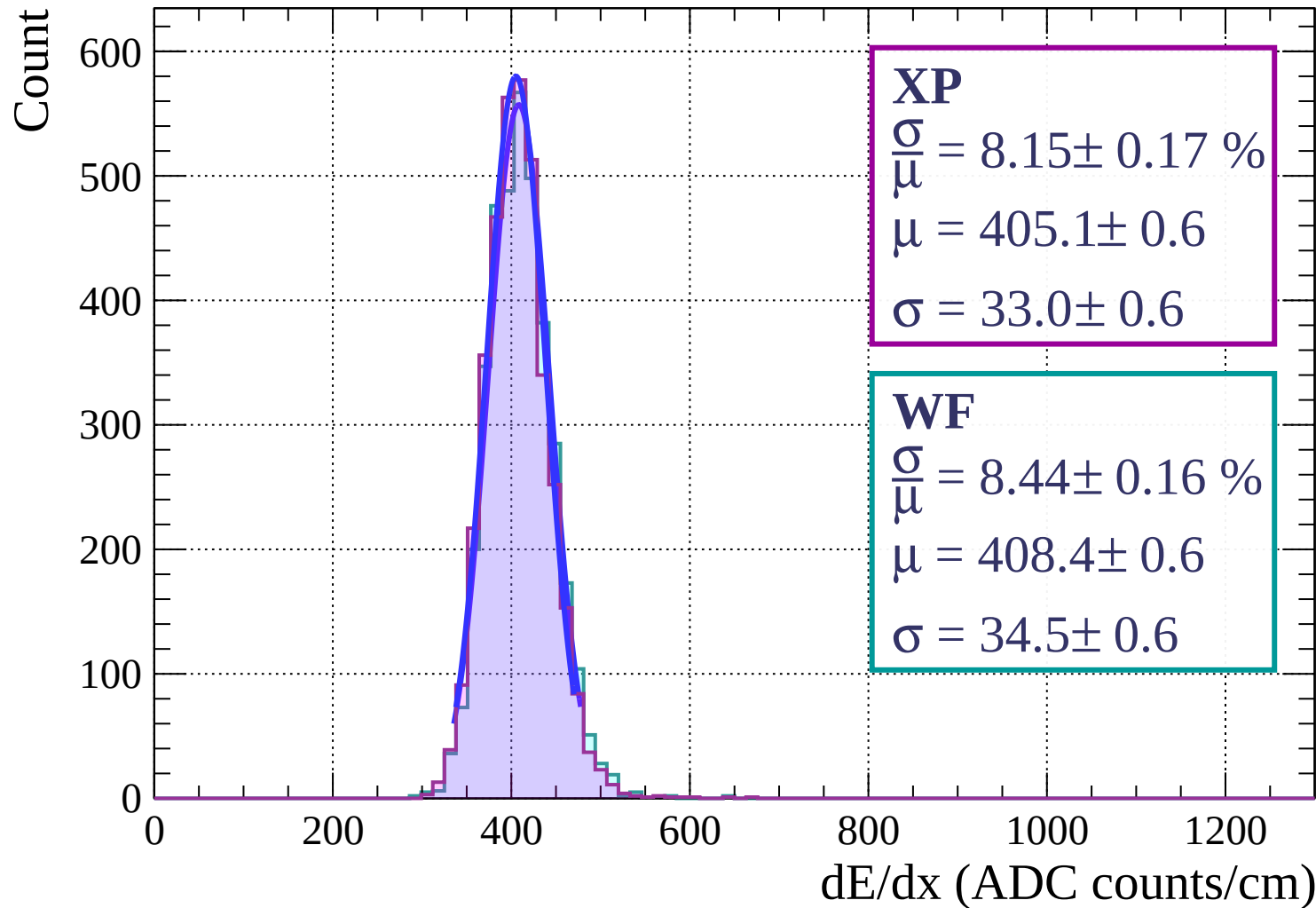
$$\sigma = 34.0 \pm 0.7$$

dE/dx (ADC counts/cm)

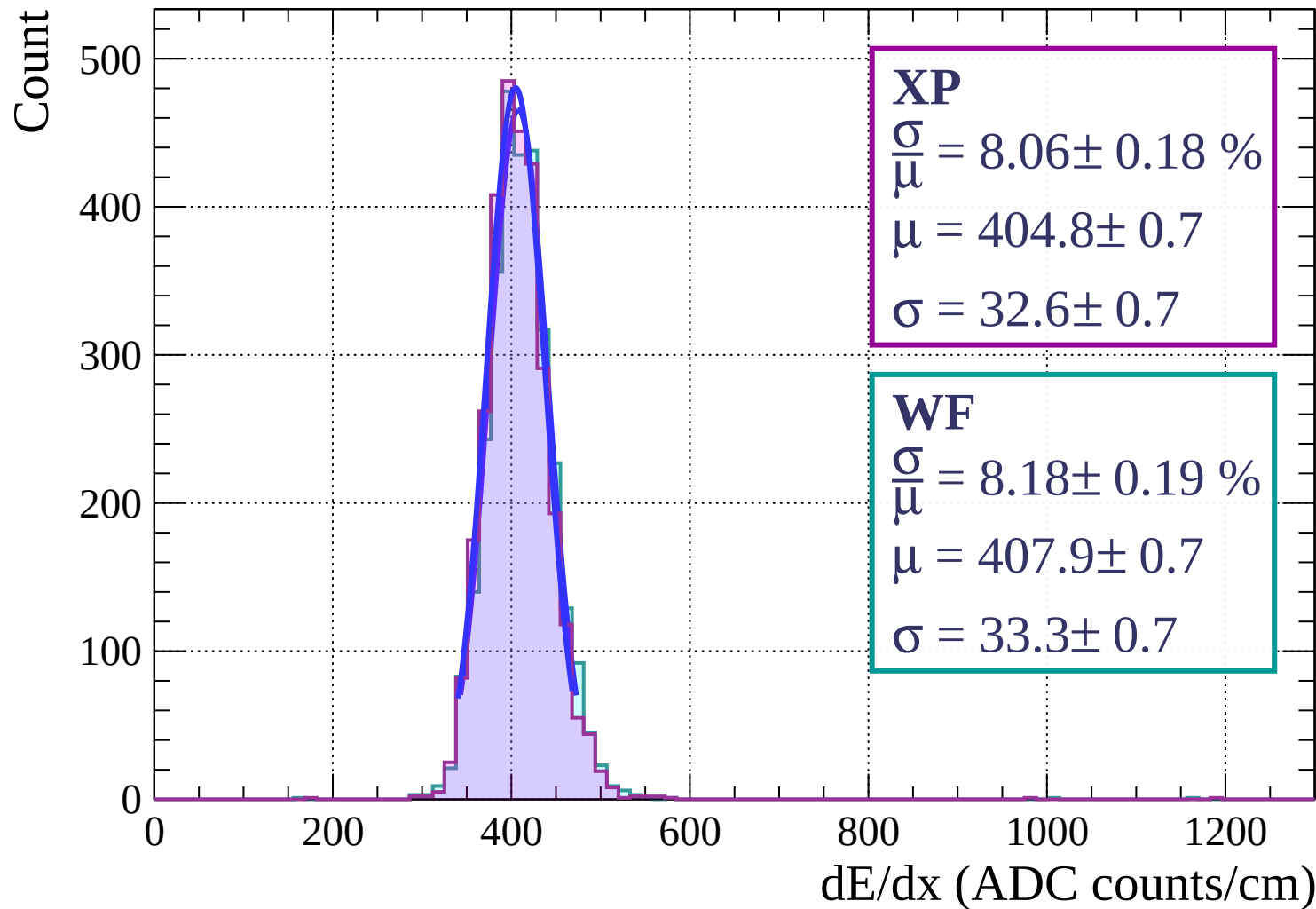
Energy loss | $0 < \theta < 6$



Energy loss | $6 < \theta < 12$

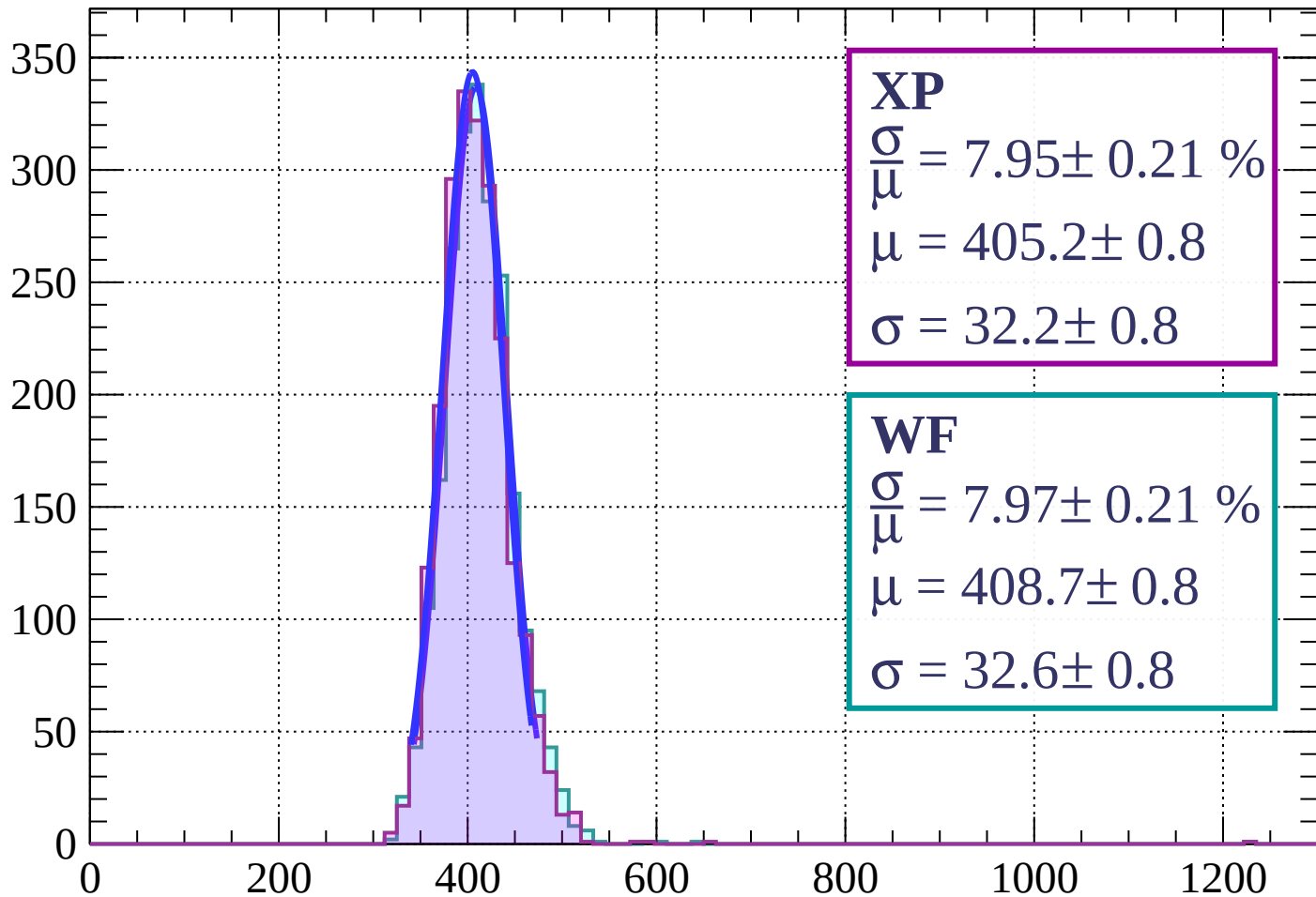


Energy loss | $12 < \theta < 18$



Energy loss | $18 < \theta < 24$

Count



XP

$$\frac{\sigma}{\mu} = 7.95 \pm 0.21 \%$$

$$\mu = 405.2 \pm 0.8$$

$$\sigma = 32.2 \pm 0.8$$

WF

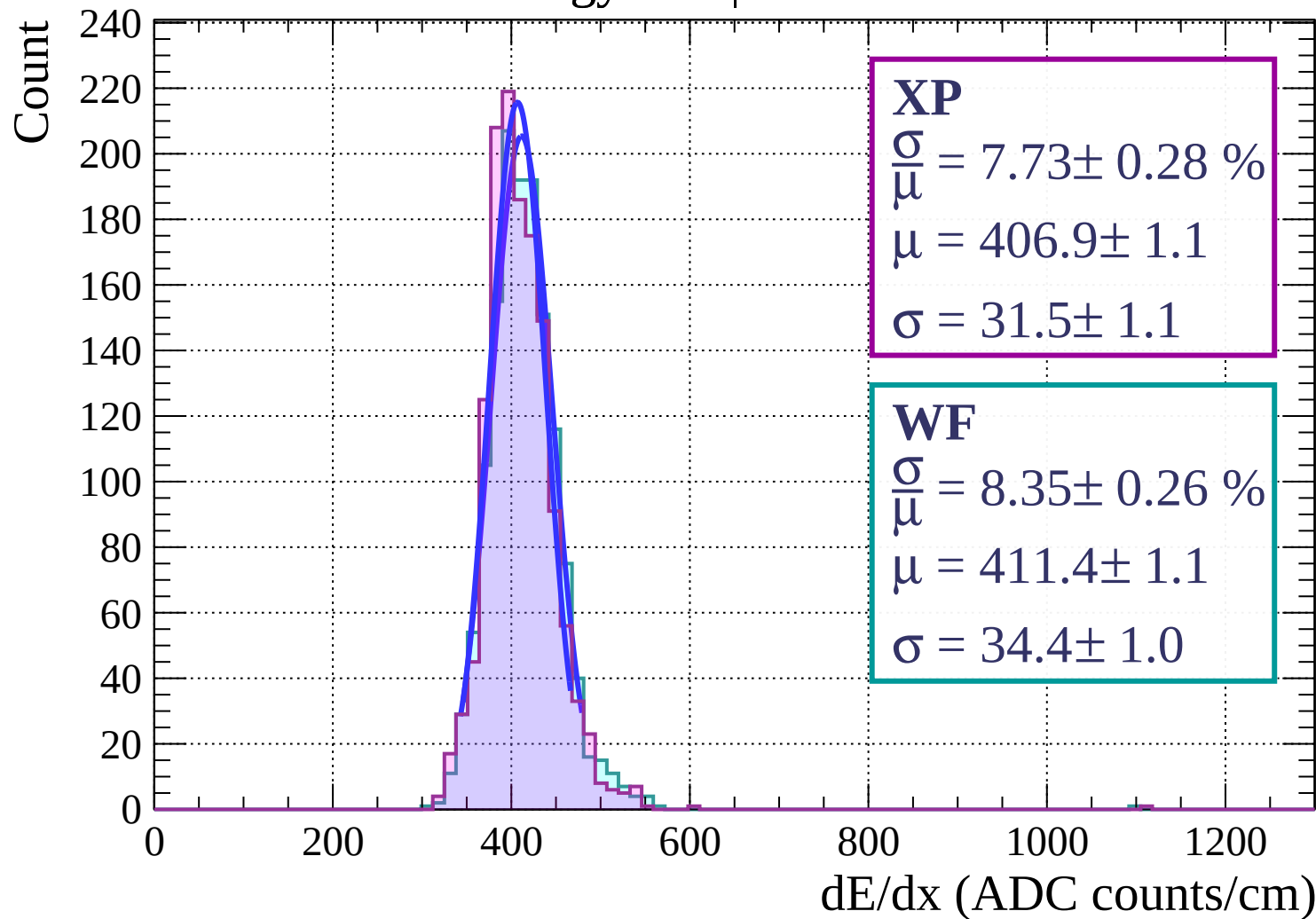
$$\frac{\sigma}{\mu} = 7.97 \pm 0.21 \%$$

$$\mu = 408.7 \pm 0.8$$

$$\sigma = 32.6 \pm 0.8$$

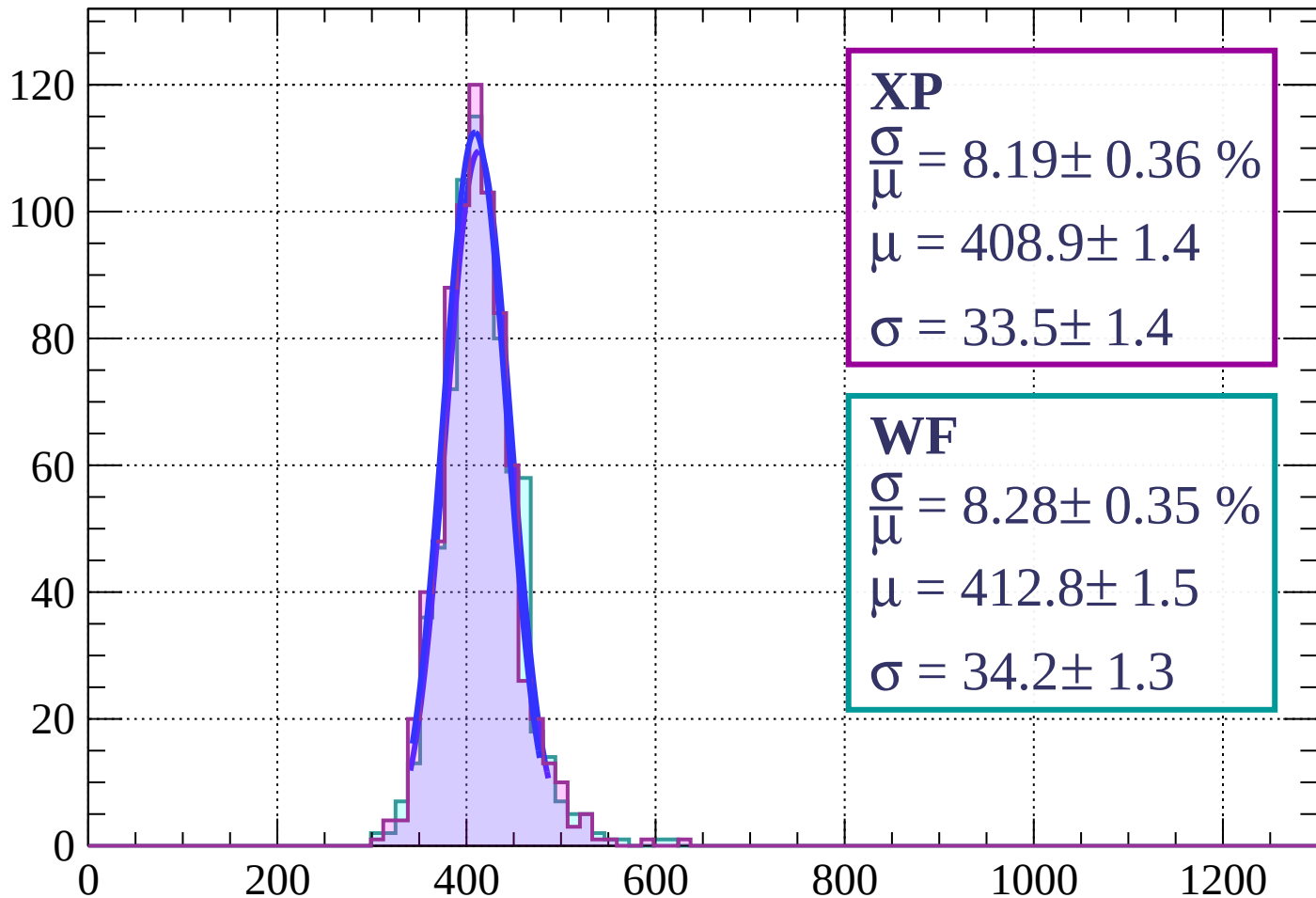
dE/dx (ADC counts/cm)

Energy loss | $24 < \theta < 30$



Energy loss | $30 < \theta < 36$

Count



XP

$$\frac{\sigma}{\mu} = 8.19 \pm 0.36 \%$$

$$\mu = 408.9 \pm 1.4$$

$$\sigma = 33.5 \pm 1.4$$

WF

$$\frac{\sigma}{\mu} = 8.28 \pm 0.35 \%$$

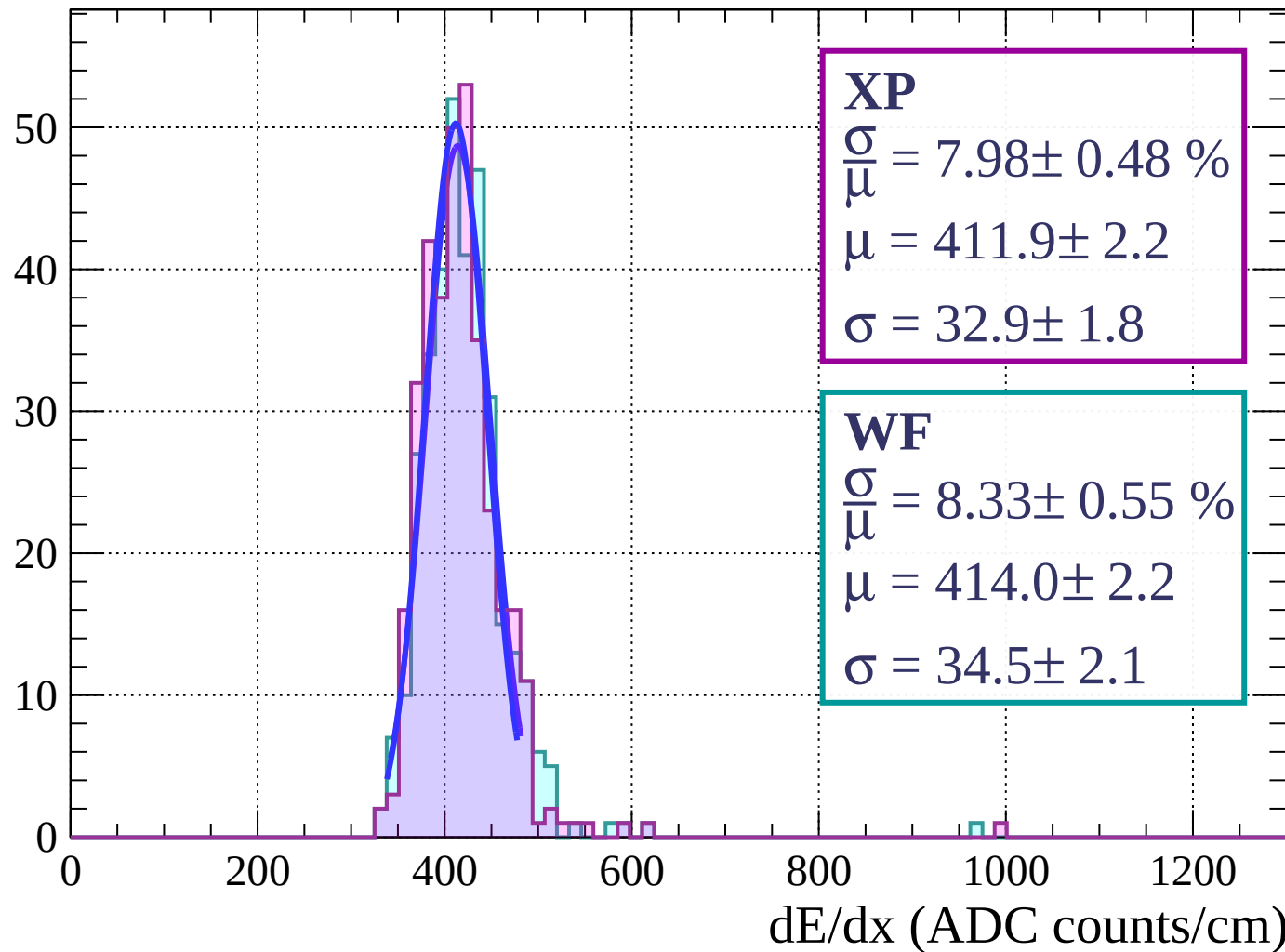
$$\mu = 412.8 \pm 1.5$$

$$\sigma = 34.2 \pm 1.3$$

dE/dx (ADC counts/cm)

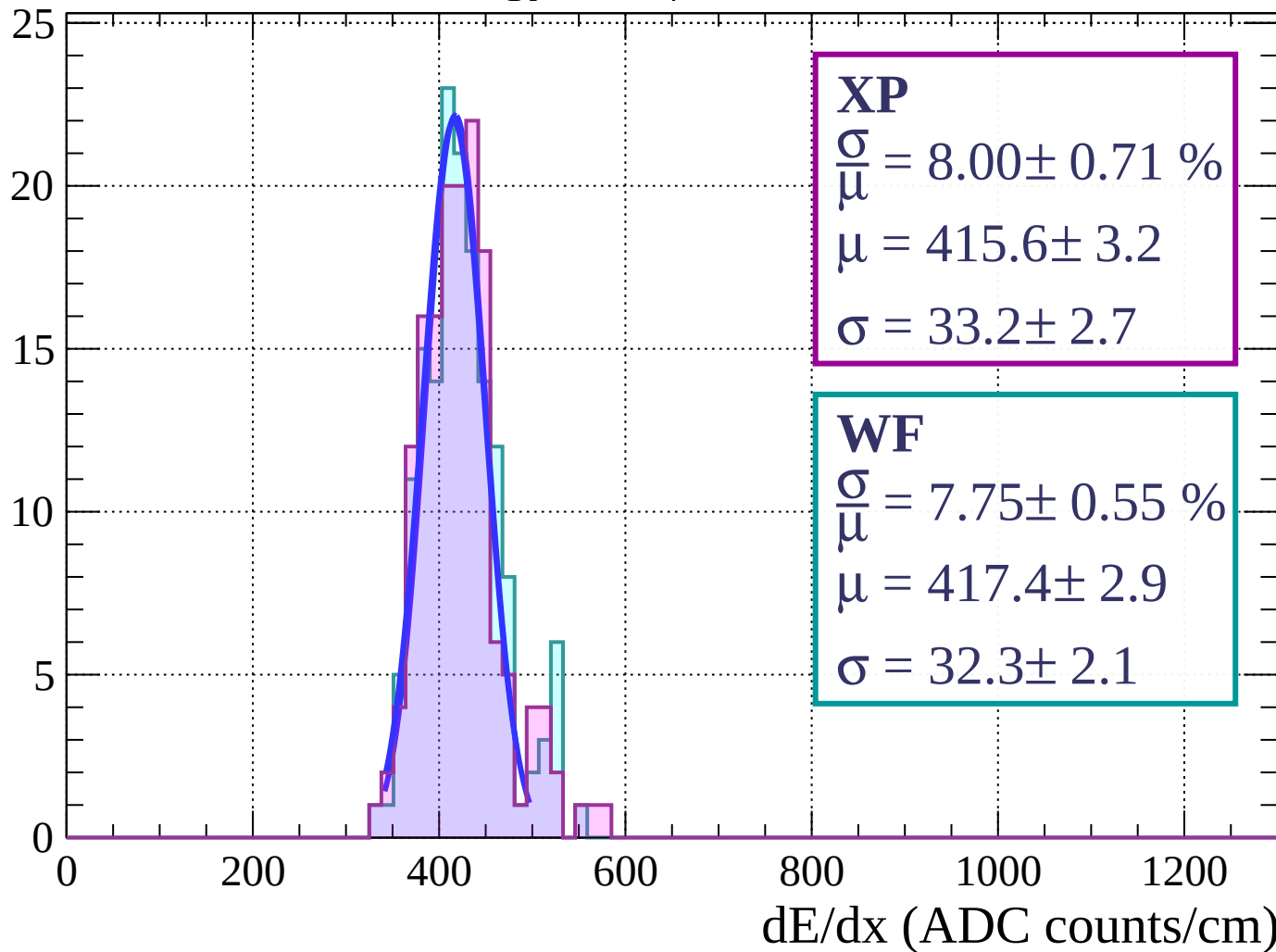
Energy loss | $36 < \theta < 42$

Count



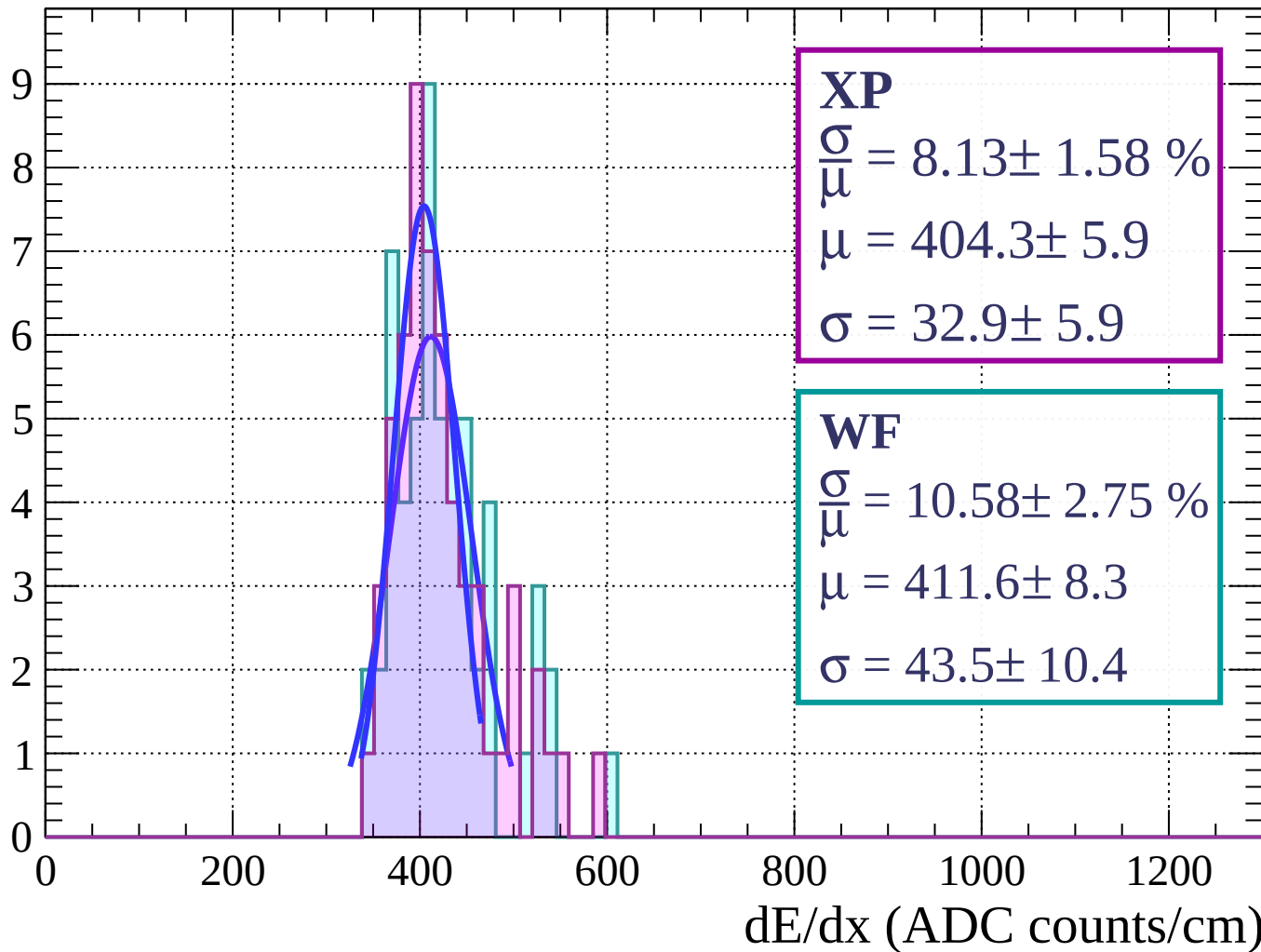
Energy loss | $42 < \theta < 48$

Count



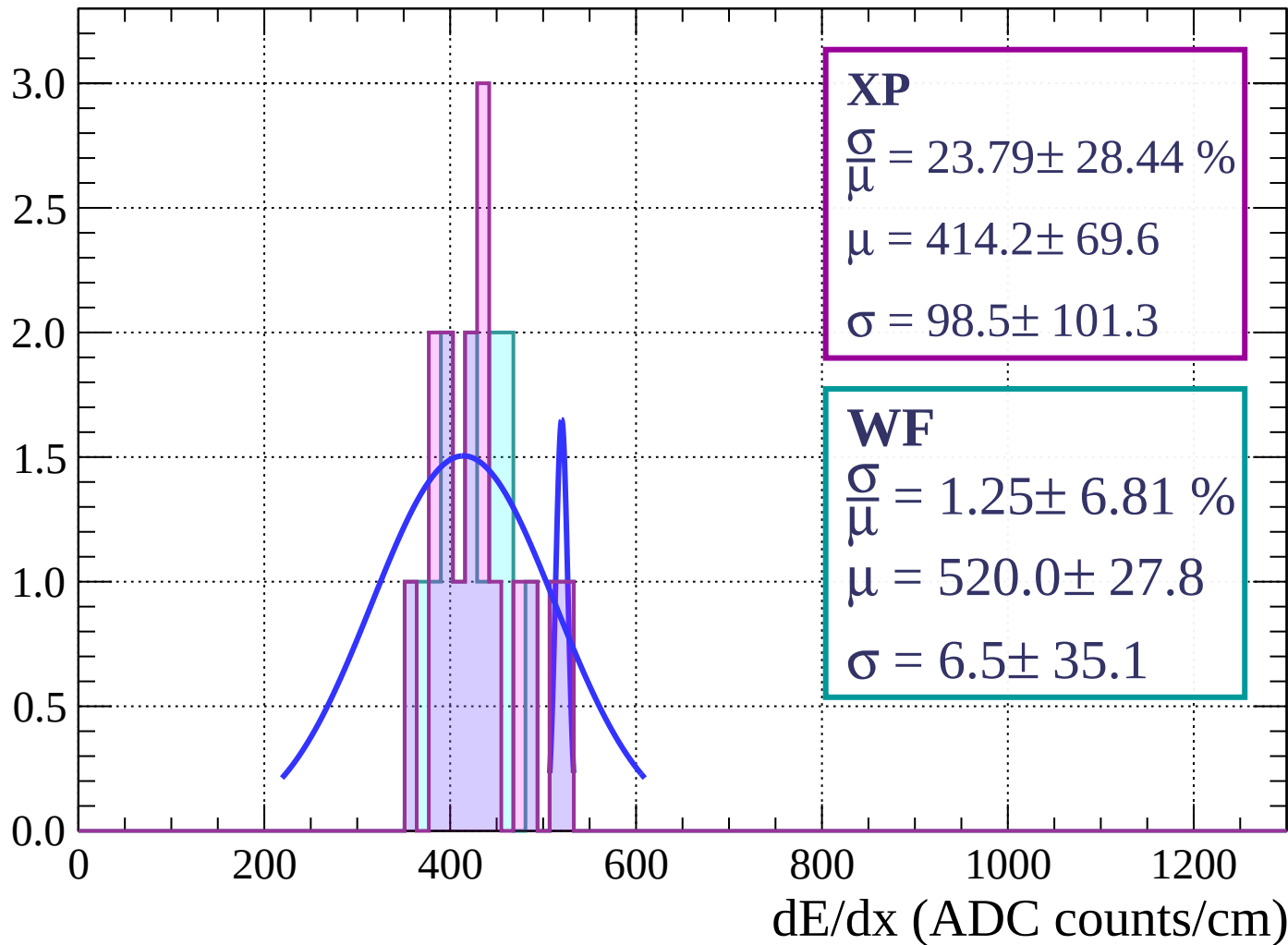
Energy loss | $48 < \theta < 54$

Count



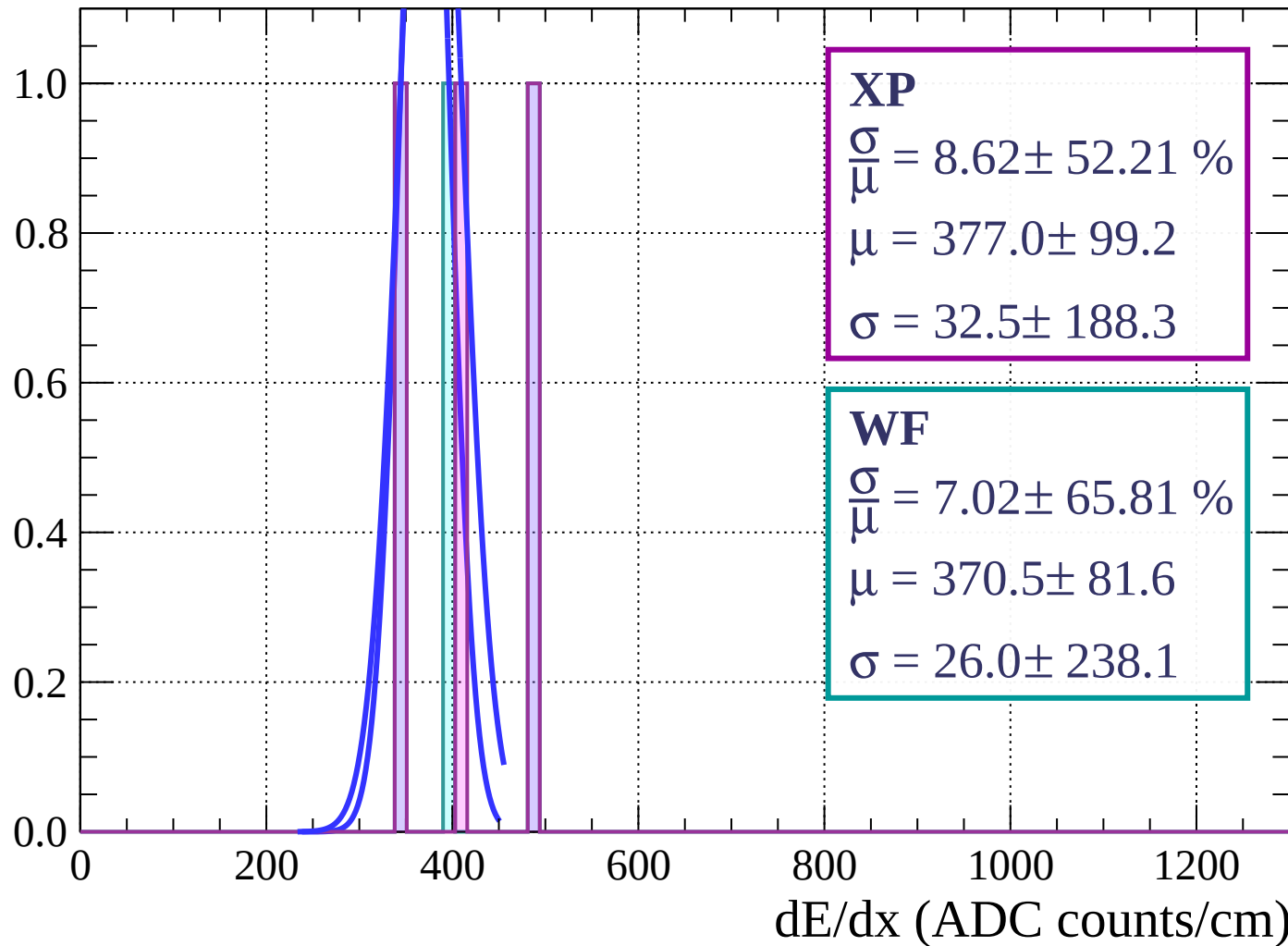
Energy loss | $54 < \theta < 60$

Count



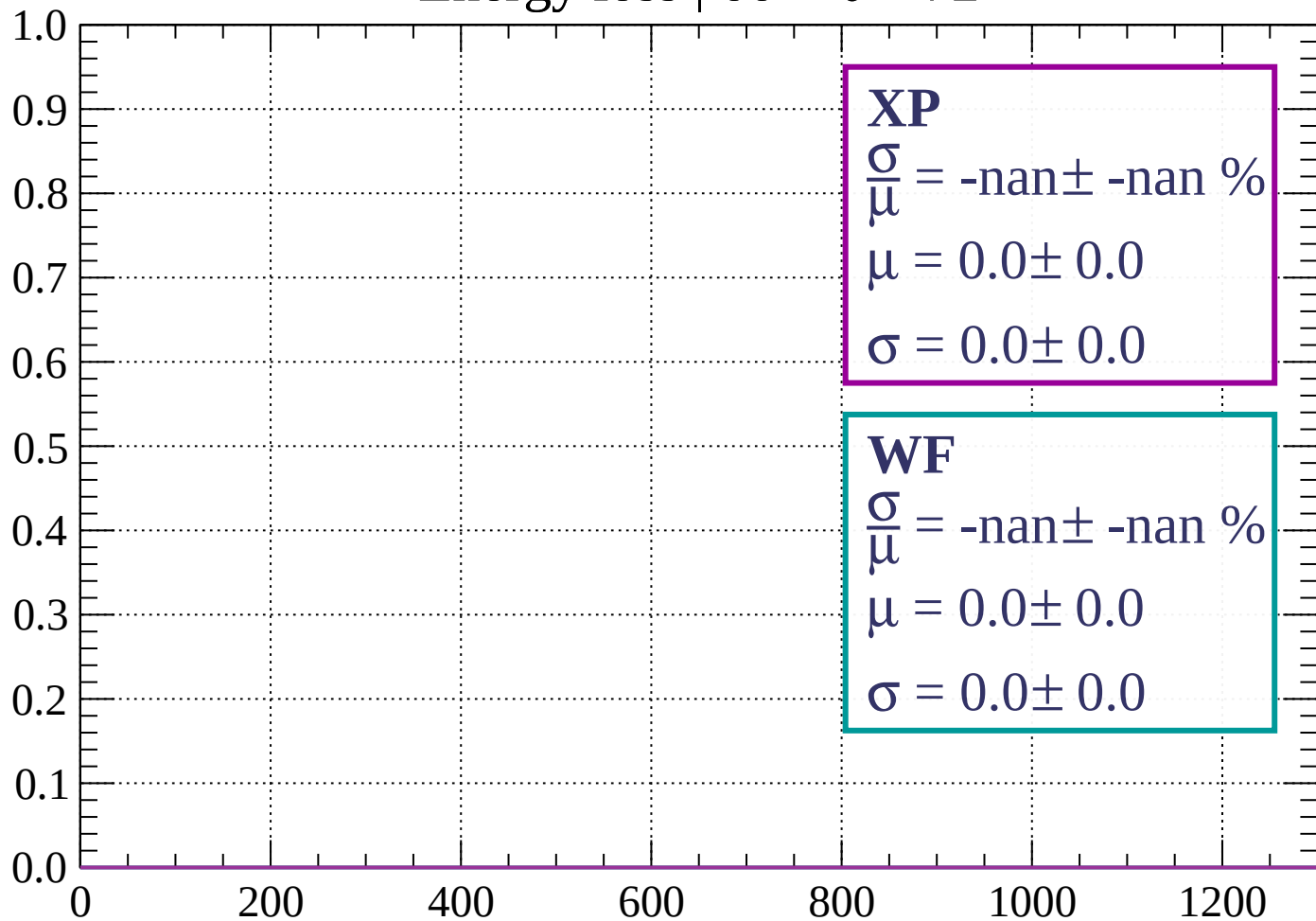
Energy loss | $60 < \theta < 66$

Count



Energy loss | $66 < \theta < 72$

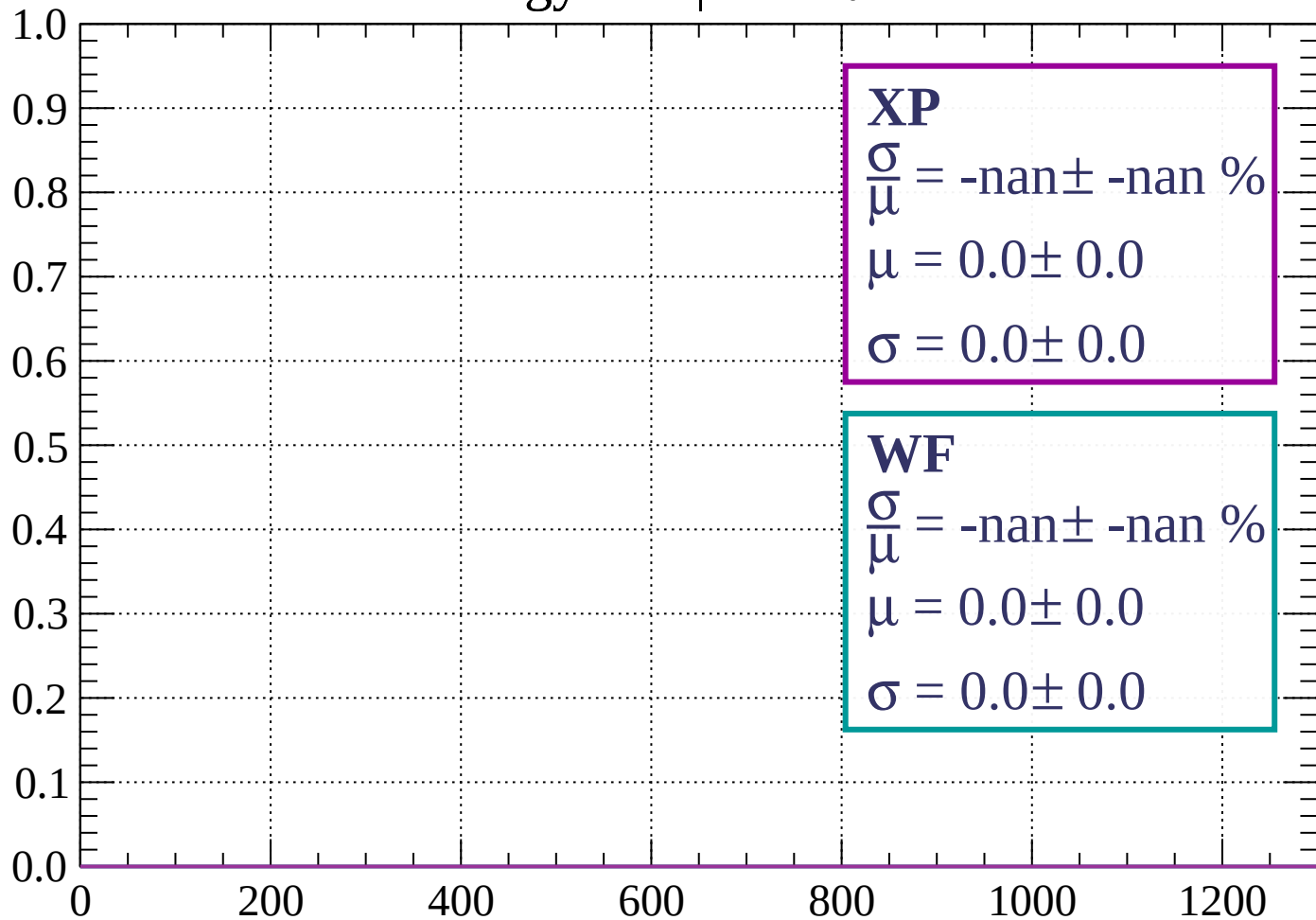
Count



dE/dx (ADC counts/cm)

Energy loss | $72 < \theta < 78$

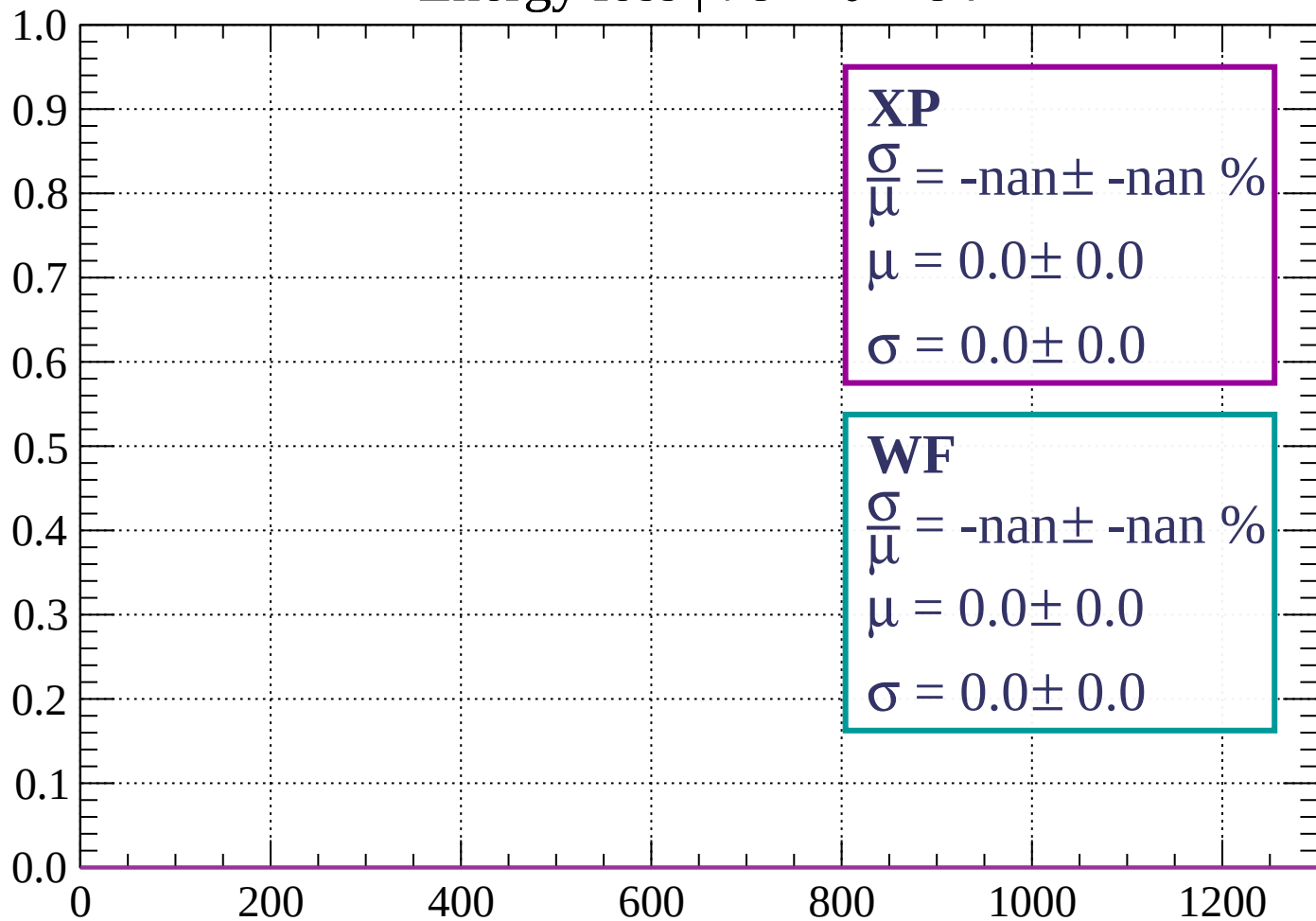
Count



dE/dx (ADC counts/cm)

Energy loss | $78 < \theta < 84$

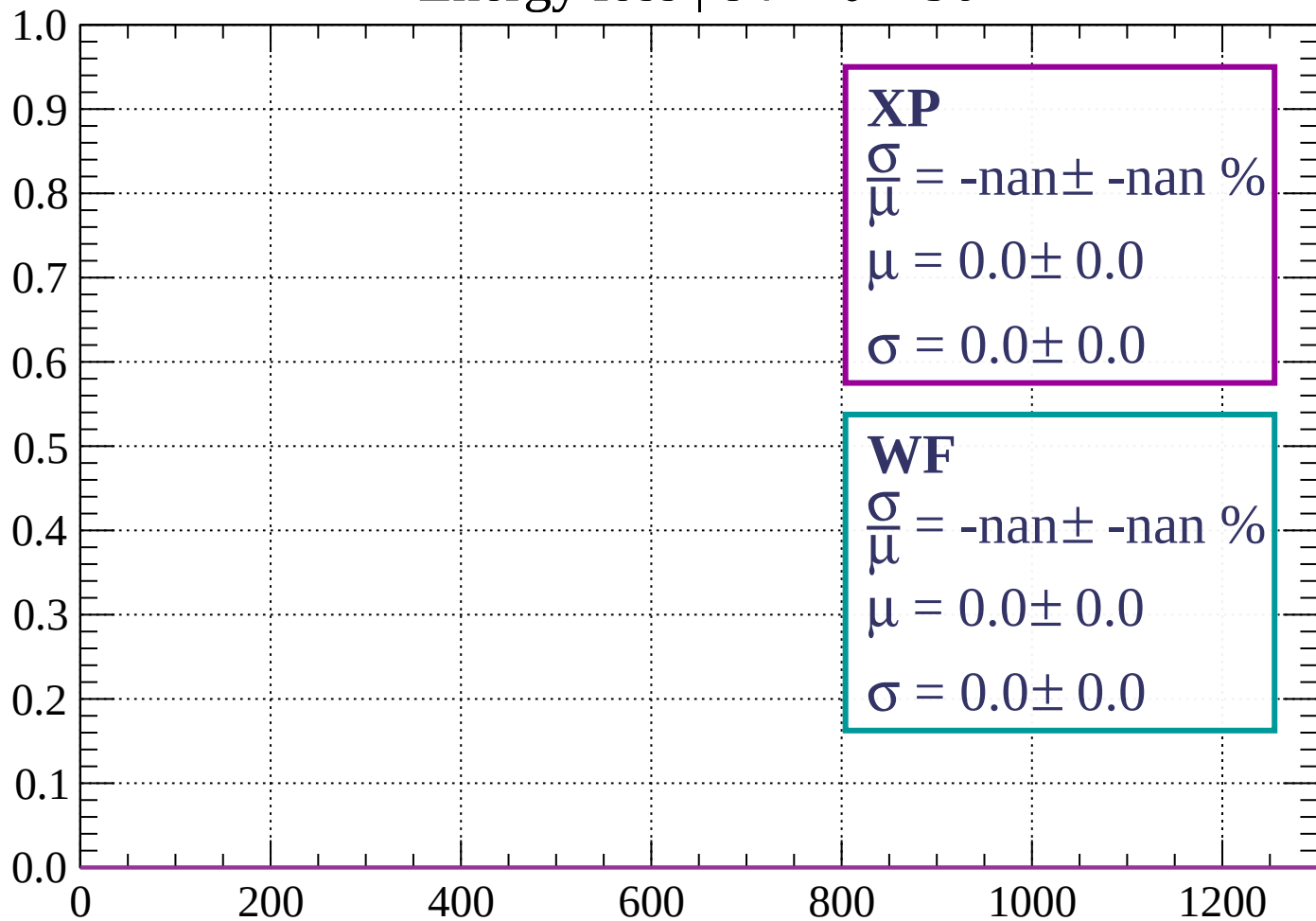
Count



dE/dx (ADC counts/cm)

Energy loss | $84 < \theta < 90$

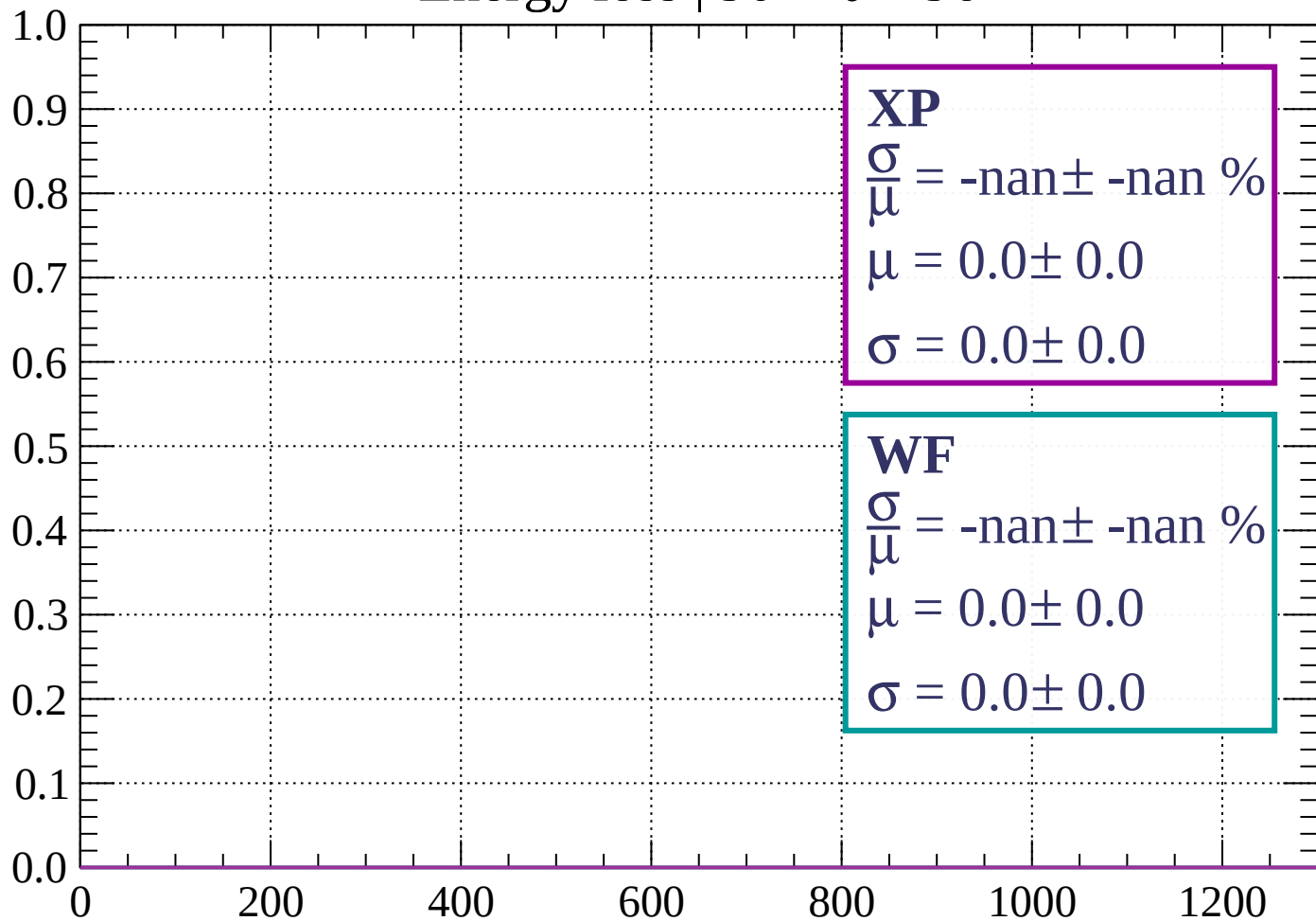
Count



dE/dx (ADC counts/cm)

Energy loss | $90 < \theta < 96$

Count



dE/dx (ADC counts/cm)

Count

